(12) INTERNATIONAL APPLICATION PUBLISHED UNDER THE PATENT COOPERATION TREATY (PCT)

(19) World Intellectual Property Organization
International Bureau

(43) International Publication Date
4 November 2004 (04.11.2004)

PCT

(10) International Publication Number
WO 2004/094623 A2

(51) International Patent Classification[7]: C12N

(21) International Application Number: PCT/US2004/008971

(22) International Filing Date: 22 March 2004 (22.03.2004)

(25) Filing Language: English

(26) Publication Language: English

(30) Priority Data:

| | | |
|---|---|---|
| 60/460,788 | 4 April 2003 (04.04.2003) | US |
| 60/470,697 | 14 May 2003 (14.05.2003) | US |
| 60/474,144 | 28 May 2003 (28.05.2003) | US |
| 60/483,286 | 27 June 2003 (27.06.2003) | US |
| 60/493,045 | 5 August 2003 (05.08.2003) | US |

(71) Applicant *(for all designated States except US)*: **INCYTE CORPORATION** [US/US]; 3160 Porter Drive, Palo Alto, CA 94304 (US).

(72) Inventors; and
(75) Inventors/Applicants *(for US only)*: **SWARNAKAR, Anita** [CA/US]; 8 Locksley Avenue #5D, San Francisco, CA 94122 (US). **BECHA, Shanya, D.** [US/US]; 1 Saint Francis #5508, San Francisco, CA 94107 (US). **TRAN, Uyen, K.** [US/US]; 2638 Mabury Square, San Jose, CA 95133 (US). **MARQUIS, Joseph, P.** [US/US]; 4428 Lazy Lane, San Jose, CA 95135 (US). **ELLIOTT, Vicki, S.** [US/US]; 3770 Polton Place Way, San Jose, CA 95121 (US). **RICHARDSON, Thomas, W.** [US/US]; 616 Canyon Road #107, Redwood City, CA 94062 (US). **CHAWLA, Narinder, K.** [US/US]; 33 Union Square, #712, Union City, CA 94587 (US). **WANG, Jonathan, T.** [US/US]; 173 Campbell Drive, Mountain View, CA 94043 (US). **FAVERO, Kristin, D.** [US/US]; 1660 Gordon Street #35, Redwood City, CA 94061 (US). **GERA, Mili** [US/US]; 4385 Crestwood Street, Fremont, CA 94538 (US). **JIANG, Xin** [US/US]; 14371 Elva Avenue, Saratoga, CA 95070 (US). **JACKSON, Alan, A.** [US/US]; 1541 Elwood Drive, Los Gatos, CA 95032 (US).

(74) Agent: **FOLEY & LARDNER LLP**; Washington Harbour, 3000 K Street, N.W., Suite 500, Washington, D.C. 20007-5143 (US).

(81) Designated States *(unless otherwise indicated, for every kind of national protection available)*: AE, AG, AL, AM, AT, AU, AZ, BA, BB, BG, BR, BW, BY, BZ, CA, CH, CN, CO, CR, CU, CZ, DE, DK, DM, DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, HR, HU, ID, IL, IN, IS, JP, KE, KG, KP, KR, KZ, LC, LK, LR, LS, LT, LU, LV, MA, MD, MG, MK, MN, MW, MX, MZ, NA, NI, NO, NZ, OM, PG, PH, PL, PT, RO, RU, SC, SD, SE, SG, SK, SL, SY, TJ, TM, TN, TR, TT, TZ, UA, UG, US, UZ, VC, VN, YU, ZA, ZM, ZW.

(84) Designated States *(unless otherwise indicated, for every kind of regional protection available)*: ARIPO (BW, GH, GM, KE, LS, MW, MZ, SD, SL, SZ, TZ, UG, ZM, ZW), Eurasian (AM, AZ, BY, KG, KZ, MD, RU, TJ, TM), European (AT, BE, BG, CH, CY, CZ, DE, DK, EE, ES, FI, FR, GB, GR, HU, IE, IT, LU, MC, NL, PL, PT, RO, SE, SI, SK, TR), OAPI (BF, BJ, CF, CG, CI, CM, GA, GN, GQ, GW, ML, MR, NE, SN, TD, TG).

**Published:**
— *without international search report and to be republished upon receipt of that report*

*For two-letter codes and other abbreviations, refer to the "Guidance Notes on Codes and Abbreviations" appearing at the beginning of each regular issue of the PCT Gazette.*

WO 2004/094623 A2

(54) **Title:** CELL ADHESION AND EXTRACELLULAR MATRIX PROTEINS

(57) **Abstract:** Various embodiments of the invention provide human cell adhesion and extracellularmatrix proteins (CADECM) and polynucleotides which identify and encode CADECM. Embodiments of the invention also provide expression vectors, host cells, antibodies, agonists, and antagonists. Other embodiments provide methods for diagnosing, treating, or preventing disorders associated with aberrant expression of CADECM.

# CELL ADHESION AND EXTRACELLULAR MATRIX PROTEINS

## TECHNICAL FIELD

The invention relates to novel nucleic acids, cell adhesion and extracellular matrix proteins encoded by these nucleic acids, and to the use of these nucleic acids and proteins in the diagnosis, treatment, and prevention of immune system disorders, neurological disorders, developmental disorders, connective tissue disorders, and cell proliferative disorders, including cancer. The invention also relates to the assessment of the effects of exogenous compounds on the expression of nucleic acids and cell adhesion and extracellular matrix proteins.

## BACKGROUND OF THE INVENTION

Cell Adhesion Proteins

The surface of a cell is rich in transmembrane proteoglycans, glycoproteins, glycolipids, and receptors. These macromolecules mediate adhesion with other cells and with components of the ECM. The interaction of the cell with its surroundings profoundly influences cell shape, strength, flexibility, motility, and adhesion. These dynamic properties are intimately associated with signal transduction pathways controlling cell proliferation and differentiation, tissue construction, and embryonic development. Families of cell adhesion molecules include the cadherins, integrins, lectins, neural cell adhesion proteins, and some members of the proline-rich proteins.

Cadherins comprise a family of calcium-dependent glycoproteins that function in mediating cell-cell adhesion in virtually all solid tissues of multicellular organisms. These proteins share multiple repeats of a cadherin-specific motif, and the repeats form the folding units of the cadherin extracellular domain. Cadherin molecules cooperate to form focal contacts, or adhesion plaques, between adjacent epithelial cells. The cadherin family includes the classical cadherins and protocadherins. Classical cadherins include the E-cadherin, N-cadherin, and P-cadherin subfamilies. E-cadherin is present on many types of epithelial cells and is especially important for embryonic development. N-cadherin is present on nerve, muscle, and lens cells and is also critical for embryonic development. P-cadherin is present on cells of the placenta and epidermis. Recent studies report that protocadherins are involved in a variety of cell-cell interactions (Suzuki, S.T. (1996) J. Cell Sci. 109:2609-2611). The intracellular anchorage of cadherins is regulated by their dynamic association with catenins, a family of cytoplasmic signal transduction proteins associated with the actin cytoskeleton. The anchorage of cadherins to the actin cytoskeleton appears to be regulated by protein tyrosine phosphorylation, and the cadherins are the target of phosphorylation-induced junctional disassembly (Aberle, H. et al. (1996) J. Cell. Biochem. 61:514-523).

Integrins are ubiquitous transmembrane adhesion molecules that link the ECM to the internal cytoskeleton. Integrins are composed of two noncovalently associated transmembrane glycoprotein subunits called α and β. At least 8 different ß subunits (ß1-ß8) and at least 12 different α subunits have been identified (α1-α8, αL, αM, αX, and αIIb). Individual α subunits are capable of associating with different ß subunits, suggesting a possible mechanism for specifying integrin function and ligand binding affinity. Members of the ß subunit family are generally of 90-110 kilodaltons (kD) in molecular weight and share about 40-48% amino acid sequence homology. About 56 cysteines distributed among four repeating units are also conserved. Some variation in these conserved features is observed among some of the more divergent ß subunit family members. Members of the α subunit family are generally 150-200 kilodaltons in molecular weight and are not as well conserved as the ß subunit family. All contain seven repeating domains of 24-45 amino acids spaced about 20-35 amino acids apart. The N-termini each contain 3-4 divalent cation binding sites. (For review, see Pigott, R. and C. Power (1994) The Adhesion Molecule Facts Book, Academic Press, San Diego, CA, pp. 9-12.)

Integrins function as receptors that specifically recognize and bind to ECM proteins such as fibronectin, fibrinogen, laminin, thrombospondin, vitronectin, von Willebrand factor, and collagen. Some integrins recognize a specific motif, the RGD sequence, at the C-termini of the ECM proteins they bind. Integrins also bind to immunoglobulin superfamily proteins such as ICAM-1, -2, and -3 and VCAM-1.

Hemopexin is a serum glycoprotein that binds heme and transports it to the liver for breakdown and iron recovery, after which the free hemopexin returns to the circulation. Structurally hemopexin consists of two similar halves of approximately two hundred amino acid residues connected by a histidine-rich hinge region. Hemopexin-like domains have been found in other types of proteins, including vitronectin (Hunt, L.T. et al. (1987) Protein Seq. Data Anal. 1:21-26; Stanley, K.K. (1986) FEBS Lett. 199:249-253). Vitronectin is a cell adhesion and spreading factor found in plasma and other tissues. Like hemopexin, vitronectin has two hemopexin-like domains. Somatomedin B is a cysteine-rich serum factor, which is proteolytically derived from the N-terminal extremity of vitronectin.

Most integrins have been shown to activate focal adhesion kinase (FAK), a protein tyrosine kinase that is linked to Ras signaling pathways that modify the cytoskeleton and stimulate the mitogen-activated protein kinase (MAPK) cascade (Hanks, S.K. and T.R. Polte (1997) BioEssays 19:137-145). Integrins can also influence growth factor signaling through direct interaction with growth factor receptor tyrosine kinases (RTKs) (Miyamoto, S. et al. (1996) J. Cell Biol. 135:1633-1642). Integrins have also been shown to play a vital role in "anoikis," a term describing programmed cell death caused by loss of cell anchorage (Frisch, S.M. and E. Ruoslahti (1997) Curr. Opin. Cell Biol. 9:701-706).

A number of diseases have been attributed to integrin defects. (See Pigott and Power, *supra*). For example, leukocyte adhesion deficiency (LAD) is an inherited disorder characterized by the impaired migration of neutrophils to sites of extravascular inflammation. LAD is caused by abnormal splicing of and a missense mutation in the RNA encoding the ß2 subunit. Additionally, defects in platelet integrin are correlated with Glanzmann's thrombasthemia, a bleeding disorder characterized by insufficient platelet aggregation.

Lectins comprise a ubiquitous family of extracellular glycoproteins which bind cell surface carbohydrates specifically and reversibly, resulting in the agglutination of cells (reviewed in Drickamer, K. and M.E. Taylor (1993) Annu. Rev. Cell Biol. 9:237-264). This function is particularly important for activation of the immune response. Lectins mediate the agglutination and mitogenic stimulation of lymphocytes at sites of inflammation (Lasky, L.A. (1991) J. Cell. Biochem. 45:139-146; Paietta, E. et al. (1989) J. Immunol. 143:2850-2857).

Lectins are further classified into subfamilies based on carbohydrate-binding specificity and other criteria. The galectin subfamily, in particular, includes lectins that bind β-galactoside carbohydrate moieties in a thiol-dependent manner (reviewed in Hadari, Y.R. et al. (1998) J. Biol. Chem. 270:3447-3453). Galectins are widely expressed and developmentally regulated. Galectins contain a characteristic carbohydrate recognition domain (CRD). The CRD comprises about 140 amino acids and contains several stretches of about 1 - 10 amino acids which are highly conserved among all galectins. A particular 6-amino acid motif within the CRD contains conserved tryptophan and arginine residues which are critical for carbohydrate binding. The CRD of some galectins also contains cysteine residues which may be important for disulfide bond formation. Secondary structure predictions indicate that the CRD forms several β-sheets.

Galectins play a number of roles in diseases and conditions associated with cell-cell and cell-matrix interactions. For example, certain galectins associate with sites of inflammation and bind to cell surface immunoglobulin E molecules. In addition, galectins may play an important role in cancer metastasis. Galectin overexpression is correlated with the metastatic potential of cancers in humans and mice. Moreover, anti-galectin antibodies inhibit processes associated with cell transformation, such as cell aggregation and anchorage-independent growth (see, for example, Su, Z.-Z. et al. (1996) Proc. Natl. Acad. Sci. USA 93:7252-7257).

Selectins, or LEC-CAMs, comprise a specialized lectin subfamily involved primarily in inflammation and leukocyte adhesion (Reviewed in Lasky, *supra*). Selectins mediate the recruitment of leukocytes from the circulation to sites of acute inflammation and are expressed on the surface of vascular endothelial cells in response to cytokine signaling. Selectins bind to specific ligands on the leukocyte cell membrane and enable the leukocyte to adhere to and migrate along the endothelial surface. Binding of selectin to its ligand leads to polarized rearrangement of the actin cytoskeleton and stimulates signal transduction within the leukocyte (Brenner, B. et al. (1997) Biochem. Biophys. Res. Commun. 231:802-807; Hidari, K. I. et al. (1997) J. Biol. Chem. 272:28750-28756). Members of the selectin family possess three characteristic motifs: a lectin or carbohydrate recognition domain; an epidermal growth factor-like domain; and a variable number of short consensus repeats (scr or "sushi" repeats) which are also present in complement regulatory proteins.

Neural cell adhesion proteins (NCAPs) play roles in the establishment of neural networks during development and regeneration of the nervous system (Uyemura, K. et al. (1996) Essays Biochem. 31:37-48; Brummendorf, T., and F.G. Rathjen (1996) Curr. Opin. Neurobiol. 6:584-593). NCAP participates in neuronal cell migration, cell adhesion, neurite outgrowth, axonal fasciculation, pathfinding, synaptic target-recognition, synaptic formation, myelination and regeneration. NCAPs are expressed on the surfaces of neurons associated with learning and memory. Mutations in genes encoding NCAPS are linked with neurological diseases, including hereditary neuropathy, Charcot-Marie-Tooth disease, Dejerine-Sottas disease, X-linked hydrocephalus, MASA syndrome (mental retardation, aphasia, shuffling gait and adducted thumbs), and spastic paraplegia type I. In some cases, expression of NCAP is not restricted to the nervous system. L1, for example, is expressed in melanoma cells and hematopoietic tumor cells where it is implicated in cell spreading and migration, and may play a role in tumor progression (Montgomery, A.M. et al. (1996) J. Cell Biol. 132:475-485).

NCAPs have at least one immunoglobulin constant or variable domain (Uyemura et al., *supra*). They are generally linked to the plasma membrane through a transmembrane domain and/or a glycosyl-phosphatidylinositol (GPI) anchor. The GPI linkage can be cleaved by GPI phospholipase C. Most NCAPs consist of an extracellular region made up of one or more immunoglobulin domains, a membrane spanning domain, and an intracellular region. Many NCAPs contain post-translational modifications including covalently attached oligosaccharide, glucuronic acid, and sulfate. NCAPs fall into three subgroups: simple-type, complex-type, and mixed-type. Simple-type NCAPs contain one or more variable or constant immunoglobulin domains, but lack other types of domains. Members of the simple-type subgroup include Schwann cell myelin protein (SMP), limbic system-associated membrane protein (LAMP), opiate-binding cell-adhesion molecule (OBCAM), and myelin-associated glycoprotein (MAG). The complex-type NCAPs contain fibronectin type III domains in addition to the immunoglobulin domains. The complex-type subgroup includes neural cell-adhesion molecule (NCAM), axonin-1, F11, Bravo, and L1. Mixed-type NCAPs contain a combination of immunoglobulin domains and other motifs such as tyrosine kinase and epidermal growth factor-like domains. This subgroup includes Trk receptors of nerve growth factors such as nerve growth factor (NGF) and neurotropin 4 (NT4), Neu differentiation factors such as glial growth factor II (GGFII) and acetylcholine receptor-inducing factor (ARIA), and the semaphorin/collapsin family such as semaphorin B and collapsin.

Semaphorins are a large group of axonal guidance molecules consisting of at least 30 different members and are found in vertebrates, invertebrates, and even certain viruses. All semaphorins contain the sema domain which is approximately 500 amino acids in length. Neuropilin, a semaphorin receptor, has been shown to promote neurite outgrowth *in vitro*. The extracellular region of neuropilins consists of three different domains: CUB, discoidin, and MAM domains. The CUB and the MAM motifs of neuropilin have been proposed to have roles in protein-protein interactions and are suggested to be involved in the binding of semaphorins through the sema and the C-terminal domains (reviewed in Raper, J.A. (2000) Curr. Opin. Neurobiol. 10:88-94).

An NCAP subfamily, the NCAP-LON subgroup, includes cell adhesion proteins expressed on distinct subpopulations of brain neurons. Members of the NCAP-LON subgroup possess three immunoglobulin domains and bind to cell membranes through GPI anchors. Kilon (a kindred of NCAP-LON), for example, is expressed in the brain cerebral cortex and hippocampus (Funatsu, N. et al. (1999) J. Biol. Chem. 274:8224-8230). Immunostaining localizes Kilon to the dendrites and soma of pyramidal neurons. Kilon has three C2 type immunoglobulin-like domains, six predicted glycosylation sites, and a GPI anchor. Expression of Kilon is developmentally regulated. It is expressed at higher levels in adult brain in comparison to embryonic and early postnatal brains. Confocal microscopy shows the presence of Kilon in dendrites of hypothalamic magnocellular neurons secreting neuropeptides, oxytocin or arginine vasopressin (Miyata, S. et al. (2000) J. Comp. Neurol. 424:74-85). Arginine vasopressin regulates body fluid homeostasis, extracellular osmolarity and intravascular volume. Oxytocin induces contractions of uterine smooth muscle during child birth and of myoepithelial cells in mammary glands during lactation. In magnocellular neurons, Kilon is proposed to play roles in the reorganization of dendritic connections during neuropeptide secretion.

The co-ordinated function of effector and accessory cells in the immune system is assisted by adhesion molecules on the cell surface that stabilize interactions between different cell types. Leukocyte function-associated antigen 1 (LFA-1) is expressed on the surface of all white blood cells and is a receptor for intercellular adhesion molecules (ICAM) 1 and 2 which are members of the immunoglobulin superfamily. The interaction of LFA-1 with ICAMs 1 and 2 provides essential accessory adhesion signals in many immune interactions, including those between T and B lymphocytes and cytotoxic T cells and their targets. In addition, both ICAMs are expressed at low levels on resting vascular endothelium. ICAM-1 is strongly upregulated by cytokine stimulation and plays a key role in the arrest of leukocytes in blood vessels at sites of inflammation and injury. A third ligand for LFA-1 expressed in resting leukocytes is ICAM-3. ICAM-3 is closely related to ICAM-1 and is constitutively expressed on all leukocytes. It consists of five immunoglobulin domains and binds LFA-1 through its two N-terminal domains (Fawcett, J. et al. (1992) Nature

360:481-484).

Cell adhesion proteins also include some members of the proline-rich proteins (PRPs). PRPs are defined by a high frequency of proline, ranging from 20-50% of the total amino acid content. Some PRPs have short domains which are rich in proline. These proline-rich regions are associated with protein-protein interactions. One family of PRPs are the proline-rich synapse-associated proteins (ProSAPs) which have been shown to bind to members of the postsynaptic density (PSD) protein family and subtypes of the somatostatin receptor (Yao, I. et al. (1999) J. Biol. Chem. 274: 27463-27466; Zitzer, H. et al. (1999) J. Biol. Chem. 274:32997-33001). Members of the ProSAP family contain six to seven ankyrin repeats at the N-terminus, followed by an SH3 domain, a PDZ domain, and seven proline-rich regions and a SAM domain at the C terminus. Several groups of ProSAPs are important structural constituents of synaptic structures in human brain (Zitzer et al., *supra*). Another member of the PRP family is the HLA-B-associated transcript 2 protein (BAT2) which is rich in proline and includes short tracts of polyproline, polyglycine, and charged amino acids. BAT2 also contains four RGD (Arg-Gly-Asp) motifs typical of integrins (Banerji, J. et al. (1990) Proc. Natl. Acad. Sci. USA 87:2374-2378).

Toposome is a cell-adhesion glycoprotein isolated from mesenchyme-blastula embryos. Toposome precursors including vitellogenin promote cell adhesion of dissociated blastula cells.

There are additional specific domains characteristic of cell adhesion proteins. One such domain is the MAM domain, a domain of about 170 amino acids found in the extracellular region of diverse proteins. These proteins all share a receptor-like architecture comprising a signal peptide, followed by a large N-terminal extracellular domain, a transmembrane region, and an intracellular domain (PROSITE document PDOC00604 MAM domain signature and profile). MAM domain proteins include zonadhesin, a sperm-specific membrane protein that binds to the zona pellucida of the egg; neuropilin, a cell adhesion molecule that functions during the formation of certain neuronal circuits, and *Xenopus laevis* thyroid hormone induced protein B, which contains four MAM domains and is involved in metamorphosis (Brown, D.D. et al. (1996) Proc. Natl. Acad. Sci. USA 93:1924-1929).

The WSC domain was originally found in the yeast WSC (cell-wall integrity and stress response component) proteins which act as sensors of environmental stress. The WSC domains are extracellular and are thought to possess a carbohydrate binding role (Ponting, C.P. et al. (1999) Curr. Biol. 9:S1-S2). A WSC domain has recently been identified in polycystin-1, a human plasma membrane protein. Mutations in polycystin-1 are the cause of the commonest form of autosomal dominant polycystic kidney disease (Ponting, C.P. et al. (1999) Curr. Biol. 9:R585-R588).

Leucine rich repeats (LRR) are short motifs found in numerous proteins from a wide range of species. LRR motifs are of variable length, most commonly 20-29 amino acids, and multiple repeats are typically present in tandem. LRR motifs are important for protein/protein interactions and cell adhesion, and LRR proteins are involved in cell/cell interactions, morphogenesis, and development (Kobe, B. and J. Deisenhofer (1995) Curr. Opin. Struct. Biol. 5:409-416). The human ISLR (immunoglobulin superfamily containing leucine-rich repeat) protein contains a C2-type immunoglobulin domain as well as LRR motifs. The ISLR gene is linked to the critical region for Bardet-Biedl syndrome, a developmental disorder of which the most common feature is retinal dystrophy (Nagasawa, A. et al. (1999) Genomics 61:37-43).

The sterile alpha motif (SAM) domain is a conserved protein binding domain, approximately 70 amino acids in length, and is involved in the regulation of many developmental processes in eukaryotes. The SAM domain can potentially function as a protein interaction module through its ability to form homo- or hetero-oligomers with other SAM domains (Schultz, J. et al. (1997) Protein Sci. 6:249-253).

Vinculin is a cellular adhesion molecule that is involved in the attachment of actin microfilaments to the plasma membrane of eukaryotic cells. This protein is composed of approximately 1000 amino acid residues and is characterized by an acidic N-terminal domain consisting of either two (in *C. elegans*) or three (in vertebrates) repeats of a 110 amino acid region. A proline-rich region is followed by a basic C-terminal domain. Two signature patterns are found in the N-terminal domain, one which seems to be involved in protein-protein interactions and one based on the repeated region (PROSITE document PDOC00568 Vinculin family signatures).

Synapsins are a family of proteins that coat synaptic vesicles and bind to actin filaments as well as other components of the cytoskeleton. Synapsins I and II each exist as two alternately spliced variants termed IA and IB or IIA and IIB and differ from each other in their C-termini. Two conserved domains within these proteins are an octapeptide containing a phosphorylated serine residue and a second stretch of 11 highly conserved residues (PROSITE document PDOC00345 Synapsin signatures).

Osteonectin domain signatures are derived from three extracellular proteins (SPARC or osteonectin, SC1, and QR1) which contain a region of 240 highly-conserved amino acid residues in their C-termini. Two signature patterns were developed based on this conserved region, one based on a cysteine-rich region and the other based on a stretch of 11 highly conserved residues (PROSITE document PDOC00535 Osteonectin domain signatures).

Extracellular Matrix Proteins

The extracellular matrix (ECM) is a complex network of glycoproteins, polysaccharides, proteoglycans, and other macromolecules that are secreted from the cell into the extracellular space. The ECM remains in close association with the cell surface and provides a supportive meshwork that profoundly influences cell shape, motility, strength, flexibility, and adhesion. In fact, adhesion of a cell to its surrounding matrix is required for cell survival except in the case of metastatic tumor cells, which have overcome the need for cell-ECM anchorage. This phenomenon suggests that the ECM plays a critical role in the molecular mechanisms of growth control and metastasis. (Reviewed in Ruoslahti, E. (1996) Sci. Am. 275:72-77.) Furthermore, the ECM determines the structure and physical properties of connective tissue and is particularly important for morphogenesis and other processes associated with embryonic development and pattern formation.

The collagens comprise a family of ECM proteins that provide structure to bone, teeth, skin, ligaments, tendons, cartilage, blood vessels, and basement membranes. Multiple collagen proteins have been identified. Three collagen molecules fold together in a triple helix stabilized by interchain disulfide bonds. Bundles of these triple helices then associate to form fibrils. Collagen primary structure consists of hundreds of (Gly-X-Y) repeats where about a third of the X and Y residues are Pro. Glycines are crucial to helix formation as the bulkier amino acid sidechains cannot fold into the triple helical conformation. Because of these strict sequence requirements, mutations in collagen genes have severe consequences. Osteogenesis imperfecta patients have brittle bones that fracture easily; in severe cases patients die *in utero* or at birth. Ehlers-Danlos syndrome patients have hyperelastic skin, hypermobile joints, and susceptibility to aortic and intestinal rupture. Chondrodysplasia patients have short stature and ocular disorders. Alport syndrome patients have hematuria, sensorineural deafness, and eye lens deformation. (Isselbacher, K.J. et al. (1994) Harrison's Principles of Internal Medicine, McGraw-Hill, Inc., New York, NY, pp. 2105-2117; and Creighton, T.E. (1984) Proteins, Structures and Molecular Principles, W.H. Freeman and Company, New York, NY, pp. 191-197.)

Elastin and related proteins confer elasticity to tissues such as skin, blood vessels, and lungs. Elastin is a highly hydrophobic protein of about 750 amino acids that is rich in proline and glycine residues. Elastin molecules are highly cross-linked, forming an extensive extracellular network of fibers and sheets. Elastin fibers are surrounded by a sheath of microfibrils which are composed of a number of glycoproteins, including fibrillin. Mutations in the gene encoding fibrillin are responsible for Marfan's syndrome, a genetic disorder characterized by defects in connective tissue. In severe cases, the aortas of afflicted individuals are prone to rupture. (Reviewed in Alberts, B. et al. (1994) Molecular Biology of the Cell, Garland Publishing, New York, NY, pp. 984-986.) The fibulin proteins connect elastic fibers and are thought to promote the formation and stabilization of the fibers. Members of the fibulin family contain epidermal growth factor-like motifs as well as an RGD cell attachment sequence (Midwood, K.S. and J.E. Schwarzbauer (2002) Current Biology 12:R279-R281).

Fibronectin is a large ECM glycoprotein found in all vertebrates. Fibronectin exists as a dimer of two subunits, each containing about 2,500 amino acids. Each subunit folds into a rod-like structure containing multiple domains. The domains each contain multiple repeated modules, the most common of which is the type III fibronectin repeat. The type III fibronectin repeat is about 90 amino acids in length and is also found in other ECM proteins and in some plasma membrane and cytoplasmic proteins. Furthermore, some type III fibronectin repeats contain a characteristic tripeptide consisting of Arginine-Glycine-Aspartic acid (RGD). The RGD sequence is recognized by the integrin family of cell surface receptors and is also found in other ECM proteins. Disruption of both copies of the gene encoding fibronectin causes early embryonic lethality in mice. The mutant embryos display extensive morphological defects, including defects in the formation of the notochord, somites, heart, blood vessels, neural tube, and extraembryonic structures. (Reviewed in Alberts et al., *supra*, pp. 986-987.)

Laminin is a major glycoprotein component of the basal lamina which underlies and supports epithelial cell sheets. Laminin is one of the first ECM proteins synthesized in the developing embryo. Laminin is an 850 kilodalton protein composed of three polypeptide chains joined in the shape of a cross by disulfide bonds. Laminin is especially important for angiogenesis and, in particular, for guiding the formation of capillaries. (Reviewed in Alberts et al., *supra*, pp. 990-991.)

Many proteinaceous ECM components are proteoglycans. Proteoglycans are composed of unbranched polysaccharide chains (glycosaminoglycans) attached to protein cores. Common proteoglycans include aggrecan, betaglycan, decorin, perlecan, serglycin, and syndecan-1. Some of these molecules not only provide mechanical support, but also bind to extracellular signaling molecules, such as fibroblast growth factor and transforming growth factor β, suggesting a role for proteoglycans in cell-cell communication. (Reviewed in Alberts et al., *supra*, pp. 973-978.) Likewise, the glycoproteins tenascin-C and tenascin-R are expressed in developing and lesioned neural tissue and provide stimulatory and anti-adhesive (inhibitory) properties, respectively, for axonal growth (Faissner, A. (1997) Cell Tissue Res. 290:331-341).

Dentin phosphoryn (DPP) is a major component of the dentin ECM. DPP is a proteoglycan that is synthesized and expressed by odontoblasts (Gu, K. et al. (1998) Eur. J. Oral Sci. 106:1043-1047). DPP is believed to nucleate or modulate the formation of hydroxyapatite crystals.

Amelogenin is an extracellular matrix protein that plays a role in the biomineralization of tooth enamel. This protein participates in the regulation of crystallite formation during tooth enamel development and thus is thought to play a major role in the structural organization and mineralization of developing tooth enamel (Li, W. et al. (2001) Matrix Biol. 19:755-60).

Mucins are highly glycosylated glycoproteins that are the major structural component of the mucus gel. The physiological functions of mucins are cytoprotection, mechanical protection, maintenance of viscosity in secretions, and cellular recognition. MUC6 is a human gastric mucin that is also found in gall bladder, pancreas, seminal vesicles, and female reproductive tract (Toribara, N.W. et al. (1997) J. Biol. Chem. 272:16398-16403). The MUC6 gene has been mapped to human chromosome 11 (Toribara, N.W. et al. (1993) J. Biol. Chem. 268:5879-5885). Hemomucin is a novel *Drosophila* surface mucin that may be involved in the induction of antibacterial effector molecules (Theopold, U. et al. (1996) J. Biol. Chem. 217:12708-12715).

Olfactomedin was originally identified as the major component of the mucus layer surrounding the chemosensory dendrites of olfactory neurons. Olfactomedin-related proteins are secreted glycoproteins with conserved C-terminal motifs. The TIGR/myocilin protein, an olfactomedin-related protein expressed in the eye, is associated with the pathogenesis of glaucoma (Kulkarni, N.H. et al. (2000) Genet. Res. 76:41-50).

Ankyrin (ANK) repeats mediate protein-protein interactions associated with diverse intracellular functions. ANK repeats are composed of about 33 amino acids that form a helix-turn-helix core preceded by a protruding "tip." These tips are of variable sequence and may play a role in protein-protein interactions. The helix-turn-helix region of the ANK repeats stack on top of one another and are stabilized by hydrophobic interactions (Yang, Y. et al. (1998) Structure 6:619-626).

Sushi repeats, also called short consensus repeats (SCR), are found in a number of proteins that share the common feature of binding to other proteins. For example, in the C-terminal domain of versican, the sushi domain is important for heparin binding. Sushi domains contain basic amino acid residues, which may play a role in binding (Oleszewski, M. et al. (2000) J. Biol. Chem. 275:34478-34485).

Link, or X-link, modules are hyaluronan-binding domains found in proteins involved in the assembly of extracellular matrix, cell adhesion, and migration. The Link module superfamily includes CD44, cartilage link protein, and aggrecan. This family also includes BEHAB (brain enriched hyaluronan-binding)/brevican, a component of the brain ECM that is dramatically upregulated in human gliomas, and appears to play a role in determining the invasive potential of brain tumor cells (Gary, S.C. et al. (1998) Curr. Opin. Neurobiol. 8:576-581). There is close similarity between the Link module and the C-type lectin domain, with the predicted hyaluronan-binding site at an analogous position to the carbohydrate-binding pocket in E-selectin (Kohda, D. et al. (1996) Cell 86:767-775).

Multidomain or mosaic proteins play an important role in the diverse functions of the extracellular matrix (Engel, J. et al. (1994) Development (Camb.):S35-S42). ECM proteins are frequently characterized by the presence of one or more domains which may contain a number of potential intracellular disulfide bridge motifs. For example, domains which match the epidermal growth factor (EGF) tandem repeat consensus are present within several known extracellular proteins that promote cell growth, development, and cell signaling. This signature sequence is about forty amino acid residues in length and includes six conserved cysteine residues, and a calcium-binding site near the N-terminus of the signature sequence. The main structure is a two-stranded beta-sheet followed by a loop to a C-terminal short two-stranded sheet. Subdomains between the conserved cysteines vary in length (Davis, C.G. (1990) New Biol. 5:410-419). Post-translational hydroxylation of aspartic acid or asparagine residues has been associated with EGF-like domains in several proteins (Prosite PDOC00010).

A number of proteins that contain calcium-binding EGF-like domain signature sequences are involved in growth and differentiation. Examples include bone morphogenic protein 1, which induces the formation of cartilage and bone; crumbs, which is a *Drosophila* epithelial development protein; Notch and a number of its homologs, which are involved in neural growth and differentiation, and transforming growth factor beta-1 binding protein (Expasy PROSITE document PDOC00913; Soler, C. and G. Carpenter, in Nicola, N.A. (1994) The Cytokine Facts Book, Oxford University Press, Oxford, UK, pp. 193-197). EGF-like domains mediate protein-protein interactions for a variety of proteins. For example, EGF-like domains in the ECM glycoprotein fibulin-1 have been shown to mediate both self-association and binding to fibronectin (Tran, H. et al. (1997) J. Biol. Chem. 272:22600-22606). Point mutations in the EGF-like domains of ECM proteins have been identified as the cause of human disorders such as Marfan syndrome and pseudochondroplasia (Maurer, P. et al. (1996) Curr. Opin. Cell Biol. 8:609-617).

The CUB domain is an extracellular domain of approximately 110 amino acid residues found mostly in developmentally regulated proteins. The CUB domain contains four conserved cysteine residues and is predicted to have a structure similar to that of immunoglobulins. Vertebrate bone morphogenic protein 1, which induces cartilage and bone formation, and fibropellins I and III from sea urchin, which form the apical lamina component of the ECM, are examples of proteins that contain both CUB and EGF domains (PROSITE PDOC00908).

Other ECM proteins are members of the type A domain of von Willebrand factor (vWFA)-like module superfamily, a diverse group of proteins with a module sharing high sequence similarity. The vWFA-like module is found not only in plasma proteins but also in plasma membrane and ECM proteins (Colombatti, A. and P. Bonaldo (1991) Blood 77:2305-2315). Crystal structure analysis of an integrin vWFA-like module shows a classic "Rossmann" fold and suggests a metal ion-dependent adhesion site for binding protein ligands (Lee, J.-O. et al. (1995) Cell 80:631-638). This family includes the protein matrilin-2, an extracellular matrix protein that is expressed in a broad range of mammalian tissues and organs. Matrilin-2 is thought to play a role in ECM assembly by bridging collagen fibrils and the aggrecan network (Deak, F. et al. (1997) J. Biol. Chem. 272:9268-9274).

The thrombospondins are multimeric, calcium-binding extracellular glycoproteins found widely in the embryonic extracellular matrix. These proteins are expressed in the developing nervous system or at specific sites in the adult nervous system after injury. Thrombospondins contain multiple EGF-type repeats, as well as a motif known as the thrombospondin type 1 repeat (TSR). The

TSR is approximately 60 amino acids in length and contains six conserved cysteine residues. Motifs within TSR domains are involved in mediating cell adhesion through binding to proteoglycans and sulfated glycolipids. Thrombospondin-1 inhibits angiogenesis and modulates endothelial cell adhesion, motility, and growth. TSR domains are found in a diverse group of other proteins, most of which are expressed in the developing nervous system and have potential roles in the guidance of cell and growth cone migration. Proteins that contain TSRs include the F-spondin gene family, the semaphorin 5 family, UNC-5, and SCO-spondin. The TSR superfamily includes the ADAMTS proteins which contain an ADAM (A Disintegrin and Metalloproteinase) domain as well as one or more TSRs. The ADAMTS proteins have roles in regulating the turnover of cartilage matrix, regulation of blood vessel growth, and possibly development of the nervous system. (Reviewed in Adams, J.C. and R.P. Tucker (2000) Dev. Dyn. 218:280-299.)

Fibrinogen, the principle protein of vertebrate blood clotting, is a hexamer consisting of two sets of three different chains (alpha, beta, and gamma). The C-terminal domain of the beta and gamma chains comprises about 270 amino acid residues and contains four cysteines involved in two disulfide bonds. This domain has also been found in mammalian tenascin-X, an ECM protein that appears to be involved in cell adhesion (Prosite PDOC00445).

<u>Expression profiling</u>

Microarrays are analytical tools used in bioanalysis. A microarray has a plurality of molecules spatially distributed over, and stably associated with, the surface of a solid support. Microarrays of polypeptides, polynucleotides, and/or antibodies have been developed and find use in a variety of applications, such as gene sequencing, monitoring gene expression, gene mapping, bacterial identification, drug discovery, and combinatorial chemistry.

One area in particular in which microarrays find use is in gene expression analysis. Array technology can provide a simple way to explore the expression of a single polymorphic gene or the expression profile of a large number of related or unrelated genes. When the expression of a single gene is examined, arrays are employed to detect the expression of a specific gene or its variants. When an expression profile is examined, arrays provide a platform for identifying genes that are tissue specific, are affected by a substance being tested in a toxicology assay, are part of a signaling cascade, carry out housekeeping functions, or are specifically related to a particular genetic predisposition, condition, disease, or disorder.

<u>Atherosclerosis</u>

Atherosclerosis and the associated coronary artery disease and cerebral stroke represent the most common cause of death in industrialized nations. Although certain key risk factors have been identified, a full molecular characterization that elucidates the causes and provide care for this complex disease has not been achieved. Molecular characterization of growth and regression of atherosclerotic vascular lesions requires identification of the genes that contribute to features of the lesion including growth, stability, dissolution, rupture and, most lethally, induction of occlusive vessel thrombus. Vascular lesions principally involve the vascular endothelium and the surrounding smooth muscle tissue.

Development of atherosclerosis is understood to be induced by the presence of circulating lipoprotein. Lipoproteins, such as the cholesterol-rich low-density lipoprotein (LDL), accumulate in the extracellular space of the vascular intima, and undergo modification. Oxidation of LDL (Ox-LDL) occurs most avidly in the sub-endothelial space where circulating antioxidant defenses are less effective. Mononuclear phagocytes enter the intima, differentiate into macrophages, and ingest modified lipids including Ox-LDL. During Ox-LDL uptake, macrophages produce cytokines (e.g. tumor necrosis factor α (TNF-α) and interleukin-1 (IL-1)) and growth factors (e.g. M-CSF, VEGF, and PDGF-BB) that elicit further cellular events that modulate atherogenesis such as smooth muscle cell proliferation and production of extracellular matrix by vascular endothelium. Additionally, these macrophages may activate genes in endothelium and smooth muscle tissue involved in inflammation and tissue differentiation, including superoxide dismutatse (SOD), IL-8, and ICAM-1.

The vascular endothelium influences not only the three classically interacting components of hemostasis: the vessel, the blood platelets and the clotting and fibrinolytic systems of plasma, but also the natural sequelae: inflammation and tissue repair. Two principal modes of endothelial behavior may be differentiated, best defined as an anti- and a prothrombotic state. Under physiological conditions endothelium mediates vascular dilatation (formation of nitric oxide (NO), $PGI_2$, adenosine, hyperpolarising factor), prevents platelet adhesion and activation (production of adenosine, NO and $PGI_2$, removal of ADP), blocks thrombin formation (tissue factor pathway inhibitor, activation of protein C via thrombomodulin, activation of antithrombin III) and mitigates fibrin deposition (t- and scuplasminogen activator production). Adhesion and transmigration of inflammatory leukocytes are attenuated, e.g. by NO and IL-10, and oxygen radicals are efficiently scavenged (urate, NO, glutathione, SOD).

When the endothelium is physically disrupted or functionally perturbed by postischemic reperfusion, acute and chronic inflammation, atherosclerosis, diabetes and chronic arterial hypertension, then completely opposing actions pertain. This prothrombotic, proinflammatory state is characterised by vaso-constriction, platelet and leukocyte activation and adhesion (externalisation, expression and upregulation of, for example, von Willebrand factor, platelet activating factor, P-selectin, ICAM-1, IL-8, MCP-1, and TNF-α), promotion of thrombin formation, coagulation and fibrin deposition at the vascular wall (expression of tissue factor, PAI-1, and phosphatidyl serine) and, in platelet-leukocyte coaggregates, additional inflammatory interactions via attachment of platelet CD40-ligand to endothelial, monocyte and B-cell CD40. Since thrombin formation and inflammatory stimulation set the stage for later tissue repair, complete abolition of such endothelial responses cannot be the goal of clinical interventions aimed at limiting procoagulatory, prothrombotic actions of a dysfunctional vascular endothelium. (See, e.g., Becker et al. (2000) Z Kardiol 89:160-167.)

Breast Cancer

More than 180,000 new cases of breast cancer are diagnosed each year, and the mortality rate for breast cancer approaches 10% of all deaths in females between the ages of 45-54 (Gish, K. (1999) AWIS Magazine 28:7-10). However the survival rate based on early diagnosis of localized breast cancer is extremely high (97%), compared with the advanced stage of the disease in which the tumor has spread beyond the breast (22%). Current procedures for clinical breast examination are lacking in sensitivity and specificity, and efforts are underway to develop comprehensive gene expression profiles for breast cancer that may be used in conjunction with conventional screening methods to improve diagnosis and prognosis of this disease (Perou, C.M. et al. (2000) Nature 406:747-752).

Mutations in two genes, BRCA1 and BRCA2, are known to greatly predispose a woman to breast cancer and may be passed on from parents to children (Gish, *supra*). However, this type of hereditary breast cancer accounts for only about 5% to 9% of breast cancers, while the vast majority of breast cancer is due to non-inherited mutations that occur in breast epithelial cells.

The relationship between expression of epidermal growth factor (EGF) and its receptor, EGFR, to human mammary carcinoma has been particularly well studied. (See Khazaie, K. et al. (1993) Cancer and Metastasis Rev. 12:255-274, and references cited therein for a review of this area.) Overexpression of EGFR, particularly coupled with down-regulation of the estrogen receptor, is a marker of poor prognosis in breast cancer patients. In addition, EGFR expression in breast tumor metastases is frequently elevated relative to the primary tumor, suggesting that EGFR is involved in tumor progression and metastasis. This is supported by accumulating evidence that EGF has effects on cell functions related to metastatic potential, such as cell motility, chemotaxis, secretion and differentiation. Changes in expression of other members of the erbB receptor family, of which EGFR is one, have also been implicated in breast cancer. The abundance of erbB receptors, such as HER-2/neu, HER-3, and HER-4, and their ligands in breast cancer points to their functional importance in the pathogenesis of the disease, and may therefore provide targets for therapy of the disease (Bacus, S.S. et al. (1994) Am. J. Clin. Pathol. 102:S13-S24). Other known markers of breast cancer include a human secreted frizzled protein mRNA that is downregulated in breast tumors; the matrix G1a protein which is overexpressed in human breast carcinoma cells; Drg1 or RTP, a gene whose expression is diminished in colon, breast, and prostate tumors; maspin, a tumor suppressor gene downregulated in invasive breast carcinomas; and CaN19, a member of the S100 protein family, all of which are down-regulated in mammary carcinoma cells relative to normal mammary epithelial cells

(Zhou, Z. et al. (1998) Int. J. Cancer 78:95-99; Chen, L. et al. (1990) Oncogene 5:1391-1395; Ulrix, W. et al (1999) FEBS Lett. 455:23-26; Sager, R. et al. (1996) Curr. Top. Microbiol. Immunol. 213:51-64; and Lee, S.W. et al. (1992) Proc. Natl. Acad. Sci. USA 89:2504-2508).

Cell lines derived from human mammary epithelial cells at various stages of breast cancer provide a useful model to study the process of malignant transformation and tumor progression as it has been shown that these cell lines retain many of the properties of their parental tumors for lengthy culture periods (Wistuba, I.I. et al. (1998) Clin. Cancer Res. 4:2931-2938). Such a model is particularly useful for comparing phenotypic and molecular characteristics of human mammary epithelial cells at various stages of malignant transformation.

Ovarian Cancer

Ovarian cancer is the leading cause of death from a gynecologic cancer. The majority of ovarian cancers are derived from epithelial cells, and 70% of patients with epithelial ovarian cancers present with late-stage disease. As a result, the long-term survival rate for this disease is very low. Identification of early-stage markers for ovarian cancer would significantly increase the survival rate. Genetic variations involved in ovarian cancer development include mutation of p53 and microsatellite instability. Gene expression patterns likely vary when normal ovary is compared to ovarian tumors.

There is a need in the art for new compositions, including nucleic acids and proteins, for the diagnosis, prevention, and treatment of immune system disorders, neurological disorders, developmental disorders, connective tissue disorders, and cell proliferative disorders, including cancer.

## SUMMARY OF THE INVENTION

Various embodiments of the invention provide purified polypeptides, cell adhesion and extracellular matrix proteins, referred to collectively as 'CADECM' and individually as 'CADECM-1,' 'CADECM-2,' 'CADECM-3,' 'CADECM-4,' 'CADECM-5,' 'CADECM-6,' 'CADECM-7,' 'CADECM-8,' 'CADECM-9,' 'CADECM-10,' 'CADECM-11,' 'CADECM-12,' 'CADECM-13,' 'CADECM-14,' 'CADECM-15,' 'CADECM-16,' 'CADECM-17,' 'CADECM-18,' 'CADECM-19,' 'CADECM-20,' 'CADECM-21,' 'CADECM-22,' 'CADECM-23,' 'CADECM-24,' 'CADECM-25,' 'CADECM-26,' 'CADECM-27,' 'CADECM-28,' 'CADECM-29,' and 'CADECM-30' and methods for using these proteins and their encoding polynucleotides for the detection, diagnosis, and treatment of diseases and medical conditions. Embodiments also provide methods for utilizing the purified cell adhesion and extracellular matrix proteins and/or their encoding polynucleotides for facilitating the drug discovery process, including determination of efficacy, dosage, toxicity, and pharmacology. Related embodiments provide methods for utilizing the purified cell adhesion and extracellular matrix proteins and/or their encoding polynucleotides for investigating the pathogenesis of diseases and medical conditions.

An embodiment provides an isolated polypeptide selected from the group consisting of a) a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, b) a polypeptide comprising a naturally occurring amino acid sequence at least 90% identical or at least about 90% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, c) a biologically active fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, and d) an immunogenic fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-30. Another embodiment provides an isolated polypeptide comprising an amino acid sequence of SEQ ID NO:1-30.

Still another embodiment provides an isolated polynucleotide encoding a polypeptide selected from the group consisting of a) a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, b) a polypeptide comprising a naturally occurring amino acid sequence at least 90% identical or at least about 90% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, c) a biologically active fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, and d) an immunogenic fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-30. In another embodiment, the polynucleotide encodes a polypeptide selected from the group consisting of SEQ ID NO:1-30. In an alternative embodiment, the polynucleotide is selected from the group consisting of SEQ ID NO:31-60.

Still another embodiment provides a recombinant polynucleotide comprising a promoter sequence operably linked to a polynucleotide encoding a polypeptide selected from the group consisting of a) a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, b) a polypeptide comprising a naturally occurring amino acid sequence at least 90% identical or at least about 90% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, c) a biologically active fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, and d) an immunogenic fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-30. Another embodiment provides a cell transformed with the recombinant polynucleotide. Yet another embodiment provides a transgenic organism comprising the recombinant polynucleotide.

Another embodiment provides a method for producing a polypeptide selected from the group consisting of a) a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, b) a polypeptide comprising a naturally occurring amino acid sequence at least 90% identical or at least about 90% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, c) a biologically active fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, and d) an immunogenic fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-30. The method comprises a) culturing a cell under conditions suitable for expression of the polypeptide, wherein said cell is transformed with a recombinant polynucleotide comprising a promoter sequence operably linked to a polynucleotide encoding the polypeptide, and b) recovering the polypeptide so expressed.

Yet another embodiment provides an isolated antibody which specifically binds to a polypeptide selected from the group consisting of a) a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, b) a polypeptide comprising a naturally occurring amino acid sequence at least 90% identical or at least about 90% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, c) a biologically active fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, and d) an immunogenic fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-30.

Still yet another embodiment provides an isolated polynucleotide selected from the group consisting of a) a polynucleotide comprising a polynucleotide sequence selected from the group consisting of SEQ ID NO:31-60, b) a polynucleotide comprising a naturally occurring polynucleotide sequence at least 90% identical or at least about 90% identical to a polynucleotide sequence selected from the group consisting of SEQ ID NO:31-60, c) a polynucleotide complementary to the polynucleotide of a), d) a polynucleotide complementary to the polynucleotide of b), and e) an RNA equivalent of a)-d). In other embodiments, the polynucleotide can comprise at least about 20, 30, 40, 60, 80, or 100 contiguous nucleotides.

Yet another embodiment provides a method for detecting a target polynucleotide in a sample, said target polynucleotide being selected from the group consisting of a) a polynucleotide comprising a polynucleotide sequence selected from the group consisting of SEQ ID NO:31-60, b) a polynucleotide comprising a naturally occurring polynucleotide sequence at least 90% identical or at least about 90% identical to a polynucleotide sequence selected from the group consisting of SEQ ID NO:31-60, c) a polynucleotide complementary to the polynucleotide of a), d) a polynucleotide complementary to the polynucleotide of b), and e) an RNA equivalent of a)-d). The method comprises a) hybridizing the sample with a probe comprising at least 20 contiguous nucleotides comprising a sequence complementary to said target polynucleotide in the sample, and which probe specifically hybridizes to said target polynucleotide, under conditions whereby a hybridization complex is formed between said probe and said target polynucleotide or fragments thereof, and b) detecting the presence or absence of said hybridization complex. In a related embodiment, the method can include detecting the amount of the hybridization complex. In still other embodiments, the probe can comprise at least about 20, 30, 40, 60, 80, or 100 contiguous nucleotides.

Still yet another embodiment provides a method for detecting a target polynucleotide in a sample, said target polynucleotide being selected from the group consisting of a) a polynucleotide comprising a polynucleotide sequence selected from the group consisting of SEQ ID NO:31-60, b) a polynucleotide comprising a naturally occurring polynucleotide sequence at least 90% identical or at least about 90% identical to a polynucleotide sequence selected from the group consisting of SEQ ID NO:31-60, c) a polynucleotide complementary to the polynucleotide of a), d) a polynucleotide complementary to the polynucleotide of b), and e) an RNA equivalent of a)-d). The method comprises a) amplifying said target polynucleotide or fragment thereof using polymerase chain reaction amplification, and b) detecting the presence or absence of said amplified target polynucleotide or fragment thereof. In a related embodiment, the method can include detecting the amount of the amplified target polynucleotide or fragment thereof.

Another embodiment provides a composition comprising an effective amount of a polypeptide selected from the group consisting of a) a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, b) a polypeptide comprising a naturally occurring amino acid sequence at least 90% identical or at least about 90% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, c) a biologically active fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, and d) an immunogenic fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, and a pharmaceutically acceptable excipient. In one embodiment, the composition can comprise an amino acid sequence selected from the group consisting of SEQ ID NO:1-30. Other embodiments provide a method of treating a disease or condition associated with decreased or abnormal expression of functional CADECM, comprising administering to a patient in need of such treatment the composition.

Yet another embodiment provides a method for screening a compound for effectiveness as an agonist of a polypeptide selected from the group consisting of a) a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, b) a polypeptide comprising a naturally occurring amino acid sequence at least 90% identical or at least about 90% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, c) a biologically active fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, and d) an immunogenic fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-30. The method comprises a) contacting a sample comprising the polypeptide with a compound, and b) detecting agonist activity in the sample. Another embodiment provides a composition comprising an agonist compound identified by the method and a pharmaceutically acceptable excipient. Yet another embodiment provides a method of treating a disease or condition associated with decreased expression of functional CADECM, comprising administering to a patient in need of such treatment the composition.

Still yet another embodiment provides a method for screening a compound for effectiveness as an antagonist of a polypeptide selected from the group consisting of a) a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, b) a polypeptide comprising a naturally occurring amino acid sequence at least 90% identical or at least about 90% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, c) a biologically active fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, and d) an immunogenic fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-30. The method comprises a) contacting a sample comprising the polypeptide with a compound, and b) detecting antagonist activity in the sample. Another embodiment provides a composition comprising an antagonist compound identified by the method and a pharmaceutically acceptable excipient. Yet another embodiment provides a method of treating a disease or condition associated with overexpression of functional CADECM, comprising administering to a patient in need of such treatment the composition.

Another embodiment provides a method of screening for a compound that specifically binds to a polypeptide selected from the group consisting of a) a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, b) a polypeptide comprising a naturally occurring amino acid sequence at least 90% identical or at least about 90% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, c) a biologically active fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, and d) an immunogenic fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-30. The method comprises a) combining the polypeptide with at least one test compound under suitable conditions, and b) detecting binding of the polypeptide to the test compound, thereby identifying a compound that specifically binds to the polypeptide.

Yet another embodiment provides a method of screening for a compound that modulates the activity of a polypeptide selected from the group consisting of a) a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, b) a polypeptide comprising a naturally occurring amino acid sequence at least 90% identical or at least about 90% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, c) a biologically active fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, and d) an immunogenic fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-30. The method comprises a) combining the polypeptide with at least one test compound under conditions permissive for the activity of the polypeptide, b) assessing the activity of the polypeptide in the presence of the test compound, and c) comparing the activity of the polypeptide in the presence of the test compound with the activity of the polypeptide in the absence of the test compound, wherein a change in the activity of the polypeptide in the presence of the test compound is indicative of a compound that modulates the activity of the polypeptide.

Still yet another embodiment provides a method for screening a compound for effectiveness in altering expression of a target polynucleotide, wherein said target polynucleotide comprises a polynucleotide sequence selected from the group consisting of SEQ ID NO:31-60, the method comprising a) contacting a sample comprising the target polynucleotide with a compound, b) detecting altered expression of the target polynucleotide, and c) comparing the expression of the target polynucleotide in the presence of varying amounts of the compound and in the absence of the compound.

Another embodiment provides a method for assessing toxicity of a test compound, said method comprising a) treating a biological sample containing nucleic acids with the test compound; b) hybridizing the nucleic acids of the treated biological sample with a probe comprising at least 20 contiguous nucleotides of a polynucleotide selected from the group consisting of i) a polynucleotide comprising a polynucleotide sequence selected from the group consisting of SEQ ID NO:31-60, ii) a polynucleotide comprising a naturally occurring polynucleotide sequence at least 90% identical or at least about 90% identical to a polynucleotide sequence selected from the group consisting of SEQ ID NO:31-60, iii) a polynucleotide having a sequence complementary to i), iv) a polynucleotide complementary to the polynucleotide of ii), and v) an RNA equivalent of i)-iv). Hybridization occurs under conditions whereby a specific hybridization complex is formed between said probe and a target polynucleotide in the biological sample, said target polynucleotide selected from the group consisting of i) a polynucleotide comprising a polynucleotide sequence selected from the group consisting of SEQ ID NO:31-60, ii) a polynucleotide comprising a naturally occurring polynucleotide sequence at least 90% identical or at least about 90% identical to a polynucleotide sequence selected from the group consisting of SEQ ID NO:31-60, iii) a polynucleotide complementary to the polynucleotide of i), iv) a polynucleotide complementary to the polynucleotide of ii), and v) an RNA equivalent of i)-iv). Alternatively, the target polynucleotide can comprise a fragment of a polynucleotide selected from the group consisting of i)-v) above; c) quantifying the amount of hybridization complex; and d) comparing the amount of hybridization complex in the treated biological sample with the amount of hybridization complex in an untreated biological sample, wherein a difference in the amount of hybridization complex in the treated biological sample is indicative of toxicity of the test compound.

## BRIEF DESCRIPTION OF THE TABLES

Table 1 summarizes the nomenclature for full length polynucleotide and polypeptide embodiments of the invention.

Table 2 shows the GenBank identification number and annotation of the nearest GenBank homolog, and the PROTEOME database identification numbers and annotations of PROTEOME database homologs, for polypeptide embodiments of the invention. The probability scores for the matches between each polypeptide and its homolog(s) are also shown.

Table 3 shows structural features of polypeptide embodiments, including predicted motifs and domains, along with the methods, algorithms, and searchable databases used for analysis of the polypeptides.

Table 4 lists the cDNA and/or genomic DNA fragments which were used to assemble polynucleotide embodiments, along with selected fragments of the polynucleotides.

Table 5 shows representative cDNA libraries for polynucleotide embodiments.

Table 6 provides an appendix which describes the tissues and vectors used for construction of the cDNA libraries shown in Table 5.

Table 7 shows the tools, programs, and algorithms used to analyze polynucleotides and polypeptides, along with applicable descriptions, references, and threshold parameters.

Table 8 shows single nucleotide polymorphisms found in polynucleotide sequences of the invention, along with allele frequencies in different human populations.

## DESCRIPTION OF THE INVENTION

Before the present proteins, nucleic acids, and methods are described, it is understood that embodiments of the invention are not limited to the particular machines, instruments, materials, and methods described, as these may vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to limit the scope of the invention.

As used herein and in the appended claims, the singular forms “a,” “an,” and “the” include plural reference unless the context clearly dictates otherwise. Thus, for example, a reference to “a host cell” includes a plurality of such host cells, and a reference to “an antibody” is a reference to one or more antibodies and equivalents thereof known to those skilled in the art, and so forth.

Unless defined otherwise, all technical and scientific terms used herein have the same meanings as commonly understood by one of ordinary skill in the art to which this invention belongs. Although any machines, materials, and methods similar or equivalent to those described herein can be used to practice or test the present invention, the preferred machines, materials and methods are now described. All publications mentioned herein are cited for the purpose of describing and disclosing the cell lines, protocols, reagents and vectors which are reported in the publications and which might be used in connection with various embodiments of the invention. Nothing herein is to be construed as an admission that the invention is not entitled to antedate such disclosure by virtue of prior invention.

**DEFINITIONS**

"CADECM" refers to the amino acid sequences of substantially purified CADECM obtained from any species, particularly a mammalian species, including bovine, ovine, porcine, murine, equine, and human, and from any source, whether natural, synthetic, semi-synthetic, or recombinant.

The term "agonist" refers to a molecule which intensifies or mimics the biological activity of CADECM. Agonists may include proteins, nucleic acids, carbohydrates, small molecules, or any other compound or composition which modulates the activity of CADECM either by directly interacting with CADECM or by acting on components of the biological pathway in which CADECM participates.

An "allelic variant" is an alternative form of the gene encoding CADECM. Allelic variants may result from at least one mutation in the nucleic acid sequence and may result in altered mRNAs or in polypeptides whose structure or function may or may not be altered. A gene may have none, one, or many allelic variants of its naturally occurring form. Common mutational changes which give rise to allelic variants are generally ascribed to natural deletions, additions, or substitutions of nucleotides. Each of these types of changes may occur alone, or in combination with the others, one or more times in a given sequence.

"Altered" nucleic acid sequences encoding CADECM include those sequences with deletions, insertions, or substitutions of different nucleotides, resulting in a polypeptide the same as CADECM or a polypeptide with at least one functional characteristic of CADECM. Included within this definition are polymorphisms which may or may not be readily detectable using a particular oligonucleotide probe of the polynucleotide encoding CADECM, and improper or unexpected hybridization to allelic variants, with a locus other than the normal chromosomal locus for the polynucleotide encoding CADECM. The encoded protein may also be "altered," and may contain deletions, insertions, or substitutions of amino acid residues which produce a silent change and result in a functionally equivalent CADECM. Deliberate amino acid substitutions may be made on the basis of one or more similarities in polarity, charge, solubility, hydrophobicity, hydrophilicity, and/or the amphipathic nature of the residues, as long as the biological or immunological activity of CADECM is retained. For example, negatively charged amino acids may include aspartic acid and glutamic acid, and positively charged amino acids may include lysine and arginine. Amino acids with uncharged polar side chains having similar hydrophilicity values may include: asparagine and glutamine; and serine and threonine. Amino acids with uncharged side chains having similar hydrophilicity values may include: leucine, isoleucine, and valine; glycine and alanine; and phenylalanine and tyrosine.

The terms "amino acid" and "amino acid sequence" can refer to an oligopeptide, a peptide, a polypeptide, or a protein sequence, or a fragment of any of these, and to naturally occurring or synthetic molecules. Where "amino acid sequence" is recited to refer to a sequence of a naturally occurring protein molecule, "amino acid sequence" and like terms are not meant to limit the amino acid sequence to the complete native amino acid sequence associated with the recited protein molecule.

"Amplification" relates to the production of additional copies of a nucleic acid. Amplification may be carried out using polymerase chain reaction (PCR) technologies or other nucleic acid amplification technologies well known in the art.

The term "antagonist" refers to a molecule which inhibits or attenuates the biological activity of CADECM. Antagonists may include proteins such as antibodies, anticalins, nucleic acids, carbohydrates, small molecules, or any other compound or composition which modulates the activity of CADECM either by directly interacting with CADECM or by acting on components of the biological pathway in which CADECM participates.

The term "antibody" refers to intact immunoglobulin molecules as well as to fragments thereof, such as Fab, $F(ab')_2$, and Fv fragments, which are capable of binding an epitopic determinant. Antibodies that bind CADECM polypeptides can be prepared using intact polypeptides or using fragments containing small peptides of interest as the immunizing antigen. The polypeptide or oligopeptide used to immunize an animal (e.g., a mouse, a rat, or a rabbit) can be derived from the translation of RNA, or synthesized chemically, and can be conjugated to a carrier protein if desired. Commonly used carriers that are chemically coupled to peptides include bovine serum albumin, thyroglobulin, and keyhole limpet hemocyanin (KLH). The coupled peptide is then used to immunize the animal.

The term "antigenic determinant" refers to that region of a molecule (i.e., an epitope) that makes contact with a particular antibody. When a protein or a fragment of a protein is used to immunize a host animal, numerous regions of the protein may induce the production of antibodies which bind specifically to antigenic determinants (particular regions or three-dimensional structures on the protein). An antigenic determinant may compete with the intact antigen (i.e., the immunogen used to elicit the immune response) for binding to an antibody.

The term "aptamer" refers to a nucleic acid or oligonucleotide molecule that binds to a specific molecular target. Aptamers are derived from an *in vitro* evolutionary process (e.g., SELEX (Systematic Evolution of Ligands by EXponential Enrichment), described in U.S. Patent No. 5,270,163), which selects for target-specific aptamer sequences from large combinatorial libraries. Aptamer compositions may be double-stranded or single-stranded, and may include deoxyribonucleotides, ribonucleotides, nucleotide derivatives, or other nucleotide-like molecules. The nucleotide components of an aptamer may have modified sugar groups (e.g., the 2'-OH group of a ribonucleotide may be replaced by 2'-F or 2'-$NH_2$), which may improve a desired property, e.g., resistance to nucleases or longer lifetime in blood. Aptamers may be conjugated to other molecules, e.g., a high molecular weight carrier to slow clearance of the aptamer from the circulatory system. Aptamers may be specifically cross-linked to their cognate ligands, e.g., by photo-activation of a cross-linker (Brody, E.N. and L. Gold (2000) J. Biotechnol. 74:5-13).

The term "intramer" refers to an aptamer which is expressed *in vivo*. For example, a vaccinia virus-based RNA expression system has been used to express specific RNA aptamers at high levels in the cytoplasm of leukocytes (Blind, M. et al. (1999) Proc. Natl. Acad. Sci. USA 96:3606-3610).

The term "spiegelmer" refers to an aptamer which includes L-DNA, L-RNA, or other left-handed nucleotide derivatives or nucleotide-like molecules. Aptamers containing left-handed nucleotides are resistant to degradation by naturally occurring enzymes, which normally act on substrates containing right-handed nucleotides.

The term "antisense" refers to any composition capable of base-pairing with the "sense" (coding) strand of a polynucleotide having a specific nucleic acid sequence. Antisense compositions may include DNA; RNA; peptide nucleic acid (PNA); oligonucleotides having modified backbone linkages such as phosphorothioates, methylphosphonates, or benzylphosphonates; oligonucleotides having modified sugar groups such as 2'-methoxyethyl sugars or 2'-methoxyethoxy sugars; or oligonucleotides having modified bases such as 5-methyl cytosine, 2'-deoxyuracil, or 7-deaza-2'-deoxyguanosine. Antisense molecules may be produced by any method including chemical synthesis or transcription. Once introduced into a cell, the complementary antisense molecule base-pairs with a naturally occurring nucleic acid sequence produced by the cell to form duplexes which block either transcription or translation. The designation "negative" or "minus" can refer to the antisense strand, and the designation "positive" or "plus" can refer to the sense strand of a reference DNA molecule.

The term "biologically active" refers to a protein having structural, regulatory, or biochemical functions of a naturally occurring molecule. Likewise, "immunologically active" or "immunogenic" refers to the capability of the natural, recombinant, or synthetic CADECM, or of any oligopeptide thereof, to induce a specific immune response in appropriate animals or cells and to bind with specific antibodies.

"Complementary" describes the relationship between two single-stranded nucleic acid sequences that anneal by base-pairing. For example, 5'-AGT-3' pairs with its complement, 3'-TCA-5'.

A "composition comprising a given polynucleotide" and a "composition comprising a given polypeptide" can refer to any composition containing the given polynucleotide or polypeptide. The composition may comprise a dry formulation or an aqueous solution. Compositions comprising polynucleotides encoding CADECM or fragments of CADECM may be employed as hybridization probes. The probes may be stored in freeze-dried form and may be associated with a stabilizing agent such as a carbohydrate. In hybridizations, the probe may be deployed in an aqueous solution containing salts (e.g., NaCl), detergents (e.g., sodium dodecyl sulfate; SDS), and other components (e.g., Denhardt's solution, dry milk, salmon sperm DNA, etc.).

"Consensus sequence" refers to a nucleic acid sequence which has been subjected to repeated DNA sequence analysis to resolve uncalled bases, extended using the XL-PCR kit (Applied Biosystems, Foster City CA) in the 5' and/or the 3' direction, and resequenced, or which has been assembled from one or more overlapping cDNA, EST, or genomic DNA fragments using a computer program for fragment assembly, such as the GELVIEW fragment assembly system (Accelrys, Burlington MA) or Phrap (University of Washington, Seattle WA). Some sequences have been both extended and assembled to produce the consensus sequence.

"Conservative amino acid substitutions" are those substitutions that are predicted to least interfere with the properties of the original protein, i.e., the structure and especially the function of the protein is conserved and not significantly changed by such substitutions. The table below shows amino acids which may be substituted for an original amino acid in a protein and which are regarded as conservative amino acid substitutions.

| Original Residue | Conservative Substitution |
|---|---|
| Ala | Gly, Ser |
| Arg | His, Lys |
| Asn | Asp, Gln, His |
| Asp | Asn, Glu |
| Cys | Ala, Ser |
| Gln | Asn, Glu, His |
| Glu | Asp, Gln, His |
| Gly | Ala |
| His | Asn, Arg, Gln, Glu |
| Ile | Leu, Val |
| Leu | Ile, Val |
| Lys | Arg, Gln, Glu |
| Met | Leu, Ile |
| Phe | His, Met, Leu, Trp, Tyr |
| Ser | Cys, Thr |
| Thr | Ser, Val |
| Trp | Phe, Tyr |
| Tyr | His, Phe, Trp |
| Val | Ile, Leu, Thr |

Conservative amino acid substitutions generally maintain (a) the structure of the polypeptide backbone in the area of the substitution, for example, as a beta sheet or alpha helical conformation, (b) the charge or hydrophobicity of the molecule at the site of the substitution, and/or (c) the bulk of the side chain.

A "deletion" refers to a change in the amino acid or nucleotide sequence that results in the absence of one or more amino acid residues or nucleotides.

The term "derivative" refers to a chemically modified polynucleotide or polypeptide. Chemical modifications of a polynucleotide can include, for example, replacement of hydrogen by an alkyl, acyl, hydroxyl, or amino group. A derivative polynucleotide encodes a polypeptide which retains at least one biological or immunological function of the natural molecule. A derivative polypeptide is one modified by glycosylation, pegylation, or any similar process that retains at least one biological or immunological function of the polypeptide from which it was derived.

A "detectable label" refers to a reporter molecule or enzyme that is capable of generating a measurable signal and is covalently or noncovalently joined to a polynucleotide or polypeptide.

"Differential expression" refers to increased or upregulated; or decreased, downregulated, or absent gene or protein expression, determined by comparing at least two different samples. Such comparisons may be carried out between, for example, a treated and an untreated sample, or a diseased and a normal sample.

"Exon shuffling" refers to the recombination of different coding regions (exons). Since an exon may represent a structural or functional domain of the encoded protein, new proteins may be assembled through the novel reassortment of stable substructures, thus allowing acceleration of the evolution of new protein functions.

A "fragment" is a unique portion of CADECM or a polynucleotide encoding CADECM which can be identical in sequence to, but shorter in length than, the parent sequence. A fragment may comprise up to the entire length of the defined sequence, minus one nucleotide/amino acid residue. For example, a fragment may comprise from about 5 to about 1000 contiguous nucleotides or amino acid residues. A fragment used as a probe, primer, antigen, therapeutic molecule, or for other purposes, may be at least 5, 10, 15, 16, 20, 25, 30, 40, 50, 60, 75, 100, 150, 250 or at least 500 contiguous nucleotides or amino acid residues in length. Fragments may be preferentially selected from certain regions of a molecule. For example, a polypeptide fragment may comprise a certain length of contiguous amino acids selected from the first 250 or 500 amino acids (or first 25% or 50%) of a polypeptide as shown in a certain defined sequence. Clearly these lengths are exemplary, and any length that is supported by the specification, including the Sequence Listing, tables, and figures, may be encompassed by the present embodiments.

A fragment of SEQ ID NO:31-60 can comprise a region of unique polynucleotide sequence that specifically identifies SEQ ID NO:31-60, for example, as distinct from any other sequence in the genome from which the fragment was obtained. A fragment of SEQ ID NO:31-60 can be employed in one or more embodiments of methods of the invention, for example, in hybridization and amplification technologies and in analogous methods that distinguish SEQ ID NO:31-60 from related polynucleotides. The precise length of a fragment of SEQ ID NO:31-60 and the region of SEQ ID NO:31-60 to which the fragment corresponds are routinely determinable by one of ordinary skill in the art based on the intended purpose for the fragment.

A fragment of SEQ ID NO:1-30 is encoded by a fragment of SEQ ID NO:31-60. A fragment of SEQ ID NO:1-30 can comprise a region of unique amino acid sequence that specifically identifies SEQ ID NO:1-30. For example, a fragment of SEQ ID NO:1-30 can be used as an immunogenic peptide for the development of antibodies that specifically recognize SEQ ID NO:1-30. The precise length of a fragment of SEQ ID NO:1-30 and the region of SEQ ID NO:1-30 to which the fragment corresponds can be determined based on the intended purpose for the fragment using one or more analytical methods described herein or otherwise known in the art.

A "full length" polynucleotide is one containing at least a translation initiation codon (e.g., methionine) followed by an open reading frame and a translation termination codon. A "full length" polynucleotide sequence encodes a "full length" polypeptide sequence.

"Homology" refers to sequence similarity or, alternatively, sequence identity, between two or more polynucleotide sequences or two or more polypeptide sequences.

The terms "percent identity" and "% identity," as applied to polynucleotide sequences, refer to the percentage of identical nucleotide matches between at least two polynucleotide sequences aligned using a standardized algorithm. Such an algorithm may insert, in a standardized and reproducible way, gaps in the sequences being compared in order to optimize alignment between two sequences, and therefore achieve a more meaningful comparison of the two sequences.

Percent identity between polynucleotide sequences may be determined using one or more computer algorithms or programs known in the art or described herein. For example, percent identity can be determined using the default parameters of the CLUSTAL V algorithm as incorporated into the MEGALIGN version 3.12e sequence alignment program. This program is part of the LASERGENE software package, a suite of molecular biological analysis programs (DNASTAR, Madison WI). CLUSTAL V is described in Higgins, D.G. and P.M. Sharp (1989; CABIOS 5:151-153) and in Higgins, D.G. et al. (1992; CABIOS 8:189-191). For pairwise alignments of polynucleotide sequences, the default parameters are set as follows: Ktuple=2, gap penalty=5, window=4, and "diagonals saved"=4. The "weighted" residue weight table is selected as the default.

Alternatively, a suite of commonly used and freely available sequence comparison algorithms which can be used is provided by the National Center for Biotechnology Information (NCBI) Basic Local Alignment Search Tool (BLAST) (Altschul, S.F. et al. (1990) J. Mol. Biol. 215:403-410), which is available from several sources, including the NCBI, Bethesda, MD, and on the Internet at ncbi.nlm.nih.gov/BLAST/. The BLAST software suite includes various sequence analysis programs including "blastn," that is used to align a known polynucleotide sequence with other polynucleotide sequences from a variety of databases. Also available is a tool called "BLAST 2 Sequences" that is used for direct pairwise comparison of two nucleotide sequences. "BLAST 2 Sequences" can be accessed and used interactively at ncbi.nlm.nih.gov/gorf/bl2.html. The "BLAST 2 Sequences" tool can be used for both blastn and blastp (discussed below). BLAST programs are commonly used with gap and other parameters set to default settings. For example, to compare two nucleotide sequences, one may use blastn with the "BLAST 2 Sequences" tool Version 2.0.12 (April-21-2000) set at default parameters. Such default parameters may be, for example:

*Matrix: BLOSUM62*

*Reward for match: 1*

*Penalty for mismatch: -2*

*Open Gap: 5 and Extension Gap: 2 penalties*

*Gap x drop-off: 50*

*Expect: 10*

*Word Size: 11*

*Filter: on*

Percent identity may be measured over the length of an entire defined sequence, for example, as defined by a particular SEQ ID number, or may be measured over a shorter length, for example, over the length of a fragment taken from a larger, defined sequence, for instance, a fragment of at least 20, at least 30, at least 40, at least 50, at least 70, at least 100, or at least 200 contiguous nucleotides. Such lengths are exemplary only, and it is understood that any fragment length supported by the sequences shown herein, in the tables, figures, or Sequence Listing, may be used to describe a length over which percentage identity may be measured.

Nucleic acid sequences that do not show a high degree of identity may nevertheless encode similar amino acid sequences due to the degeneracy of the genetic code. It is understood that changes in a nucleic acid sequence can be made using this degeneracy to produce multiple nucleic acid sequences that all encode substantially the same protein.

The phrases "percent identity" and "% identity," as applied to polypeptide sequences, refer to the percentage of identical residue matches between at least two polypeptide sequences aligned using a standardized algorithm. Methods of polypeptide sequence alignment are well-known. Some alignment methods take into account conservative amino acid substitutions. Such conservative substitutions, explained in more detail above, generally preserve the charge and hydrophobicity at the site of substitution, thus preserving the structure (and therefore function) of the polypeptide. The phrases "percent similarity" and "% similarity," as applied to polypeptide sequences, refer to the percentage of residue matches, including identical residue matches and conservative substitutions, between at least two polypeptide sequences aligned using a standardized algorithm. In contrast, conservative substitutions are not included in the calculation of percent identity between polypeptide sequences.

Percent identity between polypeptide sequences may be determined using the default parameters of the CLUSTAL V algorithm as incorporated into the MEGALIGN version 3.12e sequence alignment program (described and referenced above). For pairwise alignments of polypeptide sequences using CLUSTAL V, the default parameters are set as follows: Ktuple=1, gap penalty=3, window=5, and "diagonals saved"=5. The PAM250 matrix is selected as the default residue weight table.

Alternatively the NCBI BLAST software suite may be used. For example, for a pairwise comparison of two polypeptide sequences, one may use the "BLAST 2 Sequences" tool Version 2.0.12 (April-21-2000) with blastp set at default parameters. Such default parameters may be, for example:

*Matrix: BLOSUM62*

*Open Gap: 11 and Extension Gap: 1 penalties*

*Gap x drop-off: 50*

*Expect: 10*

*Word Size: 3*

*Filter: on*

Percent identity may be measured over the length of an entire defined polypeptide sequence, for example, as defined by a particular SEQ ID number, or may be measured over a shorter length, for example, over the length of a fragment taken from a larger, defined polypeptide sequence, for instance, a fragment of at least 15, at least 20, at least 30, at least 40, at least 50, at least 70 or at least 150 contiguous residues. Such lengths are exemplary only, and it is understood that any fragment length supported by the sequences shown herein, in the tables, figures or Sequence Listing, may be used to describe a length over which percentage identity may be measured.

"Human artificial chromosomes" (HACs) are linear microchromosomes which may contain DNA sequences of about 6 kb to 10 Mb in size and which contain all of the elements required for chromosome replication, segregation and maintenance.

The term "humanized antibody" refers to an antibody molecule in which the amino acid sequence in the non-antigen binding regions has been altered so that the antibody more closely resembles a human antibody, and still retains its original binding ability.

"Hybridization" refers to the process by which a polynucleotide strand anneals with a complementary strand through base pairing under defined hybridization conditions. Specific hybridization is an indication that two nucleic acid sequences share a high degree of complementarity. Specific hybridization complexes form under permissive annealing conditions and remain hybridized after the "washing" step(s). The washing step(s) is particularly important in determining the stringency of the hybridization process, with more stringent conditions allowing less non-specific binding, i.e., binding between pairs of nucleic acid strands that are not perfectly matched. Permissive conditions for annealing of nucleic acid sequences are routinely determinable by one of ordinary skill in the art and may be consistent among hybridization experiments, whereas wash conditions may be varied among experiments to achieve the desired stringency, and therefore hybridization specificity. Permissive annealing conditions occur, for example, at 68°C in the presence of about 6 x SSC, about 1% (w/v) SDS, and about 100 μg/ml sheared, denatured salmon sperm DNA.

Generally, stringency of hybridization is expressed, in part, with reference to the temperature under which the wash step is carried out. Such wash temperatures are typically selected to be about 5°C to 20°C lower than the thermal melting point ($T_m$) for the specific sequence at a defined ionic strength and pH. The $T_m$ is the temperature (under defined ionic strength and pH) at which 50% of the target sequence hybridizes to a perfectly matched probe. An equation for calculating $T_m$ and conditions for nucleic acid hybridization are well known and can be found in Sambrook, J. and D.W. Russell (2001; Molecular Cloning: A Laboratory Manual, 3rd ed., vol. 1-3, Cold Spring Harbor Press, Cold Spring Harbor NY, ch. 9).

High stringency conditions for hybridization between polynucleotides of the present invention include wash conditions of 68°C in the presence of about 0.2 x SSC and about 0.1% SDS, for 1 hour. Alternatively, temperatures of about 65°C, 60°C, 55°C, or 42°C may be used. SSC concentration may be varied from about 0.1 to 2 x SSC, with SDS being present at about 0.1%. Typically, blocking reagents are used to block non-specific hybridization. Such blocking reagents include, for instance, sheared and denatured salmon sperm DNA at about 100-200 μg/ml. Organic solvent, such as formamide at a concentration of about 35-50% v/v, may also be used under particular circumstances, such as for RNA:DNA hybridizations. Useful variations on these wash conditions will be readily apparent to those of ordinary skill in the art. Hybridization, particularly under high stringency conditions, may be suggestive of evolutionary similarity between the nucleotides. Such similarity is strongly indicative of a similar role for the nucleotides and their encoded polypeptides.

The term "hybridization complex" refers to a complex formed between two nucleic acids by virtue of the formation of hydrogen bonds between complementary bases. A hybridization complex may be formed in solution (e.g., $C_0t$ or $R_0t$ analysis) or formed between one nucleic acid present in solution and another nucleic acid immobilized on a solid support (e.g., paper, membranes, filters, chips, pins or glass slides, or any other appropriate substrate to which cells or their nucleic acids have been fixed).

The words "insertion" and "addition" refer to changes in an amino acid or polynucleotide sequence resulting in the addition of one or more amino acid residues or nucleotides, respectively.

"Immune response" can refer to conditions associated with inflammation, trauma, immune disorders, or infectious or genetic disease, etc. These conditions can be characterized by expression of various factors, e.g., cytokines, chemokines, and other signaling molecules, which may affect cellular and systemic defense systems.

An "immunogenic fragment" is a polypeptide or oligopeptide fragment of CADECM which is capable of eliciting an immune response when introduced into a living organism, for example, a mammal. The term "immunogenic fragment" also includes any polypeptide or oligopeptide fragment of CADECM which is useful in any of the antibody production methods disclosed herein or known in the art.

The term "microarray" refers to an arrangement of a plurality of polynucleotides, polypeptides, antibodies, or other chemical compounds on a substrate.

The terms "element" and "array element" refer to a polynucleotide, polypeptide, antibody, or other chemical compound having a unique and defined position on a microarray.

The term "modulate" refers to a change in the activity of CADECM. For example, modulation may cause an increase or a decrease in protein activity, binding characteristics, or any other biological, functional, or immunological properties of CADECM.

The phrases "nucleic acid" and "nucleic acid sequence" refer to a nucleotide, oligonucleotide, polynucleotide, or any fragment thereof. These phrases also refer to DNA or RNA of genomic or synthetic origin which may be single-stranded or double-stranded and may represent the sense or the antisense strand, to peptide nucleic acid (PNA), or to any DNA-like or RNA-like material.

"Operably linked" refers to the situation in which a first nucleic acid sequence is placed in a functional relationship with a second nucleic acid sequence. For instance, a promoter is operably linked to a coding sequence if the promoter affects the transcription or expression of the coding sequence. Operably linked DNA sequences may be in close proximity or contiguous and, where necessary to join two protein coding regions, in the same reading frame.

"Peptide nucleic acid" (PNA) refers to an antisense molecule or anti-gene agent which comprises an oligonucleotide of at least about 5 nucleotides in length linked to a peptide backbone of amino acid residues ending in lysine. The terminal lysine confers solubility to the composition. PNAs preferentially bind complementary single stranded DNA or RNA and stop transcript elongation, and may be pegylated to extend their lifespan in the cell.

"Post-translational modification" of an CADECM may involve lipidation, glycosylation, phosphorylation, acetylation, racemization, proteolytic cleavage, and other modifications known in the art. These processes may occur synthetically or biochemically. Biochemical modifications will vary by cell type depending on the enzymatic milieu of CADECM.

"Probe" refers to nucleic acids encoding CADECM, their complements, or fragments thereof, which are used to detect identical, allelic or related nucleic acids. Probes are isolated oligonucleotides or polynucleotides attached to a detectable label or reporter molecule. Typical labels include radioactive isotopes, ligands, chemiluminescent agents, and enzymes. "Primers" are short nucleic acids, usually DNA oligonucleotides, which may be annealed to a target polynucleotide by complementary base-pairing. The primer may then be extended along the target DNA strand by a DNA polymerase enzyme. Primer pairs can be used for amplification (and identification) of a nucleic acid, e.g., by the polymerase chain reaction (PCR).

Probes and primers as used in the present invention typically comprise at least 15 contiguous nucleotides of a known sequence. In order to enhance specificity, longer probes and primers may also be employed, such as probes and primers that comprise at least 20, 25, 30, 40, 50, 60, 70, 80, 90, 100, or at least 150 consecutive nucleotides of the disclosed nucleic acid sequences. Probes and primers may be considerably longer than these examples, and it is understood that any length supported by the specification, including the tables, figures, and Sequence Listing, may be used.

Methods for preparing and using probes and primers are described in, for example, Sambrook, J. and D.W. Russell (2001; <u>Molecular Cloning: A Laboratory Manual</u>, 3rd ed., vol. 1-3, Cold Spring Harbor Press, Cold Spring Harbor NY), Ausubel, F.M. et al. (1999; <u>Short Protocols in Molecular Biology</u>, 4th ed., John Wiley & Sons, New York NY), and Innis, M. et al. (1990; <u>PCR Protocols, A Guide to Methods and Applications</u>, Academic Press, San Diego CA). PCR primer pairs can be derived from a known sequence, for example, by using computer programs intended for that purpose such as Primer (Version 0.5, 1991, Whitehead Institute for Biomedical Research, Cambridge MA).

Oligonucleotides for use as primers are selected using software known in the art for such purpose. For example, OLIGO 4.06 software is useful for the selection of PCR primer pairs of up to 100 nucleotides each, and for the analysis of oligonucleotides and larger polynucleotides of up to 5,000 nucleotides from an input polynucleotide sequence of up to 32 kilobases. Similar primer selection programs have incorporated additional features for expanded capabilities. For example, the PrimOU primer selection program (available to the public from the Genome Center at University of Texas South West Medical Center, Dallas TX) is capable of choosing specific primers from megabase sequences and is thus useful for designing primers on a genome-wide scope. The Primer3 primer selection program (available to the public from the Whitehead Institute/MIT Center for Genome Research, Cambridge MA) allows the user to input a "mispriming library," in which sequences to avoid as primer binding sites are user-specified. Primer3 is useful, in particular, for the selection of oligonucleotides for microarrays. (The source code for the latter two primer selection programs may also be obtained from their respective sources and modified to meet the user's specific needs.) The PrimeGen program (available to the public from the UK Human Genome Mapping Project Resource Centre, Cambridge UK) designs primers based on multiple sequence alignments, thereby allowing selection of primers that hybridize to either the most conserved or least conserved regions of aligned nucleic acid sequences. Hence, this program is useful for identification of both unique and conserved oligonucleotides and polynucleotide fragments. The oligonucleotides and polynucleotide fragments identified by any of the above selection methods are useful in hybridization technologies, for example, as PCR or sequencing primers, microarray elements, or specific probes to identify fully or partially complementary polynucleotides in a sample of nucleic acids. Methods of oligonucleotide selection are not limited to those described above.

A "recombinant nucleic acid" is a nucleic acid that is not naturally occurring or has a sequence that is made by an artificial combination of two or more otherwise separated segments of sequence. This artificial combination is often accomplished by chemical synthesis or, more commonly, by the artificial manipulation of isolated segments of nucleic acids, e.g., by genetic engineering techniques such as those described in Sambrook and Russell (*supra*). The term recombinant includes nucleic acids that have been altered solely by addition, substitution, or deletion of a portion of the nucleic acid. Frequently, a recombinant nucleic acid may include a nucleic acid sequence operably linked to a promoter sequence. Such a recombinant nucleic acid may be part of a vector that is used, for example, to transform a cell.

Alternatively, such recombinant nucleic acids may be part of a viral vector, e.g., based on a vaccinia virus, that could be use to vaccinate a mammal wherein the recombinant nucleic acid is expressed, inducing a protective immunological response in the mammal.

A "regulatory element" refers to a nucleic acid sequence usually derived from untranslated regions of a gene and includes enhancers, promoters, introns, and 5' and 3' untranslated regions (UTRs). Regulatory elements interact with host or viral proteins which control transcription, translation, or RNA stability.

"Reporter molecules" are chemical or biochemical moieties used for labeling a nucleic acid, amino acid, or antibody. Reporter molecules include radionuclides; enzymes; fluorescent, chemiluminescent, or chromogenic agents; substrates; cofactors; inhibitors; magnetic particles; and other moieties known in the art.

An "RNA equivalent," in reference to a DNA molecule, is composed of the same linear sequence of nucleotides as the reference DNA molecule with the exception that all occurrences of the nitrogenous base thymine are replaced with uracil, and the sugar backbone is composed of ribose instead of deoxyribose.

The term "sample" is used in its broadest sense. A sample suspected of containing

CADECM, nucleic acids encoding CADECM, or fragments thereof may comprise a bodily fluid; an extract from a cell, chromosome, organelle, or membrane isolated from a cell; a cell; genomic DNA, RNA, or cDNA, in solution or bound to a substrate; a tissue; a tissue print; etc.

The terms “specific binding” and “specifically binding” refer to that interaction between a protein or peptide and an agonist, an antibody, an antagonist, a small molecule, or any natural or synthetic binding composition. The interaction is dependent upon the presence of a particular structure of the protein, e.g., the antigenic determinant or epitope, recognized by the binding molecule. For example, if an antibody is specific for epitope “A,” the presence of a polypeptide comprising the epitope A, or the presence of free unlabeled A, in a reaction containing free labeled A and the antibody will reduce the amount of labeled A that binds to the antibody.

The term “substantially purified” refers to nucleic acid or amino acid sequences that are removed from their natural environment and are isolated or separated, and are at least about 60% free, preferably at least about 75% free, and most preferably at least about 90% free from other components with which they are naturally associated.

A “substitution” refers to the replacement of one or more amino acid residues or nucleotides by different amino acid residues or nucleotides, respectively.

“Substrate” refers to any suitable rigid or semi-rigid support including membranes, filters, chips, slides, wafers, fibers, magnetic or nonmagnetic beads, gels, tubing, plates, polymers, microparticles and capillaries. The substrate can have a variety of surface forms, such as wells, trenches, pins, channels and pores, to which polynucleotides or polypeptides are bound.

A “transcript image” or “expression profile” refers to the collective pattern of gene expression by a particular cell type or tissue under given conditions at a given time.

“Transformation” describes a process by which exogenous DNA is introduced into a recipient cell. Transformation may occur under natural or artificial conditions according to various methods well known in the art, and may rely on any known method for the insertion of foreign nucleic acid sequences into a prokaryotic or eukaryotic host cell. The method for transformation is selected based on the type of host cell being transformed and may include, but is not limited to, bacteriophage or viral infection, electroporation, heat shock, lipofection, and particle bombardment. The term “transformed cells” includes stably transformed cells in which the inserted DNA is capable of replication either as an autonomously replicating plasmid or as part of the host chromosome, as well as transiently transformed cells which express the inserted DNA or RNA for limited periods of time.

A "transgenic organism," as used herein, is any organism, including but not limited to animals and plants, in which one or more of the cells of the organism contains heterologous nucleic acid introduced by way of human intervention, such as by transgenic techniques well known in the art. The nucleic acid is introduced into the cell, directly or indirectly by introduction into a precursor of the cell, by way of deliberate genetic manipulation, such as by microinjection or by infection with a recombinant virus. In another embodiment, the nucleic acid can be introduced by infection with a recombinant viral vector, such as a lentiviral vector (Lois, C. et al. (2002) Science 295:868-872). The term genetic manipulation does not include classical cross-breeding, or *in vitro* fertilization, but rather is directed to the introduction of a recombinant DNA molecule. The transgenic organisms contemplated in accordance with the present invention include bacteria, cyanobacteria, fungi, plants and animals. The isolated DNA of the present invention can be introduced into the host by methods known in the art, for example infection, transfection, transformation or transconjugation. Techniques for transferring the DNA of the present invention into such organisms are widely known and provided in references such as Sambrook and Russell (*supra*).

A "variant" of a particular nucleic acid sequence is defined as a nucleic acid sequence having at least 40% sequence identity to the particular nucleic acid sequence over a certain length of one of the nucleic acid sequences using blastn with the "BLAST 2 Sequences" tool Version 2.0.9 (May-07-1999) set at default parameters. Such a pair of nucleic acids may show, for example, at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% or greater sequence identity over a certain defined length. A variant may be described as, for example, an "allelic" (as defined above), "splice," "species," or "polymorphic" variant. A splice variant may have significant identity to a reference molecule, but will generally have a greater or lesser number of polynucleotides due to alternate splicing during mRNA processing. The corresponding polypeptide may possess additional functional domains or lack domains that are present in the reference molecule. Species variants are polynucleotides that vary from one species to another. The resulting polypeptides will generally have significant amino acid identity relative to each other. A polymorphic variant is a variation in the polynucleotide sequence of a particular gene between individuals of a given species. Polymorphic variants also may encompass "single nucleotide polymorphisms" (SNPs) in which the polynucleotide sequence varies by one nucleotide base. The presence of SNPs may be indicative of, for example, a certain population, a disease state, or a propensity for a disease state.

A "variant" of a particular polypeptide sequence is defined as a polypeptide sequence having at least 40% sequence identity or sequence similarity to the particular polypeptide sequence over a certain length of one of the polypeptide sequences using blastp with the "BLAST 2 Sequences" tool Version 2.0.9 (May-07-1999) set at default parameters. Such a pair of polypeptides may show, for example, at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% or greater sequence identity or sequence similarity over a certain defined length of one of the polypeptides.

## THE INVENTION

Various embodiments of the invention include new human cell adhesion and extracellular matrix proteins (CADECM), the polynucleotides encoding CADECM, and the use of these compositions for the diagnosis, treatment, or prevention of immune system disorders, neurological disorders, developmental disorders, connective tissue disorders, and cell proliferative disorders, including cancer.

Table 1 summarizes the nomenclature for the full length polynucleotide and polypeptide embodiments of the invention. Each polynucleotide and its corresponding polypeptide are correlated to a single Incyte project identification number (Incyte Project ID). Each polypeptide sequence is denoted by both a polypeptide sequence identification number (Polypeptide SEQ ID NO:) and an Incyte polypeptide sequence number (Incyte Polypeptide ID) as shown. Each polynucleotide sequence is denoted by both a polynucleotide sequence identification number (Polynucleotide SEQ ID NO:) and an Incyte polynucleotide consensus sequence number (Incyte Polynucleotide ID) as shown. Column 6 shows the Incyte ID numbers of physical, full length clones corresponding to the polypeptide and polynucleotide sequences of the invention. The full length clones encode polypeptides which have at least 95% sequence identity to the polypeptide sequences shown in column 3.

Table 2 shows sequences with homology to polypeptide embodiments of the invention as identified by BLAST analysis against the GenBank protein (genpept) database and the PROTEOME database. Columns 1 and 2 show the polypeptide sequence identification number (Polypeptide SEQ ID NO:) and the corresponding Incyte polypeptide sequence number (Incyte Polypeptide ID) for polypeptides of the invention. Column 3 shows the GenBank identification number (GenBank ID NO:) of the nearest GenBank homolog and the PROTEOME database identification numbers (PROTEOME ID NO:) of the nearest PROTEOME database homologs. Column 4 shows the probability scores for the matches between each polypeptide and its homolog(s). Column 5 shows the annotation of the GenBank and PROTEOME database homolog(s) along with relevant citations where applicable, all of which are expressly incorporated by reference herein.

Table 3 shows various structural features of the polypeptides of the invention. Columns 1 and 2 show the polypeptide sequence identification number (SEQ ID NO:) and the corresponding Incyte polypeptide sequence number (Incyte Polypeptide ID) for each polypeptide of the invention. Column 3 shows the number of amino acid residues in each polypeptide. Column 4 shows amino acid residues comprising signature sequences, domains, motifs, potential phosphorylation sites, and potential glycosylation sites. Column 5 shows analytical methods for protein structure/function analysis and in some cases, searchable databases to which the analytical methods were applied.

Together, Tables 2 and 3 summarize the properties of polypeptides of the invention, and these properties establish that the claimed polypeptides are cell adhesion and extracellular matrix proteins. For example, SEQ ID NO:8 is 100% identical, from residue M1 to residue S326, and 80% identical, from residue E314 to residue L363, to human vitronectin (GenBank ID g14326449) as determined by the Basic Local Alignment Search Tool (BLAST). (See Table 2.) The BLAST probability score is 1.1e-204, which indicates the probability of obtaining the observed polypeptide sequence alignment by chance. SEQ ID NO:8 also has homology to proteins that are localized to the extracellular region (excluding cell wall), mediate adhesion and migration, and are vitronectin proteins as determined by BLAST analysis using the PROTEOME database. SEQ ID NO:8 also contains hemopexin domains and somatomedin B domains as determined by searching for statistically significant matches in the hidden Markov model (HMM)-based PFAM and SMART databases of conserved protein families/domains. (See Table 3.) Data from BLIMPS, MOTIFS, and PROFILESCAN analyses, and BLAST analyses against the PRODOM and DOMO databases, provide further corroborative evidence that SEQ ID NO:8 is a vitronectin.

In another example, SEQ ID NO:14 is 99% identical, from residue M1 to residue Q525, to human cadherin-19 (GenBank ID g10803406) as determined by BLAST. (See Table 2.) The BLAST probability score is 2.3e-282. SEQ ID NO:14 also has homology to proteins that are localized to the plasma membrane, have adhesion function, and are cadherins as determined by BLAST analysis using the PROTEOME database. SEQ ID NO:14 also contains cadherin domains as determined by searching for statistically significant matches in the hidden Markov model (HMM)-based PFAM and SMART databases of conserved protein families/domains. (See Table 3.) Data from BLIMPS, MOTIFS, and PROFILESCAN analyses, and BLAST analyses against the PRODOM and DOMO databases, provide further corroborative evidence that SEQ ID NO:14 is a cadherin.

In yet another example, SEQ ID NO:18 is 99% identical, from residue I25 to residue Q135, and 99% identical, from residue A144 to residue A287, to rat ankyrin binding cell adhesion molecule neurofascin (GenBank ID g1842429) as determined by BLAST. (See Table 2.) The BLAST probability score is 1.2e-132. SEQ ID NO:18 also has homology to proteins that are localized to the plasma membrane, are members of the immunoglobulin superfamily, are neuronal cell adhesion molecules, and are overexpressed in brain tumors, as determined by BLAST analysis using the PROTEOME database. SEQ ID NO:18 also contains an immunoglobulin C-2 type domain, as determined by searching for statistically significant matches in the hidden Markov model (HMM)-based SMART database of conserved protein families/domains. SEQ ID NO:18 also contains an Ig superfamily from SCOP domain, as determined by searching for statistically significant matches in the hidden Markov model (HMM)-based INCY database of conserved protein families/domains. (See

Table 3.) Data from BLAST analyses against the PRODOM and DOMO databases provide further corroborative evidence that SEQ ID NO:18 is a neuronal cell adhesion molecule.

In an alternative example, SEQ ID NO:26 is 80% identical, from residue P16 to residue K1402, to mouse nidogen-2 (GenBank ID g23592218) as determined by BLAST. (See Table 2.) The BLAST probability score is 0.0. SEQ ID NO:26 also has homology to proteins that are localized to the extracellular matrix (cuticle and basement membrane; excluding cell wall); and specifically to nidogen-2, a basement membrane protein that localizes to elastic fibers of blood vessel walls; has an integrin-binding RGD sequence; may facilitate stabilization of basement membrane networks; binds perlecan (HSPG2), laminin-1, and collagens; and supports cell attachment in some cell lines, as determined by BLAST analysis using the PROTEOME database. SEQ ID NO:26 also contains an EGF-like, low-density lipoprotein receptor repeat class B, thyroglobulin type-1 repeat, calcium-binding EGF-like, and low-density lipoprotein-receptor YWTD domains as determined by searching for statistically significant matches in the hidden Markov model (HMM)-based PFAM/SMART databases of conserved protein families/domains. (See Table 3.) Data from BLIMPS and MOTIFS analyses, and BLAST analyses against the PRODOM and DOMO databases, provide further corroborative evidence that SEQ ID NO:26 is an extracellular matrix protein.

In a further example, SEQ ID NO:27 is 100% identical, from residue V36 to residue A288, to human mDC-SIGNIA type III isoform (GenBank ID g15281077) as determined BLAST. (See Table 2.) The BLAST probability score is 9.2e-138. SEQ ID NO:27 also has homology to proteins that are localized to the plasma membrane, possess ICAM binding affinity, and are C-type lectins, as determined by BLAST analysis using the PROTEOME database. SEQ ID NO:27 also contains a C-type lectin domain as determined by searching for statistically significant matches in the hidden Markov model (HMM)-based PFAM and SMART databases of conserved protein families/domains. (See Table 3.) Data from BLIMPS, MOTIFS, and PROFILESCAN analyses, and BLAST analyses against the PRODOM and DOMO databases, provide further corroborative evidence that SEQ ID NO:27 is a C-type lectin. SEQ ID NO:1-7, SEQ ID NO:9-13, SEQ ID NO:15-17, SEQ ID NO:19-25, and SEQ ID NO:28-30 were analyzed and annotated in a similar manner. The algorithms and parameters for the analysis of SEQ ID NO:1-30 are described in Table 7.

As shown in Table 4, the full length polynucleotide embodiments were assembled using cDNA sequences or coding (exon) sequences derived from genomic DNA, or any combination of these two types of sequences. Column 1 lists the polynucleotide sequence identification number (Polynucleotide SEQ ID NO:), the corresponding Incyte polynucleotide consensus sequence number (Incyte ID) for each polynucleotide of the invention, and the length of each polynucleotide sequence in basepairs. Column 2 shows the nucleotide start (5′) and stop (3′) positions of the cDNA and/or genomic sequences used to assemble the full length polynucleotide embodiments, and of fragments of the polynucleotides which are useful, for example, in hybridization or amplification technologies that identify SEQ ID NO:31-60 or that distinguish between SEQ ID NO:31-60 and related polynucleotides.

The polynucleotide fragments described in Column 2 of Table 4 may refer specifically, for example, to Incyte cDNAs derived from tissue-specific cDNA libraries or from pooled cDNA libraries. Alternatively, the polynucleotide fragments described in column 2 may refer to GenBank cDNAs or ESTs which contributed to the assembly of the full length polynucleotides. In addition, the polynucleotide fragments described in column 2 may identify sequences derived from the ENSEMBL (The Sanger Centre, Cambridge, UK) database (*i.e.*, those sequences including the designation "ENST"). Alternatively, the polynucleotide fragments described in column 2 may be derived from the NCBI RefSeq Nucleotide Sequence Records Database (*i.e.*, those sequences including the designation "NM" or "NT") or the NCBI RefSeq Protein Sequence Records (*i.e.*, those sequences including the designation "NP"). Alternatively, the polynucleotide fragments described in column 2 may refer to assemblages of both cDNA and Genscan-predicted exons brought together by an "exon stitching" algorithm. For example, a polynucleotide sequence identified as FL_$XXXXXX$_$N_1$_$N_2$_$YYYYY$_$N_3$_$N_4$ represents a "stitched" sequence in which *XXXXXX* is the identification number of the cluster of sequences to which the algorithm was applied, and *YYYYY* is the number of the prediction generated by the algorithm, and $N_{1,2,3...}$, if present, represent specific exons that may have been manually edited during analysis (See Example V). Alternatively, the polynucleotide fragments in column 2 may refer to assemblages of exons brought together by an "exon-stretching" algorithm. For example, a polynucleotide sequence identified as FL*XXXXXX*_g*AAAAA*_g*BBBBB*_1_*N* is a "stretched" sequence, with *XXXXXX* being the Incyte project identification number, g*AAAAA* being the GenBank identification number of the human genomic sequence to which the "exon-stretching" algorithm was applied, g*BBBBB* being the GenBank identification number or NCBI RefSeq identification number of the nearest GenBank protein homolog, and *N* referring to specific exons (See Example V). In instances where a RefSeq sequence was used as a protein homolog for the "exon-stretching" algorithm, a RefSeq identifier (denoted by "NM," "NP," or "NT") may be used in place of the GenBank identifier (*i.e.*, g*BBBBB*).

Alternatively, a prefix identifies component sequences that were hand-edited, predicted from genomic DNA sequences, or derived from a combination of sequence analysis methods. The following Table lists examples of component sequence prefixes and corresponding sequence analysis methods associated with the prefixes (see Example IV and Example V).

| Prefix | Type of analysis and/or examples of programs |
|---|---|

| | |
|---|---|
| GNN, GFG, ENST | Exon prediction from genomic sequences using, for example, GENSCAN (Stanford University, CA, USA) or FGENES (Computer Genomics Group, The Sanger Centre, Cambridge, UK). |
| GBI | Hand-edited analysis of genomic sequences. |
| FL | Stitched or stretched genomic sequences (see Example V). |
| INCY | Full length transcript and exon prediction from mapping of EST sequences to the genome. Genomic location and EST composition data are combined to predict the exons and resulting transcript. |

In some cases, Incyte cDNA coverage redundant with the sequence coverage shown in Table 4 was obtained to confirm the final consensus polynucleotide sequence, but the relevant Incyte cDNA identification numbers are not shown.

Table 5 shows the representative cDNA libraries for those full length polynucleotides which were assembled using Incyte cDNA sequences. The representative cDNA library is the Incyte cDNA library which is most frequently represented by the Incyte cDNA sequences which were used to assemble and confirm the above polynucleotides. The tissues and vectors which were used to construct the cDNA libraries shown in Table 5 are described in Table 6.

Table 8 shows single nucleotide polymorphisms (SNPs) found in polynucleotide sequences of the invention, along with allele frequencies in different human populations. Columns 1 and 2 show the polynucleotide sequence identification number (SEQ ID NO:) and the corresponding Incyte project identification number (PID) for polynucleotides of the invention. Column 3 shows the Incyte identification number for the EST in which the SNP was detected (EST ID), and column 4 shows the identification number for the SNP (SNP ID). Column 5 shows the position within the EST sequence at which the SNP is located (EST SNP), and column 6 shows the position of the SNP within the full-length polynucleotide sequence (CB1 SNP). Column 7 shows the allele found in the EST sequence. Columns 8 and 9 show the two alleles found at the SNP site. Column 10 shows the amino acid encoded by the codon including the SNP site, based upon the allele found in the EST. Columns 11-14 show the frequency of allele 1 in four different human populations. An entry of n/d (not detected) indicates that the frequency of allele 1 in the population was too low to be detected, while n/a (not available) indicates that the allele frequency was not determined for the population.

The invention also encompasses CADECM variants. Various embodiments of CADECM variants can have at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 90%, at least about 91%, at least about 92%, at least about 93%, at least about 94%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, or at least about 99% amino acid sequence identity to the CADECM amino acid sequence, and can contain at least one functional or structural characteristic of CADECM.

Various embodiments also encompass polynucleotides which encode CADECM. In a particular embodiment, the invention encompasses a polynucleotide sequence comprising a sequence selected from the group consisting of SEQ ID NO:31-60, which encodes CADECM. The polynucleotide sequences of SEQ ID NO:31-60, as presented in the Sequence Listing, embrace the equivalent RNA sequences, wherein occurrences of the nitrogenous base thymine are replaced with uracil, and the sugar backbone is composed of ribose instead of deoxyribose.

The invention also encompasses variants of a polynucleotide encoding CADECM. In particular, such a variant polynucleotide will have at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 90%, at least about 91%, at least about 92%, at least about 93%, at least about 94%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, or at least about 99% polynucleotide sequence identity to a polynucleotide encoding CADECM. A particular aspect of the invention encompasses a variant of a polynucleotide comprising a sequence selected from the group consisting of SEQ ID NO:31-60 which has at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 90%, at least about 91%, at least about 92%, at least about 93%, at least about 94%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, or at least about 99% polynucleotide sequence identity to a nucleic acid sequence selected from the group consisting of SEQ ID NO:31-60. Any one of the polynucleotide variants described above can encode a polypeptide which contains at least one functional or structural characteristic of CADECM.

In addition, or in the alternative, a polynucleotide variant of the invention is a splice variant of a polynucleotide encoding CADECM. A splice variant may have portions which have significant sequence identity to a polynucleotide encoding CADECM, but will generally have a greater or lesser number of nucleotides due to additions or deletions of blocks of sequence arising from alternate splicing during mRNA processing. A splice variant may have less than about 70%, or alternatively less than about 60%, or alternatively less than about 50% polynucleotide sequence identity to a polynucleotide encoding CADECM over its entire length; however, portions of the splice variant will have at least about 70%, or alternatively at least about 85%, or alternatively at least about 95%, or alternatively 100% polynucleotide sequence identity to portions of the polynucleotide encoding CADECM. For example, a polynucleotide comprising a sequence of SEQ ID NO:31 is a splice variant of a polynucleotide comprising a sequence of SEQ ID NO:35; and a polynucleotide comprising a sequence of SEQ ID NO:33 and a polynucleotide comprising a sequence of SEQ ID NO:34 are splice variants of one other; a polynucleotide comprising a sequence of SEQ ID NO:32, a polynucleotide comprising a sequence of SEQ ID NO:47 and a polynucleotide comprising a sequence of SEQ ID NO:50 are splice variants of each other; a polynucleotide comprising a sequence of SEQ

ID NO:42 and a polynucleotide comprising a sequence of SEQ ID NO:57 are splice variants of each other; a polynucleotide comprising a sequence of SEQ ID NO:48 and a polynucleotide comprising a sequence of SEQ ID NO:51 are splice variants of each other; and a polynucleotide comprising a sequence of SEQ ID NO:52, a polynucleotide comprising a sequence of SEQ ID NO:53 and a polynucleotide comprising a sequence of SEQ ID NO:55 are splice variants of each other; and a polynucleotide comprising a sequence of SEQ ID NO:59 and a polynucleotide comprising a sequence of SEQ ID NO:60 are splice variants of each other. Any one of the splice variants described above can encode a polypeptide which contains at least one functional or structural characteristic of CADECM. Any one of the splice variants described above can encode a polypeptide which contains at least one functional or structural characteristic of CADECM.

It will be appreciated by those skilled in the art that as a result of the degeneracy of the genetic code, a multitude of polynucleotide sequences encoding CADECM, some bearing minimal similarity to the polynucleotide sequences of any known and naturally occurring gene, may be produced. Thus, the invention contemplates each and every possible variation of polynucleotide sequence that could be made by selecting combinations based on possible codon choices. These combinations are made in accordance with the standard triplet genetic code as applied to the polynucleotide sequence of naturally occurring CADECM, and all such variations are to be considered as being specifically disclosed.

Although polynucleotides which encode CADECM and its variants are generally capable of hybridizing to polynucleotides encoding naturally occurring CADECM under appropriately selected conditions of stringency, it may be advantageous to produce polynucleotides encoding CADECM or its derivatives possessing a substantially different codon usage, e.g., inclusion of non-naturally occurring codons. Codons may be selected to increase the rate at which expression of the peptide occurs in a particular prokaryotic or eukaryotic host in accordance with the frequency with which particular codons are utilized by the host. Other reasons for substantially altering the nucleotide sequence encoding CADECM and its derivatives without altering the encoded amino acid sequences include the production of RNA transcripts having more desirable properties, such as a greater half-life, than transcripts produced from the naturally occurring sequence.

The invention also encompasses production of polynucleotides which encode CADECM and CADECM derivatives, or fragments thereof, entirely by synthetic chemistry. After production, the synthetic polynucleotide may be inserted into any of the many available expression vectors and cell systems using reagents well known in the art. Moreover, synthetic chemistry may be used to introduce mutations into a polynucleotide encoding CADECM or any fragment thereof.

Embodiments of the invention can also include polynucleotides that are capable of hybridizing to the claimed polynucleotides, and, in particular, to those having the sequences shown in

SEQ ID NO:31-60 and fragments thereof, under various conditions of stringency (Wahl, G.M. and S.L. Berger (1987) Methods Enzymol. 152:399-407; Kimmel, A.R. (1987) Methods Enzymol. 152:507-511). Hybridization conditions, including annealing and wash conditions, are described in "Definitions."

Methods for DNA sequencing are well known in the art and may be used to practice any of the embodiments of the invention. The methods may employ such enzymes as the Klenow fragment of DNA polymerase I, SEQUENASE (US Biochemical, Cleveland OH), Taq polymerase (Applied Biosystems), thermostable T7 polymerase (Amersham Biosciences, Piscataway NJ), or combinations of polymerases and proofreading exonucleases such as those found in the ELONGASE amplification system (Invitrogen, Carlsbad CA). Preferably, sequence preparation is automated with machines such as the MICROLAB 2200 liquid transfer system (Hamilton, Reno NV), PTC200 thermal cycler (MJ Research, Watertown MA) and ABI CATALYST 800 thermal cycler (Applied Biosystems). Sequencing is then carried out using either the ABI 373 or 377 DNA sequencing system (Applied Biosystems), the MEGABACE 1000 DNA sequencing system (Amersham Biosciences), or other systems known in the art. The resulting sequences are analyzed using a variety of algorithms which are well known in the art (Ausubel et al., *supra*, ch. 7; Meyers, R.A. (1995) Molecular Biology and Biotechnology, Wiley VCH, New York NY, pp. 856-853).

The nucleic acids encoding CADECM may be extended utilizing a partial nucleotide sequence and employing various PCR-based methods known in the art to detect upstream sequences, such as promoters and regulatory elements. For example, one method which may be employed, restriction-site PCR, uses universal and nested primers to amplify unknown sequence from genomic DNA within a cloning vector (Sarkar, G. (1993) PCR Methods Applic. 2:318-322). Another method, inverse PCR, uses primers that extend in divergent directions to amplify unknown sequence from a circularized template. The template is derived from restriction fragments comprising a known genomic locus and surrounding sequences (Triglia, T. et al. (1988) Nucleic Acids Res. 16:8186). A third method, capture PCR, involves PCR amplification of DNA fragments adjacent to known sequences in human and yeast artificial chromosome DNA (Lagerstrom, M. et al. (1991) PCR Methods Applic. 1:111-119). In this method, multiple restriction enzyme digestions and ligations may be used to insert an engineered double-stranded sequence into a region of unknown sequence before performing PCR. Other methods which may be used to retrieve unknown sequences are known in the art (Parker, J.D. et al. (1991) Nucleic Acids Res. 19:3055-3060). Additionally, one may use PCR, nested primers, and PROMOTERFINDER libraries (BD Clontech, Palo Alto CA) to walk genomic DNA. This procedure avoids the need to screen libraries and is useful in finding intron/exon junctions. For all PCR-based methods, primers may be designed using commercially available software, such as OLIGO 4.06 primer analysis software (National Biosciences, Plymouth MN) or another appropriate program, to be about 22 to 30 nucleotides in length, to have a GC content of about 50% or more, and to anneal to the template at temperatures of about 68°C to 72°C.

When screening for full length cDNAs, it is preferable to use libraries that have been size-selected to include larger cDNAs. In addition, random-primed libraries, which often include sequences containing the 5' regions of genes, are preferable for situations in which an oligo d(T) library does not yield a full-length cDNA. Genomic libraries may be useful for extension of sequence into 5' non-transcribed regulatory regions.

Capillary electrophoresis systems which are commercially available may be used to analyze the size or confirm the nucleotide sequence of sequencing or PCR products. In particular, capillary sequencing may employ flowable polymers for electrophoretic separation, four different nucleotide-specific, laser-stimulated fluorescent dyes, and a charge coupled device camera for detection of the emitted wavelengths. Output/light intensity may be converted to electrical signal using appropriate software (e.g., GENOTYPER and SEQUENCE NAVIGATOR, Applied Biosystems), and the entire process from loading of samples to computer analysis and electronic data display may be computer controlled. Capillary electrophoresis is especially preferable for sequencing small DNA fragments which may be present in limited amounts in a particular sample.

In another embodiment of the invention, polynucleotides or fragments thereof which encode CADECM may be cloned in recombinant DNA molecules that direct expression of CADECM, or fragments or functional equivalents thereof, in appropriate host cells. Due to the inherent degeneracy of the genetic code, other polynucleotides which encode substantially the same or a functionally equivalent polypeptides may be produced and used to express CADECM.

The polynucleotides of the invention can be engineered using methods generally known in the art in order to alter CADECM-encoding sequences for a variety of purposes including, but not limited to, modification of the cloning, processing, and/or expression of the gene product. DNA shuffling by random fragmentation and PCR reassembly of gene fragments and synthetic oligonucleotides may be used to engineer the nucleotide sequences. For example, oligonucleotide-mediated site-directed mutagenesis may be used to introduce mutations that create new restriction sites, alter glycosylation patterns, change codon preference, produce splice variants, and so forth.

The nucleotides of the present invention may be subjected to DNA shuffling techniques such as MOLECULARBREEDING (Maxygen Inc., Santa Clara CA; described in U.S. Patent No. 5,837,458; Chang, C.-C. et al. (1999) Nat. Biotechnol. 17:793-797; Christians, F.C. et al. (1999) Nat. Biotechnol. 17:259-264; and Crameri, A. et al. (1996) Nat. Biotechnol. 14:315-319) to alter or improve the biological properties of CADECM, such as its biological or enzymatic activity or its ability to bind to other molecules or compounds. DNA shuffling is a process by which a library of gene variants is produced using PCR-mediated recombination of gene fragments. The library is then subjected to selection or screening procedures that identify those gene variants with the desired properties. These preferred variants may then be pooled and further subjected to recursive rounds of DNA shuffling and selection/screening. Thus, genetic diversity is created through "artificial" breeding and rapid molecular evolution. For example, fragments of a single gene containing random point mutations may be recombined, screened, and then reshuffled until the desired properties are optimized. Alternatively, fragments of a given gene may be recombined with fragments of homologous genes in the same gene family, either from the same or different species, thereby maximizing the genetic diversity of multiple naturally occurring genes in a directed and controllable manner.

In another embodiment, polynucleotides encoding CADECM may be synthesized, in whole or in part, using one or more chemical methods well known in the art (Caruthers, M.H. et al. (1980) Nucleic Acids Symp. Ser. 7:215-223; Horn, T. et al. (1980) Nucleic Acids Symp. Ser. 7:225-232). Alternatively, CADECM itself or a fragment thereof may be synthesized using chemical methods known in the art. For example, peptide synthesis can be performed using various solution-phase or solid-phase techniques (Creighton, T. (1984) Proteins, Structures and Molecular Properties, WH Freeman, New York NY, pp. 55-60; Roberge, J.Y. et al. (1995) Science 269:202-204). Automated synthesis may be achieved using the ABI 431A peptide synthesizer (Applied Biosystems). Additionally, the amino acid sequence of CADECM, or any part thereof, may be altered during direct synthesis and/or combined with sequences from other proteins, or any part thereof, to produce a variant polypeptide or a polypeptide having a sequence of a naturally occurring polypeptide.

The peptide may be substantially purified by preparative high performance liquid chromatography (Chiez, R.M. and F.Z. Regnier (1990) Methods Enzymol. 182:392-421). The composition of the synthetic peptides may be confirmed by amino acid analysis or by sequencing (Creighton, *supra*, pp. 28-53).

In order to express a biologically active CADECM, the polynucleotides encoding CADECM or derivatives thereof may be inserted into an appropriate expression vector, i.e., a vector which contains the necessary elements for transcriptional and translational control of the inserted coding sequence in a suitable host. These elements include regulatory sequences, such as enhancers, constitutive and inducible promoters, and 5' and 3' untranslated regions in the vector and in polynucleotides encoding CADECM. Such elements may vary in their strength and specificity. Specific initiation signals may also be used to achieve more efficient translation of polynucleotides encoding CADECM. Such signals include the ATG initiation codon and adjacent sequences, e.g. the Kozak sequence. In cases where a polynucleotide sequence encoding CADECM and its initiation codon and upstream regulatory sequences are inserted into the appropriate expression vector, no additional transcriptional or translational control signals may be needed. However, in cases where only coding sequence, or a fragment thereof, is inserted, exogenous translational control signals including an in-frame ATG initiation codon should be provided by the vector. Exogenous translational elements and initiation codons may be of various origins, both natural and synthetic. The efficiency of expression may be enhanced by the inclusion of enhancers appropriate for the particular host cell system used (Scharf, D. et al. (1994) Results Probl. Cell Differ. 20:125-162).

Methods which are well known to those skilled in the art may be used to construct expression vectors containing polynucleotides encoding CADECM and appropriate transcriptional and translational control elements. These methods include *in vitro* recombinant DNA techniques, synthetic techniques, and *in vivo* genetic recombination (Sambrook and Russell, *supra*, ch. 1-4, and 8; Ausubel et al., *supra*, ch. 1, 3, and 15).

A variety of expression vector/host systems may be utilized to contain and express polynucleotides encoding CADECM. These include, but are not limited to, microorganisms such as bacteria transformed with recombinant bacteriophage, plasmid, or cosmid DNA expression vectors; yeast transformed with yeast expression vectors; insect cell systems infected with viral expression vectors (e.g., baculovirus); plant cell systems transformed with viral expression vectors (e.g., cauliflower mosaic virus, CaMV, or tobacco mosaic virus, TMV) or with bacterial expression vectors (e.g., Ti or pBR322 plasmids); or animal cell systems (Sambrook and Russell, *supra*; Ausubel et al., *supra*; Van Heeke, G. and S.M. Schuster (1989) J. Biol. Chem. 264:5503-5509; Engelhard, E.K. et al. (1994) Proc. Natl. Acad. Sci. USA 91:3224-3227; Sandig, V. et al. (1996) Hum. Gene Ther. 7:1937-1945; Takamatsu, N. (1987) EMBO J. 6:307-311; <u>The McGraw Hill Yearbook of Science and Technology</u> (1992) McGraw Hill, New York NY, pp. 191-196; Logan, J. and T. Shenk (1984) Proc. Natl. Acad. Sci. USA 81:3655-3659; Harrington, J.J. et al. (1997) Nat. Genet. 15:345-355). Expression vectors derived from retroviruses, adenoviruses, or herpes or vaccinia viruses, or from various bacterial plasmids, may be used for delivery of polynucleotides to the targeted organ, tissue, or cell population (Di Nicola, M. et al. (1998) Cancer Gen. Ther. 5:350-356; Yu, M. et al. (1993) Proc. Natl. Acad. Sci. USA 90:6340-6344; Buller, R.M. et al. (1985) Nature 317:813-815; McGregor, D.P. et al. (1994) Mol. Immunol. 31:219-226; Verma, I.M. and N. Somia (1997) Nature 389:239-242). The invention is not limited by the host cell employed.

In bacterial systems, a number of cloning and expression vectors may be selected depending upon the use intended for polynucleotides encoding CADECM. For example, routine cloning, subcloning, and propagation of polynucleotides encoding CADECM can be achieved using a multifunctional *E. coli* vector such as PBLUESCRIPT (Stratagene, La Jolla CA) or PSPORT1 plasmid (Invitrogen). Ligation of polynucleotides encoding CADECM into the vector's multiple cloning site disrupts the *lacZ* gene, allowing a colorimetric screening procedure for identification of transformed bacteria containing recombinant molecules. In addition, these vectors may be useful for *in vitro* transcription, dideoxy sequencing, single strand rescue with helper phage, and creation of nested deletions in the cloned sequence (Van Heeke, G. and S.M. Schuster (1989) J. Biol. Chem. 264:5503-5509). When large quantities of CADECM are needed, e.g. for the production of antibodies, vectors which direct high level expression of CADECM may be used. For example, vectors containing the strong, inducible SP6 or T7 bacteriophage promoter may be used.

Yeast expression systems may be used for production of CADECM. A number of vectors containing constitutive or inducible promoters, such as alpha factor, alcohol oxidase, and PGH promoters, may be used in the yeast *Saccharomyces cerevisiae* or *Pichia pastoris*. In addition, such vectors direct either the secretion or intracellular retention of expressed proteins and enable integration of foreign polynucleotide sequences into the host genome for stable propagation (Ausubel et al., *supra*; Bitter, G.A. et al. (1987) Methods Enzymol. 153:516-544; Scorer, C.A. et al. (1994) Bio/Technology 12:181-184).

Plant systems may also be used for expression of CADECM. Transcription of polynucleotides encoding CADECM may be driven by viral promoters, e.g., the 35S and 19S promoters of CaMV used alone or in combination with the omega leader sequence from TMV (Takamatsu, N. (1987) EMBO J. 6:307-311). Alternatively, plant promoters such as the small subunit of RUBISCO or heat shock promoters may be used (Coruzzi, G. et al. (1984) EMBO J. 3:1671-1680; Broglie, R. et al. (1984) Science 224:838-843; Winter, J. et al. (1991) Results Probl. Cell Differ. 17:85-105). These constructs can be introduced into plant cells by direct DNA transformation or pathogen-mediated transfection (The McGraw Hill Yearbook of Science and Technology (1992) McGraw Hill, New York NY, pp. 191-196).

In mammalian cells, a number of viral-based expression systems may be utilized. In cases where an adenovirus is used as an expression vector, polynucleotides encoding CADECM may be ligated into an adenovirus transcription/translation complex consisting of the late promoter and tripartite leader sequence. Insertion in a non-essential E1 or E3 region of the viral genome may be used to obtain infective virus which expresses CADECM in host cells (Logan, J. and T. Shenk (1984) Proc. Natl. Acad. Sci. USA 81:3655-3659). In addition, transcription enhancers, such as the Rous sarcoma virus (RSV) enhancer, may be used to increase expression in mammalian host cells. SV40 or EBV-based vectors may also be used for high-level protein expression.

Human artificial chromosomes (HACs) may also be employed to deliver larger fragments of DNA than can be contained in and expressed from a plasmid. HACs of about 6 kb to 10 Mb are constructed and delivered via conventional delivery methods (liposomes, polycationic amino polymers, or vesicles) for therapeutic purposes (Harrington, J.J. et al. (1997) Nat. Genet. 15:345-355).

For long term production of recombinant proteins in mammalian systems, stable expression of CADECM in cell lines is preferred. For example, polynucleotides encoding CADECM can be transformed into cell lines using expression vectors which may contain viral origins of replication and/or endogenous expression elements and a selectable marker gene on the same or on a separate vector. Following the introduction of the vector, cells may be allowed to grow for about 1 to 2 days in enriched media before being switched to selective media. The purpose of the selectable marker is to confer resistance to a selective agent, and its presence allows growth and recovery of cells which successfully express the introduced sequences. Resistant clones of stably transformed cells may be propagated using tissue culture techniques appropriate to the cell type.

Any number of selection systems may be used to recover transformed cell lines. These include, but are not limited to, the herpes simplex virus thymidine kinase and adenine phosphoribosyltransferase genes, for use in *tk*⁻ and *apr*⁻ cells, respectively (Wigler, M. et al. (1977) Cell 11:223-232; Lowy, I. et al. (1980) Cell 22:817-823). Also, antimetabolite, antibiotic, or herbicide resistance can be used as the basis for selection. For example, *dhfr* confers resistance to methotrexate; *neo* confers resistance to the aminoglycosides neomycin and G-418; and *als* and *pat* confer resistance to chlorsulfuron and phosphinotricin acetyltransferase, respectively (Wigler, M. et al. (1980) Proc. Natl. Acad. Sci. USA 77:3567-3570; Colbere-Garapin, F. et al. (1981) J. Mol. Biol. 150:1-14). Additional selectable genes have been described, e.g., *trpB* and *hisD*, which alter cellular requirements for metabolites (Hartman, S.C. and R.C. Mulligan (1988) Proc. Natl. Acad. Sci. USA 85:8047-8051). Visible markers, e.g., anthocyanins, green fluorescent proteins (GFP; BD Clontech), β-glucuronidase and its substrate β-glucuronide, or luciferase and its substrate luciferin may be used. These markers can be used not only to identify transformants, but also to quantify the amount of transient or stable protein expression attributable to a specific vector system (Rhodes, C.A. (1995) Methods Mol. Biol. 55:121-131).

Although the presence/absence of marker gene expression suggests that the gene of interest is also present, the presence and expression of the gene may need to be confirmed. For example, if the sequence encoding CADECM is inserted within a marker gene sequence, transformed cells containing polynucleotides encoding CADECM can be identified by the absence of marker gene function. Alternatively, a marker gene can be placed in tandem with a sequence encoding CADECM under the control of a single promoter. Expression of the marker gene in response to induction or selection usually indicates expression of the tandem gene as well.

In general, host cells that contain the polynucleotide encoding CADECM and that express CADECM may be identified by a variety of procedures known to those of skill in the art. These procedures include, but are not limited to, DNA-DNA or DNA-RNA hybridizations, PCR amplification, and protein bioassay or immunoassay techniques which include membrane, solution, or chip based technologies for the detection and/or quantification of nucleic acid or protein sequences.

Immunological methods for detecting and measuring the expression of CADECM using either specific polyclonal or monoclonal antibodies are known in the art. Examples of such techniques include enzyme-linked immunosorbent assays (ELISAs), radioimmunoassays (RIAs), and fluorescence activated cell sorting (FACS). A two-site, monoclonal-based immunoassay utilizing monoclonal antibodies reactive to two non-interfering epitopes on CADECM is preferred, but a competitive binding assay may be employed. These and other assays are well known in the art (Hampton, R. et al. (1990) Serological Methods, a Laboratory Manual, APS Press, St. Paul MN, Sect. IV; Coligan, J.E. et al. (1997) Current Protocols in Immunology, Greene Pub. Associates and Wiley-Interscience, New York NY; Pound, J.D. (1998) Immunochemical Protocols, Humana Press, Totowa NJ).

A wide variety of labels and conjugation techniques are known by those skilled in the art and may be used in various nucleic acid and amino acid assays. Means for producing labeled hybridization or PCR probes for detecting sequences related to polynucleotides encoding CADECM include oligolabeling, nick translation, end-labeling, or PCR amplification using a labeled nucleotide. Alternatively, polynucleotides encoding CADECM, or any fragments thereof, may be cloned into a vector for the production of an mRNA probe. Such vectors are known in the art, are commercially available, and may be used to synthesize RNA probes *in vitro* by addition of an appropriate RNA polymerase such as T7, T3, or SP6 and labeled nucleotides. These procedures may be conducted using a variety of commercially available kits, such as those provided by Amersham Biosciences, Promega (Madison WI), and US Biochemical. Suitable reporter molecules or labels which may be used for ease of detection include radionuclides, enzymes, fluorescent, chemiluminescent, or chromogenic agents, as well as substrates, cofactors, inhibitors, magnetic particles, and the like.

Host cells transformed with polynucleotides encoding CADECM may be cultured under conditions suitable for the expression and recovery of the protein from cell culture. The protein produced by a transformed cell may be secreted or retained intracellularly depending on the sequence and/or the vector used. As will be understood by those of skill in the art, expression vectors containing polynucleotides which encode CADECM may be designed to contain signal sequences which direct secretion of CADECM through a prokaryotic or eukaryotic cell membrane.

In addition, a host cell strain may be chosen for its ability to modulate expression of the inserted polynucleotides or to process the expressed protein in the desired fashion. Such modifications of the polypeptide include, but are not limited to, acetylation, carboxylation, glycosylation, phosphorylation, lipidation, and acylation. Post-translational processing which cleaves a "prepro" or "pro" form of the protein may also be used to specify protein targeting, folding, and/or activity. Different host cells which have specific cellular machinery and characteristic mechanisms for post-translational activities (e.g., CHO, HeLa, MDCK, HEK293, and WI38) are available from the American Type Culture Collection (ATCC, Manassas VA) and may be chosen to ensure the correct modification and processing of the foreign protein.

In another embodiment of the invention, natural, modified, or recombinant polynucleotides encoding CADECM may be ligated to a heterologous sequence resulting in translation of a fusion protein in any of the aforementioned host systems. For example, a chimeric CADECM protein containing a heterologous moiety that can be recognized by a commercially available antibody may facilitate the screening of peptide libraries for inhibitors of CADECM activity. Heterologous protein and peptide moieties may also facilitate purification of fusion proteins using commercially available affinity matrices. Such moieties include, but are not limited to, glutathione S-transferase (GST), maltose binding protein (MBP), thioredoxin (Trx), calmodulin binding peptide (CBP), 6-His, FLAG, *c-myc*, and hemagglutinin (HA). GST, MBP, Trx, CBP, and 6-His enable purification of their cognate fusion proteins on immobilized glutathione, maltose, phenylarsine oxide, calmodulin, and metal-chelate resins, respectively. FLAG, *c-myc*, and hemagglutinin (HA) enable immunoaffinity purification of fusion proteins using commercially available monoclonal and polyclonal antibodies that specifically recognize these epitope tags. A fusion protein may also be engineered to contain a proteolytic cleavage site located between the CADECM encoding sequence and the heterologous protein sequence, so that CADECM may be cleaved away from the heterologous moiety following purification. Methods for fusion protein expression and purification are discussed in Ausubel et al. (*supra*, ch. 10 and 16). A variety of commercially available kits may also be used to facilitate expression and purification of fusion proteins.

In another embodiment, synthesis of radiolabeled CADECM may be achieved *in vitro* using the TNT rabbit reticulocyte lysate or wheat germ extract system (Promega). These systems couple transcription and translation of protein-coding sequences operably associated with the T7, T3, or SP6 promoters. Translation takes place in the presence of a radiolabeled amino acid precursor, for example, $^{35}$S-methionine.

CADECM, fragments of CADECM, or variants of CADECM may be used to screen for compounds that specifically bind to CADECM. One or more test compounds may be screened for specific binding to CADECM. In various embodiments, 1, 2, 3, 4, 5, 10, 20, 50, 100, or 200 test compounds can be screened for specific binding to CADECM. Examples of test compounds can include antibodies, anticalins, oligonucleotides, proteins (e.g., ligands or receptors), or small molecules.

In related embodiments, variants of CADECM can be used to screen for binding of test compounds, such as antibodies, to CADECM, a variant of CADECM, or a combination of CADECM and/or one or more variants CADECM. In an embodiment, a variant of CADECM can be used to screen for compounds that bind to a variant of CADECM, but not to CADECM having the exact sequence of a sequence of SEQ ID NO:1-30. CADECM variants used to perform such screening can have a range of about 50% to about 99% sequence identity to CADECM, with various embodiments having 60%, 70%, 75%, 80%, 85%, 90%, and 95% sequence identity.

In an embodiment, a compound identified in a screen for specific binding to CADECM can be closely related to the natural ligand of CADECM, e.g., a ligand or fragment thereof, a natural substrate, a structural or functional mimetic, or a natural binding partner (Coligan, J.E. et al. (1991) Current Protocols in Immunology 1(2):Chapter 5). In another embodiment, the compound thus identified can be a natural ligand of a receptor CADECM (Howard, A.D. et al. (2001) Trends Pharmacol. Sci.22:132-140; Wise, A. et al. (2002) Drug Discovery Today 7:235-246).

In other embodiments, a compound identified in a screen for specific binding to CADECM can be closely related to the natural receptor to which CADECM binds, at least a fragment of the receptor, or a fragment of the receptor including all or a portion of the ligand binding site or binding pocket. For example, the compound may be a receptor for CADECM which is capable of propagating a signal, or a decoy receptor for CADECM which is not capable of propagating a signal (Ashkenazi, A. and V.M. Divit (1999) Curr. Opin. Cell Biol. 11:255-260; Mantovani, A. et al. (2001) Trends Immunol. 22:328-336). The compound can be rationally designed using known techniques. Examples of such techniques include those used to construct the compound etanercept (ENBREL; Amgen Inc., Thousand Oaks CA), which is efficacious for treating rheumatoid arthritis in humans. Etanercept is an engineered p75 tumor necrosis factor (TNF) receptor dimer linked to the Fc portion of human $IgG_1$ (Taylor, P.C. et al. (2001) Curr. Opin. Immunol. 13:611-616).

In one embodiment, two or more antibodies having similar or, alternatively, different specificities can be screened for specific binding to CADECM, fragments of CADECM, or variants of CADECM. The binding specificity of the antibodies thus screened can thereby be selected to identify particular fragments or variants of CADECM. In one embodiment, an antibody can be selected such that its binding specificity allows for preferential identification of specific fragments or variants of CADECM. In another embodiment, an antibody can be selected such that its binding specificity allows for preferential diagnosis of a specific disease or condition having increased, decreased, or otherwise abnormal production of CADECM.

In an embodiment, anticalins can be screened for specific binding to CADECM, fragments of CADECM, or variants of CADECM. Anticalins are ligand-binding proteins that have been constructed based on a lipocalin scaffold (Weiss, G.A. and H.B. Lowman (2000) Chem. Biol. 7:R177-R184; Skerra, A. (2001) J. Biotechnol. 74:257-275). The protein architecture of lipocalins can include a beta-barrel having eight antiparallel beta-strands, which supports four loops at its open end. These loops form the natural ligand-binding site of the lipocalins, a site which can be re-engineered *in vitro* by amino acid substitutions to impart novel binding specificities. The amino acid substitutions can be made using methods known in the art or described herein, and can include conservative substitutions (e.g., substitutions that do not alter binding specificity) or substitutions that modestly, moderately, or significantly alter binding specificity.

In one embodiment, screening for compounds which specifically bind to, stimulate, or inhibit CADECM involves producing appropriate cells which express CADECM, either as a secreted protein or on the cell membrane. Preferred cells can include cells from mammals, yeast, *Drosophila*, or *E. coli*. Cells expressing CADECM or cell membrane fractions which contain CADECM are then contacted with a test compound and binding, stimulation, or inhibition of activity of either CADECM or the compound is analyzed.

An assay may simply test binding of a test compound to the polypeptide, wherein binding is detected by a fluorophore, radioisotope, enzyme conjugate, or other detectable label. For example, the assay may comprise the steps of combining at least one test compound with CADECM, either in solution or affixed to a solid support, and detecting the binding of CADECM to the compound. Alternatively, the assay may detect or measure binding of a test compound in the presence of a labeled competitor. Additionally, the assay may be carried out using cell-free preparations, chemical libraries, or natural product mixtures, and the test compound(s) may be free in solution or affixed to a solid support.

An assay can be used to assess the ability of a compound to bind to its natural ligand and/or to inhibit the binding of its natural ligand to its natural receptors. Examples of such assays include radio-labeling assays such as those described in U.S. Patent No. 5,914,236 and U.S. Patent No. 6,372,724. In a related embodiment, one or more amino acid substitutions can be introduced into a polypeptide compound (such as a receptor) to improve or alter its ability to bind to its natural ligands (Matthews, D.J. and J.A. Wells. (1994) Chem. Biol. 1:25-30). In another related embodiment, one or more amino acid substitutions can be introduced into a polypeptide compound (such as a ligand) to improve or alter its ability to bind to its natural receptors (Cunningham, B.C. and J.A. Wells (1991) Proc. Natl. Acad. Sci. USA 88:3407-3411; Lowman, H.B. et al. (1991) J. Biol. Chem. 266:10982-10988).

CADECM, fragments of CADECM, or variants of CADECM may be used to screen for compounds that modulate the activity of CADECM. Such compounds may include agonists, antagonists, or partial or inverse agonists. In one embodiment, an assay is performed under conditions permissive for CADECM activity, wherein CADECM is combined with at least one test compound, and the activity of CADECM in the presence of a test compound is compared with the activity of CADECM in the absence of the test compound. A change in the activity of CADECM in the presence of the test compound is indicative of a compound that modulates the activity of CADECM. Alternatively, a test compound is combined with an *in vitro* or cell-free system comprising CADECM under conditions suitable for CADECM activity, and the assay is performed.

In either of these assays, a test compound which modulates the activity of CADECM may do so indirectly and need not come in direct contact with the test compound. At least one and up to a plurality of test compounds may be screened.

In another embodiment, polynucleotides encoding CADECM or their mammalian homologs may be "knocked out" in an animal model system using homologous recombination in embryonic stem (ES) cells. Such techniques are well known in the art and are useful for the generation of animal models of human disease (see, e.g., U.S. Patent No. 5,175,383 and U.S. Patent No. 5,767,337). For example, mouse ES cells, such as the mouse 129/SvJ cell line, are derived from the early mouse embryo and grown in culture. The ES cells are transformed with a vector containing the gene of interest disrupted by a marker gene, e.g., the neomycin phosphotransferase gene (*neo*; Capecchi, M.R. (1989) Science 244:1288-1292). The vector integrates into the corresponding region of the host genome by homologous recombination. Alternatively, homologous recombination takes place using the Cre-loxP system to knockout a gene of interest in a tissue- or developmental stage-specific manner (Marth, J.D. (1996) Clin. Invest. 97:1999-2002; Wagner, K.U. et al. (1997) Nucleic Acids Res. 25:4323-4330). Transformed ES cells are identified and microinjected into mouse cell blastocysts such as those from the C57BL/6 mouse strain. The blastocysts are surgically transferred to pseudopregnant dams, and the resulting chimeric progeny are genotyped and bred to produce heterozygous or homozygous strains. Transgenic animals thus generated may be tested with potential therapeutic or toxic agents.

Polynucleotides encoding CADECM may also be manipulated *in vitro* in ES cells derived from human blastocysts. Human ES cells have the potential to differentiate into at least eight separate cell lineages including endoderm, mesoderm, and ectodermal cell types. These cell lineages differentiate into, for example, neural cells, hematopoietic lineages, and cardiomyocytes (Thomson, J.A. et al. (1998) Science 282:1145-1147).

Polynucleotides encoding CADECM can also be used to create "knockin" humanized animals (pigs) or transgenic animals (mice or rats) to model human disease. With knockin technology, a region of a polynucleotide encoding CADECM is injected into animal ES cells, and the injected sequence integrates into the animal cell genome. Transformed cells are injected into blastulae, and the blastulae are implanted as described above. Transgenic progeny or inbred lines are studied and treated with potential pharmaceutical agents to obtain information on treatment of a human disease. Alternatively, a mammal inbred to overexpress CADECM, e.g., by secreting CADECM in its milk, may also serve as a convenient source of that protein (Janne, J. et al. (1998) Biotechnol. Annu. Rev. 4:55-74).

**THERAPEUTICS**

Chemical and structural similarity, e.g., in the context of sequences and motifs, exists between regions of CADECM and cell adhesion and extracellular matrix proteins. In addition, examples of tissues expressing CADECM can be found in Table 6 and can also be found in Example XI. Therefore, CADECM appears to play a role in immune system disorders, neurological disorders, developmental disorders, connective tissue disorders, and cell proliferative disorders, including cancer. In the treatment of disorders associated with increased CADECM expression or activity, it is desirable to decrease the expression or activity of CADECM. In the treatment of disorders associated with decreased CADECM expression or activity, it is desirable to increase the expression or activity of CADECM.

Therefore, in one embodiment, CADECM or a fragment or derivative thereof may be administered to a subject to treat or prevent a disorder associated with decreased expression or activity of CADECM. Examples of such disorders include, but are not limited to, an immune system disorder, such as acquired immunodeficiency syndrome (AIDS), X-linked agammaglobinemia of Bruton, common variable immunodeficiency (CVI), DiGeorge's syndrome (thymic hypoplasia), thymic dysplasia, isolated IgA deficiency, severe combined immunodeficiency disease (SCID), immunodeficiency with thrombocytopenia and eczema (Wiskott-Aldrich syndrome), Chediak-Higashi syndrome, chronic granulomatous diseases, hereditary angioneurotic edema, immunodeficiency associated with Cushing's disease, Addison's disease, adult respiratory distress syndrome, allergies, ankylosing spondylitis, amyloidosis, anemia, asthma, atherosclerosis, autoimmune hemolytic anemia, autoimmune thyroiditis, autoimmune polyendocrinopathy-candidiasis-ectodermal dystrophy (APECED), bronchitis, cholecystitis, contact dermatitis, Crohn's disease, atopic dermatitis, dermatomyositis, diabetes mellitus, emphysema, episodic lymphopenia with lymphocytotoxins, erythroblastosis fetalis, erythema nodosum, atrophic gastritis, glomerulonephritis, Goodpasture's syndrome, gout, Graves' disease, Hashimoto's thyroiditis, hypereosinophilia, irritable bowel syndrome, multiple sclerosis, myasthenia gravis, myocardial or pericardial inflammation, osteoarthritis, osteoporosis, pancreatitis, polymyositis, psoriasis, Reiter's syndrome, rheumatoid arthritis, scleroderma, Sjögren's syndrome, systemic anaphylaxis, systemic lupus erythematosus, systemic sclerosis, thrombocytopenic purpura, ulcerative colitis, uveitis, Werner syndrome, complications of cancer, hemodialysis, and extracorporeal circulation, viral, bacterial, fungal, parasitic, protozoal, and helminthic infections, and trauma; a neurological disorder, such as epilepsy, ischemic cerebrovascular disease, stroke, cerebral neoplasms, Alzheimer's disease, Pick's disease, Huntington's disease, dementia, Parkinson's disease and other extrapyramidal disorders, amyotrophic lateral sclerosis and other motor neuron disorders, progressive neural muscular atrophy, retinitis pigmentosa, hereditary ataxias, multiple sclerosis and other demyelinating diseases, bacterial and viral meningitis, brain abscess, subdural empyema, epidural abscess, suppurative intracranial thrombophlebitis, myelitis and radiculitis, viral central nervous system disease, prion diseases including kuru, Creutzfeldt-Jakob disease, and Gerstmann-Straussler-Scheinker syndrome, fatal familial insomnia, nutritional and metabolic diseases of the nervous system, neurofibromatosis, tuberous sclerosis, cerebelloretinal hemangioblastomatosis, encephalotrigeminal syndrome, mental retardation and other developmental disorders of the central nervous system including Down syndrome, cerebral palsy, neuroskeletal disorders, autonomic nervous system disorders, cranial nerve disorders, spinal cord diseases, muscular dystrophy and other neuromuscular disorders, peripheral nervous system disorders, dermatomyositis and polymyositis, inherited, metabolic, endocrine, and toxic myopathies, myasthenia gravis, periodic paralysis, mental disorders including mood, anxiety, and schizophrenic disorders, seasonal affective disorder (SAD), akathesia, amnesia, catatonia, diabetic neuropathy, tardive dyskinesia, dystonias, paranoid psychoses, postherpetic neuralgia, Tourette's disorder, progressive supranuclear palsy, corticobasal degeneration, and familial frontotemporal dementia; a developmental disorder, such as renal tubular acidosis, anemia, Cushing's syndrome, achondroplastic dwarfism, Duchenne and Becker muscular dystrophy, epilepsy, gonadal dysgenesis, WAGR syndrome (Wilms' tumor, aniridia, genitourinary abnormalities, and mental retardation), Smith-Magenis syndrome, myelodysplastic syndrome, hereditary mucoepithelial dysplasia, hereditary keratodermas, hereditary neuropathies such as Charcot-Marie-Tooth disease and neurofibromatosis, hypothyroidism, hydrocephalus, seizure disorders such as Syndenham's chorea and cerebral palsy, spina bifida, anencephaly, craniorachischisis, congenital glaucoma, cataract, and sensorineural hearing loss; a connective tissue disorder, such as osteogenesis imperfecta, Ehlers-Danlos syndrome, chondrodysplasias, Marfan syndrome, Alport syndrome, familial aortic aneurysm, achondroplasia, mucopolysaccharidoses, osteoporosis, osteopetrosis, Paget's disease, rickets, osteomalacia, hyperparathyroidism, renal osteodystrophy, osteonecrosis, osteomyelitis, osteoma, osteoid osteoma, osteoblastoma, osteosarcoma, osteochondroma, chondroma, chondroblastoma, chondromyxoid fibroma, chondrosarcoma, fibrous cortical defect, nonossifying fibroma, fibrous dysplasia, fibrosarcoma, malignant fibrous histiocytoma, Ewing's sarcoma, primitive neuroectodermal tumor, giant cell tumor, osteoarthritis, rheumatoid arthritis, ankylosing spondyloarthritis, Reiter's syndrome, psoriatic arthritis, enteropathic arthritis, infectious arthritis, gout, gouty arthritis, calcium pyrophosphate crystal deposition disease, ganglion, synovial cyst, villonodular synovitis, systemic sclerosis, Dupuytren's contracture, hepatic fibrosis, lupus erythematosus, mixed connective tissue disease, epidermolysis bullosa simplex, bullous congenital ichthyosiform erythroderma (epidermolytic hyperkeratosis), non-epidermolytic and epidermolytic palmoplantar keratoderma, ichthyosis bullosa of Siemens, pachyonychia congenita, and white sponge nevus; and a cell proliferative disorder, such as actinic keratosis, arteriosclerosis, atherosclerosis, bursitis, cirrhosis, hepatitis, mixed connective tissue disease (MCTD), myelofibrosis, paroxysmal nocturnal hemoglobinuria, polycythemia vera, psoriasis, primary thrombocythemia, and cancers including adenocarcinoma, leukemia, lymphoma, melanoma, myeloma, sarcoma, teratocarcinoma, and, in particular, cancers of the adrenal gland, bladder, bone, bone marrow, brain, breast, cervix, colon, gall bladder, ganglia, gastrointestinal tract, heart, kidney, liver, lung, muscle, ovary, pancreas, parathyroid, penis, prostate, salivary glands, skin, spleen, testis, thymus, thyroid, and uterus.

In another embodiment, a vector capable of expressing CADECM or a fragment or derivative thereof may be administered to a subject to treat or prevent a disorder associated with decreased expression or activity of CADECM including, but not limited to, those described above.

In a further embodiment, a composition comprising a substantially purified CADECM in conjunction with a suitable pharmaceutical carrier may be administered to a subject to treat or prevent a disorder associated with decreased expression or activity of CADECM including, but not limited to, those provided above.

In still another embodiment, an agonist which modulates the activity of CADECM may be administered to a subject to treat or prevent a disorder associated with decreased expression or activity of CADECM including, but not limited to, those listed above.

In a further embodiment, an antagonist of CADECM may be administered to a subject to treat or prevent a disorder associated with increased expression or activity of CADECM. Examples of such disorders include, but are not limited to, those immune system disorders, neurological disorders, developmental disorders, connective tissue disorders, and cell proliferative disorders, including cancer described above. In one aspect, an antibody which specifically binds CADECM may be used directly as an antagonist or indirectly as a targeting or delivery mechanism for bringing a pharmaceutical agent to cells or tissues which express CADECM.

In an additional embodiment, a vector expressing the complement of the polynucleotide encoding CADECM may be administered to a subject to treat or prevent a disorder associated with increased expression or activity of CADECM including, but not limited to, those described above.

In other embodiments, any protein, agonist, antagonist, antibody, complementary sequence, or vector embodiments may be administered in combination with other appropriate therapeutic agents. Selection of the appropriate agents for use in combination therapy may be made by one of ordinary skill in the art, according to conventional pharmaceutical principles. The combination of therapeutic agents may act synergistically to effect the treatment or prevention of the various disorders described above. Using this approach, one may be able to achieve therapeutic efficacy with lower dosages of each agent, thus reducing the potential for adverse side effects.

An antagonist of CADECM may be produced using methods which are generally known in the art. In particular, purified CADECM may be used to produce antibodies or to screen libraries of pharmaceutical agents to identify those which specifically bind CADECM. Antibodies to CADECM may also be generated using methods that are well known in the art. Such antibodies may include, but are not limited to, polyclonal, monoclonal, chimeric, and single chain antibodies, Fab fragments, and fragments produced by a Fab expression library. In an embodiment, neutralizing antibodies (i.e., those which inhibit dimer formation) can be used therapeutically. Single chain antibodies (e.g., from camels or llamas) may be potent enzyme inhibitors and may have application in the design of peptide mimetics, and in the development of immuno-adsorbents and biosensors (Muyldermans, S. (2001) J. Biotechnol. 74:277-302).

For the production of antibodies, various hosts including goats, rabbits, rats, mice, camels, dromedaries, llamas, humans, and others may be immunized by injection with CADECM or with any fragment or oligopeptide thereof which has immunogenic properties. Depending on the host species, various adjuvants may be used to increase immunological response. Such adjuvants include, but are not limited to, Freund's, mineral gels such as aluminum hydroxide, and surface active substances such as lysolecithin, pluronic polyols, polyanions, peptides, oil emulsions, KLH, and dinitrophenol. Among adjuvants used in humans, BCG (bacilli Calmette-Guerin) and *Corynebacterium parvum* are especially preferable.

It is preferred that the oligopeptides, peptides, or fragments used to induce antibodies to CADECM have an amino acid sequence consisting of at least about 5 amino acids, and generally will consist of at least about 10 amino acids. It is also preferable that these oligopeptides, peptides, or fragments are substantially identical to a portion of the amino acid sequence of the natural protein. Short stretches of CADECM amino acids may be fused with those of another protein, such as KLH, and antibodies to the chimeric molecule may be produced.

Monoclonal antibodies to CADECM may be prepared using any technique which provides for the production of antibody molecules by continuous cell lines in culture. These include, but are not limited to, the hybridoma technique, the human B-cell hybridoma technique, and the EBV-hybridoma technique (Kohler, G. et al. (1975) Nature 256:495-497; Kozbor, D. et al. (1985) J. Immunol. Methods 81:31-42; Cote, R.J. et al. (1983) Proc. Natl. Acad. Sci. USA 80:2026-2030; Cole, S.P. et al. (1984) Mol. Cell Biol. 62:109-120).

In addition, techniques developed for the production of "chimeric antibodies," such as the splicing of mouse antibody genes to human antibody genes to obtain a molecule with appropriate antigen specificity and biological activity, can be used (Morrison, S.L. et al. (1984) Proc. Natl. Acad. Sci. USA 81:6851-6855; Neuberger, M.S. et al. (1984) Nature 312:604-608; Takeda, S. et al. (1985) Nature 314:452-454). Alternatively, techniques described for the production of single chain antibodies may be adapted, using methods known in the art, to produce CADECM-specific single chain antibodies. Antibodies with related specificity, but of distinct idiotypic composition, may be generated by chain shuffling from random combinatorial immunoglobulin libraries (Burton, D.R. (1991) Proc. Natl. Acad. Sci. USA 88:10134-10137).

Antibodies may also be produced by inducing *in vivo* production in the lymphocyte population or by screening immunoglobulin libraries or panels of highly specific binding reagents as disclosed in the literature (Orlandi, R. et al. (1989) Proc. Natl. Acad. Sci. USA 86:3833-3837; Winter, G. et al. (1991) Nature 349:293-299).

Antibody fragments which contain specific binding sites for CADECM may also be generated. For example, such fragments include, but are not limited to, $F(ab')_2$ fragments produced by pepsin digestion of the antibody molecule and Fab fragments generated by reducing the disulfide bridges of the F(ab')2 fragments. Alternatively, Fab expression libraries may be constructed to allow rapid and easy identification of monoclonal Fab fragments with the desired specificity (Huse, W.D. et al. (1989) Science 246:1275-1281).

Various immunoassays may be used for screening to identify antibodies having the desired specificity. Numerous protocols for competitive binding or immunoradiometric assays using either polyclonal or monoclonal antibodies with established specificities are well known in the art. Such immunoassays typically involve the measurement of complex formation between CADECM and its specific antibody. A two-site, monoclonal-based immunoassay utilizing monoclonal antibodies reactive to two non-interfering CADECM epitopes is generally used, but a competitive binding assay may also be employed (Pound, *supra*).

Various methods such as Scatchard analysis in conjunction with radioimmunoassay techniques may be used to assess the affinity of antibodies for CADECM. Affinity is expressed as an association constant, $K_a$, which is defined as the molar concentration of CADECM-antibody complex divided by the molar concentrations of free antigen and free antibody under equilibrium conditions. The $K_a$ determined for a preparation of polyclonal antibodies, which are heterogeneous in their affinities for multiple CADECM epitopes, represents the average affinity, or avidity, of the antibodies for CADECM. The $K_a$ determined for a preparation of monoclonal antibodies, which are monospecific for a particular CADECM epitope, represents a true measure of affinity. High-affinity antibody preparations with $K_a$ ranging from about $10^9$ to $10^{12}$ L/mole are preferred for use in immunoassays in which the CADECM-antibody complex must withstand rigorous manipulations. Low-affinity antibody preparations with $K_a$ ranging from about $10^6$ to $10^7$ L/mole are preferred for use in immunopurification and similar procedures which ultimately require dissociation of CADECM, preferably in active form, from the antibody (Catty, D. (1988) Antibodies, Volume I: A Practical Approach, IRL Press, Washington DC; Liddell, J.E. and A. Cryer (1991) A Practical Guide to Monoclonal Antibodies, John Wiley & Sons, New York NY).

The titer and avidity of polyclonal antibody preparations may be further evaluated to determine the quality and suitability of such preparations for certain downstream applications. For example, a polyclonal antibody preparation containing at least 1-2 mg specific antibody/ml, preferably 5-10 mg specific antibody/ml, is generally employed in procedures requiring precipitation of CADECM-antibody complexes. Procedures for evaluating antibody specificity, titer, and avidity, and guidelines for antibody quality and usage in various applications, are generally available (Catty, *supra*; Coligan et al., *supra*).

In another embodiment of the invention, polynucleotides encoding CADECM, or any fragment or complement thereof, may be used for therapeutic purposes. In one aspect, modifications of gene expression can be achieved by designing complementary sequences or antisense molecules (DNA, RNA, PNA, or modified oligonucleotides) to the coding or regulatory regions of the gene encoding CADECM. Such technology is well known in the art, and antisense oligonucleotides or larger fragments can be designed from various locations along the coding or control regions of sequences encoding CADECM (Agrawal, S., ed. (1996) Antisense Therapeutics, Humana Press, Totawa NJ).

In therapeutic use, any gene delivery system suitable for introduction of the antisense sequences into appropriate target cells can be used. Antisense sequences can be delivered intracellularly in the form of an expression plasmid which, upon transcription, produces a sequence complementary to at least a portion of the cellular sequence encoding the target protein (Slater, J.E. et al. (1998) J. Allergy Clin. Immunol. 102:469-475; Scanlon, K.J. et al. (1995) FASEB J. 9:1288-1296). Antisense sequences can also be introduced intracellularly through the use of viral vectors, such as retrovirus and adeno-associated virus vectors (Miller, A.D. (1990) Blood 76:271-278; Ausubel et al., *supra*; Uckert, W. and W. Walther (1994) Pharmacol. Ther. 63:323-347). Other gene delivery mechanisms include liposome-derived systems, artificial viral envelopes, and other systems known in the art (Rossi, J.J. (1995) Br. Med. Bull. 51:217-225; Boado, R.J. et al. (1998) J. Pharm. Sci. 87:1308-1315; Morris, M.C. et al. (1997) Nucleic Acids Res. 25:2730-2736).

In another embodiment of the invention, polynucleotides encoding CADECM may be used for somatic or germline gene therapy. Gene therapy may be performed to (i) correct a genetic deficiency (e.g., in the cases of severe combined immunodeficiency (SCID)-X1 disease characterized by X-linked inheritance (Cavazzana-Calvo, M. et al. (2000) Science 288:669-672), severe combined immunodeficiency syndrome associated with an inherited adenosine deaminase (ADA) deficiency (Blaese, R.M. et al. (1995) Science 270:475-480; Bordignon, C. et al. (1995) Science 270:470-475), cystic fibrosis (Zabner, J. et al. (1993) Cell 75:207-216; Crystal, R.G. et al. (1995) Hum. Gene Therapy 6:643-666; Crystal, R.G. et al. (1995) Hum. Gene Therapy 6:667-703), thalassamias, familial hypercholesterolemia, and hemophilia resulting from Factor VIII or Factor IX deficiencies (Crystal, R.G. (1995) Science 270:404-410; Verma, I.M. and N. Somia (1997) Nature 389:239-242)), (ii) express a conditionally lethal gene product (e.g., in the case of cancers which result from unregulated cell proliferation), or (iii) express a protein which affords protection against intracellular parasites (e.g., against human retroviruses, such as human immunodeficiency virus (HIV) (Baltimore, D. (1988) Nature 335:395-396; Poeschla, E. et al. (1996) Proc. Natl. Acad. Sci. USA 93:11395-11399), hepatitis B or C virus (HBV, HCV); fungal parasites, such as *Candida albicans* and *Paracoccidioides brasiliensis*; and protozoan parasites such as *Plasmodium falciparum* and *Trypanosoma cruzi*). In the case where a genetic deficiency in CADECM expression or regulation causes disease, the expression of CADECM from an appropriate population of transduced cells may alleviate the clinical manifestations caused by the genetic deficiency.

In a further embodiment of the invention, diseases or disorders caused by deficiencies in CADECM are treated by constructing mammalian expression vectors encoding CADECM and introducing these vectors by mechanical means into CADECM-deficient cells. Mechanical transfer technologies for use with cells *in vivo* or *ex vitro* include (i) direct DNA microinjection into individual cells, (ii) ballistic gold particle delivery, (iii) liposome-mediated transfection, (iv) receptor-mediated gene transfer, and (v) the use of DNA transposons (Morgan, R.A. and W.F. Anderson (1993) Annu. Rev. Biochem. 62:191-217; Ivics, Z. (1997) Cell 91:501-510; Boulay, J.-L. and H. Récipon (1998) Curr. Opin. Biotechnol. 9:445-450).

Expression vectors that may be effective for the expression of CADECM include, but are not limited to, the PCDNA 3.1, EPITAG, PRCCMV2, PREP, PVAX, PCR2-TOPOTA vectors (Invitrogen, Carlsbad CA), PCMV-SCRIPT, PCMV-TAG, PEGSH/PERV (Stratagene, La Jolla CA), and PTET-OFF, PTET-ON, PTRE2, PTRE2-LUC, PTK-HYG (BD Clontech, Palo Alto CA). CADECM may be expressed using (i) a constitutively active promoter, (e.g., from cytomegalovirus (CMV), Rous sarcoma virus (RSV), SV40 virus, thymidine kinase (TK), or β-actin genes), (ii) an inducible promoter (e.g., the tetracycline-regulated promoter (Gossen, M. and H. Bujard (1992) Proc. Natl. Acad. Sci. USA 89:5547-5551; Gossen, M. et al. (1995) Science 268:1766-1769; Rossi, F.M.V. and H.M. Blau (1998) Curr. Opin. Biotechnol. 9:451-456), commercially available in the T-REX plasmid (Invitrogen)); the ecdysone-inducible promoter (available in the plasmids PVGRXR and PIND; Invitrogen); the FK506/rapamycin inducible promoter; or the RU486/mifepristone inducible promoter (Rossi, F.M.V. and H.M. Blau, *supra*)), or (iii) a tissue-specific promoter or the native promoter of the endogenous gene encoding CADECM from a normal individual.

Commercially available liposome transformation kits (e.g., the PERFECT LIPID TRANSFECTION KIT, available from Invitrogen) allow one with ordinary skill in the art to deliver polynucleotides to target cells in culture and require minimal effort to optimize experimental parameters. In the alternative, transformation is performed using the calcium phosphate method (Graham, F.L. and A.J. Eb (1973) Virology 52:456-467), or by electroporation (Neumann, E. et al. (1982) EMBO J. 1:841-845). The introduction of DNA to primary cells requires modification of these standardized mammalian transfection protocols.

In another embodiment of the invention, diseases or disorders caused by genetic defects with respect to CADECM expression are treated by constructing a retrovirus vector consisting of (i) the polynucleotide encoding CADECM under the control of an independent promoter or the retrovirus long terminal repeat (LTR) promoter, (ii) appropriate RNA packaging signals, and (iii) a Rev-responsive element (RRE) along with additional retrovirus *cis*-acting RNA sequences and coding sequences required for efficient vector propagation. Retrovirus vectors (e.g., PFB and PFBNEO) are commercially available (Stratagene) and are based on published data (Riviere, I. et al. (1995) Proc. Natl. Acad. Sci. USA 92:6733-6737), incorporated by reference herein. The vector is propagated in an appropriate vector producing cell line (VPCL) that expresses an envelope gene with a tropism for receptors on the target cells or a promiscuous envelope protein such as VSVg (Armentano, D. et al. (1987) J. Virol. 61:1647-1650; Bender, M.A. et al. (1987) J. Virol. 61:1639-1646; Adam, M.A. and A.D. Miller (1988) J. Virol. 62:3802-3806; Dull, T. et al. (1998) J. Virol. 72:8463-8471; Zufferey, R. et al. (1998) J. Virol. 72:9873-9880). U.S. Patent No. 5,910,434 to Rigg ("Method for obtaining retrovirus packaging cell lines producing high transducing efficiency retroviral supernatant") discloses a method for obtaining retrovirus packaging cell lines and is hereby incorporated by reference. Propagation of retrovirus vectors, transduction of a population of cells (e.g., $CD4^+$ T-cells), and the return of transduced cells to a patient are procedures well known to persons skilled in the art of gene therapy and have been well documented (Ranga, U. et al. (1997) J. Virol. 71:7020-7029; Bauer, G. et al. (1997) Blood 89:2259-2267; Bonyhadi, M.L. (1997) J. Virol. 71:4707-4716; Ranga, U. et al. (1998) Proc. Natl. Acad. Sci. USA 95:1201-1206; Su, L. (1997) Blood 89:2283-2290).

In an embodiment, an adenovirus-based gene therapy delivery system is used to deliver polynucleotides encoding CADECM to cells which have one or more genetic abnormalities with respect to the expression of CADECM. The construction and packaging of adenovirus-based vectors are well known to those with ordinary skill in the art. Replication defective adenovirus vectors have proven to be versatile for importing genes encoding immunoregulatory proteins into intact islets in the pancreas (Csete, M.E. et al. (1995) Transplantation 27:263-268). Potentially useful adenoviral vectors are described in U.S. Patent No. 5,707,618 to Armentano ("Adenovirus vectors for gene therapy"), hereby incorporated by reference. For adenoviral vectors, see also Antinozzi, P.A. et al. (1999; Annu. Rev. Nutr. 19:511-544) and Verma, I.M. and N. Somia (1997; Nature 18:389:239-242).

In another embodiment, a herpes-based, gene therapy delivery system is used to deliver polynucleotides encoding CADECM to target cells which have one or more genetic abnormalities with respect to the expression of CADECM. The use of herpes simplex virus (HSV)-based vectors may be especially valuable for introducing CADECM to cells of the central nervous system, for which HSV has a tropism. The construction and packaging of herpes-based vectors are well known to those with ordinary skill in the art. A replication-competent herpes simplex virus (HSV) type 1-based vector has been used to deliver a reporter gene to the eyes of primates (Liu, X. et al. (1999) Exp. Eye Res. 169:385-395). The construction of a HSV-1 virus vector has also been disclosed in detail in U.S. Patent No. 5,804,413 to DeLuca ("Herpes simplex virus strains for gene transfer"), which is hereby incorporated by reference. U.S. Patent No. 5,804,413 teaches the use of recombinant HSV d92 which consists of a genome containing at least one exogenous gene to be transferred to a cell under the control of the appropriate promoter for purposes including human gene therapy. Also taught by this patent are the construction and use of recombinant HSV strains deleted for ICP4, ICP27 and ICP22. For HSV vectors, see also Goins, W.F. et al. (1999; J. Virol. 73:519-532) and Xu, H. et al. (1994; Dev. Biol. 163:152-161). The manipulation of cloned herpesvirus sequences, the generation of recombinant virus following the transfection of multiple plasmids containing different segments of the large herpesvirus genomes, the growth and propagation of herpesvirus, and the infection of cells with herpesvirus are techniques well known to those of ordinary skill in the art.

In another embodiment, an alphavirus (positive, single-stranded RNA virus) vector is used to deliver polynucleotides encoding CADECM to target cells. The biology of the prototypic alphavirus, Semliki Forest Virus (SFV), has been studied extensively and gene transfer vectors have been based on the SFV genome (Garoff, H. and K.-J. Li (1998) Curr. Opin. Biotechnol. 9:464-469). During alphavirus RNA replication, a subgenomic RNA is generated that normally encodes the viral capsid proteins. This subgenomic RNA replicates to higher levels than the full length genomic RNA, resulting in the overproduction of capsid proteins relative to the viral proteins with enzymatic activity (e.g., protease and polymerase). Similarly, inserting the coding sequence for CADECM into the alphavirus genome in place of the capsid-coding region results in the production of a large number of CADECM-coding RNAs and the synthesis of high levels of CADECM in vector transduced cells. While alphavirus infection is typically associated with cell lysis within a few days, the ability to establish a persistent infection in hamster normal kidney cells (BHK-21) with a variant of Sindbis virus (SIN) indicates that the lytic replication of alphaviruses can be altered to suit the needs of the gene therapy application (Dryga, S.A. et al. (1997) Virology 228:74-83). The wide host range of alphaviruses will allow the introduction of CADECM into a variety of cell types. The specific transduction of a subset of cells in a population may require the sorting of cells prior to transduction. The methods of manipulating infectious cDNA clones of alphaviruses, performing alphavirus cDNA and RNA transfections, and performing alphavirus infections, are well known to those with ordinary skill in the art.

Oligonucleotides derived from the transcription initiation site, e.g., between about positions -10 and +10 from the start site, may also be employed to inhibit gene expression. Similarly, inhibition can be achieved using triple helix base-pairing methodology. Triple helix pairing is useful because it causes inhibition of the ability of the double helix to open sufficiently for the binding of polymerases, transcription factors, or regulatory molecules. Recent therapeutic advances using triplex DNA have been described in the literature (Gee, J.E. et al. (1994) in Huber, B.E. and B.I. Carr, Molecular and Immunologic Approaches, Futura Publishing, Mt. Kisco NY, pp. 163-177). A complementary sequence or antisense molecule may also be designed to block translation of mRNA by preventing the transcript from binding to ribosomes.

Ribozymes, enzymatic RNA molecules, may also be used to catalyze the specific cleavage of RNA. The mechanism of ribozyme action involves sequence-specific hybridization of the ribozyme molecule to complementary target RNA, followed by endonucleolytic cleavage. For example, engineered hammerhead motif ribozyme molecules may specifically and efficiently catalyze endonucleolytic cleavage of RNA molecules encoding CADECM.

Specific ribozyme cleavage sites within any potential RNA target are initially identified by scanning the target molecule for ribozyme cleavage sites, including the following sequences: GUA, GUU, and GUC. Once identified, short RNA sequences of between 15 and 20 ribonucleotides, corresponding to the region of the target gene containing the cleavage site, may be evaluated for secondary structural features which may render the oligonucleotide inoperable. The suitability of candidate targets may also be evaluated by testing accessibility to hybridization with complementary oligonucleotides using ribonuclease protection assays.

Complementary ribonucleic acid molecules and ribozymes may be prepared by any method known in the art for the synthesis of nucleic acid molecules. These include techniques for chemically synthesizing oligonucleotides such as solid phase phosphoramidite chemical synthesis. Alternatively, RNA molecules may be generated by *in vitro* and *in vivo* transcription of DNA molecules encoding CADECM. Such DNA sequences may be incorporated into a wide variety of vectors with suitable RNA polymerase promoters such as T7 or SP6. Alternatively, these cDNA constructs that synthesize complementary RNA, constitutively or inducibly, can be introduced into cell lines, cells, or tissues.

RNA molecules may be modified to increase intracellular stability and half-life. Possible modifications include, but are not limited to, the addition of flanking sequences at the 5' and/or 3' ends of the molecule, or the use of phosphorothioate or 2' O-methyl rather than phosphodiesterase linkages within the backbone of the molecule. This concept is inherent in the production of PNAs and can be extended in all of these molecules by the inclusion of nontraditional bases such as inosine, queosine, and wybutosine, as well as acetyl-, methyl-, thio-, and similarly modified forms of adenine, cytosine, guanine, thymine, and uracil which are not as easily recognized by endogenous endonucleases.

In other embodiments of the invention, the expression of one or more selected polynucleotides of the present invention can be altered, inhibited, decreased, or silenced using RNA interference (RNAi) or post-transcriptional gene silencing (PTGS) methods known in the art. RNAi is a post-transcriptional mode of gene silencing in which double-stranded RNA (dsRNA) introduced into a targeted cell specifically suppresses the expression of the homologous gene (i.e., the gene bearing the sequence complementary to the dsRNA). This effectively knocks out or substantially reduces the expression of the targeted gene. PTGS can also be accomplished by use of DNA or DNA fragments as well. RNAi methods are described by Fire, A. et al. (1998; Nature 391:806-811) and Gura, T. (2000; Nature 404:804-808). PTGS can also be initiated by introduction of a complementary segment of DNA into the selected tissue using gene delivery and/or viral vector delivery methods described herein or known in the art.

RNAi can be induced in mammalian cells by the use of small interfering RNA also known as siRNA. siRNA are shorter segments of dsRNA (typically about 21 to 23 nucleotides in length) that result *in vivo* from cleavage of introduced dsRNA by the action of an endogenous ribonuclease. siRNA appear to be the mediators of the RNAi effect in mammals. The most effective siRNAs appear to be 21 nucleotide dsRNAs with 2 nucleotide 3' overhangs. The use of siRNA for inducing RNAi in mammalian cells is described by Elbashir, S.M. et al. (2001; Nature 411:494-498).

siRNA can be generated indirectly by introduction of dsRNA into the targeted cell. Alternatively, siRNA can be synthesized directly and introduced into a cell by transfection methods and agents described herein or known in the art (such as liposome-mediated transfection, viral vector methods, or other polynucleotide delivery/introductory methods). Suitable siRNAs can be selected by examining a transcript of the target polynucleotide (e.g., mRNA) for nucleotide sequences downstream from the AUG start codon and recording the occurrence of each nucleotide and the 3' adjacent 19 to 23 nucleotides as potential siRNA target sites, with sequences having a 21 nucleotide length being preferred. Regions to be avoided for target siRNA sites include the 5' and 3' untranslated regions (UTRs) and regions near the start codon (within 75 bases), as these may be richer in regulatory protein binding sites. UTR-binding proteins and/or translation initiation complexes may interfere with binding of the siRNP endonuclease complex. The selected target sites for siRNA can then be compared to the appropriate genome database (e.g., human, etc.) using BLAST or other sequence comparison algorithms known in the art. Target sequences with significant homology to other coding sequences can be eliminated from consideration. The selected siRNAs can be produced by chemical synthesis methods known in the art or by *in vitro* transcription using commercially available methods and kits such as the SILENCER siRNA construction kit (Ambion, Austin TX).

In alternative embodiments, long-term gene silencing and/or RNAi effects can be induced in selected tissue using expression vectors that continuously express siRNA. This can be accomplished using expression vectors that are engineered to express hairpin RNAs (shRNAs) using methods known in the art (see, e.g., Brummelkamp, T.R. et al. (2002) Science 296:550-553; and Paddison, P.J.

et al. (2002) Genes Dev. 16:948-958). In these and related embodiments, shRNAs can be delivered to target cells using expression vectors known in the art. An example of a suitable expression vector for delivery of siRNA is the PSILENCER1.0-U6 (circular) plasmid (Ambion). Once delivered to the target tissue, shRNAs are processed *in vivo* into siRNA-like molecules capable of carrying out gene-specific silencing.

In various embodiments, the expression levels of genes targeted by RNAi or PTGS methods can be determined by assays for mRNA and/or protein analysis. Expression levels of the mRNA of a targeted gene can be determined, for example, by northern analysis methods using the NORTHERNMAX-GLY kit (Ambion); by microarray methods; by PCR methods; by real time PCR methods; and by other RNA/polynucleotide assays known in the art or described herein. Expression levels of the protein encoded by the targeted gene can be determined, for example, by microarray methods; by polyacrylamide gel electrophoresis; and by Western analysis using standard techniques known in the art.

An additional embodiment of the invention encompasses a method for screening for a compound which is effective in altering expression of a polynucleotide encoding CADECM. Compounds which may be effective in altering expression of a specific polynucleotide may include, but are not limited to, oligonucleotides, antisense oligonucleotides, triple helix-forming oligonucleotides, transcription factors and other polypeptide transcriptional regulators, and non-macromolecular chemical entities which are capable of interacting with specific polynucleotide sequences. Effective compounds may alter polynucleotide expression by acting as either inhibitors or promoters of polynucleotide expression. Thus, in the treatment of disorders associated with increased CADECM expression or activity, a compound which specifically inhibits expression of the polynucleotide encoding CADECM may be therapeutically useful, and in the treatment of disorders associated with decreased CADECM expression or activity, a compound which specifically promotes expression of the polynucleotide encoding CADECM may be therapeutically useful.

In various embodiments, one or more test compounds may be screened for effectiveness in altering expression of a specific polynucleotide. A test compound may be obtained by any method commonly known in the art, including chemical modification of a compound known to be effective in altering polynucleotide expression; selection from an existing, commercially-available or proprietary library of naturally-occurring or non-natural chemical compounds; rational design of a compound based on chemical and/or structural properties of the target polynucleotide; and selection from a library of chemical compounds created combinatorially or randomly. A sample comprising a polynucleotide encoding CADECM is exposed to at least one test compound thus obtained. The sample may comprise, for example, an intact or permeabilized cell, or an *in vitro* cell-free or reconstituted biochemical system. Alterations in the expression of a polynucleotide encoding

CADECM are assayed by any method commonly known in the art. Typically, the expression of a specific nucleotide is detected by hybridization with a probe having a nucleotide sequence complementary to the sequence of the polynucleotide encoding CADECM. The amount of hybridization may be quantified, thus forming the basis for a comparison of the expression of the polynucleotide both with and without exposure to one or more test compounds. Detection of a change in the expression of a polynucleotide exposed to a test compound indicates that the test compound is effective in altering the expression of the polynucleotide. A screen for a compound effective in altering expression of a specific polynucleotide can be carried out, for example, using a *Schizosaccharomyces pombe* gene expression system (Atkins, D. et al. (1999) U.S. Patent No. 5,932,435; Arndt, G.M. et al. (2000) Nucleic Acids Res. 28:E15) or a human cell line such as HeLa cell (Clarke, M.L. et al. (2000) Biochem. Biophys. Res. Commun. 268:8-13). A particular embodiment of the present invention involves screening a combinatorial library of oligonucleotides (such as deoxyribonucleotides, ribonucleotides, peptide nucleic acids, and modified oligonucleotides) for antisense activity against a specific polynucleotide sequence (Bruice, T.W. et al. (1997) U.S. Patent No. 5,686,242; Bruice, T.W. et al. (2000) U.S. Patent No. 6,022,691).

Many methods for introducing vectors into cells or tissues are available and equally suitable for use *in vivo*, *in vitro*, and *ex vivo*. For *ex vivo* therapy, vectors may be introduced into stem cells taken from the patient and clonally propagated for autologous transplant back into that same patient. Delivery by transfection, by liposome injections, or by polycationic amino polymers may be achieved using methods which are well known in the art (Goldman, C.K. et al. (1997) Nat. Biotechnol. 15:462-466).

Any of the therapeutic methods described above may be applied to any subject in need of such therapy, including, for example, mammals such as humans, dogs, cats, cows, horses, rabbits, and monkeys.

An additional embodiment of the invention relates to the administration of a composition which generally comprises an active ingredient formulated with a pharmaceutically acceptable excipient. Excipients may include, for example, sugars, starches, celluloses, gums, and proteins. Various formulations are commonly known and are thoroughly discussed in the latest edition of Remington's Pharmaceutical Sciences (Maack Publishing, Easton PA). Such compositions may consist of CADECM, antibodies to CADECM, and mimetics, agonists, antagonists, or inhibitors of CADECM.

In various embodiments, the compositions described herein, such as pharmaceutical compositions, may be administered by any number of routes including, but not limited to, oral, intravenous, intramuscular, intra-arterial, intramedullary, intrathecal, intraventricular, pulmonary, transdermal, subcutaneous, intraperitoneal, intranasal, enteral, topical, sublingual, or rectal means.

Compositions for pulmonary administration may be prepared in liquid or dry powder form. These compositions are generally aerosolized immediately prior to inhalation by the patient. In the case of small molecules (e.g. traditional low molecular weight organic drugs), aerosol delivery of fast-acting formulations is well-known in the art. In the case of macromolecules (e.g. larger peptides and proteins), recent developments in the field of pulmonary delivery via the alveolar region of the lung have enabled the practical delivery of drugs such as insulin to blood circulation (see, e.g., Patton, J.S. et al., U.S. Patent No. 5,997,848). Pulmonary delivery allows administration without needle injection, and obviates the need for potentially toxic penetration enhancers.

Compositions suitable for use in the invention include compositions wherein the active ingredients are contained in an effective amount to achieve the intended purpose. The determination of an effective dose is well within the capability of those skilled in the art.

Specialized forms of compositions may be prepared for direct intracellular delivery of macromolecules comprising CADECM or fragments thereof. For example, liposome preparations containing a cell-impermeable macromolecule may promote cell fusion and intracellular delivery of the macromolecule. Alternatively, CADECM or a fragment thereof may be joined to a short cationic N-terminal portion from the HIV Tat-1 protein. Fusion proteins thus generated have been found to transduce into the cells of all tissues, including the brain, in a mouse model system (Schwarze, S.R. et al. (1999) Science 285:1569-1572).

For any compound, the therapeutically effective dose can be estimated initially either in cell culture assays, e.g., of neoplastic cells, or in animal models such as mice, rats, rabbits, dogs, monkeys, or pigs. An animal model may also be used to determine the appropriate concentration range and route of administration. Such information can then be used to determine useful doses and routes for administration in humans.

A therapeutically effective dose refers to that amount of active ingredient, for example CADECM or fragments thereof, antibodies of CADECM, and agonists, antagonists or inhibitors of CADECM, which ameliorates the symptoms or condition. Therapeutic efficacy and toxicity may be determined by standard pharmaceutical procedures in cell cultures or with experimental animals, such as by calculating the $ED_{50}$ (the dose therapeutically effective in 50% of the population) or $LD_{50}$ (the dose lethal to 50% of the population) statistics. The dose ratio of toxic to therapeutic effects is the therapeutic index, which can be expressed as the $LD_{50}/ED_{50}$ ratio. Compositions which exhibit large therapeutic indices are preferred. The data obtained from cell culture assays and animal studies are used to formulate a range of dosage for human use. The dosage contained in such compositions is preferably within a range of circulating concentrations that includes the $ED_{50}$ with little or no toxicity. The dosage varies within this range depending upon the dosage form employed, the sensitivity of the patient, and the route of administration.

The exact dosage will be determined by the practitioner, in light of factors related to the subject requiring treatment. Dosage and administration are adjusted to provide sufficient levels of the active moiety or to maintain the desired effect. Factors which may be taken into account include the severity of the disease state, the general health of the subject, the age, weight, and gender of the subject, time and frequency of administration, drug combination(s), reaction sensitivities, and response to therapy. Long-acting compositions may be administered every 3 to 4 days, every week, or biweekly depending on the half-life and clearance rate of the particular formulation.

Normal dosage amounts may vary from about 0.1 $\mu$g to 100,000 $\mu$g, up to a total dose of about 1 gram, depending upon the route of administration. Guidance as to particular dosages and methods of delivery is provided in the literature and generally available to practitioners in the art. Those skilled in the art will employ different formulations for nucleotides than for proteins or their inhibitors. Similarly, delivery of polynucleotides or polypeptides will be specific to particular cells, conditions, locations, etc.

**DIAGNOSTICS**

In another embodiment, antibodies which specifically bind CADECM may be used for the diagnosis of disorders characterized by expression of CADECM, or in assays to monitor patients being treated with CADECM or agonists, antagonists, or inhibitors of CADECM. Antibodies useful for diagnostic purposes may be prepared in the same manner as described above for therapeutics. Diagnostic assays for CADECM include methods which utilize the antibody and a label to detect CADECM in human body fluids or in extracts of cells or tissues. The antibodies may be used with or without modification, and may be labeled by covalent or non-covalent attachment of a reporter molecule. A wide variety of reporter molecules, several of which are described above, are known in the art and may be used.

A variety of protocols for measuring CADECM, including ELISAs, RIAs, and FACS, are known in the art and provide a basis for diagnosing altered or abnormal levels of CADECM expression. Normal or standard values for CADECM expression are established by combining body fluids or cell extracts taken from normal mammalian subjects, for example, human subjects, with antibodies to CADECM under conditions suitable for complex formation. The amount of standard complex formation may be quantitated by various methods, such as photometric means. Quantities of CADECM expressed in subject, control, and disease samples from biopsied tissues are compared with the standard values. Deviation between standard and subject values establishes the parameters for diagnosing disease.

In another embodiment of the invention, polynucleotides encoding CADECM may be used for diagnostic purposes. The polynucleotides which may be used include oligonucleotides, complementary RNA and DNA molecules, and PNAs. The polynucleotides may be used to detect and quantify gene expression in biopsied tissues in which expression of CADECM may be correlated with disease. The diagnostic assay may be used to determine absence, presence, and excess expression of CADECM, and to monitor regulation of CADECM levels during therapeutic intervention.

In one aspect, hybridization with PCR probes which are capable of detecting polynucleotides, including genomic sequences, encoding CADECM or closely related molecules may be used to identify nucleic acid sequences which encode CADECM. The specificity of the probe, whether it is made from a highly specific region, e.g., the 5' regulatory region, or from a less specific region, e.g., a conserved motif, and the stringency of the hybridization or amplification will determine whether the probe identifies only naturally occurring sequences encoding CADECM, allelic variants, or related sequences.

Probes may also be used for the detection of related sequences, and may have at least 50% sequence identity to any of the CADECM encoding sequences. The hybridization probes of the subject invention may be DNA or RNA and may be derived from the sequence of SEQ ID NO:31-60 or from genomic sequences including promoters, enhancers, and introns of the CADECM gene.

Means for producing specific hybridization probes for polynucleotides encoding CADECM include the cloning of polynucleotides encoding CADECM or CADECM derivatives into vectors for the production of mRNA probes. Such vectors are known in the art, are commercially available, and may be used to synthesize RNA probes *in vitro* by means of the addition of the appropriate RNA polymerases and the appropriate labeled nucleotides. Hybridization probes may be labeled by a variety of reporter groups, for example, by radionuclides such as $^{32}P$ or $^{35}S$, or by enzymatic labels, such as alkaline phosphatase coupled to the probe via avidin/biotin coupling systems, and the like.

Polynucleotides encoding CADECM may be used for the diagnosis of disorders associated with expression of CADECM. Examples of such disorders include, but are not limited to, an immune system disorder, such as acquired immunodeficiency syndrome (AIDS), X-linked agammaglobinemia of Bruton, common variable immunodeficiency (CVI), DiGeorge's syndrome (thymic hypoplasia), thymic dysplasia, isolated IgA deficiency, severe combined immunodeficiency disease (SCID), immunodeficiency with thrombocytopenia and eczema (Wiskott-Aldrich syndrome), Chediak-Higashi syndrome, chronic granulomatous diseases, hereditary angioneurotic edema, immunodeficiency associated with Cushing's disease, Addison's disease, adult respiratory distress syndrome, allergies, ankylosing spondylitis, amyloidosis, anemia, asthma, atherosclerosis, autoimmune hemolytic anemia, autoimmune thyroiditis, autoimmune polyendocrinopathy-candidiasis-ectodermal dystrophy (APECED), bronchitis, cholecystitis, contact dermatitis, Crohn's disease, atopic dermatitis, dermatomyositis, diabetes mellitus, emphysema, episodic lymphopenia with lymphocytotoxins, erythroblastosis fetalis, erythema nodosum, atrophic gastritis, glomerulonephritis, Goodpasture's syndrome, gout, Graves' disease, Hashimoto's thyroiditis, hypereosinophilia, irritable bowel syndrome, multiple sclerosis, myasthenia gravis, myocardial or pericardial inflammation, osteoarthritis, osteoporosis, pancreatitis, polymyositis, psoriasis, Reiter's syndrome, rheumatoid arthritis, scleroderma, Sjögren's syndrome, systemic anaphylaxis, systemic lupus erythematosus, systemic sclerosis, thrombocytopenic purpura, ulcerative colitis, uveitis, Werner syndrome, complications of cancer, hemodialysis, and extracorporeal circulation, viral, bacterial, fungal, parasitic, protozoal, and helminthic infections, and trauma; a neurological disorder, such as epilepsy, ischemic cerebrovascular disease, stroke, cerebral neoplasms, Alzheimer's disease, Pick's disease, Huntington's disease, dementia, Parkinson's disease and other extrapyramidal disorders, amyotrophic lateral sclerosis and other motor neuron disorders, progressive neural muscular atrophy, retinitis pigmentosa, hereditary ataxias, multiple sclerosis and other demyelinating diseases, bacterial and viral meningitis, brain abscess, subdural empyema, epidural abscess, suppurative intracranial thrombophlebitis, myelitis and radiculitis, viral central nervous system disease, prion diseases including kuru, Creutzfeldt-Jakob disease, and Gerstmann-Straussler-Scheinker syndrome, fatal familial insomnia, nutritional and metabolic diseases of the nervous system, neurofibromatosis, tuberous sclerosis, cerebelloretinal hemangioblastomatosis, encephalotrigeminal syndrome, mental retardation and other developmental disorders of the central nervous system including Down syndrome, cerebral palsy, neuroskeletal disorders, autonomic nervous system disorders, cranial nerve disorders, spinal cord diseases, muscular dystrophy and other neuromuscular disorders, peripheral nervous system disorders, dermatomyositis and polymyositis, inherited, metabolic, endocrine, and toxic myopathies, myasthenia gravis, periodic paralysis, mental disorders including mood, anxiety, and schizophrenic disorders, seasonal affective disorder (SAD), akathesia, amnesia, catatonia, diabetic neuropathy, tardive dyskinesia, dystonias, paranoid psychoses, postherpetic neuralgia, Tourette's disorder, progressive supranuclear palsy, corticobasal degeneration, and familial frontotemporal dementia; a developmental disorder, such as renal tubular acidosis, anemia, Cushing's syndrome, achondroplastic dwarfism, Duchenne and Becker muscular dystrophy, epilepsy, gonadal dysgenesis, WAGR syndrome (Wilms' tumor, aniridia, genitourinary abnormalities, and mental retardation), Smith-Magenis syndrome, myelodysplastic syndrome, hereditary mucoepithelial dysplasia, hereditary keratodermas, hereditary neuropathies such as Charcot-Marie-Tooth disease and neurofibromatosis, hypothyroidism, hydrocephalus, seizure disorders such as Syndenham's chorea and cerebral palsy, spina bifida, anencephaly, craniorachischisis, congenital glaucoma, cataract, and sensorineural hearing loss; a connective tissue disorder, such as osteogenesis imperfecta, Ehlers-Danlos syndrome, chondrodysplasias, Marfan syndrome, Alport syndrome, familial aortic aneurysm, achondroplasia, mucopolysaccharidoses, osteoporosis, osteopetrosis, Paget's disease, rickets, osteomalacia, hyperparathyroidism, renal osteodystrophy, osteonecrosis, osteomyelitis, osteoma, osteoid osteoma, osteoblastoma, osteosarcoma, osteochondroma, chondroma, chondroblastoma, chondromyxoid fibroma, chondrosarcoma, fibrous cortical defect, nonossifying fibroma, fibrous dysplasia, fibrosarcoma, malignant fibrous histiocytoma, Ewing's sarcoma, primitive neuroectodermal tumor, giant cell tumor, osteoarthritis, rheumatoid arthritis, ankylosing spondyloarthritis, Reiter's syndrome, psoriatic arthritis, enteropathic arthritis, infectious arthritis, gout, gouty arthritis, calcium pyrophosphate crystal deposition disease, ganglion, synovial cyst, villonodular synovitis, systemic sclerosis, Dupuytren's contracture, hepatic fibrosis, lupus erythematosus, mixed connective tissue disease, epidermolysis bullosa simplex, bullous congenital ichthyosiform erythroderma (epidermolytic hyperkeratosis), non-epidermolytic and epidermolytic palmoplantar keratoderma, ichthyosis bullosa of Siemens, pachyonychia congenita, and white sponge nevus; and a cell proliferative disorder, such as actinic keratosis, arteriosclerosis, atherosclerosis, bursitis, cirrhosis, hepatitis, mixed connective tissue disease (MCTD), myelofibrosis, paroxysmal nocturnal hemoglobinuria, polycythemia vera, psoriasis, primary thrombocythemia, and cancers including adenocarcinoma, leukemia, lymphoma, melanoma, myeloma, sarcoma, teratocarcinoma, and, in particular, cancers of the adrenal gland, bladder, bone, bone marrow, brain, breast, cervix, colon, gall bladder, ganglia, gastrointestinal tract, heart, kidney, liver, lung, muscle, ovary, pancreas, parathyroid, penis, prostate, salivary glands, skin, spleen, testis, thymus, thyroid, and uterus. Polynucleotides encoding CADECM may be used in Southern or northern analysis, dot blot, or other membrane-based technologies; in PCR technologies; in dipstick, pin, and multiformat ELISA-like assays; and in microarrays utilizing fluids or tissues from patients to detect altered CADECM expression. Such qualitative or quantitative methods are well known in the art.

In a particular embodiment, polynucleotides encoding CADECM may be used in assays that detect the presence of associated disorders, particularly those mentioned above. Polynucleotides complementary to sequences encoding CADECM may be labeled by standard methods and added to a fluid or tissue sample from a patient under conditions suitable for the formation of hybridization complexes. After a suitable incubation period, the sample is washed and the signal is quantified and compared with a standard value. If the amount of signal in the patient sample is significantly altered in comparison to a control sample then the presence of altered levels of polynucleotides encoding CADECM in the sample indicates the presence of the associated disorder. Such assays may also be used to evaluate the efficacy of a particular therapeutic treatment regimen in animal studies, in clinical trials, or to monitor the treatment of an individual patient.

In order to provide a basis for the diagnosis of a disorder associated with expression of CADECM, a normal or standard profile for expression is established. This may be accomplished by combining body fluids or cell extracts taken from normal subjects, either animal or human, with a sequence, or a fragment thereof, encoding CADECM, under conditions suitable for hybridization or amplification. Standard hybridization may be quantified by comparing the values obtained from normal subjects with values from an experiment in which a known amount of a substantially purified polynucleotide is used. Standard values obtained in this manner may be compared with values obtained from samples from patients who are symptomatic for a disorder. Deviation from standard values is used to establish the presence of a disorder.

Once the presence of a disorder is established and a treatment protocol is initiated, hybridization assays may be repeated on a regular basis to determine if the level of expression in the patient begins to approximate that which is observed in the normal subject. The results obtained from successive assays may be used to show the efficacy of treatment over a period ranging from several days to months.

With respect to cancer, the presence of an abnormal amount of transcript (either under- or overexpressed) in biopsied tissue from an individual may indicate a predisposition for the development of the disease, or may provide a means for detecting the disease prior to the appearance of actual clinical symptoms. A more definitive diagnosis of this type may allow health professionals to employ preventative measures or aggressive treatment earlier, thereby preventing the development or further progression of the cancer.

Additional diagnostic uses for oligonucleotides designed from the sequences encoding CADECM may involve the use of PCR. These oligomers may be chemically synthesized, generated enzymatically, or produced *in vitro*. Oligomers will preferably contain a fragment of a polynucleotide encoding CADECM, or a fragment of a polynucleotide complementary to the polynucleotide encoding CADECM, and will be employed under optimized conditions for identification of a specific gene or condition. Oligomers may also be employed under less stringent conditions for detection or quantification of closely related DNA or RNA sequences.

In a particular aspect, oligonucleotide primers derived from polynucleotides encoding CADECM may be used to detect single nucleotide polymorphisms (SNPs). SNPs are substitutions, insertions and deletions that are a frequent cause of inherited or acquired genetic disease in humans. Methods of SNP detection include, but are not limited to, single-stranded conformation polymorphism (SSCP) and fluorescent SSCP (fSSCP) methods. In SSCP, oligonucleotide primers derived from polynucleotides encoding CADECM are used to amplify DNA using the polymerase chain reaction (PCR). The DNA may be derived, for example, from diseased or normal tissue, biopsy samples, bodily fluids, and the like. SNPs in the DNA cause differences in the secondary and tertiary structures of PCR products in single-stranded form, and these differences are detectable using gel electrophoresis in non-denaturing gels. In fSCCP, the oligonucleotide primers are fluorescently labeled, which allows detection of the amplimers in high-throughput equipment such as DNA sequencing machines. Additionally, sequence database analysis methods, termed in silico SNP (isSNP), are capable of identifying polymorphisms by comparing the sequence of individual overlapping DNA fragments which assemble into a common consensus sequence. These computer-based methods filter out sequence variations due to laboratory preparation of DNA and sequencing errors using statistical models and automated analyses of DNA sequence chromatograms. In the alternative, SNPs may be detected and characterized by mass spectrometry using, for example, the high throughput MASSARRAY system (Sequenom, Inc., San Diego CA).

SNPs may be used to study the genetic basis of human disease. For example, at least 16 common SNPs have been associated with non-insulin-dependent diabetes mellitus. SNPs are also useful for examining differences in disease outcomes in monogenic disorders, such as cystic fibrosis, sickle cell anemia, or chronic granulomatous disease. For example, variants in the mannose-binding lectin, MBL2, have been shown to be correlated with deleterious pulmonary outcomes in cystic fibrosis. SNPs also have utility in pharmacogenomics, the identification of genetic variants that influence a patient's response to a drug, such as life-threatening toxicity. For example, a variation in N-acetyl transferase is associated with a high incidence of peripheral neuropathy in response to the anti-tuberculosis drug isoniazid, while a variation in the core promoter of the ALOX5 gene results in diminished clinical response to treatment with an anti-asthma drug that targets the 5-lipoxygenase pathway. Analysis of the distribution of SNPs in different populations is useful for investigating genetic drift, mutation, recombination, and selection, as well as for tracing the origins of populations and their migrations (Taylor, J.G. et al. (2001) Trends Mol. Med. 7:507-512; Kwok, P.-Y. and Z. Gu (1999) Mol. Med. Today 5:538-543; Nowotny, P. et al. (2001) Curr. Opin. Neurobiol. 11:637-641).

Methods which may also be used to quantify the expression of CADECM include radiolabeling or biotinylating nucleotides, coamplification of a control nucleic acid, and interpolating results from standard curves (Melby, P.C. et al. (1993) J. Immunol. Methods 159:235-244; Duplaa, C. et al. (1993) Anal. Biochem. 212:229-236). The speed of quantitation of multiple samples may be accelerated by running the assay in a high-throughput format where the oligomer or polynucleotide of interest is presented in various dilutions and a spectrophotometric or colorimetric response gives rapid quantitation.

In further embodiments, oligonucleotides or longer fragments derived from any of the polynucleotides described herein may be used as elements on a microarray. The microarray can be used in transcript imaging techniques which monitor the relative expression levels of large numbers of genes simultaneously as described below. The microarray may also be used to identify genetic variants, mutations, and polymorphisms. This information may be used to determine gene function, to understand the genetic basis of a disorder, to diagnose a disorder, to monitor progression/regression of disease as a function of gene expression, and to develop and monitor the activities of therapeutic agents in the treatment of disease. In particular, this information may be used to develop a pharmacogenomic profile of a patient in order to select the most appropriate and effective treatment regimen for that patient. For example, therapeutic agents which are highly effective and display the fewest side effects may be selected for a patient based on his/her pharmacogenomic profile.

In another embodiment, CADECM, fragments of CADECM, or antibodies specific for CADECM may be used as elements on a microarray. The microarray may be used to monitor or measure protein-protein interactions, drug-target interactions, and gene expression profiles, as described above.

A particular embodiment relates to the use of the polynucleotides of the present invention to generate a transcript image of a tissue or cell type. A transcript image represents the global pattern of gene expression by a particular tissue or cell type. Global gene expression patterns are analyzed by quantifying the number of expressed genes and their relative abundance under given conditions and at a given time (Seilhamer et al., "Comparative Gene Transcript Analysis," U.S. Patent No. 5,840,484; hereby expressly incorporated by reference herein). Thus a transcript image may be generated by hybridizing the polynucleotides of the present invention or their complements to the totality of transcripts or reverse transcripts of a particular tissue or cell type. In one embodiment, the hybridization takes place in high-throughput format, wherein the polynucleotides of the present invention or their complements comprise a subset of a plurality of elements on a microarray. The resultant transcript image would provide a profile of gene activity.

Transcript images may be generated using transcripts isolated from tissues, cell lines, biopsies, or other biological samples. The transcript image may thus reflect gene expression *in vivo*, as in the case of a tissue or biopsy sample, or *in vitro*, as in the case of a cell line.

Transcript images which profile the expression of the polynucleotides of the present invention may also be used in conjunction with *in vitro* model systems and preclinical evaluation of pharmaceuticals, as well as toxicological testing of industrial and naturally-occurring environmental compounds. All compounds induce characteristic gene expression patterns, frequently termed molecular fingerprints or toxicant signatures, which are indicative of mechanisms of action and toxicity (Nuwaysir, E.F. et al. (1999) Mol. Carcinog. 24:153-159; Steiner, S. and N.L. Anderson (2000) Toxicol. Lett. 112-113:467-471). If a test compound has a signature similar to that of a compound with known toxicity, it is likely to share those toxic properties. These fingerprints or signatures are most useful and refined when they contain expression information from a large number of genes and gene families. Ideally, a genome-wide measurement of expression provides the highest quality signature. Even genes whose expression is not altered by any tested compounds are important as well, as the levels of expression of these genes are used to normalize the rest of the expression data. The normalization procedure is useful for comparison of expression data after treatment with different compounds. While the assignment of gene function to elements of a toxicant signature aids in interpretation of toxicity mechanisms, knowledge of gene function is not necessary for the statistical matching of signatures which leads to prediction of toxicity (see, for example, Press Release 00-02 from the National Institute of Environmental Health Sciences, released February 29, 2000, available at niehs.nih.gov/oc/news/toxchip.htm). Therefore, it is important and desirable in toxicological screening using toxicant signatures to include all expressed gene sequences.

In an embodiment, the toxicity of a test compound can be assessed by treating a biological sample containing nucleic acids with the test compound. Nucleic acids that are expressed in the treated biological sample are hybridized with one or more probes specific to the polynucleotides of the present invention, so that transcript levels corresponding to the polynucleotides of the present invention may be quantified. The transcript levels in the treated biological sample are compared with levels in an untreated biological sample. Differences in the transcript levels between the two samples are indicative of a toxic response caused by the test compound in the treated sample.

Another embodiment relates to the use of the polypeptides disclosed herein to analyze the proteome of a tissue or cell type. The term proteome refers to the global pattern of protein expression in a particular tissue or cell type. Each protein component of a proteome can be subjected individually to further analysis. Proteome expression patterns, or profiles, are analyzed by quantifying the number of expressed proteins and their relative abundance under given conditions and at a given time. A profile of a cell's proteome may thus be generated by separating and analyzing the polypeptides of a particular tissue or cell type. In one embodiment, the separation is achieved using two-dimensional gel electrophoresis, in which proteins from a sample are separated by isoelectric focusing in the first dimension, and then according to molecular weight by sodium dodecyl sulfate slab gel electrophoresis in the second dimension (Steiner and Anderson, *supra*). The proteins are visualized in the gel as discrete and uniquely positioned spots, typically by staining the gel with an agent such as Coomassie Blue or silver or fluorescent stains. The optical density of each protein spot is generally proportional to the level of the protein in the sample. The optical densities of equivalently positioned protein spots from different samples, for example, from biological samples either treated or untreated with a test compound or therapeutic agent, are compared to identify any changes in protein spot density related to the treatment. The proteins in the spots are partially sequenced using, for example, standard methods employing chemical or enzymatic cleavage followed by mass spectrometry. The identity of the protein in a spot may be determined by comparing its partial sequence, preferably of at least 5 contiguous amino acid residues, to the polypeptide sequences of interest. In some cases, further sequence data may be obtained for definitive protein identification.

A proteomic profile may also be generated using antibodies specific for CADECM to quantify the levels of CADECM expression. In one embodiment, the antibodies are used as elements on a microarray, and protein expression levels are quantified by contacting the microarray with the sample and detecting the levels of protein bound to each array element (Lueking, A. et al. (1999) Anal. Biochem. 270:103-111; Mendoze, L.G. et al. (1999) Biotechniques 27:778-788). Detection may be performed by a variety of methods known in the art, for example, by reacting the proteins in the sample with a thiol- or amino-reactive fluorescent compound and detecting the amount of fluorescence bound at each array element.

Toxicant signatures at the proteome level are also useful for toxicological screening, and should be analyzed in parallel with toxicant signatures at the transcript level. There is a poor correlation between transcript and protein abundances for some proteins in some tissues (Anderson, N.L. and J. Seilhamer (1997) Electrophoresis 18:533-537), so proteome toxicant signatures may be useful in the analysis of compounds which do not significantly affect the transcript image, but which alter the proteomic profile. In addition, the analysis of transcripts in body fluids is difficult, due to rapid degradation of mRNA, so proteomic profiling may be more reliable and informative in such cases.

In another embodiment, the toxicity of a test compound is assessed by treating a biological sample containing proteins with the test compound. Proteins that are expressed in the treated biological sample are separated so that the amount of each protein can be quantified. The amount of each protein is compared to the amount of the corresponding protein in an untreated biological sample. A difference in the amount of protein between the two samples is indicative of a toxic response to the test compound in the treated sample. Individual proteins are identified by sequencing the amino acid residues of the individual proteins and comparing these partial sequences to the polypeptides of the present invention.

In another embodiment, the toxicity of a test compound is assessed by treating a biological sample containing proteins with the test compound. Proteins from the biological sample are incubated with antibodies specific to the polypeptides of the present invention. The amount of protein recognized by the antibodies is quantified. The amount of protein in the treated biological sample is compared with the amount in an untreated biological sample. A difference in the amount of protein between the two samples is indicative of a toxic response to the test compound in the treated sample.

Microarrays may be prepared, used, and analyzed using methods known in the art (Brennan, T.M. et al. (1995) U.S. Patent No. 5,474,796; Schena, M. et al. (1996) Proc. Natl. Acad. Sci. USA 93:10614-10619; Baldeschweiler et al. (1995) PCT application WO95/25116; Shalon, D. et al. (1995) PCT application WO95/35505; Heller, R.A. et al. (1997) Proc. Natl. Acad. Sci. USA 94:2150-2155; Heller, M.J. et al. (1997) U.S. Patent No. 5,605,662). Various types of microarrays are well known and thoroughly described in Schena, M., ed. (1999; DNA Microarrays: A Practical Approach, Oxford

University Press, London).

In another embodiment of the invention, nucleic acid sequences encoding CADECM may be used to generate hybridization probes useful in mapping the naturally occurring genomic sequence. Either coding or noncoding sequences may be used, and in some instances, noncoding sequences may be preferable over coding sequences. For example, conservation of a coding sequence among members of a multi-gene family may potentially cause undesired cross hybridization during chromosomal mapping. The sequences may be mapped to a particular chromosome, to a specific region of a chromosome, or to artificial chromosome constructions, e.g., human artificial chromosomes (HACs), yeast artificial chromosomes (YACs), bacterial artificial chromosomes (BACs), bacterial P1 constructions, or single chromosome cDNA libraries (Harrington, J.J. et al. (1997) Nat. Genet. 15:345-355; Price, C.M. (1993) Blood Rev. 7:127-134; Trask, B.J. (1991) Trends Genet. 7:149-154). Once mapped, the nucleic acid sequences may be used to develop genetic linkage maps, for example, which correlate the inheritance of a disease state with the inheritance of a particular chromosome region or restriction fragment length polymorphism (RFLP) (Lander, E.S. and D. Botstein (1986) Proc. Natl. Acad. Sci. USA 83:7353-7357).

Fluorescent *in situ* hybridization (FISH) may be correlated with other physical and genetic map data (Heinz-Ulrich, et al. (1995) in Meyers, *supra*, pp. 965-968). Examples of genetic map data can be found in various scientific journals or at the Online Mendelian Inheritance in Man (OMIM) World Wide Web site. Correlation between the location of the gene encoding CADECM on a physical map and a specific disorder, or a predisposition to a specific disorder, may help define the region of DNA associated with that disorder and thus may further positional cloning efforts.

*In situ* hybridization of chromosomal preparations and physical mapping techniques, such as linkage analysis using established chromosomal markers, may be used for extending genetic maps. Often the placement of a gene on the chromosome of another mammalian species, such as mouse, may reveal associated markers even if the exact chromosomal locus is not known. This information is valuable to investigators searching for disease genes using positional cloning or other gene discovery techniques. Once the gene or genes responsible for a disease or syndrome have been crudely localized by genetic linkage to a particular genomic region, e.g., ataxia-telangiectasia to 11q22-23, any sequences mapping to that area may represent associated or regulatory genes for further investigation (Gatti, R.A. et al. (1988) Nature 336:577-580). The nucleotide sequence of the instant invention may also be used to detect differences in the chromosomal location due to translocation, inversion, etc., among normal, carrier, or affected individuals.

In another embodiment of the invention, CADECM, its catalytic or immunogenic fragments, or oligopeptides thereof can be used for screening libraries of compounds in any of a variety of drug screening techniques. The fragment employed in such screening may be free in solution, affixed to a solid support, borne on a cell surface, or located intracellularly. The formation of binding complexes between CADECM and the agent being tested may be measured.

Another technique for drug screening provides for high throughput screening of compounds having suitable binding affinity to the protein of interest (Geysen, et al. (1984) PCT application WO84/03564). In this method, large numbers of different small test compounds are synthesized on a solid substrate. The test compounds are reacted with CADECM, or fragments thereof, and washed. Bound CADECM is then detected by methods well known in the art. Purified CADECM can also be coated directly onto plates for use in the aforementioned drug screening techniques. Alternatively, non-neutralizing antibodies can be used to capture the peptide and immobilize it on a solid support.

In another embodiment, one may use competitive drug screening assays in which neutralizing antibodies capable of binding CADECM specifically compete with a test compound for binding CADECM. In this manner, antibodies can be used to detect the presence of any peptide which shares one or more antigenic determinants with CADECM.

In additional embodiments, the nucleotide sequences which encode CADECM may be used in any molecular biology techniques that have yet to be developed, provided the new techniques rely on properties of nucleotide sequences that are currently known, including, but not limited to, such properties as the triplet genetic code and specific base pair interactions.

Without further elaboration, it is believed that one skilled in the art can, using the preceding description, utilize the present invention to its fullest extent. The following embodiments are, therefore, to be construed as merely illustrative, and not limitative of the remainder of the disclosure in any way whatsoever.

The disclosures of all patents, applications, and publications mentioned above and below, including U.S. Ser. No. 60/460,788, U.S. Ser. No. 60/470,697, U.S. Ser. No. 60/474,144, U.S. Ser. No. 60/483,286, and U.S. Ser. No. 60/493,045, are hereby expressly incorporated by reference.

## EXAMPLES

### I. Construction of cDNA Libraries

Incyte cDNAs are derived from cDNA libraries described in the LIFESEQ database (Incyte, Palo Alto CA). Some tissues are homogenized and lysed in guanidinium isothiocyanate, while others are homogenized and lysed in phenol or in a suitable mixture of denaturants, such as TRIZOL (Invitrogen), a monophasic solution of phenol and guanidine isothiocyanate. The resulting lysates are centrifuged over CsCl cushions or extracted with chloroform. RNA is precipitated from the lysates with either isopropanol or sodium acetate and ethanol, or by other routine methods.

Phenol extraction and precipitation of RNA are repeated as necessary to increase RNA purity. In some cases, RNA is treated with DNase. For most libraries, poly(A)+ RNA is isolated using oligo d(T)-coupled paramagnetic particles (Promega), OLIGOTEX latex particles (QIAGEN, Chatsworth CA), or an OLIGOTEX mRNA purification kit (QIAGEN). Alternatively, RNA is isolated directly from tissue lysates using other RNA isolation kits, e.g., the POLY(A)PURE mRNA purification kit (Ambion, Austin TX).

In some cases, Stratagene is provided with RNA and constructs the corresponding cDNA libraries. Otherwise, cDNA is synthesized and cDNA libraries are constructed with the UNIZAP vector system (Stratagene) or SUPERSCRIPT plasmid system (Invitrogen), using the recommended procedures or similar methods known in the art (Ausubel et al., *supra*, ch. 5). Reverse transcription is initiated using oligo d(T) or random primers. Synthetic oligonucleotide adapters are ligated to double stranded cDNA, and the cDNA is digested with the appropriate restriction enzyme or enzymes. For most libraries, the cDNA is size-selected (300-1000 bp) using SEPHACRYL S1000, SEPHAROSE CL2B, or SEPHAROSE CL4B column chromatography (Amersham Biosciences) or preparative agarose gel electrophoresis. cDNAs are ligated into compatible restriction enzyme sites of the polylinker of a suitable plasmid, e.g., PBLUESCRIPT plasmid (Stratagene), PSPORT1 plasmid (Invitrogen, Carlsbad CA), PCDNA2.1 plasmid (Invitrogen), PBK-CMV plasmid (Stratagene), PCR2-TOPOTA plasmid (Invitrogen), PCMV-ICIS plasmid (Stratagene), pIGEN (Incyte, Palo Alto CA), pRARE (Incyte), or pINCY (Incyte), or derivatives thereof. Recombinant plasmids are transformed into competent *E. coli* cells including XL1-Blue, XL1-BlueMRF, or SOLR from Stratagene or DH5$\alpha$, DH10B, or ElectroMAX DH10B from Invitrogen.

## II. Isolation of cDNA Clones

Plasmids obtained as described in Example I are recovered from host cells by *in vivo* excision using the UNIZAP vector system (Stratagene) or by cell lysis. Plasmids are purified using at least one of the following: a Magic or WIZARD Minipreps DNA purification system (Promega); an AGTC Miniprep purification kit (Edge Biosystems, Gaithersburg MD); and QIAWELL 8 Plasmid, QIAWELL 8 Plus Plasmid, QIAWELL 8 Ultra Plasmid purification systems or the R.E.A.L. PREP 96 plasmid purification kit from QIAGEN. Following precipitation, plasmids are resuspended in 0.1 ml of distilled water and stored, with or without lyophilization, at 4°C.

Alternatively, plasmid DNA is amplified from host cell lysates using direct link PCR in a high-throughput format (Rao, V.B. (1994) Anal. Biochem. 216:1-14). Host cell lysis and thermal cycling steps are carried out in a single reaction mixture. Samples are processed and stored in 384-well plates, and the concentration of amplified plasmid DNA is quantified fluorometrically using PICOGREEN dye (Molecular Probes, Eugene OR) and a FLUOROSKAN II fluorescence scanner (Labsystems Oy, Helsinki, Finland).

## III. Sequencing and Analysis

Incyte cDNA recovered in plasmids as described in Example II are sequenced as follows.

Sequencing reactions are processed using standard methods or high-throughput instrumentation such as the ABI CATALYST 800 (Applied Biosystems) thermal cycler or the PTC-200 thermal cycler (MJ Research) in conjunction with the HYDRA microdispenser (Robbins Scientific) or the MICROLAB 2200 (Hamilton) liquid transfer system. cDNA sequencing reactions are prepared using reagents provided by Amersham Biosciences or supplied in ABI sequencing kits such as the ABI PRISM BIGDYE Terminator cycle sequencing ready reaction kit (Applied Biosystems). Electrophoretic separation of cDNA sequencing reactions and detection of labeled polynucleotides are carried out using the MEGABACE 1000 DNA sequencing system (Amersham Biosciences); the ABI PRISM 373 or 377 sequencing system (Applied Biosystems) in conjunction with standard ABI protocols and base calling software; or other sequence analysis systems known in the art. Reading frames within the cDNA sequences are identified using standard methods (Ausubel et al., *supra*, ch. 7). Some of the cDNA sequences are selected for extension using the techniques disclosed in Example VIII.

Polynucleotide sequences derived from Incyte cDNAs are validated by removing vector, linker, and poly(A) sequences and by masking ambiguous bases, using algorithms and programs based on BLAST, dynamic programming, and dinucleotide nearest neighbor analysis. The Incyte cDNA sequences or translations thereof are then queried against a selection of public databases such as the GenBank primate, rodent, mammalian, vertebrate, and eukaryote databases, and BLOCKS, PRINTS, DOMO, PRODOM; PROTEOME databases with sequences from *Homo sapiens*, *Rattus norvegicus*, *Mus musculus*, *Caenorhabditis elegans*, *Saccharomyces cerevisiae*, *Schizosaccharomyces pombe*, and *Candida albicans* (Incyte, Palo Alto CA); hidden Markov model (HMM)-based protein family databases such as PFAM, INCY, and TIGRFAM (Haft, D.H. et al. (2001) Nucleic Acids Res. 29:41-43); and HMM-based protein domain databases such as SMART (Schultz, J. et al. (1998) Proc. Natl. Acad. Sci. USA 95:5857-5864; Letunic, I. et al. (2002) Nucleic Acids Res. 30:242-244). (HMM is a probabilistic approach which analyzes consensus primary structures of gene families; see, for example, Eddy, S.R. (1996) Curr. Opin. Struct. Biol. 6:361-365.) The queries are performed using programs based on BLAST, FASTA, BLIMPS, and HMMER. The Incyte cDNA sequences are assembled to produce full length polynucleotide sequences. Alternatively, GenBank cDNAs, GenBank ESTs, stitched sequences, stretched sequences, or Genscan-predicted coding sequences (see Examples IV and V) are used to extend Incyte cDNA assemblages to full length. Assembly is performed using programs based on Phred, Phrap, and Consed, and cDNA assemblages are screened for open reading frames using programs based on GeneMark, BLAST, and FASTA. The full length polynucleotide sequences are translated to derive the corresponding full length polypeptide sequences. Alternatively, a polypeptide may begin at any of the methionine residues of the full length translated polypeptide. Full length polypeptide sequences are subsequently analyzed by querying against databases such as the GenBank protein databases (genpept), SwissProt, the PROTEOME databases, BLOCKS, PRINTS, DOMO, PRODOM, Prosite, hidden Markov model (HMM)-based protein family databases such as PFAM, INCY, and TIGRFAM; and HMM-based protein domain databases such as SMART. Full length polynucleotide sequences are also analyzed using MACDNASIS PRO software (MiraiBio, Alameda CA) and LASERGENE software (DNASTAR). Polynucleotide and polypeptide sequence alignments are generated using default parameters specified by the CLUSTAL algorithm as incorporated into the MEGALIGN multisequence alignment program (DNASTAR), which also calculates the percent identity between aligned sequences.

Table 7 summarizes tools, programs, and algorithms used for the analysis and assembly of Incyte cDNA and full length sequences and provides applicable descriptions, references, and threshold parameters. The first column of Table 7 shows the tools, programs, and algorithms used, the second column provides brief descriptions thereof, the third column presents appropriate references, all of which are incorporated by reference herein in their entirety, and the fourth column presents, where applicable, the scores, probability values, and other parameters used to evaluate the strength of a match between two sequences (the higher the score or the lower the probability value, the greater the identity between two sequences).

The programs described above for the assembly and analysis of full length polynucleotide and polypeptide sequences are also used to identify polynucleotide sequence fragments from SEQ ID NO:31-60. Fragments from about 20 to about 4000 nucleotides which are useful in hybridization and amplification technologies are described in Table 4, column 2.

## IV. Identification and Editing of Coding Sequences from Genomic DNA

Putative cell adhesion and extracellular matrix proteins are initially identified by running the Genscan gene identification program against public genomic sequence databases (e.g., gbpri and gbhtg). Genscan is a general-purpose gene identification program which analyzes genomic DNA sequences from a variety of organisms (Burge, C. and S. Karlin (1997) J. Mol. Biol. 268:78-94; Burge, C. and S. Karlin (1998) Curr. Opin. Struct. Biol. 8:346-354). The program concatenates predicted exons to form an assembled cDNA sequence extending from a methionine to a stop codon. The output of Genscan is a FASTA database of polynucleotide and polypeptide sequences. The maximum range of sequence for Genscan to analyze at once is set to 30 kb. To determine which of these Genscan predicted cDNA sequences encode cell adhesion and extracellular matrix proteins, the encoded polypeptides are analyzed by querying against PFAM models for cell adhesion and extracellular matrix proteins. Potential cell adhesion and extracellular matrix proteins are also identified by homology to Incyte cDNA sequences that have been annotated as cell adhesion and extracellular matrix proteins. These selected Genscan-predicted sequences are then compared by BLAST analysis to the genpept and gbpri public databases. Where necessary, the Genscan-predicted sequences are then edited by comparison to the top BLAST hit from genpept to correct errors in the sequence predicted by Genscan, such as extra or omitted exons. BLAST analysis is also used to find any Incyte cDNA or public cDNA coverage of the Genscan-predicted sequences, thus providing evidence for transcription. When Incyte cDNA coverage is available, this information is used to correct or confirm the Genscan predicted sequence. Full length polynucleotide sequences are obtained by assembling Genscan-predicted coding sequences with Incyte cDNA sequences and/or public cDNA sequences using the assembly process described in Example III. Alternatively, full length polynucleotide sequences are derived entirely from edited or unedited Genscan-predicted coding sequences.

**V. Assembly of Genomic Sequence Data with cDNA Sequence Data**

**"Stitched" Sequences**

Partial cDNA sequences are extended with exons predicted by the Genscan gene identification program described in Example IV. Partial cDNAs assembled as described in Example III are mapped to genomic DNA and parsed into clusters containing related cDNAs and Genscan exon predictions from one or more genomic sequences. Each cluster is analyzed using an algorithm based on graph theory and dynamic programming to integrate cDNA and genomic information, generating possible splice variants that are subsequently confirmed, edited, or extended to create a full length sequence. Sequence intervals in which the entire length of the interval is present on more than one sequence in the cluster are identified, and intervals thus identified are considered to be equivalent by transitivity. For example, if an interval is present on a cDNA and two genomic sequences, then all three intervals are considered to be equivalent. This process allows unrelated but consecutive genomic sequences to be brought together, bridged by cDNA sequence. Intervals thus identified are then "stitched" together by the stitching algorithm in the order that they appear along their parent sequences to generate the longest possible sequence, as well as sequence variants. Linkages between intervals which proceed along one type of parent sequence (cDNA to cDNA or genomic sequence to genomic sequence) are given preference over linkages which change parent type (cDNA to genomic sequence). The resultant stitched sequences are translated and compared by BLAST analysis to the genpept and gbpri public databases. Incorrect exons predicted by Genscan are corrected by comparison to the top BLAST hit from genpept. Sequences are further extended with additional cDNA sequences, or by inspection of genomic DNA, when necessary.

**"Stretched" Sequences**

Partial DNA sequences are extended to full length with an algorithm based on BLAST analysis. First, partial cDNAs assembled as described in Example III are queried against public databases such as the GenBank primate, rodent, mammalian, vertebrate, and eukaryote databases using the BLAST program. The nearest GenBank protein homolog is then compared by BLAST analysis to either Incyte cDNA sequences or GenScan exon predicted sequences described in

Example IV. A chimeric protein is generated by using the resultant high-scoring segment pairs (HSPs) to map the translated sequences onto the GenBank protein homolog. Insertions or deletions may occur in the chimeric protein with respect to the original GenBank protein homolog. The GenBank protein homolog, the chimeric protein, or both are used as probes to search for homologous genomic sequences from the public human genome databases. Partial DNA sequences are therefore "stretched" or extended by the addition of homologous genomic sequences. The resultant stretched sequences are examined to determine whether they contain a complete gene.

**VI. Chromosomal Mapping of CADECM Encoding Polynucleotides**

The sequences used to assemble SEQ ID NO:31-60 are compared with sequences from the Incyte LIFESEQ database and public domain databases using BLAST and other implementations of the Smith-Waterman algorithm. Sequences from these databases that matched SEQ ID NO:31-60 are assembled into clusters of contiguous and overlapping sequences using assembly algorithms such as Phrap (Table 7). Radiation hybrid and genetic mapping data available from public resources such as the Stanford Human Genome Center (SHGC), Whitehead Institute for Genome Research (WIGR), and Généthon are used to determine if any of the clustered sequences have been previously mapped. Inclusion of a mapped sequence in a cluster results in the assignment of all sequences of that cluster, including its particular SEQ ID NO:, to that map location.

Map locations are represented by ranges, or intervals, of human chromosomes. The map position of an interval, in centiMorgans, is measured relative to the terminus of the chromosome's p-arm. (The centiMorgan (cM) is a unit of measurement based on recombination frequencies between chromosomal markers. On average, 1 cM is roughly equivalent to 1 megabase (Mb) of DNA in humans, although this can vary widely due to hot and cold spots of recombination.) The cM distances are based on genetic markers mapped by Généthon which provide boundaries for radiation hybrid markers whose sequences were included in each of the clusters. Human genome maps and other resources available to the public, such as the NCBI "GeneMap'99" World Wide Web site (ncbi.nlm.nih.gov/genemap/), can be employed to determine if previously identified disease genes map within or in proximity to the intervals indicated above.

**VII. Analysis of Polynucleotide Expression**

Northern analysis is a laboratory technique used to detect the presence of a transcript of a gene and involves the hybridization of a labeled nucleotide sequence to a membrane on which RNAs from a particular cell type or tissue have been bound (Sambrook and Russell, *supra*, ch. 7; Ausubel et al., *supra*, ch. 4).

Analogous computer techniques applying BLAST are used to search for identical or related molecules in databases such as GenBank or LIFESEQ (Incyte). This analysis is much faster than multiple membrane-based hybridizations. In addition, the sensitivity of the computer search can be modified to determine whether any particular match is categorized as exact or similar. The basis of the search is the product score, which is defined as:

$$\frac{\text{BLAST Score x Percent Identity}}{5 \text{ x minimum } \{\text{length(Seq. 1), length(Seq. 2)}\}}$$

The product score takes into account both the degree of similarity between two sequences and the length of the sequence match. The product score is a normalized value between 0 and 100, and is calculated as follows: the BLAST score is multiplied by the percent nucleotide identity and the product is divided by (5 times the length of the shorter of the two sequences). The BLAST score is calculated by assigning a score of +5 for every base that matches in a high-scoring segment pair (HSP), and -4 for every mismatch. Two sequences may share more than one HSP (separated by gaps). If there is more than one HSP, then the pair with the highest BLAST score is used to calculate the product score. The product score represents a balance between fractional overlap and quality in a BLAST alignment. For example, a product score of 100 is produced only for 100% identity over the entire length of the shorter of the two sequences being compared. A product score of 70 is produced either by 100% identity and 70% overlap at one end, or by 88% identity and 100% overlap at the other. A product score of 50 is produced either by 100% identity and 50% overlap at one end, or 79% identity and 100% overlap.

Alternatively, polynucleotides encoding CADECM are analyzed with respect to the tissue sources from which they are derived. For example, some full length sequences are assembled, at least in part, with overlapping Incyte cDNA sequences (see Example III). Each cDNA sequence is derived from a cDNA library constructed from a human tissue. Each human tissue is classified into one of the following organ/tissue categories: cardiovascular system; connective tissue; digestive system; embryonic structures; endocrine system; exocrine glands; genitalia, female; genitalia, male; germ cells; hemic and immune system; liver; musculoskeletal system; nervous system; pancreas; respiratory system; sense organs; skin; stomatognathic system; unclassified/mixed; or urinary tract. The number of libraries in each category is counted and divided by the total number of libraries across all categories. Similarly, each human tissue is classified into one of the following disease/condition categories: cancer, cell line, developmental, inflammation, neurological, trauma, cardiovascular, pooled, and other, and the number of libraries in each category is counted and divided by the total number of libraries across all categories. The resulting percentages reflect the tissue- and disease-specific expression of cDNA encoding CADECM. cDNA sequences and cDNA library/tissue information are found in the LIFESEQ database (Incyte, Palo Alto CA).

## VIII. Extension of CADECM Encoding Polynucleotides

Full length polynucleotides are produced by extension of an appropriate fragment of the full length molecule using oligonucleotide primers designed from this fragment. One primer is synthesized to initiate 5' extension of the known fragment, and the other primer is synthesized to initiate 3' extension of the known fragment. The initial primers are designed using OLIGO 4.06 software (National Biosciences), or another appropriate program, to be about 22 to 30 nucleotides in length, to have a GC content of about 50% or more, and to anneal to the target sequence at temperatures of about 68 °C to about 72 °C. Any stretch of nucleotides which would result in hairpin structures and primer-primer dimerizations is avoided.

Selected human cDNA libraries are used to extend the sequence. If more than one extension is necessary or desired, additional or nested sets of primers are designed.

High fidelity amplification is obtained by PCR using methods well known in the art. PCR is performed in 96-well plates using the PTC-200 thermal cycler (MJ Research, Inc.). The reaction mix contains DNA template, 200 nmol of each primer, reaction buffer containing $Mg^{2+}$, $(NH_4)_2SO_4$, and 2-mercaptoethanol, Taq DNA polymerase (Amersham Biosciences), ELONGASE enzyme (Invitrogen), and Pfu DNA polymerase (Stratagene), with the following parameters for primer pair PCI A and PCI B: Step 1: 94 °C, 3 min; Step 2: 94 °C, 15 sec; Step 3: 60 °C, 1 min; Step 4: 68 °C, 2 min; Step 5: Steps 2, 3, and 4 repeated 20 times; Step 6: 68 °C, 5 min; Step 7: storage at 4 °C. In the alternative, the parameters for primer pair T7 and SK+ are as follows: Step 1: 94 °C, 3 min; Step 2: 94 °C, 15 sec; Step 3: 57 °C, 1 min; Step 4: 68 °C, 2 min; Step 5: Steps 2, 3, and 4 repeated 20 times; Step 6: 68 °C, 5 min; Step 7: storage at 4 °C.

The concentration of DNA in each well is determined by dispensing 100 $\mu$l PICOGREEN quantitation reagent (0.25% (v/v) PICOGREEN; Molecular Probes, Eugene OR) dissolved in 1X TE and 0.5 $\mu$l of undiluted PCR product into each well of an opaque fluorimeter plate (Corning Costar, Acton MA), allowing the DNA to bind to the reagent. The plate is scanned in a Fluoroskan II (Labsystems Oy, Helsinki, Finland) to measure the fluorescence of the sample and to quantify the concentration of DNA. A 5 $\mu$l to 10 $\mu$l aliquot of the reaction mixture is analyzed by electrophoresis on a 1 % agarose gel to determine which reactions are successful in extending the sequence.

The extended nucleotides are desalted and concentrated, transferred to 384-well plates, digested with CviJI cholera virus endonuclease (Molecular Biology Research, Madison WI), and sonicated or sheared prior to religation into pUC 18 vector (Amersham Biosciences). For shotgun sequencing, the digested nucleotides are separated on low concentration (0.6 to 0.8%) agarose gels, fragments are excised, and agar digested with Agar ACE (Promega). Extended clones were religated using T4 ligase (New England Biolabs, Beverly MA) into pUC 18 vector (Amersham Biosciences), treated with Pfu DNA polymerase (Stratagene) to fill-in restriction site overhangs, and transfected into competent *E. coli* cells. Transformed cells are selected on antibiotic-containing media, and individual colonies are picked and cultured overnight at 37 °C in 384-well plates in LB/2x carb liquid media.

The cells are lysed, and DNA is amplified by PCR using Taq DNA polymerase (Amersham Biosciences) and Pfu DNA polymerase (Stratagene) with the following parameters: Step 1: 94°C, 3 min; Step 2: 94°C, 15 sec; Step 3: 60°C, 1 min; Step 4: 72°C, 2 min; Step 5: steps 2, 3, and 4 repeated 29 times; Step 6: 72°C, 5 min; Step 7: storage at 4°C. DNA is quantified by PICOGREEN reagent (Molecular Probes) as described above. Samples with low DNA recoveries are reamplified using the same conditions as described above. Samples are diluted with 20% dimethysulfoxide (1:2, v/v), and sequenced using DYENAMIC energy transfer sequencing primers and the DYENAMIC DIRECT kit (Amersham Biosciences) or the ABI PRISM BIGDYE Terminator cycle sequencing ready reaction kit (Applied Biosystems).

In like manner, full length polynucleotides are verified using the above procedure or are used to obtain 5' regulatory sequences using the above procedure along with oligonucleotides designed for such extension, and an appropriate genomic library.

**IX. Identification of Single Nucleotide Polymorphisms in CADECM Encoding Polynucleotides**

Common DNA sequence variants known as single nucleotide polymorphisms (SNPs) are identified in SEQ ID NO:31-60 using the LIFESEQ database (Incyte). Sequences from the same gene are clustered together and assembled as described in Example III, allowing the identification of all sequence variants in the gene. An algorithm consisting of a series of filters is used to distinguish SNPs from other sequence variants. Preliminary filters remove the majority of basecall errors by requiring a minimum Phred quality score of 15, and remove sequence alignment errors and errors resulting from improper trimming of vector sequences, chimeras, and splice variants. An automated procedure of advanced chromosome analysis is applied to the original chromatogram files in the vicinity of the putative SNP. Clone error filters use statistically generated algorithms to identify errors introduced during laboratory processing, such as those caused by reverse transcriptase, polymerase, or somatic mutation. Clustering error filters use statistically generated algorithms to identify errors resulting from clustering of close homologs or pseudogenes, or due to contamination by non-human sequences. A final set of filters removes duplicates and SNPs found in immunoglobulins or T-cell receptors.

Certain SNPs are selected for further characterization by mass spectrometry using the high throughput MASSARRAY system (Sequenom, Inc.) to analyze allele frequencies at the SNP sites in four different human populations. The Caucasian population comprises 92 individuals (46 male, 46 female), including 83 from Utah, four French, three Venezualan, and two Amish individuals. The African population comprises 194 individuals (97 male, 97 female), all African Americans. The Hispanic population comprises 324 individuals (162 male, 162 female), all Mexican Hispanic. The

Asian population comprises 126 individuals (64 male, 62 female) with a reported parental breakdown of 43% Chinese, 31% Japanese, 13% Korean, 5% Vietnamese, and 8% other Asian. Allele frequencies are first analyzed in the Caucasian population; in some cases those SNPs which show no allelic variance in this population are not further tested in the other three populations.

**X. Labeling and Use of Individual Hybridization Probes**

Hybridization probes derived from SEQ ID NO:31-60 are employed to screen cDNAs, genomic DNAs, or mRNAs. Although the labeling of oligonucleotides, consisting of about 20 base pairs, is specifically described, essentially the same procedure is used with larger nucleotide fragments. Oligonucleotides are designed using state-of-the-art software such as OLIGO 4.06 software (National Biosciences) and labeled by combining 50 pmol of each oligomer, 250 $\mu$Ci of [$\gamma$-$^{32}$P] adenosine triphosphate (Amersham Biosciences), and T4 polynucleotide kinase (DuPont NEN, Boston MA). The labeled oligonucleotides are substantially purified using a SEPHADEX G-25 superfine size exclusion dextran bead column (Amersham Biosciences). An aliquot containing $10^7$ counts per minute of the labeled probe is used in a typical membrane-based hybridization analysis of human genomic DNA digested with one of the following endonucleases: Ase I, Bgl II, Eco RI, Pst I, Xba I, or Pvu II (DuPont NEN).

The DNA from each digest is fractionated on a 0.7% agarose gel and transferred to NYTRAN PLUS nylon membranes (Schleicher & Schuell, Durham NH). Hybridization is carried out for 16 hours at 40°C. To remove nonspecific signals, blots are sequentially washed at room temperature under conditions of up to, for example, 0.1 x saline sodium citrate and 0.5% sodium dodecyl sulfate. Hybridization patterns are visualized using autoradiography or an alternative imaging means and compared.

**XI. Microarrays**

The linkage or synthesis of array elements upon a microarray can be achieved utilizing photolithography, piezoelectric printing (ink-jet printing; see, e.g., Baldeschweiler et al., *supra*), mechanical microspotting technologies, and derivatives thereof. The substrate in each of the aforementioned technologies should be uniform and solid with a non-porous surface (Schena, M., ed. (1999) DNA Microarrays: A Practical Approach, Oxford University Press, London). Suggested substrates include silicon, silica, glass slides, glass chips, and silicon wafers. Alternatively, a procedure analogous to a dot or slot blot may also be used to arrange and link elements to the surface of a substrate using thermal, UV, chemical, or mechanical bonding procedures. A typical array may be produced using available methods and machines well known to those of ordinary skill in the art and may contain any appropriate number of elements (Schena, M. et al. (1995) Science 270:467-470; Shalon, D. et al. (1996) Genome Res. 6:639-645; Marshall, A. and J. Hodgson (1998) Nat. Biotechnol. 16:27-31).

Full length cDNAs, Expressed Sequence Tags (ESTs), or fragments or oligomers thereof may comprise the elements of the microarray. Fragments or oligomers suitable for hybridization can be selected using software well known in the art such as LASERGENE software (DNASTAR). The array elements are hybridized with polynucleotides in a biological sample. The polynucleotides in the biological sample are conjugated to a fluorescent label or other molecular tag for ease of detection. After hybridization, nonhybridized nucleotides from the biological sample are removed, and a fluorescence scanner is used to detect hybridization at each array element. Alternatively, laser desorbtion and mass spectrometry may be used for detection of hybridization. The degree of complementarity and the relative abundance of each polynucleotide which hybridizes to an element on the microarray may be assessed. In one embodiment, microarray preparation and usage is described in detail below.

**Tissue or Cell Sample Preparation**

Total RNA is isolated from tissue samples using the guanidinium thiocyanate method and poly(A)$^+$ RNA is purified using the oligo-(dT) cellulose method. Each poly(A)$^+$ RNA sample is reverse transcribed using MMLV reverse-transcriptase, 0.05 pg/$\mu$l oligo-(dT) primer (21mer), 1X first strand buffer, 0.03 units/$\mu$l RNase inhibitor, 500 $\mu$M dATP, 500 $\mu$M dGTP, 500 $\mu$M dTTP, 40 $\mu$M dCTP, 40 $\mu$M dCTP-Cy3 (BDS) or dCTP-Cy5 (Amersham Biosciences). The reverse transcription reaction is performed in a 25 ml volume containing 200 ng poly(A)$^+$ RNA with GEMBRIGHT kits (Incyte). Specific control poly(A)$^+$ RNAs are synthesized by *in vitro* transcription from non-coding yeast genomic DNA. After incubation at 37°C for 2 hr, each reaction sample (one with Cy3 and another with Cy5 labeling) is treated with 2.5 ml of 0.5M sodium hydroxide and incubated for 20 minutes at 85°C to the stop the reaction and degrade the RNA. Samples are purified using two successive CHROMA SPIN 30 gel filtration spin columns (BD Clontech, Palo Alto CA) and after combining, both reaction samples are ethanol precipitated using 1 ml of glycogen (1 mg/ml), 60 ml sodium acetate, and 300 ml of 100% ethanol. The sample is then dried to completion using a SpeedVAC (Savant Instruments Inc., Holbrook NY) and resuspended in 14 $\mu$l 5X SSC/0.2% SDS.

**Microarray Preparation**

Sequences of the present invention are used to generate array elements. Each array element is amplified from bacterial cells containing vectors with cloned cDNA inserts. PCR amplification uses primers complementary to the vector sequences flanking the cDNA insert. Array elements are amplified in thirty cycles of PCR from an initial quantity of 1-2 ng to a final quantity greater than 5 $\mu$g. Amplified array elements are then purified using SEPHACRYL-400 (Amersham Biosciences).

Purified array elements are immobilized on polymer-coated glass slides. Glass microscope slides (Corning) are cleaned by ultrasound in 0.1% SDS and acetone, with extensive distilled water washes between and after treatments. Glass slides are etched in 4% hydrofluoric acid (VWR Scientific Products Corporation (VWR), West Chester PA), washed extensively in distilled water, and coated with 0.05% aminopropyl silane (Sigma-Aldrich, St. Louis MO) in 95% ethanol. Coated slides are cured in a 110°C oven.

Array elements are applied to the coated glass substrate using a procedure described in U.S. Patent No. 5,807,522, incorporated herein by reference. 1 $\mu$l of the array element DNA, at an average concentration of 100 ng/$\mu$l, is loaded into the open capillary printing element by a high-speed robotic apparatus. The apparatus then deposits about 5 nl of array element sample per slide.

Microarrays are UV-crosslinked using a STRATALINKER UV-crosslinker (Stratagene). Microarrays are washed at room temperature once in 0.2% SDS and three times in distilled water. Non-specific binding sites are blocked by incubation of microarrays in 0.2% casein in phosphate buffered saline (PBS) (Tropix, Inc., Bedford MA) for 30 minutes at 60°C followed by washes in 0.2% SDS and distilled water as before.

**Hybridization**

Hybridization reactions contain 9 $\mu$l of sample mixture consisting of 0.2 $\mu$g each of Cy3 and Cy5 labeled cDNA synthesis products in 5X SSC, 0.2% SDS hybridization buffer. The sample mixture is heated to 65°C for 5 minutes and is aliquoted onto the microarray surface and covered with an 1.8 cm$^2$ coverslip. The arrays are transferred to a waterproof chamber having a cavity just slightly larger than a microscope slide. The chamber is kept at 100% humidity internally by the addition of 140 $\mu$l of 5X SSC in a corner of the chamber. The chamber containing the arrays is incubated for about 6.5 hours at 60°C. The arrays are washed for 10 min at 45°C in a first wash buffer (1X SSC, 0.1% SDS), three times for 10 minutes each at 45°C in a second wash buffer (0.1X SSC), and dried.

**Detection**

Reporter-labeled hybridization complexes are detected with a microscope equipped with an Innova 70 mixed gas 10 W laser (Coherent, Inc., Santa Clara CA) capable of generating spectral lines at 488 nm for excitation of Cy3 and at 632 nm for excitation of Cy5. The excitation laser light is focused on the array using a 20X microscope objective (Nikon, Inc., Melville NY). The slide containing the array is placed on a computer-controlled X-Y stage on the microscope and raster-scanned past the objective. The 1.8 cm x 1.8 cm array used in the present example is scanned with a resolution of 20 micrometers.

In two separate scans, a mixed gas multiline laser excites the two fluorophores sequentially. Emitted light is split, based on wavelength, into two photomultiplier tube detectors (PMT R1477, Hamamatsu Photonics Systems, Bridgewater NJ) corresponding to the two fluorophores. Appropriate filters positioned between the array and the photomultiplier tubes are used to filter the signals. The emission maxima of the fluorophores used are 565 nm for Cy3 and 650 nm for Cy5. Each array is typically scanned twice, one scan per fluorophore using the appropriate filters at the laser source, although the apparatus is capable of recording the spectra from both fluorophores simultaneously.

The sensitivity of the scans is typically calibrated using the signal intensity generated by a cDNA control species added to the sample mixture at a known concentration. A specific location on the array contains a complementary DNA sequence, allowing the intensity of the signal at that location to be correlated with a weight ratio of hybridizing species of 1:100,000. When two samples from different sources (e.g., representing test and control cells), each labeled with a different fluorophore, are hybridized to a single array for the purpose of identifying genes that are differentially expressed, the calibration is done by labeling samples of the calibrating cDNA with the two fluorophores and adding identical amounts of each to the hybridization mixture.

The output of the photomultiplier tube is digitized using a 12-bit RTI-835H analog-to-digital (A/D) conversion board (Analog Devices, Inc., Norwood MA) installed in an IBM-compatible PC computer. The digitized data are displayed as an image where the signal intensity is mapped using a linear 20-color transformation to a pseudocolor scale ranging from blue (low signal) to red (high signal). The data is also analyzed quantitatively. Where two different fluorophores are excited and measured simultaneously, the data are first corrected for optical crosstalk (due to overlapping emission spectra) between the fluorophores using each fluorophore's emission spectrum.

A grid is superimposed over the fluorescence signal image such that the signal from each spot is centered in each element of the grid. The fluorescence signal within each element is then integrated to obtain a numerical value corresponding to the average intensity of the signal. The software used for signal analysis is the GEMTOOLS gene expression analysis program (Incyte). Array elements that exhibit at least about a two-fold change in expression, a signal-to-background ratio of at least about 2.5, and an element spot size of at least about 40%, are considered to be differentially expressed.

**Expression**

For example, expression of SEQ ID NO:40 was down-regulated in treated breast carcinoma cells versus untreated human mammary epithelial cells (HMEC) as determined by microarray analysis. The gene expression profile of this nonmalignant mammary epithelial cell line HMEC was compared to the gene expression profiles of seven breast carcinoma lines at different stages of tumor progression. Cell lines compared included: a) MCF-10A, a breast mammary gland (luminal ductal characteristics) cell line isolated from a female with fibrocystic breast disease, b) BT-20, a breast carcinoma cell line derived *in vitro* from the cells emigrating out of thin slices of tumor mass isolated from a female, c) BT-474, a breast ductal carcinoma cell line that was isolated from a solid, invasive ductal carcinoma of the breast obtained from a female, d) BT-483, a breast ductal carcinoma cell line that was isolated from a papillary invasive ductal tumor obtained from a normal, menstruating, parous female with a family history of breast cancer, e) Hs-578T, a breast ductal carcinoma cell line isolated from a female with breast carcinoma, f) MCF-7, a nonmalignant breast adenocarcinoma cell line isolated from the pleural effusion of a female, and g) MDA-MB-468, a breast adenocarcinoma cell line isolated from the pleural effusion of a female with metastatic adenocarcinoma of the breast. All cells were grown under optimal growth conditions in the presence of growth factors and nutrients, with the seven breast carcinoma cell lines being treated with mammary epithelium growth medium (MEGM). In this experiment, the expression of SEQ ID NO:40 was decreased at least two-fold in one out of the seven breast carcinoma cell lines tested, the BT-474, and this same result was obtained in two separate experiments.

In another example, expression of SEQ ID NO:40 was down-regulated in breast carcinoma cells versus untreated human mammary epithelial cells (HMEC) as determined by microarray analysis. The gene expression profile of this nonmalignant mammary epithelial cell line HMEC was compared to the gene expression profiles of the same seven breast carcinoma cell lines listed above. All cells were grown in basal media in the absence of growth factors and hormones for 24 hours prior to comparison The expression of SEQ ID NO:40 was decreased at least two-fold in one of the seven breast carcinoma cell lines tested, the BT-474. Therefore, in various embodiments, SEQ ID NO:40 can be used for one or more of the following: i) monitoring treatment of breast cancer, ii) diagnostic assays for breast cancer, and iii) developing therapeutics and/or other treatments for breast cancer.

In an alternative example, SEQ ID NO:44 showed tissue-specific expression as determined by microarray analysis. RNA samples isolated from a variety of normal human tissues were compared to a common reference sample. Tissues contributing to the reference sample were selected for their ability to provide a complete distribution of RNA in the human body and include brain (4%), heart (7%), kidney (3%), lung (8%), placenta (46%), small intestine (9%), spleen (3%), stomach (6%), testis (9%), and uterus (5%). The normal tissues assayed were obtained from at least three different donors. RNA from each donor was separately isolated and individually hybridized to the microarray. Since these hybridization experiments were conducted using a common reference sample, differential expression values are directly comparable from one tissue to another. The expression of SEQ ID NO:44 was increased by at least two-fold in ventricular heart tissue as compared to the reference sample. Therefore, SEQ ID NO:44 can be used as a tissue marker for ventricular heart tissue.

In an alternative example, expression of SEQ ID NO:47 was down-regulated in breast carcinoma cells versus human mammary epithelial cells (HMEC) as determined by microarray analysis. The gene expression profile of this nonmalignant mammary epithelial cell line HMEC was compared to the gene expression profiles of breast carcinoma lines at different stages of tumor progression. Cell lines compared included:, a) BT-20, a breast carcinoma cell line derived in vitro from the cells emigrating out of thin slices of tumor mass isolated from a female, b) BT-474, a breast ductal carcinoma cell line that was isolated from a solid, invasive ductal carcinoma of the breast obtained from a female, c) BT-483, a breast ductal carcinoma cell line that was isolated from a papillary invasive ductal tumor obtained from a normal, menstruating, parous female with a family history of breast cancer, d) Hs-578T, a breast ductal carcinoma cell line isolated from a female with breast carcinoma, e) MCF-7, a nonmalignant breast adenocarcinoma cell line isolated from the pleural effusion of a female, f) MCF-10A, a breast mammary gland (luminal ductal characteristics) cell line isolated from a female with fibrocystic breast disease, and g) MDA-MB-468, a breast adenocarcinoma cell line isolated from the pleural effusion of a female with metastatic adenocarcinoma of the breast. All cells were grown in basal media in the absence of growth factors and hormones for 24 hours prior to comparison. The expression of SEQ ID NO:47 was decreased at least two-fold in five out of seven breast carcinoma cell lines tested (BT-20, BT-483, MCF-7, MCF-10A, and MDA-MB-468).

In another example, the expression of SEQ ID NO:47 was down-regulated in breast carcinoma cells versus untreated human mammary epithelial cells (HMEC) as determined by microarray analysis. The gene expression profile of this nonmalignant mammary epithelial cell line HMEC was compared to the gene expression profiles of seven breast carcinoma lines at different stages of tumor progression. Cell lines compared included: a) MCF-10A, a breast mammary gland cell line isolated from a female with fibrocystic breast disease, b) MCF-7, a nonmalignant breast adenocarcinoma cell line isolated from the pleural effusion of a female, c) T-47D, a breast carcinoma cell line isolated from a pleural effusion obtained from a female with an infiltrating ductal carcinoma of the breast, d) Sk-BR-3, a breast adenocarcinoma cell line isolated from a malignant pleural effusion of a female, e) BT-20, a breast carcinoma cell line derived *in vitro* from cells emigrating out of thin slices of the tumor mass isolated from a female, f) MDA-mb-435S, a spindle-shaped strain that evolved from the parent line (435) from pleural effusion of a female with metastatic, ductal adenocarcinoma of the breast, and g) MDA-mb-231, a breast tumor cell line isolated from the pleural effusion of a female. All cells were grown in defined serum-free H14 medium to 70-80% confluence prior to RNA harvest. The expression of SEQ ID NO:47 was decreased at least two-fold in five out of seven breast carcinoma cell lines tested (MCF-7, T-47D, Sk-BR-3, BT-20, and MDA-mb-435S). Therefore, in various embodiments, SEQ ID NO:47 can be used for one or more of the following: i) monitoring treatment of breast cancer, ii) diagnostic assays for breast cancer, and iii) developing therapeutics and/or other treatments for breast cancer.

In an alternative example, SEQ ID NO:50 showed differential expression in a number of breast cancer cell lines, as determined by microarray analysis. The gene expression profile of a nonmalignant mammary epithelial cell line (HMEC) was compared to the gene expression profiles of breast carcinoma lines at different stages of tumor progression. Cell lines compared included: a) MCF-10A, a breast mammary gland (luminal ductal characteristics) cell line isolated from a 36-year-old woman with fibrocystic breast disease; b) MCF-7, a nonmalignant breast adenocarcinoma cell line isolated from the pleural effusion of a 69-year-old female; c) BT-20, a breast carcinoma cell line derived *in vitro* from the cells emigrating out of thin slices of tumor mass isolated from a 74-year-old female; d) BT-474, a breast ductal carcinoma cell line that was isolated from a solid, invasive ductal carcinoma of the breast obtained from a 60-year-old woman; e) BT-483, a breast ductal carcinoma cell line that was isolated from a papillary invasive ductal tumor obtained from a 23-year-old normal, menstruating, parous female with a family history of breast cancer; f) Hs 578T, a breast ductal carcinoma cell line isolated from a 74-year-old female with breast carcinoma; and g) MDA-MB-468, a breast adenocarcinoma cell line isolated from the pleural effusion of a 51-year-old female with metastatic adenocarcinoma of the breast. In the first set of experiments, all cells were grown under optimal growth conditions, in the presence of growth factors and nutrients; breast carcinoma cell lines were also treated with mammary epithelium growth medium (MEGM). The expression level of SEQ ID NO:50 was decreased by at least 2-fold in MCF7, BT-474, BT-483 and MDA-MB-468 cell lines, when compared to the expression levels of SEQ ID NO:50 in HMEC cells. In another set of experiments, all cells were grown in basal media in the absence of growth factors and hormones for 24 hours prior to comparison. The expression level of SEQ ID NO:50 was decreased by at least 2-fold in MCF-10A, MCF7, BT-20, BT-483 and MDA-MB-468 cell lines, when compared to the expression levels of SEQ ID NO:50 in HMEC cells.

In another set of experiments, the cell lines examined included the above listed cell lines and the following additional cell lines: a) T-47D, a breast carcinoma cell line isolated from a pleural effusion obtained from a 54-year-old female with an infiltrating ductal carcinoma of the breast; b) Sk-BR-3, a breast adenocarcinoma cell line isolated from a malignant pleural effusion of a 43-year-old female; c) MDA-mb-231, a breast tumor cell line isolated from the pleural effusion of a 51-year-old female; and d) MDA-mb-435S, a spindle-shaped strain that evolved from the parent line (435) isolated by R. Cailleau from pleural effusion of a 31-year-old female with metastatic, ductal adenocarcinoma of the breast. Untreated HMEC control cells were compared to untreated breast cancer cell lines. The expression of SEQ ID NO:50 decreased by at least 2-fold in T-47D cells, at least 5-fold in MCF7 cells, at least 4-fold in BT-20 and Sk-BR-3 cells, and at least 3-fold in MDA-MD-435S and MCF-10A cells, when compared to the expression levels detected in the untreated HMEC cells. In another set of experiments, expression of SEQ ID NO:50 was examined in several breast cancer lines, detailed above, and compared to the expression level detected in MCF-10A fibroblastic cells. The expression of SEQ ID NO:50 decreased at least 3-fold in T-47D cells, in comparison to the gene expression levels detected in MCF-10A cells. Therefore, in various embodiments, SEQ ID NO:50 can be used for one or more of the following: i) monitoring treatment of breast cancer, ii) diagnostic assays for breast, and iii) developing therapeutics and/or other treatments breast cancer.

In yet another example, expression of SEQ ID NO:59 and SEQ ID NO:60 was up-regulated in diseased tissue versus normal tissue as determined by microarray analysis. A normal ovary from a female donor was compared to an ovarian tumor from the same donor (Huntsman Cancer Institute, Salt Lake City, UT). Expression of SEQ ID NO:59 and SEQ ID NO:60 was increased at least two-fold in one out of two samples examined for SEQ ID NO:59 and SEQ ID NO:60. Therefore, in various embodiments, SEQ ID NO:59 and SEQ ID NO:60 can be used for one or more of the following: i) monitoring treatment of ovarian cancer, ii) diagnostic assays for ovarian cancer, and iii) developing therapeutics and/or other treatments for ovarian cancer.

**XII. Complementary Polynucleotides**

Sequences complementary to the CADECM-encoding sequences, or any parts thereof, are used to detect, decrease, or inhibit expression of naturally occurring CADECM. Although use of oligonucleotides comprising from about 15 to 30 base pairs is described, essentially the same procedure is used with smaller or with larger sequence fragments. Appropriate oligonucleotides are designed using OLIGO 4.06 software (National Biosciences) and the coding sequence of CADECM. To inhibit transcription, a complementary oligonucleotide is designed from the most unique 5' sequence and used to prevent promoter binding to the coding sequence. To inhibit translation, a complementary oligonucleotide is designed to prevent ribosomal binding to the CADECM-encoding transcript.

**XIII. Expression of CADECM**

Expression and purification of CADECM is achieved using bacterial or virus-based expression systems. For expression of CADECM in bacteria, cDNA is subcloned into an appropriate vector containing an antibiotic resistance gene and an inducible promoter that directs high levels of cDNA transcription. Examples of such promoters include, but are not limited to, the *trp-lac* (*tac*) hybrid promoter and the T5 or T7 bacteriophage promoter in conjunction with the *lac* operator regulatory element. Recombinant vectors are transformed into suitable bacterial hosts, e.g., BL21(DE3). Antibiotic resistant bacteria express CADECM upon induction with isopropyl beta-D-thiogalactopyranoside (IPTG). Expression of CADECM in eukaryotic cells is achieved by infecting insect or mammalian cell lines with recombinant *Autographica californica* nuclear polyhedrosis virus (AcMNPV), commonly known as baculovirus. The nonessential polyhedrin gene of baculovirus is replaced with cDNA encoding CADECM by either homologous recombination or bacterial-mediated transposition involving transfer plasmid intermediates. Viral infectivity is maintained and the strong polyhedrin promoter drives high levels of cDNA transcription. Recombinant baculovirus is used to infect *Spodoptera frugiperda* (Sf9) insect cells in most cases, or human hepatocytes, in some cases. Infection of the latter requires additional genetic modifications to baculovirus (Engelhard, E.K. et al. (1994) Proc. Natl. Acad. Sci. USA 91:3224-3227; Sandig, V. et al. (1996) Hum. Gene Ther. 7:1937-1945).

In most expression systems, CADECM is synthesized as a fusion protein with, e.g., glutathione S-transferase (GST) or a peptide epitope tag, such as FLAG or 6-His, permitting rapid, single-step, affinity-based purification of recombinant fusion protein from crude cell lysates. GST, a 26-kilodalton enzyme from *Schistosoma japonicum*, enables the purification of fusion proteins on immobilized glutathione under conditions that maintain protein activity and antigenicity (Amersham Biosciences). Following purification, the GST moiety can be proteolytically cleaved from CADECM at specifically engineered sites. FLAG, an 8-amino acid peptide, enables immunoaffinity purification using commercially available monoclonal and polyclonal anti-FLAG antibodies (Eastman Kodak). 6-His, a stretch of six consecutive histidine residues, enables purification on metal-chelate resins (QIAGEN). Methods for protein expression and purification are discussed in Ausubel et al. (*supra*, ch. 10 and 16). Purified CADECM obtained by these methods can be used directly in the assays shown in Examples XVII and XVIII, where applicable.

**XIV. Functional Assays**

CADECM function is assessed by expressing the sequences encoding CADECM at physiologically elevated levels in mammalian cell culture systems. cDNA is subcloned into a mammalian expression vector containing a strong promoter that drives high levels of cDNA expression. Vectors of choice include PCMV SPORT plasmid (Invitrogen, Carlsbad CA) and PCR3.1 plasmid (Invitrogen), both of which contain the cytomegalovirus promoter. 5-10 $\mu$g of recombinant vector are transiently transfected into a human cell line, for example, an endothelial or hematopoietic cell line, using either liposome formulations or electroporation. 1-2 $\mu$g of an additional plasmid containing sequences encoding a marker protein are co-transfected. Expression of a marker protein provides a means to distinguish transfected cells from nontransfected cells and is a reliable predictor of cDNA expression from the recombinant vector. Marker proteins of choice include, e.g., Green Fluorescent Protein (GFP; BD Clontech), CD64, or a CD64-GFP fusion protein. Flow cytometry (FCM), an automated, laser optics-based technique, is used to identify transfected cells expressing GFP or CD64-GFP and to evaluate the apoptotic state of the cells and other cellular properties. FCM detects and quantifies the uptake of fluorescent molecules that diagnose events preceding or coincident with cell death. These events include changes in nuclear DNA content as measured by staining of DNA with propidium iodide; changes in cell size and granularity as measured by forward light scatter and 90 degree side light scatter; down-regulation of DNA synthesis as measured by decrease in bromodeoxyuridine uptake; alterations in expression of cell surface and intracellular proteins as measured by reactivity with specific antibodies; and alterations in plasma membrane composition as measured by the binding of fluorescein-conjugated Annexin V protein to the cell surface. Methods in flow cytometry are discussed in Ormerod, M.G. (1994; Flow Cytometry, Oxford, New York NY).

The influence of CADECM on gene expression can be assessed using highly purified populations of cells transfected with sequences encoding CADECM and either CD64 or CD64-GFP. CD64 and CD64-GFP are expressed on the surface of transfected cells and bind to conserved regions of human immunoglobulin G (IgG). Transfected cells are efficiently separated from nontransfected cells using magnetic beads coated with either human IgG or antibody against CD64 (DYNAL, Lake Success NY). mRNA can be purified from the cells using methods well known by those of skill in the art. Expression of mRNA encoding CADECM and other genes of interest can be analyzed by northern analysis or microarray techniques.

**XV. Production of CADECM Specific Antibodies**

CADECM substantially purified using polyacrylamide gel electrophoresis (PAGE; see, e.g., Harrington, M.G. (1990) Methods Enzymol. 182:488-495), or other purification techniques, is used to immunize animals (e.g., rabbits, mice, etc.) and to produce antibodies using standard protocols.

Alternatively, the CADECM amino acid sequence is analyzed using LASERGENE software (DNASTAR) to determine regions of high immunogenicity, and a corresponding oligopeptide is synthesized and used to raise antibodies by means known to those of skill in the art. Methods for selection of appropriate epitopes, such as those near the C-terminus or in hydrophilic regions are well described in the art (Ausubel et al., *supra*, ch. 11).

Typically, oligopeptides of about 15 residues in length are synthesized using an ABI 431A peptide synthesizer (Applied Biosystems) using FMOC chemistry and coupled to KLH (Sigma-Aldrich, St. Louis MO) by reaction with N-maleimidobenzoyl-N-hydroxysuccinimide ester (MBS) to increase immunogenicity (Ausubel et al., *supra*). Rabbits are immunized with the oligopeptide-KLH complex in complete Freund's adjuvant. Resulting antisera are tested for antipeptide and anti-CADECM activity by, for example, binding the peptide or CADECM to a substrate, blocking with 1% BSA, reacting with rabbit antisera, washing, and reacting with radio-iodinated goat anti-rabbit IgG.

**XVI. Purification of Naturally Occurring CADECM Using Specific Antibodies**

Naturally occurring or recombinant CADECM is substantially purified by immunoaffinity chromatography using antibodies specific for CADECM. An immunoaffinity column is constructed by covalently coupling anti-CADECM antibody to an activated chromatographic resin, such as CNBr-activated SEPHAROSE (Amersham Biosciences). After the coupling, the resin is blocked and washed according to the manufacturer's instructions.

Media containing CADECM are passed over the immunoaffinity column, and the column is washed under conditions that allow the preferential absorbance of CADECM (e.g., high ionic strength buffers in the presence of detergent). The column is eluted under conditions that disrupt antibody/CADECM binding (e.g., a buffer of pH 2 to pH 3, or a high concentration of a chaotrope, such as urea or thiocyanate ion), and CADECM is collected.

**XVII. Identification of Molecules Which Interact with CADECM**

CADECM, or biologically active fragments thereof, are labeled with $^{125}$I Bolton-Hunter reagent (Bolton, A.E. and W.M. Hunter (1973) Biochem. J. 133:529-539). Candidate molecules previously arrayed in the wells of a multi-well plate are incubated with the labeled CADECM, washed, and any wells with labeled CADECM complex are assayed. Data obtained using different concentrations of CADECM are used to calculate values for the number, affinity, and association of CADECM with the candidate molecules.

Alternatively, molecules interacting with CADECM are analyzed using the yeast two-hybrid system as described in Fields, S. and O. Song (1989; Nature 340:245-246), or using commercially available kits based on the two-hybrid system, such as the MATCHMAKER system (BD Clontech).

CADECM may also be used in the PATHCALLING process (CuraGen Corp., New Haven CT) which employs the yeast two-hybrid system in a high-throughput manner to determine all interactions between the proteins encoded by two large libraries of genes (Nandabalan, K. et al. (2000) U.S. Patent No. 6,057,101).

**XVIII. Demonstration of CADECM Activity**

An assay for CADECM activity measures the expression of CADECM on the cell surface. cDNA encoding CADECM is transfected into a non-leukocytic cell line. Cell surface proteins are labeled with biotin (de la Fuente, M.A. et al. (1997) Blood 90:2398-2405). Immunoprecipitations are performed using CADECM-specific antibodies, and immunoprecipitated samples are analyzed using SDS-PAGE and immunoblotting techniques. The ratio of labeled immunoprecipitant to unlabeled immunoprecipitant is proportional to the amount of CADECM expressed on the cell surface.

Alternatively, an assay for CADECM activity measures the amount of cell aggregation induced by overexpression of CADECM. In this assay, cultured cells such as NIH3T3 are transfected with cDNA encoding CADECM contained within a suitable mammalian expression vector under control of a strong promoter. Cotransfection with cDNA encoding a fluorescent marker protein, such as Green Fluorescent Protein (CLONTECH), is useful for identifying stable transfectants. The amount of cell agglutination, or clumping, associated with transfected cells is compared with that associated with untransfected cells. The amount of cell agglutination is a direct measure of CADECM activity.

Alternatively, an assay for CADECM activity measures the disruption of cytoskeletal filament networks upon overexpression of CADECM in cultured cell lines (Rezniczek, G. A. et al. (1998) J. Cell Biol. 141:209-225). cDNA encoding CADECM is subcloned into a mammalian expression vector that drives high levels of cDNA expression. This construct is transfected into cultured cells, such as rat kangaroo PtK2 or rat bladder carcinoma 804G cells. Actin filaments and intermediate filaments such as keratin and vimentin are visualized by immunofluorescence microscopy using antibodies and techniques well known in the art. The configuration and abundance of cyoskeletal filaments can be assessed and quantified using confocal imaging techniques. In particular, the bundling and collapse of cytoskeletal filament networks is indicative of CADECM activity.

Alternatively, cell adhesion activity in CADECM is measured in a 96-well plate in which wells are first coated with CADECM by adding solutions of CADECM of varying concentrations to the wells. Excess CADECM is washed off with saline, and the wells incubated with a solution of 1% bovine serum albumin to block non-specific cell binding. Aliquots of a cell suspension of a suitable cell type are then added to the wells and incubated for a period of time at 37 °C. Non-adherent cells are washed off with saline and the cells stained with a suitable cell stain such as Coomassie blue. The intensity of staining is measured using a variable wavelength multi-well plate reader and compared to a standard curve to determine the number of cells adhering to the CADECM coated plates. The degree of cell staining is proportional to the cell adhesion activity of CADECM in the sample.

Various modifications and variations of the described compositions, methods, and systems of the invention will be apparent to those skilled in the art without departing from the scope and spirit of the invention. It will be appreciated that the invention provides novel and useful proteins, and their encoding polynucleotides, which can be used in the drug discovery process, as well as methods for using these compositions for the detection, diagnosis, and treatment of diseases and conditions. Although the invention has been described in connection with certain embodiments, it should be understood that the invention as claimed should not be unduly limited to such specific embodiments. Nor should the description of such embodiments be considered exhaustive or limit the invention to the precise forms disclosed. Furthermore, elements from one embodiment can be readily recombined with elements from one or more other embodiments. Such combinations can form a number of embodiments within the scope of the invention. It is intended that the scope of the invention be defined by the following claims and their equivalents.

# Table 1

| Incyte Project ID | Polypeptide SEQ ID NO: | Incyte Polypeptide ID | Polynucleotide SEQ ID NO: | Incyte Polynucleotide ID | Incyte Full Length Clones |
|---|---|---|---|---|---|
| 7514594 | 1 | 7514594CD1 | 31 | 7514594CB1 | 95037202CA2, 95048617CA2, 95048773CA2 |
| 7515764 | 2 | 7515764CD1 | 32 | 7515764CB1 | |
| 7514575 | 3 | 7514575CD1 | 33 | 7514575CB1 | 95028214CA2, 95028486CA2 |
| 7514576 | 4 | 7514576CD1 | 34 | 7514576CB1 | 95028314CA2, 95028406CA2, 95028454CA2, 95028462CA2, 95028586CA2 |
| 7515593 | 5 | 7515593CD1 | 35 | 7515593CB1 | 4163744CA2, 95048425CA2, 95048509CA2, 95048541CA2 |
| 7517754 | 6 | 7517754CD1 | 36 | 7517754CB1 | |
| 7517819 | 7 | 7517819CD1 | 37 | 7517819CB1 | |
| 7518460 | 8 | 7518460CD1 | 38 | 7518460CB1 | 95077761CA2 |
| 7518734 | 9 | 7518734CD1 | 39 | 7518734CB1 | 95075872CA2 |
| 7513469 | 10 | 7513469CD1 | 40 | 7513469CB1 | |
| 7518836 | 11 | 7518836CD1 | 41 | 7518836CB1 | 95067129CA2 |
| 7519868 | 12 | 7519868CD1 | 42 | 7519868CB1 | |
| 7520048 | 13 | 7520048CD1 | 43 | 7520048CB1 | |
| 7520157 | 14 | 7520157CD1 | 44 | 7520157CB1 | 90069946CA2, 90070038CA2, 90070046CA2, 95130377CA2 |
| 7520485 | 15 | 7520485CD1 | 45 | 7520485CB1 | 4938634CA2, 90143737CA2 |
| 7525723 | 16 | 7525723CD1 | 46 | 7525723CB1 | 95056568CA2 |
| 7525966 | 17 | 7525966CD1 | 47 | 7525966CB1 | |
| 7519667 | 18 | 7519667CD1 | 48 | 7519667CB1 | |
| 7520638 | 19 | 7520638CD1 | 49 | 7520638CB1 | 90196529CA2 |
| 7518947 | 20 | 7518947CD1 | 50 | 7518947CB1 | 90156696CA2 |
| 7519652 | 21 | 7519652CD1 | 51 | 7519652CB1 | 90097633CA2 |
| 7520381 | 22 | 7520381CD1 | 52 | 7520381CB1 | 90198764CA2 |
| 7520409 | 23 | 7520409CD1 | 53 | 7520409CB1 | |
| 7520538 | 24 | 7520538CD1 | 54 | 7520538CB1 | |
| 7520667 | 25 | 7520667CD1 | 55 | 7520667CB1 | |
| 7517567 | 26 | 7517567CD1 | 56 | 7517567CB1 | |
| 7519867 | 27 | 7519867CD1 | 57 | 7519867CB1 | |
| 7519911 | 28 | 7519911CD1 | 58 | 7519911CB1 | 90127163CA2 |

# Table 1

| Incyte Project ID | Polypeptide SEQ ID NO: | Incyte Polypeptide ID | Polynucleotide SEQ ID NO: | Incyte Polynucleotide ID | Incyte Full Length Clones |
|---|---|---|---|---|---|
| 7520005 | 29 | 7520005CD1 | 59 | 7520005CB1 | |
| 7520484 | 30 | 7520484CD1 | 60 | 7520484CB1 | |

# Table 2

| Polypeptide SEQ ID NO: | Incyte Polypeptide ID | GenBank ID NO: or PROTEOME ID NO: | Probability Score | Annotation |
|---|---|---|---|---|
| 1 | 7514594CD1 | g5821288 | 1.6E-89 | [Homo sapiens] macrophage C-type lectin Mincle |
| | | | | Matsumoto, M. et al. A novel LPS-inducible C-type lectin is a transcriptional target of NF-IL6 in macrophages J. Immunol. 163, 5039-5048 (1999) |
| | | 569282\|CLECSF9 | 1.3E-90 | [Homo sapiens][Small molecule-binding protein] C-type lectin (calcium dependent, carbohydrate-recognition domain) superfamily member 9, a macrophage-inducible lectin that may be a downstream target of the transcription factor NF-IL6 (CEBPB) and may be induced by inflammatory stimuli |
| | | | | Matsumoto, M. et al. A novel LPS-inducible C-type lectin is a transcriptional target of NF-IL6 in macrophages J. Immunol. 163, 5039-5048 (1999) |
| | | 609080\|Clecsf9 | 3.2E-51 | [Mus musculus][Small molecule-binding protein] C-type lectin (calcium dependent, carbohydrate-recognition domain) superfamily member 9, a group 2 C-type lectin and downstream target of the transcription factor NF-IL6 (Cebpb) and may be involved in the inflammatory response of activated macrophages |
| | | | | Balch, S. G. et al. Organization of the mouse macrophage C-type lectin (Mcl) gene and identification of a subgroup of related lectin molecules Eur. J. Immunogene.t 29, 61-64. (2002) |
| 2 | 7515764CD1 | g3510460 | 8.0E-17 | [Homo sapiens] collagen type IX alpha 2 chain |
| | | | | Pihlajamaa, T. et al. Human COL9A1 and COL9A2 genes. Two genes of 90 and 15 kb code for similar polypeptides of the same collagen molecule Matrix Biol. 17, 237-241 (1998) |
| | | 782507\|FLJ30681 | 1.9E-149 | [Homo sapiens] Protein containing an epidermal growth factor (EGF)-like domain, has a region of high similarity to a region of collagen type V alpha 2 subunit (human COL5A2), which is an extracellular matrix structural protein involved in connective tissue maintenance |

# Table 2

| Polypeptide SEQ ID NO: | Incyte Polypeptide ID | GenBank ID NO: or PROTEOME ID NO: | Probability Score | Annotation |
|---|---|---|---|---|
| | | 625863\|COL9A2 | 6.2E-18 | [Homo sapiens][Structural protein][Extracellular matrix (cuticle and basement membrane); Extracellular (excluding cell wall)] Alpha 2 subunit of type IX collagen, member of the family of fibril-associated collagens that are structural components of cartilage extracellular matrix; mutation of the corresponding gene causes epiphyseal dysplasia and invertebral disc disease |
| | | | | Holden, P. et al. Identification of novel pro-alpha2(IX) collagen gene mutations in two families with distinctive oligo-epiphyseal forms of multiple epiphyseal dysplasia Am. J. Hum. Genet. 65, 31-38 (1999) |
| | | | | Bonnemann, C. G. et al. A mutation in the alpha 3 chain of type IX collagen causes autosomal dominant multiple epiphyseal dysplasia with mild myopathy Proc. Natl. Acad. Sci. U. S. A. 97, 1212-1217 (2000) |
| 3 | 7514575CD1 | g3483011 | 1.9E-39 | [Homo sapiens] uroplakin II |
| | | | | Lobban, E. D. et al. Uroplakin gene expression by normal and neoplastic human urothelium Am. J. Pathol. 153, 1957-1967 (1998) |
| | | 568028\|UPK2 | 1.5E-40 | [Homo sapiens][Plasma membrane] Uroplakin II, integral membrane component of asymmetric unit membranes in superficial urothelial cells of urothelium; expressed in differentiated transitional cell carcinoma and is detected in peripheral blood of patients with metastatic disease |
| | | | | Wu, X. R. et al. Mammalian uroplakins. A group of highly conserved urothelial differentiation-related membrane proteins J. Biol. Chem. 269, 13716-13724 (1994) |
| | | 581801\|Upk2 | 2.1E-36 | [Mus musculus][Plasma membrane] Uroplakin II, integral membrane component of asymmetric unit membranes in suprabasal cells of the bladder urothelium; human UPK2 is expressed in differentiated transitional cell carcinoma and is detected in peripheral blood during metastatic disease |
| | | | | Lin, J. H. et al. A tissue-specific promoter that can drive a foreign gene to express in the suprabasal urothelial cells of transgenic mice Proc. Nat. Aca. Sci. U. S. A. 92, 679-683 (1995) |
| 4 | 7514576CD1 | g3483011 | 4.4E-49 | [Homo sapiens] uroplakin II |

# Table 2

| Polypeptide SEQ ID NO: | Incyte Polypeptide ID | GenBank ID NO: or PROTEOME ID NO: | Probability Score | Annotation |
|---|---|---|---|---|
| | | | | Lobban, E. D. et al. (1998) *supra* |
| | | 568028\|UPK2 | 3.5E-50 | [Homo sapiens][Plasma membrane] Uroplakin II, integral membrane component of asymmetric unit membranes in superficial urothelial cells of urothelium; expressed in differentiated transitional cell carcinoma and is detected in peripheral blood of patients with metastatic disease |
| | | | | Seraj, M. J. et al. Molecular determination of perivesical and lymph node metastasis after radical cystectomy for urothelial carcinoma of the bladder Clin. Cancer Res. 7, 1516-1522 (2001) |
| | | 581801\|Upk2 | 2.0E-45 | [Mus musculus][Plasma membrane] Uroplakin II, integral membrane component of asymmetric unit membranes in suprabasal cells of the bladder urothelium; human UPK2 is expressed in differentiated transitional cell carcinoma and is detected in peripheral blood during metastatic disease |
| | | | | Hu, P. et al. (2000) *supra* |
| 5 | 7515593CD1 | g5821288 | 6.9E-37 | [Homo sapiens] macrophage C-type lectin Mincle |
| | | | | Matsumoto, M. et al. A novel LPS-inducible C-type lectin is a transcriptional target of NF-IL6 in macrophages J. Immunol. 163, 5039-5048 (1999) |
| | | 569282\|CLECSF 9 | 5.3E-38 | [Homo sapiens][Small molecule-binding protein] C-type lectin (calcium dependent, carbohydrate-recognition domain) superfamily member 9, a macrophage-inducible lectin that may be a downstream target of the transcription factor NF-IL6 (CEBPB) and may be induced by inflammatory stimuli |
| | | | | Matsumoto, M. et al. (1999) *supra* |
| | | 609080\|Clecsf9 | 1.4E-16 | [Mus musculus][Small molecule-binding protein] C-type lectin (calcium dependent, carbohydrate-recognition domain) superfamily member 9, a group 2 C-type lectin and downstream target of the transcription factor NF-IL6 (Cebpb) and may be involved in the inflammotory response of activated macrophages |
| | | | | Matsumoto, M. et al. (1999) *supra* |
| 6 | 7517754CD1 | g7634793 | 4.2E-69 | [Homo sapiens] EGF-containing fibulin-like extracellular matrix protein 2 |

# Table 2

| Polypeptide SEQ ID NO: | Incyte Polypeptide ID | GenBank ID NO: or PROTEOME ID NO: | Probability Score | Annotation |
|---|---|---|---|---|
| | | | | Katsanis, N. et al. Isolation of a paralog of the Doyne honeycomb retinal dystrophy gene from the multiple retinopathy critical region on 11q13 Hum. Genet. 106, 66-72 (2000) |
| | | 571186\|EFEMP2 | 3.3E-70 | [Homo sapiens][Structural protein][Extracellular matrix (cuticle and basement membrane); Basement membrane (extracellular matrix); Extracellular (excluding cell wall)] Egf-containing fibulin-like extracellular matrix protein 2 (Fibulin 4), contains four EGF domains and six calcium-binding EGF domains and may be involved in tumor progression |
| | | | | Giltay, R. et al. Sequence, recombinant expression and tissue localization of two novel extracellular matrix proteins, fibulin-3 and fibulin-4 Matrix Biol. 18, 469-480 (1999) |
| | | 619022\|Efemp2 | 1.3E-61 | [Mus musculus] Egf-containing fibulin-like extracellular matrix protein 2 (Fibulin 4), contains four EGF domains and six calcium-binding EGF domains, binds preferentially to a mutant form of p53 (Trp53) and may be involved in tumor progression |
| | | | | Gallagher, W. M. et al. MBP1: a novel mutant p53-specific protein partner with oncogenic properties Oncogene 18, 3608-3616 (1999) |
| 7 | 7517819CD1 | g186497 | 1.4E-272 | [Homo sapiens] integrin alpha-3 chain |
| | | | | Takada, Y. et al. Molecular cloning and expression of the cDNA for alpha 3 subunit of human alpha 3 beta 1 (VLA-3), an integrin receptor for fibronectin, laminin, and collagen J. Cell Biol. 115, 257-266 (1991) |
| | | 568212\|ITGA3 | 1.1E-273 | [Homo sapiens][Adhesin/agglutinin; Receptor (signalling)][Extracellular matrix (cuticle and basement membrane); Basement membrane (extracellular matrix); Plasma membrane; Cell junction] Integrin alpha 3, subunit of VLA-3 integrin, a receptor for collagen, laminin and fibronectin, plays a role in cell adhesion and migration, may play a role in the migration and invasion of melanoma cells |

# Table 2

| Polypeptide SEQ ID NO: | Incyte Polypeptide ID | GenBank ID NO: or PROTEOME ID NO: | Probability Score | Annotation |
|---|---|---|---|---|
| | | | | Puzon-McLaughlin, W. et al. Multiple discontinuous ligand-mimetic antibody binding sites define a ligand binding pocket in integrin alpha(IIb)beta(3) J. Biol. Chem. 275, 7795-7802. (2000) |
| | | 587097\|Itga3 | 8.0E-241 | [Mus musculus][Receptor (signalling); Small molecule-binding protein][Plasma membrane] Integrin alpha 3, subunit of VLA-3 integrin, a receptor for collagen, laminin, fibronectin and epiligrin, plays a role in cell adhesion and migration, may play a role in basement membrane organization and branching morphogenesis |
| | | | | Takeuchi, K. et al. cDNA cloning of mouse VLA-3 alpha subunit J. Cell Biochem. 57, 371-377 (1995) |
| 8 | 7518460CD1 | g14326449 | 1.1E-204 | [Homo sapiens] vitronectin |
| | | 618414\|VTN | 6.0E-205 | [Homo sapiens][Small molecule-binding protein][Extracellular (excluding cell wall)] Vitronectin (S-protein, epibolin), an integrin ligand that mediates cell-to-substrate adhesion and migration, contributes to tissue remodelling, inflammation, and possibly blood coagulation, may have roles in cancer cell invasion and atherosclerosis |
| | | | | Mink, M. et al. A novel human gene (SARM) at chromosome 17q11 encodes a protein with a SAM motif and structural similarity to armadillo/beta-catenin that is conserved in mouse, drosophila, and Caenorhabditis elegans Genomics 74, 234-244 (2001) |
| | | 586139\|Vtn | 1.3E-154 | [Mus musculus][Small molecule-binding protein][Extracellular (excluding cell wall)] Vitronectin, an integrin ligand that mediates cell adhesion, has roles in hemostasis, angiogenesis, and acute-phase and inflammatory responses, human VTN may have roles in cancer cell invasion and atherosclerosis |
| | | | | Seiffert, D. et al. Detection of vitronectin mRNA in tissues and cells of the mouse Proc. Natl. Acad. Sci. U. S. A. 88, 9402-9406 (1991) |
| 9 | 7518734CD1 | g15593994 | 3.6E-127 | [Homo sapiens] junctional adhesion molecule-2 |

## Table 2

| Polypeptide SEQ ID NO: | Incyte Polypeptide ID | GenBank ID NO: or PROTEOME ID NO: | Probability Score | Annotation |
|---|---|---|---|---|
| | | 768992\|Jcam2 | 1.3E-34 | [Mus musculus][Plasma membrane; Cell junction] Junctional cell adhesion molecule 2, member of the immunoglobulin superfamily, localizes to tight junctional complexes and is involved in transmigration of lymphocytes across high endothelial venules |
| | | | | Palmeri, D. et al. Vascular endothelial junction-associated molecule, a novel member of the immunoglobulin superfamily, is localized to intercellular boundaries of endothelial cells J. Biol. Chem. 275, 19139-19145 (2000) |
| | | 618048\|JAM2 | 2.1E-34 | [Homo sapiens][Plasma membrane] Junctional cell adhesion molecule 2, a member of the immunoglobulin superfamily, localized to tight junctional complexes, may help in neutrophil and monocyte transendothelial migration |
| | | | | Cunningham, S. A. et al. A novel protein with homology to the junctional adhesion molecule. Characterization of leukocyte interactions J. Biol. Chem. 275, 34750-34756 (2000) |
| 10 | 7513469CD1 | g904223 | 3.7E-250 | [Homo sapiens] polycystic kidney disease 1 protein |
| | | | | Ward, C. J. et al. The polycystic kidney disease 1 gene encodes a 14 kb transcript and lies within a duplicated region on chromosome 16. The European Polycystic Kidney Disease Consortium [published erratum appears in Cell 1994 Aug 26; 78: following 724], Cell 77, 881-894 (1994) |
| | | | | Hughes, J. et al., The polycystic kidney disease 1 (PKD1) gene encodes a novel protein with multiple cell recognition domains, Nat. Genet. 10, 151-160 (1995) |
| | | 336982\|PKD1 | 2.9E-251 | [Homo sapiens][Inhibitor or repressor;Channel (passive transporter);Transporter][Cytoplasmic;Plasma membrane;] Polycystic kidney disease 1 (Polycystin), an integral membrane protein that forms a non-selective cation channel with PDK2, involved in cell adhesion; genetic mutations are responsible for most cases of autosomal dominant polycystic kidney disease |
| | | | | Hughes, J. et al. The polycystic kidney disease 1 (PKD1) gene encodes a novel protein with multiple cell recognition domains. Nat. Genet. 10, 151-160 (1995) |

# Table 2

| Polypeptide SEQ ID NO: | Incyte Polypeptide ID | GenBank ID NO: or PROTEOME ID NO: | Probability Score | Annotation |
|---|---|---|---|---|
| | | 587249\|Pkd1 | 3.9E-195 | [Mus musculus][Adhesin/agglutinin][Plasma membrane] Polycystic kidney disease 1 homolog, required for morphogenesis of tubular structures in the kidney and pancreas, and for the structural integrity of blood vessels; human PDK1 mutations result in autosomal dominant polycystic kidney disease |
| 11 | 7518836CD1 | g950417 | 1.5E-82 | [Homo sapiens] cell surface glycoprotein CD44 |
| | | | | Shtivelman, E. et al. Expression of CD44 is repressed in neuroblastoma cells. Mol. Cell. Biol. 11, 5446-5453 (1991) |
| | | 661102\|CD44 | 4.9E-83 | [Homo sapiens][Adhesin/agglutinin;Receptor (signalling);Small molecule-binding protein][Cytoplasmic;Cytoskeletal;Plasma membrane] Transmembrane glycoprotein CD44, receptor for hyaluronate, involved in extracellular matrix attachment and tumor metastasis, regulates cytokine gene expression in inflammatory diseases, binds osteopontin (SPP1), which stimulates cellular chemotaxis |
| | | | | Zhou, D. F. et al. Molecular cloning and expression of Pgp-1. The mouse homolog of the human H-CAM (Hermes) lymphocyte homing receptor. J. Immunol. 143, 3390-3395 (1989) |
| | | 790385\|Cd44 | 1.4E-64 | [Rattus norvegicus][Receptor (signalling)][Plasma membrane] Transmembrane glycoprotein CD44, receptor for hyaluronate, involved in extracellular matrix attachment and tumor metastasis, regulates cytokine gene expression in inflammatory diseases, binds osteopontin (SPP1), which stimulates cellular chemotaxis |
| | | | | Gunthert, U. et al. A new variant of glycoprotein CD44 confers metastatic potential to rat carcinoma cells. Cell 65, 13-24 (1991) |
| 12 | 7519868CD1 | g15281077 | 4.7E-117 | [Homo sapiens] mDC-SIGN1A type III isoform |
| | | | | Mummidi, S. et al. Extensive repertoire of membrane-bound and soluble dendritic cell-specific ICAM-3-grabbing nonintegrin 1 (DC-SIGN1) and DC-SIGN2 isoforms. Inter-individual variation in expression of DC-SIGN transcripts J. Biol. Chem. 276, 33196-33212 (2001) |

# Table 2

| Polypeptide SEQ ID NO: | Incyte Polypeptide ID | GenBank ID NO: or PROTEOME ID NO: | Probability Score | Annotation |
|---|---|---|---|---|
| | | 617980\|CD209 | 4.7E-117 | [Homo sapiens][Small molecule-binding protein][Plasma membrane] Dendritic cell-specific ICAM3-grabbing nonintegrin, a dendritic cell specific, C-type lectin that binds ICAM3, promotes dendritic cell adhesion to T cells and stimulates T cell proliferation during an immune response; may promote infection by HIV-1 |
| | | | | Curtis, B. M. et al. Sequence and expression of a membrane-associated C-type lectin that exhibits CD4-independent binding of human immunodeficiency virus envelope glycoprotein gp120. Proc. Natl. Acad. Sci. U. S. A. 89, 8356-8360 (1992) |
| | | 777260\|Cd209a | 5.1E-73 | [Mus musculus] Protein containing a C-type lectin domain, which mediate calcium-dependent carbohydrate recognition, has moderate similarity to c-type lectin (calcium dependent, carbohydrate-recognition domain) superfamily member 9 (mouse Clecsf9) |
| | | | | Park, C. G. et al., Five mouse homologues of the human dendritic cell C-type lectin, DC-SIGN. Int. Immunol. 13, 1283-1290 (2001) |
| 13 | 7520048CD1 | g18252658 | 2.4E-106 | [Mus musculus] Jedi-736 protein |
| | | 716305\|MEGF10 | 1.4E-50 | [Homo sapiens] MEGF10 protein, contains multiple EGF-like domains |
| | | | | Nagase, T. et al. Prediction of the coding sequences of unidentified human genes. XX. The complete sequences of 100 new cDNA clones from brain which code for large proteins in vitro. DNA Res. 8, 85-95 (2001) |
| | | 662841\|Egfl3 | 7.4E-26 | [Rattus norvegicus] Multiple epidermal growth factor (EGF)-like domains 6, contains 30 epidermal growth factor-like motifs, putative secreted protein, predicted to bind calcium |
| | | | | Nakayama, M. et al. Identification of high-molecular-weight proteins with multiple EGF-like motifs by motif-trap screening. Genomics 51, 27-34 (1998) |
| 14 | 7520157CD1 | g10803406 | 2.3E-282 | [Homo sapiens] cadherin-19 |
| | | | | Kools, P. et al. Characterization of three novel human cadherin genes (CDH7, CDH19, and CDH20) clustered on chromosome 18q22-q23 and with high homology to chicken cadherin-7. Genomics 68, 283-295 (2000) |

# Table 2

| Polypeptide SEQ ID NO: | Incyte Polypeptide ID | GenBank ID NO: or PROTEOME ID NO: | Probability Score | Annotation |
|---|---|---|---|---|
| | | 742392\|CDH19 | 1.6E-283 | [Homo sapiens] Cadherin 19 type 2, a cell adhesion molecule that may also be a tumor suppressor |
| | | | | Kremmidiotis, G. et al. Localization of human cadherin genes to chromosome regions exhibiting cancer-related loss of heterozygosity. Genomics 49, 467-471 (1998) |
| | | 752241\|CDH10 | 6.8E-169 | [Homo sapiens][Adhesin/agglutinin][Plasma membrane] Cadherin 10 (T2-cadherin), a member of the cadherin family of calcium-dependent glycoproteins that mediate cell-cell interactions |
| | | | | Shimoyama, Y. et al. Identification of three human type-II classic cadherins and frequent heterophilic interactions between different subclasses of type-II classic cadherins. Biochem. J. 349, 159-167 (2000) |
| 15 | 7520485CD1 | g15383610 | 3.5E-135 | [Homo sapiens] mDC-SIGN2 type IV isoform |
| | | | | Mummidi, S. et al. Extensive repertoire of membrane-bound and soluble dendritic cell-specific ICAM-3-grabbing nonintegrin 1 (DC-SIGN1) and DC-SIGN2 isoforms. Inter-individual variation in expression of DC-SIGN transcripts. J. Biol. Chem. 276, 33196-33212 (2001) *supra* |
| | | 800231\|CD209L | 4.0E-113 | [Homo sapiens][Plasma membrane] CD209 antigen-like, a member of the C-type lectin family, acts as a receptor for ICAM3 and binds to strains of HIV-1, HIV-2 and simian immunodeficiency virus, promoting their pathogenesis |
| | | | | Yokoyama-Kobayashi, M. et al. Selection of cDNAs encoding putative type II membrane proteins on the cell surface from a human full-length cDNA bank. Gene 228, 161-167 (1999) |
| | | 777260\|Cd209a | 1.8E-14 | [Mus musculus] Protein containing a C-type lectin domain, which mediate calcium-dependent carbohydrate recognition, has moderate similarity to c-type lectin (calcium dependent, carbohydrate-recognition domain) superfamily member 9 (mouse Clecsf 9) |
| | | | | Baribaud, F. et al. Functional and antigenic characterization of human, rhesus macaque, pigtailed macaque, and murine DC-SIGN. J. Virol. 75, 10281-10289 (2001) |

# Table 2

| Polypeptide SEQ ID NO: | Incyte Polypeptide ID | GenBank ID NO: or PROTEOME ID NO: | Probability Score | Annotation |
|---|---|---|---|---|
| 16 | 7525723CD1 | g1699162 | 1.6E-244 | [Homo sapiens] X-linked retintis pigmentosa candidate gene |
| | | | | Dry, K. L. et al. Identification of a novel gene, ETX1 from Xp21.1, a candidate gene for X-linked retintis pigmentosa (RP3). Hum. Mol. Genet. 4, 2347-2353 (1995) |
| | | 343818\|SRPX | 1.1E-245 | [Homo sapiens] Sushi-repeat-containing protein (X chromosome), may be involved in negative regulation of cell growth, expression is reduced in several cancer cell lines; mutation of the corresponding gene may be associated with X-linked retinitis pigmentosa (XLRP) |
| | | | | Meindl, A. et al. A gene (SRPX) encoding a sushi-repeat-containing protein is deleted in patients with X-linked retinitis pigmentosa. Hum. Mol. Genet. 4, 2339-2346 (1995) |
| | | 629524\|Srpx | 1.0E-231 | [Rattus norvegicus][Plasma membrane] Sushi-repeat-containing protein (X chromosome), protein suppressed in v-src-transformed cells, may be involved in inhibition of transformation; mutation of the human SRPX may be associated with X-linked retinitis pigmentosa (XLRP) |
| | | | | Pan, J. et al. Isolation of a novel gene down-regulated by v-src. FEBS Lett. 383, 21-25 (1996) |
| 17 | 7525966CD1 | g6018421 | 2.0E-17 | [Caenorhabditis elegans] (AL110498) contains similarity to Pfam domain: PF00008 (EGF-like domain), Score=380.3, E-value=6.1e-111, N=34~cDNA EST yk14g6.3 comes from this gene~cDNA EST yk14g6.5 comes from this gene~cDNA EST yk172f7.5 comes from this gene~cDNA EST yk145h2.5 comes from this gene~cDNA EST yk273b3.3 comes from this gene~cDNA EST yk335h11.3 comes from this gene~cDNA EST yk354a10.3 comes from this gene~cDNA EST yk471g4.3 comes from this gene~cDNA EST yk273b3.5 comes from this gene~cDNA EST yk335h11.5 comes from this gene~cDNA EST y> |
| | | | | Genome sequence of the nematode C. elegans: a platform for investigating biology. The C. elegans Sequencing Consortium. Science 282, 2012-2018 (1998) |

# Table 2

| Polypeptide SEQ ID NO: | Incyte Polypeptide ID | GenBank ID NO: or PROTEOME ID NO: | Probability Score | Annotation |
|---|---|---|---|---|
| | | 782507\|FLJ30681 | 7.6E-126 | [Homo sapiens] Protein containing an epidermal growth factor (EGF)-like domain, has a region of high similarity to a region of collagen type V alpha 2 subunit (human COL5A2), which is an extracellular matrix structural protein involved in connective tissue maintenance |
| | | | | Nagase, T. et al. Prediction of the coding sequences of unidentified human genes. XXII. The complete sequences of 50 new cDNA clones which code for large proteins. DNA Res. 8, 319-327 (2001) |
| | | 662841\|Egfl3 | 2.1E-14 | [Rattus norvegicus] Multiple epidermal growth factor (EGF)-like domains 6, contains 30 epidermal growth factor-like motifs, putative secreted protein, predicted to bind calcium |
| | | | | Nakayama, M. et al. Identification of high-molecular-weight proteins with multiple EGF-like motifs by motif-trap screening. Genomics 51, 27-34 (1998) *supra* |
| 18 | 7519667CD1 | g1842429 | 1.2E-132 | [Rattus norvegicus] ankyrin binding cell adhesion molecule neurofascin |
| | | | | Davis, J.Q. et al. Molecular composition of the node of Ranvier: identification of ankyrin-binding cell adhesion molecules neurofascin (mucin+/third FNIII domain-) and NrCAM at nodal axon segments. J. Cell Biol. 135:1355-1367 (1996). |
| | | 341732\|NRCAM | 1.8E-85 | [Homo sapiens] [Adhesin/agglutinin] [Plasma membrane] Neuronal cell adhesion molecule, a member of the immunoglobulin superfamily, predicted to have a role in neuronal cell adhesion |
| | | | | Lane, R.P. et al. Characterization of a highly conserved human homolog to the chicken neural cell surface protein Bravo/Nr-CAM that maps to chromosome band 7q31. Genomics 35:456-465 (1996). |
| | | | | Sehgal, A. et al. Cell adhesion molecule Nr-CAM is over-expressed in human brain tumors. Int. J. Cancer 76:451-458 (1998). |
| | | 790459\|Nfasc | 2.1E-82 | [Rattus norvegicus] [Axon; Plasma membrane; Cell junction] Neurofascin, an ankyrin-binding cell adhesion molecule of the L1 subgroup of the immunoglobulin G superfamily, implicated in neurite outgrowth, fasciculation, interneuronal adhesion, and axo-glial signaling |

# Table 2

| Polypeptide SEQ ID NO: | Incyte Polypeptide ID | GenBank ID NO: or PROTEOME ID NO: | Probability Score | Annotation |
|---|---|---|---|---|
| | | | | Davis, J.Q. et al. (1996), *supra.* |
| | | | | Tait, S. et al. An oligodendrocyte cell adhesion molecule at the site of assembly of the paranodal axo-glial junction. J. Cell Biol. 150:657-666 (2000). |
| | | | | Davis, J.Q. et al. Ankyrin binding activity shared by the neurofascin/L1/NrCAM family of nervous system cell adhesion molecules. J. Biol. Chem. 269:27163-27166 (1994). |
| | | | | Lustig, M. et al. Nr-CAM and neurofascin interactions regulate ankyrin G and sodium channel clustering at the node of Ranvier. Curr. Biol. 11:1864-1869 (2001). |
| 19 | 7520638CD1 | g3068592 | 3.2E-253 | [Mus musculus] punc |
| | | | | Salbaum, J.M. Punc, a novel mouse gene of the immunoglobulin superfamily, is expressed predominantly in the developing nervous system. Mech. Dev. 71:201-204 (1998). |
| | | | | Salbaum, J.M. Genomic structure and chromosomal localization of the mouse gene Punc. Mamm. Genome 10:107-111 (1999). |
| | | 582679\|Punc | 2.2E-254 | [Mus musculus] [Adhesin/agglutinin] [Plasma membrane] Putative neuronal cell adhesion molecule, a member of the immunoglobulin superfamily of cell surface proteins, may play a role in cerebellar control of motor coordination and early embryogenesis |
| | | | | Salbaum, J.M. (1998), *supra.* |
| | | | | Salbaum, J.M. (1999), *supra.* |
| | | | | Yang, W. et al. Impaired motor coordination in mice that lack punc. Mol. Cell. Biol. 21:6031-6043 (2001). |
| | | 789395\|DDM36 | 3.1E-102 | [Homo sapiens] Protein containing five fibronectin type III domains, which are involved in cell surface binding, and an immunoglobulin (Ig) domain, has a region of low similarity to a region of deleted in colorectal cancer (netrin 1 receptor, rat Dcc) |
| | | | | Keino-Masu, K. et al. Deleted in Colorectal Cancer (DCC) encodes a netrin receptor. Cell 87:175-185 (1996). |

# Table 2

| Polypeptide SEQ ID NO: | Incyte Polypeptide ID | GenBank ID NO: or PROTEOME ID NO: | Probability Score | Annotation |
|---|---|---|---|---|
| 20 | 7518947CD1 | g5305673 | 4.9E-13 | [Homo sapiens] developmental arteries and neural crest EGF-like protein |
| | | | | Nakamura, T. et al. DANCE, a novel secreted RGD protein expressed in developing, atherosclerotic, and balloon-injured arteries. J. Biol. Chem. 274:22476-22483 (1999). |
| | | | | Nakamura, T. et al. Fibulin-5/DANCE is essential for elastogenesis in vivo. Nature 415:171-175 (2002). |
| | | | | Schiemann, W.P. et al. Context-specific effects of fibulin-5 (DANCE/EVEC) on cell proliferation, motility, and invasion. Fibulin-5 is induced by transforming growth factor-beta and affects protein kinase cascades. J. Biol. Chem. 277:27367-27377 (2002). |
| | | 782507\|FLJ30681 | 3.1E-177 | [Homo sapiens] Protein containing an epidermal growth factor (EGF)-like domain, has a region of high similarity to a region of collagen type V alpha 2 subunit (human COL5A2), which is an extracellular matrix structural protein involved in connective tissue maintenance |
| | | 609072\|Cegf1 | 1.8E-14 | [Mus musculus] Protein containing nine epidermal growth factor (EGF)-like domains and a CUB domain, which often occur extracellularly in developmentally regulated proteins, has a region of moderate similarity to a region of matrilin 2 (mouse Matn2) |
| | | | | Grimmond, S. et al. Expression of a novel mammalian epidermal growth factor-related gene during mouse neural development. Mech. Dev. 102:209-211 (2001). |
| 21 | 7519652CD1 | g1842427 | 2.7E-81 | [Rattus norvegicus] ankyrin binding cell adhesion molecule neurofascin |
| | | | | Davis, J.Q. et al. (1996), *supra.* |
| | | 341732\|NRCAM | 1.6E-82 | [Homo sapiens] [Adhesin/agglutinin] [Plasma membrane] Neuronal cell adhesion molecule, a member of the immunoglobulin superfamily, predicted to have a role in neuronal cell adhesion |
| | | | | Lane, R.P. et al. (1996), *supra.* |
| | | | | Sehgal, A. et al. (1998), *supra.* |

# Table 2

| Polypeptide SEQ ID NO: | Incyte Polypeptide ID | GenBank ID NO: or PROTEOME ID NO: | Probability Score | Annotation |
|---|---|---|---|---|
| | | 790459\|Nfasc | 2.1E-82 | [Rattus norvegicus] [Axon; Plasma membrane; Cell junction] Neurofascin, an ankyrin-binding cell adhesion molecule of the L1 subgroup of the immunoglobulin G superfamily, implicated in neurite outgrowth, fasciculation, interneuronal adhesion, and axo-glial signaling |
| | | | | Davis, J.Q. et al. (1996), *supra.* |
| | | | | Tait, S. et al. (2000), *supra.* |
| | | | | Davis, J.Q. et al. (1994), *supra.* |
| | | | | Lustig, M. et al. (2001), *supra.* |
| 22 | 7520381CD1 | g10334772 | 2.4E-230 | [Homo sapiens] MUCDHL-ALT |
| | | 659060\|MUCDHL | 2.4E-232 | [Homo sapiens] Mucin and cadherin like, has a region of low similarity to proline-rich repeat proteins; corresponding gene is in region associated with many cancers |
| | | | | Paris, M.J. and Williams, B.R. Characterization of a 500-kb contig spanning the region between c-Ha-Ras and MUC2 on chromosome 11p15.5. Genomics 69:196-202 (2000). |
| | | 629545\|Mucdhl | 4.0E-147 | [Rattus norvegicus] [Receptor (signaling)] [Basolateral plasma membrane] Mu-Protocadherin, a glycosylated cell-cell adhesion molecule that is present in two developmentally regulated forms, may participate in kidney and lung development by aiding in the regulation of branching morphogenesis |
| | | | | Goldberg, M. et al. micro-Protocadherin, a novel developmentally regulated protocadherin with mucin-like domains. J. Biol. Chem. 275:24622-24629 (2000). |
| 23 | 7520409CD1 | g37181861 | 0.0 | MUCDHL [Homo sapiens] |
| | | | | Clark, H.F. et al. The Secreted Protein Discovery Initiative (SPDI), a Large-Scale Effort to Identify Novel Human Secreted and Transmembrane Proteins: A Bioinformatics Assessment. Genome Res. 13:2265-2270 (2003) |
| | | g10334774 | 1.1E-225 | [Homo sapiens] MUCDHL-FL |

# Table 2

| Polypeptide SEQ ID NO: | Incyte Polypeptide ID | GenBank ID NO: or PROTEOME ID NO: | Probability Score | Annotation |
|---|---|---|---|---|
| | | 659060\|MUCDHL | 7.8E-227 | [Homo sapiens] Mucin and cadherin like, has a region of low similarity to proline-rich repeat proteins; corresponding gene is in region associated with many cancers |
| | | | | Paris, M.J. and Williams, B.R. (2000), *supra.* |
| | | 629545\|Mucdhl | 1.2E-149 | [Rattus norvegicus][Receptor (signalling)][Basolateral plasma membrane] Mu-Protocadherin, a glycosylated cell-cell adhesion molecule that is present in two developmentally regulated forms, may participate in kidney and lung development by aiding in the regulation of branching morphogenesis |
| | | | | Goldberg, M. et al. (2000), *supra.* |
| 24 | 7520538CD1 | g20269129 | 0.0 | [Homo sapiens] MEGF6 |
| | | | | Nakayama, M. et al. Identification of high-molecular-weight proteins with multiple EGF-like motifs by motif-trap screening. Genomics 51:27-34 (1998). |
| | | 662841\|Egfl3 | 1.4E-277 | [Rattus norvegicus] Multiple epidermal growth factor (EGF)-like domains 6, contains 30 epidermal growth factor-like motifs, putative secreted protein, predicted to bind calcium |
| | | | | Zhou, Z. et al. CED-1 Is a Transmembrane Receptor that Mediates Cell Corpse Engulfment in C. elegans. Cell 104:43-56 (2001). |
| | | | | Nakayama, M. et al. (1998), *supra.* |
| | | 716683\|MEGF11 | 1.0E-120 | [Homo sapiens] Protein containing fifteen epidermal growth factor (EGF)-like domains, has moderate similarity to a region of C. elegans CED-1, which is a putative cell surface phagocytic receptor that recognizes cell corpses in apoptosis |
| 25 | 7520667CD1 | g10334774 | 4.2E-257 | [Homo sapiens] MUCDHL-FL |
| | | 659060\|MUCDHL | 2.9E-258 | [Homo sapiens] Mucin and cadherin like, has a region of low similarity to proline-rich repeat proteins; corresponding gene is in region associated with many cancers |
| | | | | Paris, M.J. and Williams, B.R. (2000), *supra.* |

# Table 2

| Polypeptide SEQ ID NO: | Incyte Polypeptide ID | GenBank ID NO: or PROTEOME ID NO: | Probability Score | Annotation |
|---|---|---|---|---|
| | | 629545\|Mucdhl | 6.2E-163 | [Rattus norvegicus] [Receptor (signaling)] [Basolateral plasma membrane] Mu-Protocadherin, a glycosylated cell-cell adhesion molecule that is present in two developmentally regulated forms, may participate in kidney and lung development by aiding in the regulation of branching morphogenesis |
| | | | | Goldberg, M. et al. (2000), *supra.* |
| 26 | 7517567CD1 | g23592218 | 0.0 | [Mus musculus] nidogen-2 |
| | | | | Schymeinsky, J. et al., Gene structure and functional analysis of the mouse nidogen-2 gene: nidogen-2 is not essential for basement membrane formation in mice, Mol. Cell. Biol. 22, 6820-6830 (2002) |
| | | 582237\|Nid2 | 0.0 | [Mus musculus][Extracellular matrix (cuticle and basement membrane); Basement membrane (extracellular matrix); Extracellular (excluding cell wall)] Nidogen 2, a basement membrane protein that localizes to elastic fibers of blood vessel walls; has integrin-binding RGD sequence; may facilitate stabilization of basement membrane networks |
| | | | | Kimura, N. et al., Entactin-2: a new member of basement membrane protein with high homology to entactin/nidogen., Exp Cell Res 241, 36-45 (1998). |
| | | 429052\|NID2 | 0.0 | [Homo sapiens][Extracellular matrix (cuticle and basement membrane); Basement membrane (extracellular matrix); Extracellular (excluding cell wall)] Nidogen 2, a basement membrane protein that binds perlecan (HSPG2), laminin-1, and collagens; supports cell attachment in some cell lines |
| | | | | Kohfeldt, E. et al., Nidogen-2: a new basement membrane protein with diverse binding properties., J Mol Biol 282, 99-109 (1998). |
| 27 | 7519867CD1 | g15281077 | 9.2E-138 | mDC-SIGN1A type III isoform [Homo sapiens] |
| | | | | Mummidi, S. et al. Extensive repertoire of membrane-bound and soluble dendritic cell-specific ICAM-3-grabbing nonintegrin 1 (DC-SIGN1) and DC-SIGN2 isoforms. Inter-individual variation in expression of DC-SIGN transcripts. J. Biol. Chem. 276, 33196-33212 (2001) |

# Table 2

| Polypeptide SEQ ID NO: | Incyte Polypeptide ID | GenBank ID NO: or PROTEOME ID NO: | Probability Score | Annotation |
|---|---|---|---|---|
| | | 617980\|CD209 | 3.5E-128 | [Homo sapiens][Small molecule-binding protein][Plasma membrane] Dendritic cell-specific ICAM3-grabbing nonintegrin, a dendritic cell-specific, C-type lectin that binds ICAM3, promotes dendritic cell adhesion to T cells and stimulates T cell proliferation during an immune response; may promote infection by HIV-1 |
| | | | | Curtis, B. M. et al. Sequence and expression of a membrane-associated C-type lectin that exhibits CD4-independent binding of human immunodeficiency virus envelope glycoprotein gp120. Proc. Natl. Acad. Sci. U. S. A. 89, 8356-8360 (1992) |
| | | 800231\|CD209L | 2.3E-101 | [Homo sapiens][Plasma membrane] CD209 antigen-like, a member of the C-type lectin family, acts as a receptor for ICAM3 and binds to strains of HIV-1, HIV-2 and simian immunodeficiency virus, promoting their pathogenesis |
| | | | | Pohlmann, S. et al. DC-SIGNR, a DC-SIGN homologue expressed in endothelial cells, binds to human and simian immunodeficiency viruses and activates infection in trans. Proc. Natl. Acad. Sci. U. S. A. 98, 2670-2675 (2001) |
| 28 | 7519911CD1 | g11935122 | 1.3E-53 | papilin [Mus musculus] |
| | | | | Kramerova, I. A. et al. Papilin in development; a pericellular protein with a homology to the ADAMTS metalloproteinases. Development 127, 5475-5485 (2000) |
| | | 568954\|ADAMTS6 | 3.0E-53 | [Homo sapiens][Hydrolase; Protease (other than proteasomal)] Disintegrin like metalloprotease with thrombospondin type 1 motif 6, a member of the ADAMTS family of disintegrin metalloproteases with thrombospondin motifs, has a cysteine-rich and a reprolysin-type catalytic domain |
| | | | | Hurskainen, T. L. et al. ADAM-TS5, ADAM-TS6, and ADAM-TS7, novel members of a new family of zinc metalloproteases. General features and genomic distribution of the ADAM-TS family. J. Biol. Chem. 274, 25555-25563 (1999) |
| | | 703727\|ADAMTS12 | 4.7E-43 | [Homo sapiens] Disintegrin-like and metalloprotease thrombospondin type 1 motif 12, member of the ADAMTS family of disintegrin metalloproteases with thrombospondin type 1 motifs, may play a role in cell adhesion during fetal lung development |

# Table 2

| Polypeptide SEQ ID NO: | Incyte Polypeptide ID | GenBank ID NO: or PROTEOME ID NO: | Probability Score | Annotation |
|---|---|---|---|---|
| | | | | Cal, S. et al. Identification, characterization, and intracellular processing of adam-ts12, a novel human disintegrin with a complex structural organization involving multiple thrombospondin-1 repeats. J. Biol. Chem. 276, 17932-17940 (2001) |
| 29 | 7520005CD1 | g16197710 | 2.3E-203 | emu1 protein [Mus musculus] |
| | | | | Leimeister, C. et al. Developmental expression and biochemical characterization of Emu family members. Dev. Biol. 249, 204-218 (2002) |
| | | 768994\|Emu1 | 1.4E-204 | [Mus musculus] Protein containing two collagen triple helix repeats, which are found in some extracellular proteins, has a region of moderate similarity to a region of collagen type VI alpha 1 (human COL6A1), which may be involved in maintaining muscle fiber integrity |
| | | | | Leimeister, C. et al. Dev. Biol. 249, 204-218 (2002) *supra* |
| | | 753687\|COL5A1 | 2.7E-45 | [Homo sapiens][Structural protein][Extracellular matrix (cuticle and basement membrane); Extracellular (excluding cell wall)] Alpha 1 subunit of type V collagen, involved in skeletal development and contributes to the structure and stability of skin; mutation of corresponding gene causes Ehlers-Danlos syndrome types I and II |
| | | | | Burrows, N. P. et al. A point mutation in an intronic branch site results in aberrant splicing of COL5A1 and in Ehlers-Danlos syndrome type II in two British families. Am. J. Hum. Genet. 63, 390-398 (1998) |
| 30 | 7520484CD1 | g16191616 | 5.0E-178 | emu1 protein [Homo sapiens] |
| | | | | Leimeister, C. et al. Dev. Biol. 249, 204-218 (2002) *supra* |
| | | 768994\|Emu1 | 2.2E-142 | [Mus musculus] Protein containing two collagen triple helix repeats, which are found in some extracellular proteins, has a region of moderate similarity to a region of collagen type VI alpha 1 (human COL6A1), which may be involved in maintaining muscle fiber integrity |
| | | | | Leimeister, C. et al. Dev. Biol. 249, 204-218 (2002) *supra* |

# Table 2

| Polypeptide SEQ ID NO: | Incyte Polypeptide ID | GenBank ID NO: or PROTEOME ID NO: | Probability Score | Annotation |
|---|---|---|---|---|
| 30 | | 578546\|Col1a1 | 1.9E-47 | [Mus musculus][Structural protein][Extracellular matrix (cuticle and basement membrane); Extracellular (excluding cell wall)] Alpha 1 subunit of type I collagen, a structural protein; mutations in human COL1A1 cause Ehlers-Danlos syndrome type VII, osteoporosis, and osteogenesis imperfecta |
| | | | | Lanske, B. et al. The parathyroid hormone (PTH)/PTH-related peptide receptor mediates actions of both ligands in murine bone. Endocrinology 139, 5194-5204 (1998) |

# Table 3

| SEQ ID NO: | Incyte Polypeptide ID | Amino Acid Residues | Signature Sequences, Domains and Motifs | Analytical Methods and Databases |
|---|---|---|---|---|
| 1 | 7514594CD1 | 174 | signal_cleavage: M1-C42 | SPSCAN |
| | | | Signal Peptide: M21-C42, M21-V44, M21-T45 | HMMER |
| | | | C-type lectin (CTL) or carbohydrate-recogn: C42-E161 | HMMER_SMART |
| | | | Lectin C-type domain: L59-M162 | HMMER_PFAM |
| | | | Cytosolic domain: T40-L174<br>Transmembrane domain: Q20-I39<br>Non-cytosolic domain: M1-S19 | TMHMMER |
| | | | C-type lectin domain IPB001304: F93-W105 | BLIMPS_BLOCKS |
| | | | C-type lectin domain signature and profile: S116-K172 | PROFILESCAN |
| | | | Type II antifreeze protein signature PR00356: F93-D109, W147-C160 | BLIMPS_PRINTS |
| | | | C-TYPE LECTIN DM00035<br>\|P02707\|74-202: K79-E161<br>\|A46274\|248-377: S67-E161 | BLAST_DOMO |
| | | | Potential Phosphorylation Sites: S3 S83 S118 T12 T45 T51 T130 T136 | MOTIFS |
| | | | Potential Glycosylation Sites: N2 N62 | MOTIFS |
| 2 | 7515764CD1 | 323 | signal_cleavage: M1-T34 | SPSCAN |
| | | | Signal Peptide: M1-T34, M1-R36 | HMMER |
| | | | Epidermal growth factor-like domain.: E137-T175, V92-L133 | HMMER_SMART |
| | | | Calcium-binding EGF-like domain: D134-T175, D89-L133 | HMMER_SMART |
| | | | EGF-like domain: C138-C174, C93-K126 | HMMER_PFAM |
| | | | Calcium-binding EGF-like domain IPB001881:C111-R121, C150-C161 | BLIMPS_BLOCKS |

# Table 3

| SEQ ID NO: | Incyte Polypeptide ID | Amino Acid Residues | Signature Sequences, Domains and Motifs | Analytical Methods and Databases |
|---|---|---|---|---|
| | | | COLLAGEN ALPHA PRECURSOR CHAIN REPEAT SIGNAL CONNECTIVE TISSUE EXTRACELLULAR MATRIX PD000007: P181-G256, P181-P257, P181-E245 | BLAST_PRODOM |
| | | | SIMILAR TO CUTICULAR COLLAGEN PD067228: D169-G256 | BLAST_PRODOM |
| | | | PRECOLLAGEN P PRECURSOR SIGNAL PD072959: G177-P255, G188-G256, P181-G256 | BLAST_PRODOM |
| | | | PRECURSOR SIGNAL COLLAGEN ALPHA 3IX CHAIN EXTRACELLULAR MATRIX CONNECTIVE TISSUE PD028299: D183-R252 | BLAST_PRODOM |
| | | | FIBRILLAR COLLAGEN CARBOXYL-TERMINAL DM00019<br>\|S09646\|482-667: G177-F300, G171-P257<br>\|S21369\|493-678: G177-F300, G171-P257<br>\|S05378\|482-667: G177-F300, G171-P257<br>\|Q02388\|1373-1547: P3-G17 G171-P257, T173-L258, I166-G256, G185-G256 | BLAST_DOMO |
| | | | Potential Phosphorylation Sites: S62 S156 S242 S277 S283 T34 T54 T175 T303 T314 Y131 | MOTIFS |
| | | | Potential Glycosylation Sites: N142 N182 | MOTIFS |
| | | | Cell attachment sequence: R176-D178 | MOTIFS |
| | | | EGF-like domain signature 2: C159-C174 | MOTIFS |
| | | | Calcium-binding EGF-like domain pattern signature: D134-C159 | MOTIFS |
| 3 | 7514575CD1 | 97 | signal_cleavage: M1-A25 | SPSCAN |
| | | | Signal Peptide: I7-A25, P6-A25, M1-G23, M1-A25, M1-F27 | HMMER |

# Table 3

| SEQ ID NO: | Incyte Polypeptide ID | Amino Acid Residues | Signature Sequences, Domains and Motifs | Analytical Methods and Databases |
|---|---|---|---|---|
| | | | UROPLAKIN II PRECURSOR UPII GLYCOPROTEIN TRANSMEMBRANE SIGNAL PD022112: P22-L86 | BLAST_PRODOM |
| | | | Potential Glycosylation Sites: N28 N57 N66 | MOTIFS |
| 4 | 7514576CD1 | 116 | signal_cleavage: M1-A25 | SPSCAN |
| | | | Signal Peptide: I7-A25, P6-A25, M1-G23, M1-A25, M1-F27 | HMMER |
| | | | UROPLAKIN II PRECURSOR UPII GLYCOPROTEIN TRANSMEMBRANE SIGNAL PD022112: P22-Q105 | BLAST_PRODOM |
| | | | Potential Phosphorylation Sites: S88 | MOTIFS |
| | | | Potential Glycosylation Sites: N28 N57 N66 | MOTIFS |
| 5 | 7515593CD1 | 81 | signal_cleavage: M1-C42 | SPSCAN |
| | | | Signal Peptide: M21-C42, M21-V44, M21-T45 | HMMER |
| | | | Cytosolic domain: T40-T81<br>Transmembrane domain: Q20-I39<br>Non-cytosolic domain: M1-S19 | TMHMMER |
| | | | Potential Phosphorylation Sites: S3 T12 T45 T51 | MOTIFS |
| | | | Potential Glycosylation Sites: N2 N62 | MOTIFS |
| 6 | 7517754CD1 | 196 | signal_cleavage: M1-P27 | SPSCAN |
| | | | Signal Peptide: S6-P27, M1-A25, M1-P27, G10-P27, M1-S30, S6-P27 | HMMER |
| | | | Calcium-binding EGF-like domain IPB001881:C71-P82 | BLIMPS_BLOCKS |
| | | | GLYCOPROTEIN EGF-LIKE DOMAIN T16 H411 PRECURSOR SIGNAL UPH1 UP50 PD030337: Q28-V122 | BLAST_PRODOM |
| | | | Potential Phosphorylation Sites: S26 S35 T60 T149 | MOTIFS |
| | | | Calcium-binding EGF-like domain pattern signature: D54-C80 | MOTIFS |
| 7 | 7517819CD1 | 544 | signal_cleavage: M1-A32 | SPSCAN |
| | | | Signal Peptide: M1-G26, M1-A32 | HMMER |

# Table 3

| SEQ ID NO: | Incyte Polypeptide ID | Amino Acid Residues | Signature Sequences, Domains and Motifs | Analytical Methods and Databases |
|---|---|---|---|---|
| 7 | | | Integrin alpha (beta-propellor repeats: S365-H418, V303-L360, P48-V110, G246-E299, G426-P482 | HMMER_SMART |
| | | | FG-GAP repeat: G366-G423, G304-P364, G49-P114, L427-V479, N247-V303 | HMMER_PFAM |
| | | | Cytosolic domain: M1-P12<br>Transmembrane domain: R13-L35<br>Non-cytosolic domain: D36-G544 | TMHMMER |
| | | | Integrins alpha chain IPB000413: R87-A98, G213-G222, G300-P329, A368-P392 | BLIMPS_BLOCKS |
| | | | Integrin alpha subunit signature PR01185: N247-F259, T267-M278, G300-G320, A368-P392, F430-G451, H456-P475 | BLIMPS_PRINTS |
| | | | INTEGRIN PRECURSOR SIGNAL GLYCOPROTEIN ALPHA CELL ADHESION TRANSMEMBRANE EXTRACELLULAR MATRIX PD001221: E193-E287, G300-I468 | BLAST_PRODOM |
| | | | INTEGRIN ALPHA3 PRECURSOR GALACTOPROTEIN B3 GAPB3 VLA3 ALPHA CHAIN CD49C PD038098: V147-L192 | BLAST_PRODOM |
| | | | INTEGRIN PRECURSOR CELL GLYCOPROTEIN SIGNAL ADHESION TRANSMEMBRANE EXTRACELLULAR MATRIX CYTOSKELETON PD001587: A32-Y144 | BLAST_PRODOM |
| | | | INTEGRIN ALPHA3 PRECURSOR GALACTOPROTEIN B3 GAPB3 VLA3 ALPHA CHAIN CD49C CELL ADHESION GLYCOPROTEIN TRANSMEMBRANE SIGNAL EXTRACELLULAR MATRIX CYTOSKELETON REPEAT PD106957: M1-S31 | BLAST_PRODOM |

# Table 3

| SEQ ID NO: | Incyte Polypeptide ID | Amino Acid Residues | Signature Sequences, Domains and Motifs | Analytical Methods and Databases |
|---|---|---|---|---|
| | | | INTEGRINS ALPHA CHAIN DM00458 \|P17852\|42-236: V42-Y237 \|P26007\|27-223: G46-G222 \|P23229\|34-229: V42-G222 | BLAST_DOMO |
| | | | INTEGRINS ALPHA CHAIN DM01028\|P17852\|408-907: G408-E492 | BLAST_DOMO |
| | | | Potential Phosphorylation Sites: S150 S173 S187 S238 S375 S405 S452 S489 S502 T62 T85 T109 Y237 | MOTIFS |
| | | | Potential Glycosylation Sites: N86 N107 N265 | MOTIFS |
| 8 | 7518460CD1 | 363 | signal_cleavage: M1-A19 | SPSCAN |
| | | | Signal Peptide: L4-A19, M1-A19, M1-Q21, M1-S23, M1-C24, M1-C28, M1-G31 | HMMER |
| | | | Hemopexin-like repeats.: I206-D252, F161-G204, V254-H304, W313-C357 | HMMER_SMART |
| | | | Somatomedin B -like domains: D20-V62 | HMMER_SMART |
| | | | Somatomedin B domain: D20-T63 | HMMER_PFAM |
| | | | Hemopexin: I206-D252, F161-G204, V254-H304, F317-C357 | HMMER_PFAM |
| | | | Somatomedin B domain IPB001212: C38-C58, E203-S244, N253-Q290, L320-P358 | BLIMPS_BLOCKS |
| | | | Hemopexin domain signature: L221-Q283 F175-D232 | PROFILESCAN |
| | | | Somatomedin B domain signature: L18-Y78 | PROFILESCAN |
| | | | Somatomedin B signature PR00022: A19-F32, K36-Y47, Q48-K59 | BLIMPS_PRINTS |
| | | | VITRONECTIN PRECURSOR HEPARIN BINDING CELL ADHESION GLYCOPROTEIN SULFATATION SIGNAL SERUM SPREADING PD007442: T63-D162 | BLAST_PRODOM |

# Table 3

| SEQ ID NO: | Incyte Polypeptide ID | Amino Acid Residues | Signature Sequences, Domains and Motifs | Analytical Methods and Databases |
|---|---|---|---|---|
| | | | VITRONECTIN PRECURSOR HEPARIN BINDING CELL ADHESION GLYCOPROTEIN SULFATATION SIGNAL SERUM SPREADING PD150319: D252-F302 | BLAST_PRODOM |
| | | | VITRONECTIN DM03553<br>\|P04004\|1-147: M1-Q148<br>\|P48819\|1-124: M1-P125<br>\|P22458\|1-147: M1-E126<br>\|P29788\|1-146: M1-P147 | BLAST_DOMO |
| | | | Potential Phosphorylation Sites: S23 S130 S137 S157 S266 S289 S312 S326 T63 T69 T76 T88 T336 Y330 | MOTIFS |
| | | | Potential Glycosylation Sites: N86 N169 N242 | MOTIFS |
| | | | Cell attachment sequence: R64-D66 | MOTIFS |
| | | | Hemopexin domain signature: I196-F210 | MOTIFS |
| | | | Somatomedin B domain signature: C38-C58 | MOTIFS |
| 9 | 7518734CD1 | 265 | signal_cleavage: M1-G30 | SPSCAN |
| | | | Signal Peptide: M1-A31 | HMMER |
| | | | Immunoglobulin: R145-F238, N38-Q136 | HMMER_SMART |
| | | | Immunoglobulin C-2 Type: P151-G226, Q44-D122 | HMMER_SMART |
| | | | Immunoglobulin domain: G153-A221, F46-V117 | HMMER_PFAM |
| | | | V type Ig domains from SCOP: V32-P142 | HMMER_INCY |
| | | | Ig superfamily from SCOP: L34-T141, V143-A228 | HMMER_INCY |
| | | | Potential Phosphorylation Sites: S37 S99 S110 S227 S262 T106 T254 T258 Y77 | MOTIFS |
| | | | Potential Glycosylation Sites: N104 N192 | MOTIFS |
| 10 | 7513469CD1 | 511 | Signal Peptide: M1-G25, M1-G24, M1-G29, M1-A23, P6-A23 | HMMER |
| | | | signal_cleavage: M1-A23 | SPSCAN |
| | | | Leucine Rich Repeat:D68-S91, A92-F115, N116-P137 | HMMER_PFAM |
| | | | Leucine rich repeat C-terminal domain: N125-G177 | HMMER_PFAM |

# Table 3

| SEQ ID NO: | Incyte Polypeptide ID | Amino Acid Residues | Signature Sequences, Domains and Motifs | Analytical Methods and Databases |
|---|---|---|---|---|
| 10 | | | Leucine rich repeat N-terminal domain: P32-P66 | HMMER_PFAM |
| | | | PKD domain: P270-V352 | HMMER_PFAM |
| | | | WSC domain: Y180-G261 | HMMER_PFAM |
| | | | Leucine rich repeat C-terminal domain: N125-G177 | HMMER_SMART |
| | | | Leucine rich repeat N-terminal domain: P32-A71 | HMMER_SMART |
| | | | Leucine-rich repeats, typical: T70-N89, L90-N113 | HMMER_SMART |
| | | | Repeats in polycystic kidney disease 1: S272-A355 | HMMER_SMART |
| | | | WSC: present in yeast cell wall integrity: G177-A271 | HMMER_SMART |
| | | | Leucine-rich repeat signature PR00019: L93-L106, L90-I103 | BLIMPS_PRINTS |
| | | | GLYCOPROTEIN PRECURSOR SIGNAL PROTEIN LEUCINE REPEAT EGF-LIKE DOMAIN TRANSMEMBRANE REPEAT PLATELET PD070590: F115-P185 | BLAST_PRODOM |
| | | | POLYCYSTIC KIDNEY PROTEIN DISEASE POLYCYSTIN PRECURSOR AUTOSOMAL DOMINANT SIGNAL LEUCINE REPEAT PD150716: P26-A69 | BLAST_PRODOM |
| | | | POLYCYSTIC KIDNEY PROTEIN DISEASE POLYCYSTIN PRECURSOR AUTOSOMAL DOMINANT SIGNAL LEUCINE REPEAT PD150789: A354-Q435 | BLAST_PRODOM |
| | | | POLYCYSTIC KIDNEY DISEASE PROTEIN POLYCYSTIN PRECURSOR AUTOSOMAL DOMINANT SIGNAL LEUCINE REPEAT PD151307: A182-P283 | BLAST_PRODOM |
| | | | PKD REPEAT DM00690\|P98161\|317-395: A317-E396 | BLAST_DOMO |
| | | | PROTEOGLYCAN CARBOXYL-TERMINAL DM01130\|P98161\|119-193: E119-A194 | BLAST_DOMO |

# Table 3

| SEQ ID NO: | Incyte Polypeptide ID | Amino Acid Residues | Signature Sequences, Domains and Motifs | Analytical Methods and Databases |
|---|---|---|---|---|
| | | | Potential Phosphorylation Sites: S52, S104, S322, S383, S408, T70, T105, T258, T303 | MOTIFS |
| | | | Potential Glycosylation Sites: N50, N89, N116, N121, N187 | MOTIFS |
| | | | Leucine zipper pattern: L72-L93 | MOTIFS |
| 11 | 7518836CD1 | 161 | Signal Peptide: M1-A20, M1-D24 | HMMER |
| | | | signal_cleavage: M1-A20 | SPSCAN |
| | | | Cytosolic domain: R92-V161<br>Transmembrane domain: W69-S91<br>Non-cytosolic domain: M1-E68 | TMHMMER |
| | | | Link domain IPB000538: T57-A109 | BLIMPS_BLOCKS |
| | | | CD44 antigen precursor signature PR00658: E51-L70, I71-R93, C95-K114, D141-G160 | BLIMPS_PRINTS |
| | | | CD44 GLYCOPROTEIN ANTIGEN PRECURSOR PHAGOCYTIC I PGP-1 HUTCH-I EXTRACELLULAR MATRIX PD004309: S18-W69 | BLAST_PRODOM |
| | | | CD44 GLYCOPROTEIN ANTIGEN PRECURSOR PHAGOCYTIC I PGP-1 HUTCH-I EXTRACELLULAR MATRIX PD006761: V85-V161 | BLAST_PRODOM |
| | | | CD44; LONG; SURFACE; ADHESION;<br>DM03720\|A53286\|153-365: D24-V161<br>DM03720\|B38745\|157-502: H29-V161<br>DM03720\|P20944\|155-361: D24-V161<br>DM03720\|S24631\|462-698: G23-V161 | BLAST_DOMO |
| | | | Potential Phosphorylation Sites: S40, S91, S116, T145 | MOTIFS |
| | | | Potential Glycosylation Sites: N55 | MOTIFS |
| 12 | 7519868CD1 | 289 | signal_cleavage: M1-K62 | SPSCAN |
| | | | Lectin C-type domain: S158-K264 | HMMER_PFAM |

# Table 3

| SEQ ID NO: | Incyte Polypeptide ID | Amino Acid Residues | Signature Sequences, Domains and Motifs | Analytical Methods and Databases |
|---|---|---|---|---|
| | | | C-type lectin (CTL) or carbohydrate-recognition: C141-K263 | HMMER_SMART |
| | | | C-type lectin domain IPB001304: W145-C169, W200-W212 | BLIMPS_BLOCKS |
| | | | C-type lectin domain signature and profile: G210-T283 | PROFILESCAN |
| | | | Type II antifreeze protein signature PR00356: P140-C152, C152-C169, K170-F187, W200-D216, W249-C262 | BLIMPS_PRINTS |
| | | | LECTIN PROTEIN PRECURSOR GLYCOPROTEIN SIGNAL RECEPTOR TRANSMEMBRANE REPEAT CELL DOMAIN PD000254:C169-C254 S245-F276 | BLAST_PRODOM |
| | | | C-TYPE LECTIN DM00035\|A46274\|248-377: A133-K263 DM00035\|P02707\|74-202: L137-K264 DM00035\|P20693\|179-304: C138-C262 DM00035\|P34927\|145-276: E135-K263 | BLAST_DOMO |
| | | | Potential Phosphorylation Sites: S72, S92, S158, S181, S192, S195, S223, S270 | MOTIFS |
| | | | Potential Glycosylation Sites: N80 | MOTIFS |
| | | | C-type lectin domain signature: C241-C262 | MOTIFS |
| 13 | 7520048CD1 | 368 | Signal Peptide: M1-T20, M1-P23, M1-A18, M1-N22, M1-R16 | HMMER |
| | | | signal_cleavage: M1-T20 | SPSCAN |
| | | | Laminin-type EGF-like (LE) domain IPB002049: A204-P214, G195-C205, C135-G151 | BLIMPS_BLOCKS |
| | | | EGF-like domain IPB000561: C120-G128 | BLIMPS_BLOCKS |
| | | | Type III EGF-like signature PR00011: C198-A216, S133-G161 | BLIMPS_PRINTS |
| | | | Potential Phosphorylation Sites: S30, S154, S233, S276, S298, T38, T313 | MOTIFS |

# Table 3

| SEQ ID NO: | Incyte Polypeptide ID | Amino Acid Residues | Signature Sequences, Domains and Motifs | Analytical Methods and Databases |
|---|---|---|---|---|
| | | | Potential Glycosylation Sites: N152, N153 | MOTIFS |
| | | | Egf_1 :C120-C131 | MOTIFS |
| | | | Egf_2 :C120-C131 | MOTIFS |
| 14 | 7520157CD1 | 688 | Signal Peptide: M1-G21, M1-A22, M10-N25 | HMMER |
| | | | signal_cleavage: M1-G21 | SPSCAN |
| | | | Cadherin cytoplasmic region: K535-A683 | HMMER_PFAM |
| | | | Cadherin domain: N48-S139, Y153-S247, Y261-E362, Y375-Q459 | HMMER_PFAM |
| | | | Cadherin repeats: Q65-P146, V170-P254, I278-L369, V392-P468 | HMMER_SMART |
| | | | Cadherin domain IPB002126: H38-L70, G85-E108, A235-I244, A622-G670 | BLIMPS_BLOCKS |
| | | | Cadherins extracellular repeated domain signature: V222-I275, I114-V167 | PROFILESCAN |
| | | | Cadherin signature PR00205: S88-R107, F148-D177, E219-G231, A235-P254, P254-E267 | BLIMPS_PRINTS |
| | | | GLYCOPROTEIN CELL ADHESION TRANSMEMBRANE CALCIUM-BINDING REPEAT PRECURSOR SIGNAL PHOSPHORYLATION CYTOSKELETON PD001401: K535-G681 | BLAST_PRODOM |
| | | | CADHERIN REPEAT<br>DM00697\|S55391\|516-785:E501-Q525 I528-S682<br>DM00697\|P55285\|517-785:H503-Q525 K535-G681<br>DM00697\|P55287\|521-790:F505-F543 I528-S682<br>DM00697\|I48277\|521-790:V157-G193 I528-S682<br>DM00697\|P55287\|521-790: V157-G193<br>DM00697\|S55391\|516-785: P158-A184 | BLAST_DOMO |

# Table 3

| SEQ ID NO: | Incyte Polypeptide ID | Amino Acid Residues | Signature Sequences, Domains and Motifs | Analytical Methods and Databases |
|---|---|---|---|---|
| | | | Potential Phosphorylation Sites: S68, S173, S209, S210, S294, S393, S492, S499, S510, S650, S657, T28, T124, T171, T264, T307, T395, T441, T443, T520, T586, T590, Y153 | MOTIFS |
| | | | Potential Glycosylation Sites: N57, N74, N419, N437, N508, N515, N516 | MOTIFS |
| | | | Cadherins extracellular repeated domain signature: I136-P146, I244-P254 | MOTIFS |
| 15 | 7520485CD1 | 272 | Signal Peptide: M59-S77, M59-S80 | HMMER |
| | | | Cytosolic domain: M1-G49<br>Transmembrane domain: A50-V72<br>Non-cytosolic domain: S73-A272 | TMHMMER |
| | | | GP120; C-TYPE; LECTIN; HIV;<br>DM01861\|A46274\|205-246:E125-K166 E148-A190<br>DM01861\|A46274\|205-246: E171-A213, E194-A236, E102-A144, E217-K258, E82-A121 | BLAST_DOMO |
| | | | Potential Phosphorylation Sites: S84, S104 | MOTIFS |
| | | | Potential Glycosylation Sites: N92 | MOTIFS |
| 16 | 7525723CD1 | 444 | Signal Peptide: M1-S30, P8-S30, H6-S30, L13-S30 | HMMER |
| | | | signal_cleavage: M1-S30 | SPSCAN |
| | | | HYR domain: V155-V237 | HMMER_PFAM |
| | | | Sushi domain (SCR repeat):C37-C95, C100-C154, C242-C297 | HMMER_PFAM |
| | | | Domain abundant in complement control protein: C37-C95, C100-C154, C242-C297 | HMMER_SMART |
| | | | Complement C9 signature PR00764: V93-C113, C49-C69 | BLIMPS_PRINTS |
| | | | Sushi domain proteins (SCR repeat proteins) PF00084: C49-Q53, N120-Y131, L288-C297 | BLIMPS_PFAM |

Table 3

| SEQ ID NO: | Incyte Polypeptide ID | Amino Acid Residues | Signature Sequences, Domains and Motifs | Analytical Methods and Databases |
|---|---|---|---|---|
| | | | PROTEIN SUSHI REPEAT CONTAINING SRPX PRECURSOR SIGNAL REPEAT SUSH REPEAT POLYMORPHISM ALTERNATIVE PD020056:V25-P35 F31-R99 | BLAST_PRODOM |
| | | | PROTEIN SUSHI REPEAT CONTAINING SRPX PRECURSOR SIGNAL REPEAT SUSHI REPEAT POLYMORPHISM ALTERNATIVE PD019555: V155-R241 | BLAST_PRODOM |
| | | | PROTEIN SUSHI REPEAT CONTAINING SRPX PRECURSOR SIGNAL REPEAT SUSHI REPEAT POLYMORPHISM ALTERNATIVE PD021630: A298-C442 | BLAST_PRODOM |
| | | | SUSHI REPEAT<br>DM04887\|P33730\|1-610:G45-E169 Y262-A299<br>DM04887\|P27113\|1-551:G45-V167 R238-V306<br>DM04887\|P16581\|1-609: C37-M300<br>DM04887\|P27113\|1-551: I40-C297, C37-C154<br>DM04887\|P33730\|1-610: K43-M300, G45-C154, R241-C297 | BLAST_DOMO |
| | | | Potential Phosphorylation Sites: S30, S85, S90, S121, S148, S153, S166, S257, S291, T63, T132, T140, T178, T220, T230, T329, T355, Y118, Y262, Y409 | MOTIFS |
| 17 | 7525966CD1 | 357 | Signal Peptide: M1-R36, M1-T34 | HMMER |
| | | | signal_cleavage: M1-T34 | SPSCAN |
| | | | EGF-like domain: C93-K126, C138-C174 | HMMER_PFAM |
| | | | Epidermal growth factor-like domain: V92-L133, E137-T175 | HMMER_SMART |
| | | | Calcium-binding EGF-like domain: D89-L133, D134-T175 | HMMER_SMART |

# Table 3

| SEQ ID NO: | Incyte Polypeptide ID | Amino Acid Residues | Signature Sequences, Domains and Motifs | Analytical Methods and Databases |
|---|---|---|---|---|
| | | | EGF-like domain IPB000561: C111-Y119 | BLIMPS_BLOCKS |
| | | | Calcium-binding EGF-like domain IPB001881: C111-R121, C150-C161 | BLIMPS_BLOCKS |
| | | | Type I EGF signature PR00009: L133-H148, L154-Y165 | BLIMPS_PRINTS |
| | | | Type II EGF-like signature PR00010: D89-Q100, G171-D178, G155-Y165 | BLIMPS_PRINTS |
| | | | Potential Phosphorylation Sites: S62, S156, S311, S317, T34, T54, T175, T184, T201, T337, T348, Y131 | MOTIFS |
| | | | Potential Glycosylation Sites: N142, N182 | MOTIFS |
| | | | EGF-like domain signature 2: C159-C174 | MOTIFS |
| | | | Calcium-binding EGF-like domain pattern signature: D134-C159 | MOTIFS |
| 18 | 7519667CD1 | 287 | Signal Peptide: M1-G22, M1-A24, M1-I27, P6-A24 | HMMER |
| | | | signal_cleavage: M1-A24 | SPSCAN |
| | | | Immunoglobulin C-2 Type: D54-G125 (E-value = 0.011) | HMMER_SMART |
| | | | Ig superfamily from SCOP: T44-N142 | HMMER_INCY |
| | | | Cytosolic domains: M1-P6, K179-A287<br>Transmembrane domains: P7-M29, G156-I178<br>Non-cytosolic domain: D30-Q155 | TMHMMER |
| | | | CELL ADHESION PRECURSOR SIGNAL MOLECULE IMMUNOGLOBULIN GLYCOPROTEIN TRANSMEMBRANE REPEAT FOLD PD003273: I174-S278 | BLAST_PRODOM |
| | | | CELL ADHESION MOLECULE PRECURSOR SIGNAL GLYCOPROTEIN TRANSMEMBRANE REPEAT IMMUNOGLOBULIN FOLD PD150530: N35-F119 | BLAST_PRODOM |
| | | | IMMUNOGLOBULIN DM00001\|S26180\|45-129: T44-S129 | BLAST_DOMO |

# Table 3

| SEQ ID NO: | Incyte Polypeptide ID | Amino Acid Residues | Signature Sequences, Domains and Motifs | Analytical Methods and Databases |
|---|---|---|---|---|
| | | | NEURAL CELL ADHESION MOLECULE L1 DM02463\|P32004\|987-1232: A150-K263 DM02463\|P35331\|1009-1259: A144-K263 DM02463\|S26180\|1027-1247: A144-K263 | BLAST_DOMO |
| | | | Potential Phosphorylation Sites: S47, S91, S96, S129, S211, S237, T223, T228, T260 | MOTIFS |
| | | | Potential Glycosylation Sites: N271 | MOTIFS |
| 19 | 7520638CD1 | 572 | Signal Peptide: M1-G35, R11-G35, P13-G35, A14-G35, L16-G35, P18-G35, P13-G37 | HMMER |
| | | | signal_cleavage: M1-G35 | SPSCAN |
| | | | Fibronectin type III domain: P424-S510 | HMMER_PFAM |
| | | | Immunoglobulin domain: G56-A119, G153-A211, H252-A309, G344-A402 | HMMER_PFAM |
| | | | Fibronectin type 3 domain: P424-S507 | HMMER_SMART |
| | | | Immunoglobulin: P48-A136, P145-S228, P244-Q327, P336-L418 | HMMER_SMART |
| | | | Immunoglobulin C-2 Type: V54-G124, E151-S216, T250-T314, P342-G407 | HMMER_SMART |
| | | | I type Ig domains from SCOP: E41-D140, K236-F332, H335-G425 | HMMER_INCY |
| | | | Ig superfamily from SCOP: F44-I133, F141-G231, I240-A330, F332-L423 | HMMER_INCY |
| | | | IMMUNOGLOBULIN DM00001\|P43146\|328-410: P329-Q410 | BLAST_DOMO |
| | | | Potential Phosphorylation Sites: S8, S108, S128, S216, S272, S412, S436, S443, S486, S527, T74, T137, T148, T181, T502, T517 | MOTIFS |
| | | | Potential Glycosylation Sites: N93, N246, N381, N382 | MOTIFS |
| 20 | 7518947CD1 | 403 | Signal Peptide: M1-T34, M1-R36 | HMMER |

# Table 3

<table>
<tr><th>SEQ ID NO:</th><th>Incyte Polypeptide ID</th><th>Amino Acid Residues</th><th>Signature Sequences, Domains and Motifs</th><th>Analytical Methods and Databases</th></tr>
<tr><td>20</td><td></td><td></td><td>signal_cleavage: M1-T34</td><td>SPSCAN</td></tr>
<tr><td></td><td></td><td></td><td>Collagen triple helix repeat (20 copies): G235-L294</td><td>HMMER_PFAM</td></tr>
<tr><td></td><td></td><td></td><td>EGF-like domain: C93-K126, C138-C174</td><td>HMMER_PFAM</td></tr>
<tr><td></td><td></td><td></td><td>Epidermal growth factor-like domain: V92-L133, E137-T175</td><td>HMMER_SMART</td></tr>
<tr><td></td><td></td><td></td><td>Calcium-binding EGF-like domain: D89-L133, D134-T175</td><td>HMMER_SMART</td></tr>
<tr><td></td><td></td><td></td><td>Calcium-binding EGF-like domain<br>IPB001881: C111-R121, C150-C161 (P < 0.0041)</td><td>BLIMPS_BLOCKS</td></tr>
<tr><td></td><td></td><td></td><td>Type I EGF signature<br>PR00009: L154-Y165</td><td>BLIMPS_PRINTS</td></tr>
<tr><td></td><td></td><td></td><td>Type II EGF-like signature<br>PR00010: D89-Q100, G155-Y165 (P < 0.0017)</td><td>BLIMPS_PRINTS</td></tr>
<tr><td></td><td></td><td></td><td>COLLAGEN ALPHA PRECURSOR CHAIN REPEAT SIGNAL CONNECTIVE TISSUE EXTRACELLULAR MATRIX<br>PD000007: P246-P304, P246-P290, P246-G289, G247-G289</td><td>BLAST_PRODOM</td></tr>
<tr><td></td><td></td><td></td><td>FIBRILLAR COLLAGEN CARBOXYL-TERMINAL<br>DM00019|Q02388|1373-1547: P3-G17, L245-Q335, P246-L307<br>DM00019|P20908|1300-1524: P223-Q335, P246-Q335, G247-Q335<br>DM00019|P39061|455-643: G230-V305, S241-V305<br>DM00019|S18803|1304-1528: P223-Q335, P246-Q335, G247-Q335</td><td>BLAST_DOMO</td></tr>
<tr><td></td><td></td><td></td><td>Potential Phosphorylation Sites: S62, S156, S357, S363, T34, T54, T175, T184, T201, T383, T394, Y131</td><td>MOTIFS</td></tr>
<tr><td></td><td></td><td></td><td>Potential Glycosylation Sites: N142, N182</td><td>MOTIFS</td></tr>
</table>

# Table 3

<table>
<tr><th>SEQ ID NO:</th><th>Incyte Polypeptide ID</th><th>Amino Acid Residues</th><th>Signature Sequences, Domains and Motifs</th><th>Analytical Methods and Databases</th></tr>
<tr><td></td><td></td><td></td><td>Cell attachment sequence: R176-D178</td><td>MOTIFS</td></tr>
<tr><td></td><td></td><td></td><td>Aspartic acid and asparagine hydroxylation site: C150-C161</td><td>MOTIFS</td></tr>
<tr><td></td><td></td><td></td><td>EGF-like domain signature 2: C159-C174</td><td>MOTIFS</td></tr>
<tr><td></td><td></td><td></td><td>Calcium-binding EGF-like domain pattern signature: D134-C159</td><td>MOTIFS</td></tr>
<tr><td>21</td><td>7519652CD1</td><td>281</td><td>Signal Peptide: M1-G22, M1-A24, M1-I27, P6-A24</td><td>HMMER</td></tr>
<tr><td></td><td></td><td></td><td>signal_cleavage: M1-A24</td><td>SPSCAN</td></tr>
<tr><td></td><td></td><td></td><td>Immunoglobulin C-2 Type: D48-G119 (E-value = 0.011)</td><td>HMMER_SMART</td></tr>
<tr><td></td><td></td><td></td><td>Ig superfamily from SCOP: T38-N136</td><td>HMMER_INCY</td></tr>
<tr><td></td><td></td><td></td><td>Cytosolic domains: M1-P6, K173-A281<br>Transmembrane domains: P7-M29, G150-I172<br>Non-cytosolic domain: D30-Q149</td><td>TMHMMER</td></tr>
<tr><td></td><td></td><td></td><td>CELL ADHESION PRECURSOR SIGNAL MOLECULE IMMUNOGLOBULIN GLYCOPROTEIN TRANSMEMBRANE REPEAT FOLD PD003273: I168-S272</td><td>BLAST_PRODOM</td></tr>
<tr><td></td><td></td><td></td><td>CELL ADHESION MOLECULE PRECURSOR SIGNAL GLYCOPROTEIN TRANSMEMBRANE REPEAT IMMUNOGLOBULIN FOLD PD150530: L31-F113</td><td>BLAST_PRODOM</td></tr>
<tr><td></td><td></td><td></td><td>IMMUNOGLOBULIN DM00001|S26180|45-129: T38-S123</td><td>BLAST_DOMO</td></tr>
<tr><td></td><td></td><td></td><td>NEURAL CELL ADHESION MOLECULE L1<br>DM02463|P32004|987-1232: A144-K257<br>DM02463|P35331|1009-1259: A138-K257<br>DM02463|S26180|1027-1247: A138-K257</td><td>BLAST_DOMO</td></tr>
<tr><td></td><td></td><td></td><td>Potential Phosphorylation Sites: S41, S85, S90, S123, S205, S231, T217, T222, T254</td><td>MOTIFS</td></tr>
<tr><td></td><td></td><td></td><td>Potential Glycosylation Sites: N265</td><td>MOTIFS</td></tr>
</table>

# Table 3

| SEQ ID NO: | Incyte Polypeptide ID | Amino Acid Residues | Signature Sequences, Domains and Motifs | Analytical Methods and Databases |
|---|---|---|---|---|
| 22 | 7520381CD1 | 520 | Signal Peptide: M1-G22, M1-A25, M1-A27 | HMMER |
| | | | Cadherin repeats: V32-P122, L149-P235, I274-P352 | HMMER_SMART |
| | | | Cadherin signature PR00205: T106-E123 | BLIMPS_PRINTS |
| | | | Potential Phosphorylation Sites: S90, S113, T163, T208, T241, T400, T456, Y162 | MOTIFS |
| | | | Potential Glycosylation Sites: N44, N140, N198, N297, N308, N405 | MOTIFS |
| | | | Cadherins extracellular repeated domain signature: V112-P122 | MOTIFS |
| 23 | 7520409CD1 | 674 | Signal Peptide: M1-A25, M1-A27 | HMMER |
| | | | Cadherin repeats: Y29-P122, I245-P323 | HMMER_SMART |
| | | | Non-cytosolic domain: M1-D496<br>Transmembrane domain: M497-V519<br>Cytosolic domain: H520-I674 | TMHMMER |
| | | | Cadherin signature PR00205: Q74-S100, T106-E123 (P < 0.0071) | BLIMPS_PRINTS |
| | | | Potential Phosphorylation Sites: S90, S113, S423, S493, S531, S650, T163, T179, T212, T371, T416, T628, Y162, Y617 | MOTIFS |
| | | | Potential Glycosylation Sites: N44, N140, N268, N279, N376 | MOTIFS |
| | | | Cadherins extracellular repeated domain signature: V112-P122 | MOTIFS |
| 24 | 7520538CD1 | 743 | Signal Peptide: M1-A22, M1-V24, M1-G26, M1-Y29 | HMMER |
| | | | signal_cleavage: M1-G26 | SPSCAN |
| | | | EGF-like domain: C60-C95, C101-C137, C206-C237, C250-C280, C293-C323, C327-C368, C381-C413, C426-C455, C468-C499, C512-C542, C555-C586 | HMMER_PFAM |

# Table 3

| SEQ ID NO: | Incyte Polypeptide ID | Amino Acid Residues | Signature Sequences, Domains and Motifs | Analytical Methods and Databases |
|---|---|---|---|---|
| | | | Epidermal growth factor-like domain: E59-L96, S100-V138, D205-N238, N249-E281, H292-H324, T326-Q369, E371-G414, N425-E456, G467-Q500, S511-Q543, D554-E587, H598-G636 | HMMER_SMART |
| | | | Laminin-type epidermal growth factor-like domain: C254-C293, C297-C336, C340-C381, C385-C426, C430-C468, C472-C512, C516-C555, C559-C599 | HMMER_SMART |
| | | | Calcium-binding EGF-like domain IPB001881: C71-C82 | BLIMPS_BLOCKS |
| | | | Laminin-type EGF-like (LE) domain IPB002049: S207-G217, G259-C269, C271-G287 | BLIMPS_BLOCKS |
| | | | Type III EGF-like signature PR00011: C262-C280, C481-C499 | BLIMPS_PRINTS |
| | | | MEGF6 GLYCOPROTEIN EGF-LIKE DOMAIN PD169326: L140-D205 PD165309: N298-C327 | BLAST_PRODOM |
| | | | Potential Phosphorylation Sites: S49, S358, S712, T190, T192, T209, T278, T451 | MOTIFS |
| | | | Potential Glycosylation Sites: N238, N249, N425, N560 | MOTIFS |
| | | | Aspartic acid and asparagine hydroxylation site: C71-C82 | MOTIFS |
| | | | EGF-like domain signature 1: C226-C237, C269-C280, C312-C323, C357-C368, C444-C455, C488-C499, C531-C542, C575-C586 | MOTIFS |
| | | | EGF-like domain signature 2: C80-C95, C122-C137, C226-C237, C357-C368, C488-C499, C531-C542, C575-C586 | MOTIFS |
| | | | Calcium-binding EGF-like domain pattern signature: D56-C80 | MOTIFS |
| 25 | 7520667CD1 | 710 | Signal Peptide: M1-A27, M1-A25 | HMMER |

# Table 3

| SEQ ID NO: | Incyte Polypeptide ID | Amino Acid Residues | Signature Sequences, Domains and Motifs | Analytical Methods and Databases |
|---|---|---|---|---|
| | | | Cadherin repeats: Y29-P122, L149-P235, I274-P352 | HMMER_SMART |
| | | | Non-cytosolic domain: M1-D532<br>Transmembrane domain: M533-V555<br>Cytosolic domain: H556-I710 | TMHMMER |
| | | | Cadherin signature<br>PR00205: T106-E123 | BLIMPS_PRINTS |
| | | | Potential Phosphorylation Sites:<br>S90, S113, S452, S529, S567, S686, T163, T208, T241, T400, T445, T664, Y162, Y653 | MOTIFS |
| | | | Potential Glycosylation Sites:<br>N44, N140, N198, N297, N308, N405 | MOTIFS |
| | | | Cadherins extracellular repeated domain signature:<br>V112-P122 | MOTIFS |
| 26 | 7517567CD1 | 1402 | Signal Peptide: M1-L31, M1-A30 | HMMER |
| | | | signal_cleavage: M1-A30 | SPSCAN |
| | | | EGF-like domain: C515-C550, C790-C826, C832-C869, C879-C917, C923-C956 | HMMER_PFAM |
| | | | Low-density lipoprotein receptor repeat class B: R1181-I1223, R1225-I1266, G1268-F1311 | HMMER_PFAM |
| | | | Thyroglobulin type-1 repeat: C967-C1032, C1046-C1111 | HMMER_PFAM |
| | | | Epidermal growth factor-like domain: T514-L551, P789-V827, E831-I870, P878-T918, E922-I957 | HMMER_SMART |
| | | | Calcium-binding EGF-like domain: N511-L551, P786-V827, D828-I870, N877-T918, D919-I957 | HMMER_SMART |
| | | | Low-density lipoprotein-receptor YWTD domain: K1161-E1203, E1205-R1247, K1248-R1292, R1293-R1335, R1336-T1376 | HMMER_SMART |

# Table 3

| SEQ ID NO: | Incyte Polypeptide ID | Amino Acid Residues | Signature Sequences, Domains and Motifs | Analytical Methods and Databases |
|---|---|---|---|---|
| 26 | | | Thyroglobulin type-I repeats: I987-P1036, V1067-V1115 | HMMER_SMART |
| | | | Thyroglobulin type-1 repeat IPB000716: H986-C1001, C1008-Q1022 | BLIMPS_BLOCKS |
| | | | Integrin beta, C-terminus IPB001169: H804-C826 | BLIMPS_BLOCKS |
| | | | Calcium-binding EGF-like domain IPB001881: C813-G823, C845-C856 | BLIMPS_BLOCKS |
| | | | Type II EGF-like signature PR00010: G939-Y949 | BLIMPS_PRINTS |
| | | | GLYCOPROTEIN EGF-LIKE DOMAIN PRECURSOR BASEMENT MEMBRANE EXTRACELLULAR MATRIX SIGNAL CALCIUM BINDING PD014162: P552-P789 P1030-P1043 E553-L714 | BLAST_PRODOM |
| | | | GLYCOPROTEIN EGFLIKE DOMAIN PROTEIN PRECURSOR BASEMENT MEMBRANE EXTRACELLULAR MATRIX SIGNAL PD007677: V77-G285 | BLAST_PRODOM |
| | | | GLYCOPROTEIN EGF-LIKE DOMAIN NIDOGEN2 PRECURSOR NID2 OSTEONIDOGEN BASEMENT MEMBRANE EXTRACELLULAR PD132287: S272-T514 | BLAST_PRODOM |
| | | | GLYCOPROTEIN EGF-LIKE DOMAIN NIDOGEN2 PRECURSOR NID2 OSTEONIDOGEN BASEMENT MEMBRANE EXTRACELLULAR PD153979: F1355-K1402 | BLAST_PRODOM |
| | | | THYROGLOBULIN TYPE I REPEAT DM00280\|P04233\|211-272: T965-T1023 DM00280\|P04441\|194-255: T965-T1023 DM00280\|P10247\|195-256: T965-T1023 DM00280\|P31226\|108-174: H986-C1032 | BLAST_DOMO |

# Table 3

| SEQ ID NO: | Incyte Polypeptide ID | Amino Acid Residues | Signature Sequences, Domains and Motifs | Analytical Methods and Databases |
|---|---|---|---|---|
| | | | Potential Phosphorylation Sites: S42, S84, S147, S250, S274, S340, S348, S360, S386, S397, S458, S500, S670, S708, S724, S851, S858, S942, S1232, S1245, S1285, S1350, T245, T304, T577, T653, T800, T807, T918, T965, T1039, T1044, T1060, T1123, T1322, T1328, T1333, T1399 | MOTIFS |
| | | | Potential Glycosylation Sites: N444, N685, N720, N730, N1151 | MOTIFS |
| | | | Aspartic acid and asparagine hydroxylation site: C528-C539, C845-C856, C934-C945 | MOTIFS |
| | | | EGF-like domain signature 2: C813-C826, C854-C869, C904-C917, C943-C956 | MOTIFS |
| | | | Calcium-binding EGF-like domain pattern signature: D828-C854, D919-C943 | MOTIFS |
| | | | Thyroglobulin type-1 repeat signature: H986-G1015, Y1066-G1095 | MOTIFS |
| 27 | 7519867CD1 | 288 | Lectin C-type domain: S157-K263 | HMMER_PFAM |
| | | | C-type lectin (CTL) or carbohydrate-recognition domain: C140-K262 | HMMER_SMART |
| | | | C-type lectin domain IPB001304: W144-C168, W199-W211 | BLIMPS_BLOCKS |
| | | | C-type lectin domain signature and profile: G209-T282 | PROFILESCAN |
| | | | Type II antifreeze protein signature PR00356: P139-C151, C151-C168, K169-F186, W199-D215, W248-C261 | BLIMPS_PRINTS |
| | | | LECTIN PROTEIN PRECURSOR GLYCOPROTEIN SIGNAL RECEPTOR TRANSMEMBRANE REPEAT CELL DOMAIN PD000254: C168-C253 S244-F275 | BLAST_PRODOM |

# Table 3

<table>
<tr><th>SEQ ID NO:</th><th>Incyte Polypeptide ID</th><th>Amino Acid Residues</th><th>Signature Sequences, Domains and Motifs</th><th>Analytical Methods and Databases</th></tr>
<tr><td></td><td></td><td></td><td>C-TYPE LECTIN DM00035<br>|A46274|248-377: A132-K262<br>|P02707|74-202: L136-K263<br>|P20693|179-304: C137-C261<br>|P34927|145-276: E134-K262</td><td>BLAST_DOMO</td></tr>
<tr><td></td><td></td><td></td><td>Potential Phosphorylation Sites: S48, S68, S157, S180, S191, S194, S222, S269</td><td>MOTIFS</td></tr>
<tr><td></td><td></td><td></td><td>Potential Glycosylation Sites: N56</td><td>MOTIFS</td></tr>
<tr><td></td><td></td><td></td><td>C-type lectin domain signature: C240-C261</td><td>MOTIFS</td></tr>
<tr><td>28</td><td>7519911CD1</td><td>481</td><td>Signal Peptide: M1-A42, M1-G39, T17-G39</td><td>HMMER</td></tr>
<tr><td></td><td></td><td></td><td>signal_cleavage: M1-G39</td><td>SPSCAN</td></tr>
<tr><td></td><td></td><td></td><td>NTR/C345C module: L373-C479</td><td>HMMER_PFAM</td></tr>
<tr><td></td><td></td><td></td><td>Thrombospondin type 1 domain: T49-C96</td><td>HMMER_PFAM</td></tr>
<tr><td></td><td></td><td></td><td>Thrombospondin type 1 repeats: W48-P97</td><td>HMMER_SMART</td></tr>
<tr><td></td><td></td><td></td><td>PROTEIN PROCOLLAGEN THROMBOSPONDIN MOTIFS N-PROTEINASE A DISINTEGRIN METALLOPROTEASE WITH ADAMTS1 PD011654: Q134-C200</td><td>BLAST_PRODOM</td></tr>
<tr><td></td><td></td><td></td><td>Potential Phosphorylation Sites: S66, S69, S465, T17, T119, T287, T397, T475</td><td>MOTIFS</td></tr>
<tr><td></td><td></td><td></td><td>Potential Glycosylation Sites: N197, N218</td><td>MOTIFS</td></tr>
<tr><td>29</td><td>7520005CD1</td><td>460</td><td>Signal Peptide: M1-A22, M1-S24, M1-A21, P4-A22, M1-A28</td><td>HMMER</td></tr>
<tr><td></td><td></td><td></td><td>signal_cleavage: M1-A21</td><td>SPSCAN</td></tr>
<tr><td></td><td></td><td></td><td>Collagen triple helix repeat (20 copies): G225-P284, G316-R375</td><td>HMMER_PFAM</td></tr>
<tr><td></td><td></td><td></td><td>Cytosolic domain: M1-A6<br>Transmembrane domain: W7-P29<br>Non-cytosolic domain: F30-G460</td><td>TMHMMER</td></tr>
</table>

# Table 3

<table>
<tr><th>SEQ ID NO:</th><th>Incyte Polypeptide ID</th><th>Amino Acid Residues</th><th>Signature Sequences, Domains and Motifs</th><th>Analytical Methods and Databases</th></tr>
<tr><td>29</td><td></td><td></td><td>COLLAGEN ALPHA PRECURSOR CHAIN REPEAT SIGNAL CONNECTIVE TISSUE EXTRACELLULAR MATRIX PD000007: G240-G349, G310-E392, P308-E386, G310-E392, G216-P294, P183-G276, G310-G385, P165-G267</td><td>BLAST_PRODOM</td></tr>
<tr><td></td><td></td><td></td><td>SIMILAR TO CUTICULAR COLLAGEN PD067228: G234-G331, P241-G334, N212-G322, P308-K387, P308-G385, P314-G391, P274-Q374, G252-G349, E163-G276, P323-E392</td><td>BLAST_PRODOM</td></tr>
<tr><td></td><td></td><td></td><td>ZK265.2 PROTEIN PD068401: G243-P354, P311-P384</td><td>BLAST_PRODOM</td></tr>
<tr><td></td><td></td><td></td><td>PRECOLLAGEN P PRECURSOR SIGNAL PD072959: G310-G393, G310-H389, G240-G346, G187-P294, G264-P366, G246-G358, G273-P384</td><td>BLAST_PRODOM</td></tr>
<tr><td></td><td></td><td></td><td>FIBRILLAR COLLAGEN CARBOXYL-TERMINAL DM00019<br>|P20908|1300-1524:G267-L394 G310-G442<br>|P20908|1300-1524: G181-G391, P167-G385<br>|S18803|1304-1528: P178-G391, P167-G385, G222-G385, G310-G442, P308-L394, G313-G391<br>DM00042<br>|P28481|702-911:P173-E386 G427-G442<br>|A41182|662-871:P173-E386 G427-G442<br>|A41182|662-871: P167-G370, P226-G391, G216-G391, G252-E392, P165-G282, P268-G442<br>|P28481|702-911: P167-G370, P226-G391, G216-G391, G252-E392, P165-G282, P268-G442</td><td>BLAST_DOMO</td></tr>
<tr><td></td><td></td><td></td><td>Potential Phosphorylation Sites: S31, S72, S103, S188, S203, S455, T80, T89, T122, T146, T160, T172, T259, T299</td><td>MOTIFS</td></tr>
</table>

# Table 3

| SEQ ID NO: | Incyte Polypeptide ID | Amino Acid Residues | Signature Sequences, Domains and Motifs | Analytical Methods and Databases |
|---|---|---|---|---|
| | | | Potential Glycosylation Sites: N51, N138 | MOTIFS |
| 30 | 7520484CD1 | 415 | Signal Peptide: M1-A22, M1-S24, M1-A21, P4-A22, M1-A28 | HMMER |
| | | | signal_cleavage: M1-A21 | SPSCAN |
| | | | Collagen triple helix repeat (20 copies):G180-P239, G271-R330 | HMMER_PFAM |
| | | | Cytosolic domain: M1-A6<br>Transmembrane domain: W7-P29<br>Non-cytosolic domain: F30-G415 | TMHMMER |
| | | | COLLAGEN ALPHA PRECURSOR CHAIN REPEAT SIGNAL CONNECTIVE TISSUE EXTRACELLULAR MATRIX PD000007: G195-G304, G265-E347, P263-E341, G265-E347, G171-P249, G174-P249, P137-G228, G265-G340, Q136-G210 | BLAST_PRODOM |
| | | | SIMILAR TO CUTICULAR COLLAGEN PD067228: G189-G286, D163-G277, E135-R242, P263-K342, P263-G340, P269-G346, P229-Q329, G207-G304, P278-E347, P137-G231 | BLAST_PRODOM |
| | | | ZK265.2 PROTEIN PD068401: G198-P309, P266-P339, P145-P239, G162-P282 | BLAST_PRODOM |
| | | | PRECOLLAGEN P PRECURSOR SIGNAL PD072959: G265-G348, G265-H344, G219-P321, G195-G301, G153-P249, G228-P339, G201-G313 | BLAST_PRODOM |

# Table 3

| SEQ ID NO: | Incyte Polypeptide ID | Amino Acid Residues | Signature Sequences, Domains and Motifs | Analytical Methods and Databases |
|---|---|---|---|---|
| 30 | | | FIBRILLAR COLLAGEN CARBOXYL-TERMINAL DM00019<br>\|P20908\|1300-1524:G222-L349 G265-G397<br>\|P20908\|1300-1524: P148-G346, P140-G340, G174-G340, P139-K342, P139-G280<br>\|S18803\|1304-1528: P148-G346, G153-G340, P140-G340, G177-G340, G265-G397, P137-G286, P263-L349, G268-G346<br>DM00042<br>\|A41182\|662-871:P145-E341 G382-G397<br>\|B41182\|730-939:P145-E341 G382-G397<br>\|A41182\|662-871: P139-G334, P154-G346, P181-G397, P137-G237<br>\|B41182\|730-939: P139-G334, P154-G346, P181-G397, P137-G237 | BLAST_DOMO |
| | | | Potential Phosphorylation Sites: S31, S72, S103, S160, S410, T80, T89, T118, T132, T144, T214, T254 | MOTIFS |
| | | | Potential Glycosylation Sites: N51, N110 | MOTIFS |

# Table 4

| Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length | Sequence Fragments |
|---|---|
| 31/7514594CB1<br>833 | 1-833, 162-710 |
| 32/7515764CB1<br>1026 | 1-463, 1-464, 1-530, 1-563, 1-566, 1-576, 2-1026, 313-1026, 574-1026 |
| 33/7514575CB1<br>809 | 1-474, 1-476, 2-476, 2-809 |
| 34/7514576CB1<br>1058 | 1-725, 3-725, 7-723, 193-454, 193-723, 193-725, 194-454, 194-1058, 203-725, 511-723, 511-725 |
| 35/7515593CB1<br>534 | 1-222, 1-534, 2-533, 2-534, 335-533 |
| 36/7517754CB1<br>2370 | 1-284, 1-2370, 117-343, 117-344, 117-345, 117-348, 117-350, 117-368, 117-389, 117-407, 118-444, 119-395, 122-365, 124-359, 132-369, 133-363, 135-440, 136-359, 146-383, 153-358, 162-357, 162-362, 188-443, 199-535, 240-476, 278-547, 279-551, 305-535, 313-510, 557-1071, 666-937, 877-1090, 877-1231, 877-1266, 915-1266, 934-1333, 934-1416, 937-1559, 939-1201, 939-1220, 939-1233, 943-1557, 947-1164, 957-1558, 961-1659, 962-1508, 963-1422, 966-1229, 970-1183, 1000-1490, 1002-1414, 1005-1123, 1006-1276, 1009-1817, 1030-1662, 1063-1332, 1073-1540, 1076-1537, 1097-1563, 1098-1528, 1104-1258, 1105-1374, 1120-1317, 1120-1366, 1121-1395, 1125-1377, 1130-1801, 1138-1307, 1138-1706, 1146-1737, 1151-1429, 1156-1414, 1157-1390, 1159-1781, 1161-1436, 1161-1439, 1161-1448, 1162-1597, 1162-1600, 1162-1920, 1164-1454, 1166-1277, 1168-1632, 1177-1421, 1179-1414, 1179-1466, 1179-1865, 1187-1443, 1187-1449, 1187-1580, 1189-1450, 1194-1463, 1194-1496, 1195-1414, 1207-1471, 1219-1489, 1239-1527, 1243-1530, 1249-1679, 1249-1803, |
| | 1261-1991, 1262-1565, 1267-1749, 1272-1793, 1273-1562, 1274-1470, 1277-1542, 1280-1886, 1287-1539, 1287-1754, 1291-1546, 1294-1752, 1296-1909, 1307-1904, 1309-1539, 1309-1566, 1310-1589, 1311-1583, 1338-1626, 1339-1560, 1339-1618, 1344-1597, 1355-1659, 1367-1887, 1372-1650, 1376-1647, 1376-1650, 1378-1670, 1378-1672, 1387-1612, 1389-1624, 1389-1631, 1400-1677, 1400-1684, 1413-1557, 1415-1995, 1423-1541, 1428-1682, 1433-1707, 1441-1705, 1441-1977, 1452-2069, 1457-2276, 1463-1740, 1468-1732, 1471-1924, 1473-1709, 1473-1747, 1487-1570, 1487-1772, 1489-1788, 1489-1938, 1490-1850, 1503-1980, 1507-1984, 1510-1885, 1519-1841, 1528-1992, 1535-1728, 1535-1858, 1535-1870, 1535-1885, 1536-1992, 1538-1992, 1543-1994, 1550-1971, 1551-1992, 1553-1800, 1555-1862, 1555-1995, 1557-1989, 1558-1992, 1560-2068, 1574-1868, 1582-1992, 1583-1721, 1584-1838, 1585-1830, 1585-1928, 1585-1934, 1592-1883, 1593-1922, 1594-1871, 1602-1995, 1602-2248, 1603-1983, 1603-2177, 1614-1989, 1617-1872, 1617-2095, 1626-1859, 1626-1884, 1626-2148, |

Table 4

| Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length | Sequence Fragments |
|---|---|
| | 1627-1992, 1632-1883, 1637-1885, 1647-1892, 1650-2370, 1651-2335, 1660-2273, 1665-2208, 1667-1991, 1670-2335, 1685-1995, 1688-1991, 1702-1961, 1702-2195, 1704-1987, 1706-1991, 1706-2317, 1707-1989, 1707-2335, 1708-1930, 1708-1956, 1708-1962, 1708-1995, 1715-1883, 1715-2100, 1723-1995, 1727-1990, 1736-2009, 1743-1989, 1743-2057, 1745-1970, 1747-1985, 1748-2253, 1750-1990, 1752-2025, 1752-2040, 1754-1995, 1757-1992, 1760-2020, 1761-2005, 1761-2068, 1761-2100, 1770-1995, 1784-2363, 1786-1992, 1787-2324, 1789-2304, 1789-2329, 1792-1936, 1799-2339, 1807-1992, 1808-2340, 1811-2310, 1827-2084, 1827-2094, 1837-2122, 1839-2069, 1841-2085, 1842-2225, 1867-2340, 1882-2131, 1889-2176, 1893-2132, 1905-1995, 1906-2178, 1914-1995, 1925-2370, 1936-2304, 1945-2370, 1951-2369, 1953-2369, 1960-2370, 1961-2370, 1965-2370, 1966-2176, 1967-2370, 1981-2257, 1996-2219, 1996-2321, 2003-2119, 2003-2245, 2003-2253, 2005-2257, 2010-2370, 2014-2369, 2015-2261, 2018-2370, 2035-2369, 2038-2370, 2039-2343, 2042-2369, 2049-2363, 2063-2370, |
| | 2064-2339, 2068-2324, 2068-2362, 2068-2369, 2075-2254, 2078-2282, 2108-2370, 2111-2356, 2126-2370, 2183-2369, 2194-2369, 2198-2369, 2200-2370 |
| 37/7517819CB1 4695 | 1-4695, 122-614, 201-476, 487-936, 747-1311, 768-1036, 768-1365, 776-1073, 821-1463, 855-1533, 966-1419, 1053-1499, 1121-1392, 1218-1518, 1498-2260, 1615-2171, 1615-2217, 1656-1724, 1672-1914, 1672-1963, 1672-2079, 1672-2479, 1674-2291, 1675-2285, 1731-2109, 1731-2218, 1731-2291, 1732-2509, 1733-2355, 1743-2509, 1759-2237, 1764-2164, 1769-2171, 1809-2255, 1842-2390, 1870-2509, 1886-2183, 1886-2509, 1896-2390, 1917-2449, 1949-2380, 1981-2351, 1994-2506, 2035-2351, 2072-2357, 2077-2710, 2120-2380, 2146-3028, 2148-2768, 2163-2503, 2164-2404, 2183-2269, 2183-2402, 2219-2525, 2226-2830, 2232-2535, 2246-2509, 2264-2508, 2274-2537, 2336-3034, 2352-2793, 2352-2885, 2353-2755, 2366-2774, 2414-2902, 2445-2686, 2449-2722, 2452-2509, 2453-2764, 2453-2892, 2514-2737, 2575-2872, 2592-3006, 2596-2730, 2613-2854, 2632-2818, 2634-3056, 2634-3075, 2635-3101, 2635-3119, 2648-3216, 2651-3476, 2655-2840, 2700-3082, 2736-3152, 2741-3212, 2752-3474, 2757-3280, 2757-3296, 2766-3332, 2768-3289, 2768-3315, 2769-3306, 2799-3371, |

## Table 4

| Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length | Sequence Fragments |
|---|---|
| | 2812-3048, 2824-3324, 2857-3124, 2858-3504, 2902-3150, 2921-3180, 2936-3209, 2937-3151, 2938-3261, 2938-3306, 2947-3058, 2950-3184, 2968-3276, 2974-3256, 2974-3320, 2974-3437, 2985-3163, 2985-3243, 2988-3267, 2992-3293, 3000-3257, 3028-3298, 3052-3235, 3072-3269, 3095-3381, 3097-3315, 3098-3354, 3104-3369, 3104-3381, 3113-3403, 3124-3381, 3126-3395, 3128-3668, 3131-3322, 3134-3702, 3140-3528, 3140-3803, 3166-3432, 3168-3542, 3168-3575, 3177-3481, 3177-3608, 3196-3424, 3196-3451, 3197-3498, 3197-3811, 3211-3748, 3213-3500, 3235-3700, 3241-3826, 3241-3865, 3249-3846, 3263-3791, 3283-3462, 3283-3518, 3284-3906, 3290-3470, 3290-3852, 3293-3551, 3293-3903, 3293-4003, 3302-4004, 3306-3561, 3312-3656, 3312-3779, 3313-3590, 3325-3461, 3328-3610, 3329-3579, 3350-3969, 3355-3596, 3355-3609, 3366-3780, 3367-3643, 3379-3622, 3384-3626, 3384-3978, 3393-4004, 3408-3605, 3408-3652, 3410-3986, 3418-3670, 3418-3692, 3425-3697, 3437-4126, 3459-3695, 3460-3734, 3460-3762, 3460-3847, 3460-3870, 3460-3904, 3460-3913, 3460-3934, |
| | 3460-3968, 3460-3971, 3460-3976, 3460-3984, 3460-3986, 3460-3987, 3460-3990, 3460-4029, 3470-3735, 3470-3738, 3483-3695, 3486-3916, 3503-3809, 3504-3787, 3505-3787, 3506-3763, 3506-4041, 3508-3816, 3508-4213, 3516-3762, 3518-3956, 3524-3773, 3526-3785, 3529-4079, 3534-3768, 3545-3763, 3552-3706, 3552-3799, 3557-3763, 3557-3796, 3558-4282, 3559-3839, 3560-3826, 3560-4116, 3560-4128, 3561-3821, 3565-3808, 3565-3820, 3567-4147, 3569-3871, 3570-4111, 3572-3710, 3580-4097, 3582-3812, 3585-3990, 3588-4378, 3595-3898, 3602-4295, 3603-3979, 3603-4048, 3609-3956, 3612-3871, 3615-3862, 3616-4195, 3616-4201, 3621-3879, 3626-3852, 3626-3991, 3632-3909, 3644-3806, 3652-4500, 3654-3706, 3665-3932, 3667-3986, 3670-3921, 3671-3917, 3686-4026, 3686-4031, 3688-4167, 3690-4334, 3701-3902, 3701-3914, 3702-3790, 3708-4270, 3716-4330, 3724-3920, 3724-3987, 3724-4050, 3725-3987, 3725-3996, 3725-4018, 3728-4265, 3729-3989, 3731-3985, 3762-4009, 3762-4029, 3764-3986, 3776-3985, 3776-4215, 3781-4224, 3781-4226, 3788-4026, 3789-4021, |
| | 3790-3866, 3792-4134, 3792-4413, 3794-4361, 3795-4054, 3807-4058, 3816-4317, 3822-4411, 3823-4080, 3824-4069, 3825-4098, 3850-4337, 3851-3984, 3851-4409, 3855-4309, 3862-4055, 3863-4183, 3864-4082, 3881-4114, 3881-4294, 3887-4119, 3887-4134, 3901-4124, 3904-4151, 3911-4167, 3911-4175, 3912-4204, 3918-4195, 3930-4194, 3930-4208, 3935-4188, 3965-4237, 3987-4287, 4014-4221, 4021-4195, 4046-4296, 4047-4213, 4315-4401, 4319-4375, 4320-4377, 4323-4377 |
| 38/7518460CB1 1110 | 1-662, 156-1110, 352-1109, 980-1110 |
| 39/7518734CB1 1787 | 1-515, 2-1787, 72-917, 315-916 |

# Table 4

| Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length | Sequence Fragments |
|---|---|
| 40/7513469CB1<br>13426 | 1-317, 1-13426, 17-236, 223-630, 250-660, 254-660, 257-287, 281-766, 281-852, 295-925, 296-724, 351-768, 476-995, 503-875, 583-999, 614-870, 658-875, 814-1350, 895-1345, 939-1513, 1024-1377, 1448-2082, 1468-2082, 1475-2078, 1510-2004, 1513-2138, 1514-2086, 1516-2082, 1541-2062, 1543-2082, 1547-2120, 1586-2082, 1587-2171, 1597-2071, 1638-2103, 1657-2106, 1742-2082, 1749-1994, 1774-2466, 1786-2401, 1796-2407, 1837-2380, 1964-2390, 1978-2653, 2093-2764, 2127-2753, 2140-2736, 2141-2502, 2157-2788, 2408-2890, 2492-2852, 2652-3132, 2685-3127, 5077-5540, 8568-8893, 10313-10809, 10315-10809, 11630-11776, 11638-11776, 11656-11805, 11656-11808, 11809-12193, 11815-12485, 11817-12133, 11817-12432, 11818-12241, 11827-12175, 11827-12277, 11849-12253, 11853-12095, 11853-12100, 11856-12162, 11948-12192, 11967-12187, 11967-12256, 11967-12362, 11967-12403, 11967-12424, 11967-12439, 11967-12494, 11967-12505, 11967-12558, 11967-12565, 11967-12639, 11967-12652, 11967-12677, 11967-12766, 11967-12779, 11967-12793, 11970-12324, |
| | 12026-12245, 12106-12375, 12200-12486, 12212-12840, 12236-12530, 12248-12437, 12319-12829, 12335-12931, 12346-12842, 12386-13002, 12421-13074, 12464-12601, 12493-13161, 12534-12813, 12558-13187, 12562-13192, 12572-13266, 12583-13203, 12590-13218, 12607-13191, 12609-13279, 12682-13211, 12732-12948, 12742-13281, 12761-13178, 12831-13363, 12839-13425, 12849-13135, 12857-13105, 12861-13184, 12862-13426, 12875-13143, 12875-13175, 12875-13225, 12875-13288, 12881-13143, 12883-13362, 12904-13057, 12904-13326, 12904-13417, 12912-13420, 12917-13254, 12930-13193, 12938-13177, 12939-13174, 12939-13186, 12939-13192, 12939-13224, 12943-13214, 12944-13300, 12945-13276, 12946-13310, 12970-13414, 12988-13256, 12988-13268, 12990-13247, 12991-13159, 12996-13242, 13007-13288, 13027-13325, 13040-13290, 13043-13296, 13049-13297, 13051-13312, 13053-13263, 13075-13314, 13075-13404, 13096-13340, 13097-13338, 13116-13425, 13141-13354, 13178-13339, 13185-13334, 13210-13294 |
| 41/7518836CB1<br>699 | 1-699, 193-698, 195-699 |
| 42/7519868CB1<br>1135 | 1-497, 26-1111, 325-405, 325-431, 325-474, 325-497, 362-431, 394-500, 431-1135 |
| 43/7520048CB1<br>3228 | 1-989, 34-875, 35-915, 36-739, 57-809, 65-3226, 66-3228, 387-1188, 395-1172, 674-1374, 674-1565, 783-1609, 804-1633, 916-1821, 937-1828, 1170-1864, 1171-1948, 1354-2168, 1357-2234, 1727-2571, 1891-2325, 2000-2600, 2000-2677, 2332-3228, 2343-3228, 2408-3228, 2520-3228, 2522-3228, 2529-3223, 2617-3228, 2866-3228, 2898-3228, 3083-3228 |
| 44/7520157CB1<br>2265 | 1-827, 2-765, 25-873, 54-732, 54-788, 54-824, 54-967, 102-2265, 691-1471, 691-1475, 1325-2146, 1338-2176, 1391-2265, 1448-2265, 1468-2265, 1501-2265, 1514-2265, 1562-2265, 1616-2265 |

# Table 4

| Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length | Sequence Fragments |
|---|---|
| 45/7520485CB1 2093 | 1-772, 3-2037, 295-1124, 319-1124, 371-483, 371-496, 371-552, 371-565, 371-621, 371-634, 371-690, 371-703, 371-759, 371-828, 440-834, 440-897, 443-897, 509-858, 509-897, 512-897, 578-871, 578-897, 581-897, 647-897, 650-897, 716-897, 762-897, 785-897, 931-1850, 938-1852, 1406-2093 |
| 46/7525723CB1 1384 | 1-904, 2-1383, 494-1384, 600-1384, 667-1384 |
| 47/7525966CB1 1128 | 1-689, 2-1127, 384-1128 |
| 48/7519667CB1 1066 | 1-798, 1-1066, 326-1066 |
| 49/7520638CB1 2870 | 1-651, 1-662, 1-750, 1-827, 1-837, 1-874, 2-2818, 5-839, 770-1595, 770-1673, 775-1615, 777-1562, 1450-2203, 1479-2194, 1985-2870, 2103-2870, 2135-2870 |
| 50/7518947CB1 2280 | 1-611, 1-704, 1-727, 1-798, 1-807, 1-809, 1-838, 1-2216, 663-1428, 674-1354, 752-1649, 798-1649, 1308-2280, 1397-2280, 1518-2280, 1594-2280 |
| 51/7519652CB1 1050 | 1-248, 1-744, 1-870, 1-1050, 187-1049, 251-1050, 308-1050 |
| 52/7520381CB1 2055 | 1-749, 1-762, 1-775, 1-779, 1-844, 2-1984, 532-1357, 575-1357, 616-1268, 619-1397, 1275-2055, 1300-2055, 1673-2055 |
| 53/7520409CB1 2120 | 1-778, 1-801, 1-975, 2-2048, 627-1412, 660-1502, 738-1493, 868-1524, 1333-1376, 1384-2120, 1426-1469, 1489-2120, 1697-2120 |
| 54/7520538CB1 2852 | 1-657, 1-789, 5-2804, 459-1361, 527-956, 626-1372, 629-1399, 1024-1378, 1157-1733, 1263-2194, 1268-1962, 1297-1949, 1321-2163, 1433-2162, 1518-2194, 1985-2849, 2184-2852 |
| 55/7520667CB1 2227 | 1-826, 2-2225, 674-1547, 708-1509, 818-1601, 844-1629, 861-1629, 869-1626, 1425-1484, 1518-1577, 1561-1882, 1592-1872, 1592-2227 |
| 56/7517567CB1 4953 | 1-687, 1-838, 1-841, 53-4928, 769-1579, 769-1623, 769-1754, 771-1619, 774-1497, 774-1579, 777-1620, 795-1576, 1530-2370, 1556-2470, 1561-2298, 2210-3113, 2213-2911, 2376-3181, 2398-3132, 2905-3730, 2908-3573, 3540-4314, 3541-4306, 4356-4953, 4364-4952 |
| 57/7519867CB1 1133 | 1-496, 22-1105, 252-325, 252-358, 252-394, 252-427, 252-463, 291-427, 321-496, 360-1133, 390-496, 429-496 |

# Table 4

| Polynucleotide SEQ ID NO:/ Incyte ID/ Sequence Length | Sequence Fragments |
|---|---|
| 58/7519911CB1 1786 | 1-615, 1-1784, 907-1786, 1138-1786 |
| 59/7520005CB1 1483 | 1-739, 1-1457, 1-1459, 58-817, 214-817, 688-1310, 692-1012, 717-1483 |
| 60/7520484CB1 1306 | 1-536, 1-849, 1-855, 1-874, 1-1305, 1-1306, 229-827, 379-691, 384-755, 539-1306, 782-1306 |

# Table 5

| Polynucleotide SEQ ID NO: | Incyte Project ID: | Representative Library |
|---|---|---|
| 36 | 7517754CB1 | LUNGFET03 |
| 37 | 7517819CB1 | BLADTUT04 |
| 40 | 7513469CB1 | BRAUNOR01 |

# Table 6

| Library | Vector | Library Description |
|---|---|---|
| BLADTUT04 | pINCY | Library was constructed using RNA isolated from bladder tumor tissue removed from a 60-year-old Caucasian male during a radical cystectomy, prostatectomy, and vasectomy. Pathology indicated grade 3 transitional cell carcinoma in the left bladder wall. Carcinoma in-situ was identified in the dome and trigone. Patient history included tobacco use. Family history included type I diabetes, malignant neoplasm of the stomach, atherosclerotic coronary artery disease, and acute myocardial infarction. |
| BRAUNOR01 | pINCY | This random primed library was constructed using RNA isolated from striatum, globus pallidus and posterior putamen tissue removed from an 81-year-old Caucasian female who died from a hemorrhage and ruptured thoracic aorta due to atherosclerosis. Pathology indicated moderate atherosclerosis involving the internal carotids, bilaterally; microscopic infarcts of the frontal cortex and hippocampus, and scattered diffuse amyloid plaques and neurofibrillary tangles, consistent with age. Grossly, the leptomeninges showed only mild thickening and hyalinization along the superior sagittal sinus. The remainder of the leptomeninges was thin and contained some congested blood vessels. Mild atrophy was found mostly in the frontal poles and lobes, and temporal lobes, bilaterally. Microscopically, there were pairs of Alzheimer type II astrocytes within the deep layers of the neocortex. There was increased satellitosis around neurons in the deep gray matter in the middle frontal cortex. The amygdala contained rare diffuse plaques and neurofibrillary tangles. The posterior hippocampus contained |
| | | a microscopic area of cystic cavitation with hemosiderin-laden macrophages surrounded by reactive gliosis. Patient history included sepsis, cholangitis, post-operative atelectasis, pneumonia CAD, cardiomegaly due to left ventricular hypertrophy, splenomegaly, arteriolonephrosclerosis, nodular colloidal goiter, emphysema, CHF, hypothyroidism, and peripheral vascular disease. |
| LUNGFET03 | pINCY | Library was constructed using RNA isolated from lung tissue removed from a Caucasian female fetus, who died at 20 weeks' gestation. |

## Table 7

| Program | Description | Reference | Parameter Threshold |
|---|---|---|---|
| ABI FACTURA | A program that removes vector sequences and masks ambiguous bases in nucleic acid sequences. | Applied Biosystems, Foster City, CA. | |
| ABI/PARACEL FDF | A Fast Data Finder useful in comparing and annotating amino acid or nucleic acid sequences. | Applied Biosystems, Foster City, CA; Paracel Inc., Pasadena, CA. | Mismatch <50% |
| ABI AutoAssembler | A program that assembles nucleic acid sequences. | Applied Biosystems, Foster City, CA. | |
| BLAST | A Basic Local Alignment Search Tool useful in sequence similarity search for amino acid and nucleic acid sequences. BLAST includes five functions: blastp, blastn, blastx, tblastn, and tblastx. | Altschul, S.F. et al. (1990) J. Mol. Biol. 215:403-410; Altschul, S.F. et al. (1997) Nucleic Acids Res. 25:3389-3402. | *ESTs:* Probability value=1.0E-8 or less<br>*Full Length sequences*: Probability value= 1.0E-10 or less |
| FASTA | A Pearson and Lipman algorithm that searches for similarity between a query sequence and a group of sequences of the same type. FASTA comprises as least five functions: fasta, tfasta, fastx, tfastx, and ssearch. | Pearson, W.R. and D.J. Lipman (1988) Proc. Natl. Acad Sci. USA 85:2444-2448; Pearson, W.R. (1990) Methods Enzymol. 183:63-98; and Smith, T.F. and M.S. Waterman (1981) Adv. Appl. Math. 2:482-489. | *ESTs:* fasta E value=1.06E-6<br>*Assembled ESTs*: fasta Identity= 95% or greater and<br>Match length=200 bases or greater; fastx E value=1.0E-8 or less<br>*Full Length sequences:*<br>fastx score=100 or greater |
| BLIMPS | A BLocks IMProved Searcher that matches a sequence against those in BLOCKS, PRINTS, DOMO, PRODOM, and PFAM databases to search for gene families, sequence homology, and structural fingerprint regions. | Henikoff, S. and J.G. Henikoff (1991) Nucleic Acids Res. 19:6565-6572; Henikoff, J.G. & S. Henikoff (1996) Methods Enzymol. 266:88-105; and Attwood, T.K. et al. (1997) J. Chem. Inf. Comput. Sci. 37:417-424. | Probability value=1.0E-3 or less |
| HMMER | An algorithm for searching a query sequence against hidden Markov model (HMM)-based databases of protein family consensus sequences, such as PFAM, INCY, SMART, and TIGRFAM. | Krogh, A. et al. (1994) J. Mol. Biol. 235:1501-1531; Sonnhammer, E.L.L. et al. (1988) Nucleic Acids Res. 26:320-322; Durbin, R. et al. (1998) Our World View, in a Nutshell, Cambridge Univ. Press, p. 1-350 | *PFAM, INCY, SMART, or TIGRFAM hits:* Probability value=1.0E-3 or less<br>*Signal peptide hits:* Score= 0 or greater |

## Table 7 (cont.)

| Program | Description | Reference | Parameter Threshold |
|---|---|---|---|
| ProfileScan | An algorithm that searches for structural and sequence motifs in protein sequences that match sequence patterns defined in Prosite. | Gribskov, M. et al. (1988) CABIOS 4:61-66; Gribskov, M. et al. (1989) Methods Enzymol. 183:146-159; Bairoch, A. et al. (1997) Nucleic Acids Res. 25:217-221. | Normalized quality score≥GCG-specified "HIGH" value for that particular Prosite motif. Generally, score=1.4-2.1. |
| Phred | A base-calling algorithm that examines automated sequencer traces with high sensitivity and probability. | Ewing, B. et al. (1998) Genome Res. 8:175-185; Ewing, B. and P. Green (1998) Genome Res. 8:186-194. | |
| Phrap | A Phils Revised Assembly Program including SWAT and CrossMatch, programs based on efficient implementation of the Smith-Waterman algorithm, useful in searching sequence homology and assembling DNA sequences. | Smith, T.F. and M.S. Waterman (1981) Adv. Appl. Math. 2:482-489; Smith, T.F. and M.S. Waterman (1981) J. Mol. Biol. 147:195-197; and Green, P., University of Washington, Seattle, WA. | Score=120 or greater; Match length=56 or greater |
| Consed | A graphical tool for viewing and editing Phrap assemblies. | Gordon, D. et al. (1998) Genome Res. 8:195-202. | |
| SPScan | A weight matrix analysis program that scans protein sequences for the presence of secretory signal peptides. | Nielson, H. et al. (1997) Protein Engineering 10:1-6; Claverie, J.M. and S. Audic (1997) CABIOS 12:431-439. | Score=3.5 or greater |
| TMAP | A program that uses weight matrices to delineate transmembrane segments on protein sequences and determine orientation. | Persson, B. and P. Argos (1994) J. Mol. Biol. 237:182-192; Persson, B. and P. Argos (1996) Protein Sci. 5:363-371. | |
| TMHMMER | A program that uses a hidden Markov model (HMM) to delineate transmembrane segments on protein sequences and determine orientation. | Sonnhammer, E.L. et al. (1998) Proc. Sixth Intl. Conf. on Intelligent Systems for Mol. Biol., Glasgow et al., eds., The Am. Assoc. for Artificial Intelligence Press, Menlo Park, CA, pp. 175-182. | |
| Motifs | A program that searches amino acid sequences for patterns that matched those defined in Prosite. | Bairoch, A. et al. (1997) Nucleic Acids Res. 25:217-221; Wisconsin Package Program Manual, version 9, page M51-59, Genetics Computer Group, Madison, WI. | |

# Table 8

| SEQ ID NO: | PID | EST ID | SNP ID | EST SNP | CB1 SNP | EST Allele | Allele 1 | Allele 2 | Amino Acid | Caucasian Allele 1 frequency | African Allele 1 frequency | Asian Allele 1 frequency | Hispanic Allele 1 frequency |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 31 | 7514594 | 2875524H1 | SNP00131200 | 2 | 137 | G | A | G | noncoding | n/a | n/a | n/a | n/a |
| 36 | 7517754 | 1002205H1 | SNP00004748 | 37 | 148 | G | G | A | noncoding | n/a | n/a | n/a | n/a |
| 36 | 7517754 | 1253069H1 | SNP00154954 | 208 | 1364 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 36 | 7517754 | 1442033H1 | SNP00154955 | 167 | 1475 | T | T | C | noncoding | n/a | n/a | n/a | n/a |
| 36 | 7517754 | 1569549H1 | SNP00004749 | 169 | 481 | C | C | T | H92 | n/a | n/a | n/a | n/a |
| 36 | 7517754 | 2098028H1 | SNP00025640 | 117 | 1063 | G | G | T | noncoding | n/a | n/a | n/a | n/a |
| 36 | 7517754 | 2207032H1 | SNP00025641 | 181 | 1186 | T | T | C | noncoding | n/a | n/a | n/a | n/a |
| 36 | 7517754 | 7735718J1 | SNP00025641 | 372 | 1185 | T | T | C | noncoding | n/a | n/a | n/a | n/a |
| 37 | 7517819 | 1239656H1 | SNP00001234 | 224 | 4158 | G | G | A | noncoding | 0.84 | 0.74 | 0.54 | 0.83 |
| 37 | 7517819 | 1285569H1 | SNP00001235 | 185 | 4222 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 37 | 7517819 | 1313103H1 | SNP00001236 | 52 | 4435 | C | C | T | noncoding | 0.95 | n/a | n/a | n/a |
| 37 | 7517819 | 1452622H1 | SNP00001233 | 226 | 3840 | C | C | T | noncoding | 0.96 | n/d | 0.78 | n/a |
| 37 | 7517819 | 1553026H1 | SNP00076261 | 169 | 3153 | C | C | G | noncoding | n/d | n/d | n/d | n/d |
| 37 | 7517819 | 1923152T6 | SNP00001235 | 402 | 4245 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 37 | 7517819 | 1987045T6 | SNP00001234 | 466 | 4182 | G | G | A | noncoding | 0.84 | 0.74 | 0.54 | 0.83 |
| 37 | 7517819 | 1987045T6 | SNP00001235 | 402 | 4246 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 37 | 7517819 | 2672574H1 | SNP00017022 | 138 | 2301 | G | G | A | noncoding | n/a | n/a | n/a | n/a |
| 37 | 7517819 | 2719353F6 | SNP00076260 | 294 | 1069 | T | C | T | F332 | n/d | n/a | 0.58 | n/a |
| 37 | 7517819 | 6775428J1 | SNP00001233 | 331 | 3822 | C | C | T | noncoding | 0.96 | n/d | 0.78 | n/a |
| 37 | 7517819 | 7318428H1 | SNP00100997 | 280 | 248 | C | C | T | H59 | n/d | n/a | n/a | n/a |
| 37 | 7517819 | 7422115T1 | SNP00001236 | 167 | 4442 | C | C | T | noncoding | 0.95 | n/a | n/a | n/a |
| 37 | 7517819 | 7617176J1 | SNP00001236 | 97 | 4406 | C | C | T | noncoding | 0.95 | n/a | n/a | n/a |
| 37 | 7517819 | 7639867J2 | SNP00001233 | 544 | 3841 | C | C | T | noncoding | 0.96 | n/d | 0.78 | n/a |
| 37 | 7517819 | 7699766H1 | SNP00001234 | 465 | 4159 | G | G | A | noncoding | 0.84 | 0.74 | 0.54 | 0.83 |
| 37 | 7517819 | 7699766H1 | SNP00001235 | 401 | 4223 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 37 | 7517819 | 7721346H2 | SNP00017022 | 200 | 2309 | G | G | A | noncoding | n/a | n/a | n/a | n/a |
| 37 | 7517819 | 7756596H1 | SNP00076260 | 104 | 1058 | C | C | T | P329 | n/d | n/a | 0.58 | n/a |
| 38 | 7518460 | 1306340H1 | SNP00124030 | 51 | 372 | C | C | T | A124 | n/d | n/d | n/d | n/d |

# Table 8

| SEQ ID NO: | PID | EST ID | SNP ID | EST SNP | CB1 SNP | EST Allele | Allele 1 | Allele 2 | Amino Acid | Caucasian Allele 1 frequency | African Allele 1 frequency | Asian Allele 1 frequency | Hispanic Allele 1 frequency |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 38 | 7518460 | 1438027F6 | SNP00124030 | 423 | 374 | C | C | T | P125 | n/d | n/d | n/d | n/d |
| 38 | 7518460 | 1540491H1 | SNP00016903 | 135 | 553 | C | C | A | D184 | n/a | n/a | n/a | n/a |
| 38 | 7518460 | 1813381T6 | SNP00001135 | 129 | 1000 | C | C | T | N333 | 0.11 | n/a | n/a | n/a |
| 38 | 7518460 | 2515469T6 | SNP00001135 | 89 | 1031 | C | C | T | P344 | 0.11 | n/a | n/a | n/a |
| 38 | 7518460 | 6714275H1 | SNP00016904 | 459 | 1011 | G | G | C | R337 | n/d | n/a | n/a | n/a |
| 38 | 7518460 | 7716459J1 | SNP00016903 | 275 | 532 | C | C | A | G177 | n/a | n/a | n/a | n/a |
| 38 | 7518460 | 7716459J1 | SNP00124030 | 93 | 352 | C | C | T | N117 | n/d | n/d | n/d | n/d |
| 39 | 7518734 | 1272038F1 | SNP00025350 | 66 | 1475 | T | T | G | noncoding | n/a | n/a | n/a | n/a |
| 39 | 7518734 | 1272038F1 | SNP00100719 | 7 | 1416 | T | T | C | noncoding | n/a | n/a | n/a | n/a |
| 39 | 7518734 | 1904218T6 | SNP00025350 | 71 | 1494 | T | T | G | noncoding | n/a | n/a | n/a | n/a |
| 39 | 7518734 | 1904218T6 | SNP00100719 | 130 | 1435 | T | T | C | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 1300067H1 | SNP00051775 | 123 | 9658 | C | C | A | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 1300067H1 | SNP00051776 | 180 | 9715 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 1312735H1 | SNP00105966 | 42 | 9276 | G | G | C | noncoding | n/d | n/d | n/d | n/d |
| 40 | 7513469 | 1318165H1 | SNP00069037 | 198 | 12732 | C | C | T | noncoding | n/d | n/a | n/d | n/d |
| 40 | 7513469 | 1318165H1 | SNP00101119 | 11 | 12544 | C | C | T | noncoding | n/d | n/a | n/a | n/a |
| 40 | 7513469 | 1366472H1 | SNP00009676 | 92 | 12039 | G | G | A | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 1366472H1 | SNP00009677 | 135 | 12082 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 1419359H1 | SNP00037930 | 196 | 12325 | G | G | C | noncoding | n/d | n/a | n/a | n/a |
| 40 | 7513469 | 1481839H1 | SNP00010403 | 87 | 8780 | A | G | A | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 1481839H1 | SNP00093429 | 27 | 8720 | T | T | C | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 1572606H1 | SNP00101117 | 72 | 10674 | C | C | T | noncoding | n/d | n/d | n/d | n/d |
| 40 | 7513469 | 1623175F6 | SNP00009676 | 72 | 12040 | G | G | A | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 1623175F6 | SNP00132541 | 348 | 12316 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 1904455H1 | SNP00049782 | 195 | 12393 | T | T | C | noncoding | 0.96 | 0.97 | 0.98 | 0.97 |
| 40 | 7513469 | 1904455H1 | SNP00069036 | 159 | 12357 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 1998106R6 | SNP00141290 | 269 | 13143 | A | A | C | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 2020032H1 | SNP00114917 | 23 | 9393 | A | A | G | noncoding | n/d | n/d | n/d | n/d |

# Table 8

| SEQ ID NO: | PID | EST ID | SNP ID | EST SNP | CB1 SNP | EST Allele | Allele 1 | Allele 2 | Amino Acid | Caucasian Allele 1 frequency | African Allele 1 frequency | Asian Allele 1 frequency | Hispanic Allele 1 frequency |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 40 | 7513469 | 2049978H1 | SNP00101118 | 19 | 12536 | T | T | C | noncoding | n/d | 0.42 | 0.86 | 0.68 |
| 40 | 7513469 | 2078786H1 | SNP00065015 | 33 | 9669 | T | T | C | noncoding | 0.77 | 0.68 | 0.73 | 0.7 |
| 40 | 7513469 | 2152163H1 | SNP00049781 | 45 | 11991 | G | G | C | noncoding | n/d | n/a | n/a | n/a |
| 40 | 7513469 | 2199929H1 | SNP00037927 | 70 | 7598 | T | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 2199929H1 | SNP00054887 | 169 | 7697 | G | G | A | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 2360678H1 | SNP00092406 | 211 | 9833 | G | G | A | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 2460907H1 | SNP00040498 | 97 | 9075 | G | A | G | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 2689609H1 | SNP00127996 | 52 | 9805 | A | A | G | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 2953460H1 | SNP00040499 | 52 | 9532 | A | A | G | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 3172439H1 | SNP00127179 | 83 | 9183 | G | A | G | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 3360170T6 | SNP00141290 | 47 | 13161 | A | A | C | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 3450931H1 | SNP00143083 | 92 | 1900 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 3539617H1 | SNP00069035 | 69 | 10210 | G | G | A | noncoding | n/d | n/d | n/d | n/d |
| 40 | 7513469 | 3746553H1 | SNP00069034 | 85 | 9531 | T | C | T | noncoding | 0.94 | 0.94 | 0.92 | 0.94 |
| 40 | 7513469 | 3776409H1 | SNP00059686 | 95 | 8270 | G | G | A | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 3825875H1 | SNP00093071 | 83 | 7715 | C | C | T | noncoding | 0.9 | 0.9 | 0.9 | 0.95 |
| 40 | 7513469 | 5577084F9 | SNP00062231 | 419 | 9939 | T | T | C | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 5643861H1 | SNP00071879 | 53 | 1578 | C | T | C | D484 | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 5643861H1 | SNP00093275 | 97 | 1622 | G | G | C | R499 | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 5643861H1 | SNP00093277 | 70 | 1595 | G | G | T | G490 | 0.51 | 0.3 | 0.49 | 0.58 |
| 40 | 7513469 | 6001971H1 | SNP00113564 | 184 | 10025 | G | A | G | noncoding | 0.46 | n/a | n/a | n/a |
| 40 | 7513469 | 6034571H1 | SNP00049410 | 401 | 699 | C | C | T | T191 | n/d | n/a | n/a | n/a |
| 40 | 7513469 | 6034571H1 | SNP00092492 | 304 | 602 | G | G | A | G159 | n/d | n/d | n/d | n/d |
| 40 | 7513469 | 6034571H1 | SNP00132539 | 503 | 801 | G | G | C | S225 | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 6427491H1 | SNP00049409 | 424 | 921 | G | C | G | L265 | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 6427491H1 | SNP00121185 | 298 | 1047 | C | C | T | F307 | 0.81 | n/a | n/a | n/a |
| 40 | 7513469 | 6491238H1 | SNP00092196 | 103 | 9441 | A | G | A | noncoding | 0.94 | 0.94 | 0.93 | 0.93 |
| 40 | 7513469 | 6547486H2 | SNP00138082 | 15 | 2015 | G | G | A | noncoding | n/a | n/a | n/a | n/a |

# Table 8

| SEQ ID NO: | PID | EST ID | SNP ID | EST SNP | CB1 SNP | EST Allele | Allele 1 | Allele 2 | Amino Acid | Caucasian Allele 1 frequency | African Allele 1 frequency | Asian Allele 1 frequency | Hispanic Allele 1 frequency |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 40 | 7513469 | 6616633H1 | SNP00059686 | 439 | 8272 | G | G | A | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 6754725J1 | SNP00124968 | 61 | 7457 | T | T | C | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 6765492H1 | SNP00118577 | 413 | 9947 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 6765728H1 | SNP00072947 | 317 | 2085 | C | T | C | noncoding | 0.49 | 0.52 | 0.48 | 0.44 |
| 40 | 7513469 | 6765728H1 | SNP00072948 | 465 | 1937 | G | T | G | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 6765728H1 | SNP00092462 | 147 | 2255 | C | C | T | noncoding | 0.45 | 0.23 | 0.53 | 0.45 |
| 40 | 7513469 | 6765728H1 | SNP00092830 | 470 | 1932 | T | T | C | noncoding | 0.84 | n/d | 0.81 | 0.84 |
| 40 | 7513469 | 6767972J1 | SNP00092465 | 197 | 840 | C | C | T | S238 | 0.87 | 0.9 | 0.84 | 0.92 |
| 40 | 7513469 | 6769131H1 | SNP00141288 | 205 | 2754 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 6769131H1 | SNP00141289 | 89 | 2870 | A | G | A | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 6770216J1 | SNP00132541 | 494 | 12315 | T | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 6774178H1 | SNP00092792 | 287 | 1815 | C | C | T | noncoding | 0.24 | 0.17 | 0.23 | 0.25 |
| 40 | 7513469 | 6774178H1 | SNP00141286 | 266 | 1794 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 6774696H1 | SNP00101115 | 265 | 7506 | G | A | G | noncoding | 0.46 | 0.52 | 0.55 | 0.64 |
| 40 | 7513469 | 6774696H1 | SNP00132540 | 106 | 7347 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 6774696J1 | SNP00069033 | 140 | 8029 | C | C | T | noncoding | n/d | n/d | n/d | n/d |
| 40 | 7513469 | 6774956H1 | SNP00046593 | 136 | 2483 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 6774956H1 | SNP00092308 | 208 | 2555 | G | G | A | noncoding | 0.43 | 0.39 | 0.47 | 0.46 |
| 40 | 7513469 | 6775308J1 | SNP00139895 | 255 | 2363 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 6799977J1 | SNP00139897 | 480 | 881 | C | C | T | P252 | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 6800766H1 | SNP00132538 | 149 | 97 | A | C | A | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 6805391H1 | SNP00101109 | 8 | 5193 | T | T | C | noncoding | 0.31 | 0.36 | 0.34 | 0.34 |
| 40 | 7513469 | 6805391H1 | SNP00101110 | 28 | 5213 | T | T | C | noncoding | 0.36 | 0.33 | 0.32 | 0.28 |
| 40 | 7513469 | 6805391H1 | SNP00101111 | 158 | 5342 | C | C | G | noncoding | 0.19 | 0.19 | 0.2 | 0.24 |
| 40 | 7513469 | 6806243H1 | SNP00101116 | 336 | 10441 | T | C | T | noncoding | 0.76 | n/a | n/d | n/a |
| 40 | 7513469 | 6811588J1 | SNP00124916 | 543 | 2959 | T | T | C | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 6811588J1 | SNP00141287 | 284 | 2697 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 6811588J1 | SNP00141289 | 433 | 2849 | G | G | A | noncoding | n/a | n/a | n/a | n/a |

# Table 8

| SEQ ID NO: | PID | EST ID | SNP ID | EST SNP | CB1 SNP | EST Allele | Allele 1 | Allele 2 | Amino Acid | Caucasian Allele 1 frequency | African Allele 1 frequency | Asian Allele 1 frequency | Hispanic Allele 1 frequency |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 40 | 7513469 | 6813021J1 | SNP00124916 | 324 | 2980 | T | T | C | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 6813021J1 | SNP00141287 | 60 | 2718 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 6882944J1 | SNP00101107 | 128 | 4965 | C | T | C | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 6882944J1 | SNP00101108 | 192 | 5029 | T | C | T | noncoding | 0.38 | 0.3 | 0.35 | 0.37 |
| 40 | 7513469 | 6887815H1 | SNP00092236 | 304 | 7272 | G | G | A | noncoding | 0.85 | 0.85 | 0.85 | 0.88 |
| 40 | 7513469 | 6887815H1 | SNP00100130 | 132 | 7100 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 6888523H1 | SNP00135365 | 298 | 7335 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 6889409H1 | SNP00048931 | 324 | 7376 | G | G | C | noncoding | n/d | n/a | n/a | n/a |
| 40 | 7513469 | 6889922H1 | SNP00092644 | 87 | 9412 | A | A | G | noncoding | 0.64 | 0.67 | 0.65 | 0.68 |
| 40 | 7513469 | 6889922J1 | SNP00054887 | 78 | 7671 | G | G | A | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 6889922J1 | SNP00093071 | 96 | 7689 | C | C | T | noncoding | 0.9 | 0.9 | 0.9 | 0.95 |
| 40 | 7513469 | 6891982J1 | SNP00101106 | 406 | 4802 | G | G | A | noncoding | 0.94 | 0.86 | 0.91 | 0.85 |
| 40 | 7513469 | 6892244H1 | SNP00145761 | 289 | 8063 | T | T | C | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 6893323H1 | SNP00101104 | 69 | 1245 | C | C | T | L373 | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 6893855J1 | SNP00092915 | 595 | 2801 | C | C | T | noncoding | 0.8 | 0.84 | 0.84 | 0.81 |
| 40 | 7513469 | 6893855J1 | SNP00135051 | 264 | 3133 | A | G | A | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 7032093H1 | SNP00097063 | 48 | 7720 | G | C | G | noncoding | n/d | n/a | n/a | n/a |
| 40 | 7513469 | 7032093H1 | SNP00097064 | 67 | 7739 | G | C | G | noncoding | n/d | n/a | n/a | n/a |
| 40 | 7513469 | 7120532H1 | SNP00145563 | 38 | 8701 | A | A | G | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 7151922H1 | SNP00111052 | 180 | 6665 | C | T | C | noncoding | 0.15 | 0.34 | 0.4 | 0.25 |
| 40 | 7513469 | 7322604H1 | SNP00037927 | 136 | 7599 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 7322604H1 | SNP00054887 | 235 | 7698 | G | G | A | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 7322604H1 | SNP00093071 | 253 | 7716 | C | C | T | noncoding | 0.9 | 0.9 | 0.9 | 0.95 |
| 40 | 7513469 | 7360622H1 | SNP00114914 | 145 | 3107 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 7361246H1 | SNP00037930 | 101 | 12285 | G | G | C | noncoding | n/d | n/a | n/a | n/a |
| 40 | 7513469 | 7361246H1 | SNP00049782 | 169 | 12353 | T | T | C | noncoding | 0.96 | 0.97 | 0.98 | 0.97 |
| 40 | 7513469 | 7361246H1 | SNP00069036 | 133 | 12317 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 7361246H1 | SNP00101118 | 311 | 12496 | T | T | C | noncoding | n/d | 0.42 | 0.86 | 0.68 |

# Table 8

| SEQ ID NO: | PID | EST ID | SNP ID | EST SNP | CB1 SNP | EST Allele | Allele 1 | Allele 2 | Amino Acid | Caucasian Allele 1 frequency | African Allele 1 frequency | Asian Allele 1 frequency | Hispanic Allele 1 frequency |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 40 | 7513469 | 7361246H1 | SNP00101119 | 319 | 12504 | C | C | T | noncoding | n/d | n/a | n/a | n/a |
| 40 | 7513469 | 7372380H2 | SNP00101110 | 16 | 5214 | T | T | C | noncoding | 0.36 | 0.33 | 0.32 | 0.28 |
| 40 | 7513469 | 7372380H2 | SNP00101111 | 144 | 5343 | C | C | G | noncoding | 0.19 | 0.19 | 0.2 | 0.24 |
| 40 | 7513469 | 7382018H1 | SNP00092357 | 136 | 9722 | T | T | G | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 7382018H1 | SNP00097251 | 130 | 9716 | A | A | G | noncoding | 0.37 | 0.57 | 0.41 | 0.38 |
| 40 | 7513469 | 7382018H1 | SNP00112001 | 28 | 9614 | T | T | C | noncoding | 0.19 | n/a | n/a | n/a |
| 40 | 7513469 | 7384807H1 | SNP00046593 | 162 | 2484 | T | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 7384807H1 | SNP00092308 | 234 | 2556 | G | G | A | noncoding | 0.43 | 0.39 | 0.47 | 0.46 |
| 40 | 7513469 | 7394162H1 | SNP00143083 | 178 | 1901 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 7394552H1 | SNP00143083 | 192 | 1889 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 7433008H1 | SNP00040498 | 275 | 9074 | G | A | G | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 7625022H1 | SNP00105967 | 254 | 9510 | A | A | G | noncoding | 0.13 | 0.31 | 0.23 | 0.12 |
| 40 | 7513469 | 7625675J1 | SNP00037930 | 288 | 12295 | G | G | C | noncoding | n/d | n/a | n/a | n/a |
| 40 | 7513469 | 7625675J1 | SNP00049782 | 356 | 12363 | T | T | C | noncoding | 0.96 | 0.97 | 0.98 | 0.97 |
| 40 | 7513469 | 7625675J1 | SNP00069036 | 320 | 12327 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 7625675J1 | SNP00101118 | 499 | 12506 | T | T | C | noncoding | n/d | 0.42 | 0.86 | 0.68 |
| 40 | 7513469 | 7625675J1 | SNP00101119 | 507 | 12514 | C | C | T | noncoding | n/d | n/a | n/a | n/a |
| 40 | 7513469 | 7638610J1 | SNP00069035 | 230 | 10211 | G | G | A | noncoding | n/d | n/d | n/d | n/d |
| 40 | 7513469 | 7638610J1 | SNP00101116 | 462 | 10442 | C | C | T | noncoding | 0.76 | n/a | n/d | n/a |
| 40 | 7513469 | 7651724J1 | SNP00092236 | 296 | 7267 | A | G | A | noncoding | 0.85 | 0.85 | 0.85 | 0.88 |
| 40 | 7513469 | 7651724J1 | SNP00100130 | 119 | 7095 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 7652417J1 | SNP00093071 | 13 | 7720 | C | C | T | noncoding | 0.9 | 0.9 | 0.9 | 0.95 |
| 40 | 7513469 | 7661943H1 | SNP00133663 | 446 | 4352 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 7687279H1 | SNP00101115 | 96 | 7512 | A | A | G | noncoding | 0.46 | 0.52 | 0.55 | 0.64 |
| 40 | 7513469 | 7687507J1 | SNP00092899 | 14 | 9773 | T | T | C | noncoding | 0.39 | 0.33 | 0.47 | 0.37 |
| 40 | 7513469 | 7689336H1 | SNP00040498 | 41 | 9076 | G | A | G | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 7689336H1 | SNP00114917 | 360 | 9394 | A | A | G | noncoding | n/d | n/d | n/d | n/d |
| 40 | 7513469 | 7718729H1 | SNP00047298 | 58 | 8198 | G | C | G | noncoding | 0.6 | 0.77 | 0.6 | 0.67 |

# Table 8

| SEQ ID NO: | PID | EST ID | SNP ID | EST SNP | CB1 SNP | EST Allele | Allele 1 | Allele 2 | Amino Acid | Caucasian Allele 1 frequency | African Allele 1 frequency | Asian Allele 1 frequency | Hispanic Allele 1 frequency |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 40 | 7513469 | 7730004J1 | SNP00037930 | 95 | 12302 | G | G | C | noncoding | n/d | n/a | n/a | n/a |
| 40 | 7513469 | 7730004J1 | SNP00049782 | 164 | 12370 | T | T | C | noncoding | 0.96 | 0.97 | 0.98 | 0.97 |
| 40 | 7513469 | 7730004J1 | SNP00069036 | 128 | 12335 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 8630504J1 | SNP00092357 | 256 | 9741 | T | T | G | noncoding | n/a | n/a | n/a | n/a |
| 40 | 7513469 | 8630504J1 | SNP00097251 | 250 | 9735 | A | A | G | noncoding | 0.37 | 0.57 | 0.41 | 0.38 |
| 40 | 7513469 | 8630504J1 | SNP00112001 | 148 | 9633 | T | T | C | noncoding | 0.19 | n/a | n/a | n/a |
| 41 | 7518836 | 1356886H1 | SNP00069281 | 160 | 632 | C | C | T | A146 | n/d | n/d | n/d | n/d |
| 41 | 7518836 | 1664687T6 | SNP00069280 | 325 | 432 | C | C | T | F79 | n/a | n/a | n/a | n/a |
| 41 | 7518836 | 1664687T6 | SNP00069281 | 116 | 641 | C | C | T | T149 | n/d | n/d | n/d | n/d |
| 41 | 7518836 | 1872489T6 | SNP00069281 | 125 | 633 | C | C | T | A146 | n/d | n/d | n/d | n/d |
| 41 | 7518836 | 2054236T6 | SNP00069280 | 335 | 423 | C | C | T | L76 | n/a | n/a | n/a | n/a |
| 43 | 7520048 | 7000840F8 | SNP00064275 | 148 | 2486 | G | G | C | noncoding | 0.81 | n/a | n/a | n/a |
| 43 | 7520048 | 7427017F8 | SNP00103835 | 569 | 1736 | T | T | C | noncoding | n/a | n/a | n/a | n/a |
| 44 | 7520157 | 3044873F6 | SNP00044192 | 259 | 1278 | G | G | A | V391 | 0.71 | 0.89 | n/d | 0.7 |
| 45 | 7520485 | 138687F1 | SNP00034927 | 215 | 1827 | C | T | C | noncoding | n/a | n/a | n/a | n/a |
| 45 | 7520485 | 138687T6 | SNP00034927 | 161 | 1824 | C | T | C | noncoding | n/a | n/a | n/a | n/a |
| 46 | 7525723 | 1599649F6 | SNP00020716 | 271 | 974 | G | G | A | E318 | n/a | n/a | n/a | n/a |
| 46 | 7525723 | 1599649F6 | SNP00020717 | 495 | 1198 | C | C | T | S393 | n/a | n/a | n/a | n/a |
| 46 | 7525723 | 2277516T6 | SNP00020717 | 481 | 1199 | C | C | T | S393 | n/a | n/a | n/a | n/a |
| 46 | 7525723 | 7623291J1 | SNP00020716 | 99 | 987 | G | G | A | V323 | n/a | n/a | n/a | n/a |
| 49 | 7520638 | 4224128F6 | SNP00112063 | 360 | 2582 | T | T | G | noncoding | 0.45 | 0.35 | 0.77 | 0.45 |
| 49 | 7520638 | 4710260H1 | SNP00112063 | 200 | 2583 | G | T | G | noncoding | 0.45 | 0.35 | 0.77 | 0.45 |
| 50 | 7518947 | 7203719H1 | SNP00104550 | 64 | 2239 | C | C | T | noncoding | 0.78 | 0.84 | 0.84 | 0.83 |
| 52 | 7520381 | 1428210F6 | SNP00067699 | 189 | 999 | G | G | C | R318 | n/a | n/a | n/a | n/a |
| 52 | 7520381 | 1466614H1 | SNP00067699 | 137 | 996 | G | G | C | S317 | n/a | n/a | n/a | n/a |
| 52 | 7520381 | 1841407R6 | SNP00052496 | 130 | 1616 | G | G | A | noncoding | n/a | n/a | n/a | n/a |
| 52 | 7520381 | 2680928F6 | SNP00019836 | 457 | 1117 | A | C | A | R357 | 0.57 | n/a | n/a | n/a |
| 52 | 7520381 | 2921753H1 | SNP00099984 | 93 | 751 | G | G | A | P235 | n/d | n/a | n/a | n/a |

# Table 8

| SEQ ID NO: | PID | EST ID | SNP ID | EST SNP | CB1 SNP | EST Allele | Allele 1 | Allele 2 | Amino Acid | Caucasian Allele 1 frequency | African Allele 1 frequency | Asian Allele 1 frequency | Hispanic Allele 1 frequency |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 52 | 7520381 | 3700038H1 | SNP00128246 | 44 | 1208 | C | C | T | Q388 | n/a | n/a | n/a | n/a |
| 52 | 7520381 | 5655746H1 | SNP00067697 | 39 | 481 | C | C | T | P145 | n/a | n/a | n/a | n/a |
| 52 | 7520381 | 5655746H1 | SNP00067698 | 89 | 532 | C | C | A | Y162 | n/a | n/a | n/a | n/a |
| 52 | 7520381 | 6827471H1 | SNP00052491 | 198 | 1010 | T | C | T | F322 | n/a | n/a | n/a | n/a |
| 52 | 7520381 | 6827976H1 | SNP00052495 | 444 | 1527 | G | G | A | R494 | n/a | n/a | n/a | n/a |
| 52 | 7520381 | 7637455H1 | SNP00099984 | 104 | 752 | A | G | A | R236 | n/d | n/a | n/a | n/a |
| 52 | 7520381 | 7637455J1 | SNP00067697 | 265 | 452 | C | C | T | H136 | n/a | n/a | n/a | n/a |
| 52 | 7520381 | 7637455J1 | SNP00067698 | 316 | 503 | C | C | A | H153 | n/a | n/a | n/a | n/a |
| 53 | 7520409 | 1428210F6 | SNP00067699 | 189 | 912 | G | G | C | R289 | n/a | n/a | n/a | n/a |
| 53 | 7520409 | 1466614H1 | SNP00067699 | 137 | 909 | G | G | C | S288 | n/a | n/a | n/a | n/a |
| 53 | 7520409 | 1801147H1 | SNP00052494 | 136 | 1352 | C | C | G | P436 | n/a | n/a | n/a | n/a |
| 53 | 7520409 | 1841407R6 | SNP00052496 | 130 | 1681 | G | G | A | Q545 | n/a | n/a | n/a | n/a |
| 53 | 7520409 | 2680928F6 | SNP00019836 | 457 | 1030 | A | C | A | R328 | 0.57 | n/a | n/a | n/a |
| 53 | 7520409 | 2921753H1 | SNP00099984 | 93 | 664 | G | G | A | P206 | n/d | n/a | n/a | n/a |
| 53 | 7520409 | 3512553H1 | SNP00052493 | 68 | 1418 | G | A | G | G458 | n/a | n/a | n/a | n/a |
| 53 | 7520409 | 5655746H1 | SNP00067697 | 39 | 481 | C | C | T | P145 | n/a | n/a | n/a | n/a |
| 53 | 7520409 | 5655746H1 | SNP00067698 | 89 | 532 | C | C | A | Y162 | n/a | n/a | n/a | n/a |
| 53 | 7520409 | 5865257H1 | SNP00128246 | 19 | 1121 | T | C | T | stop359 | n/a | n/a | n/a | n/a |
| 53 | 7520409 | 6497659H1 | SNP00052492 | 86 | 1321 | G | G | A | Q425 | n/a | n/a | n/a | n/a |
| 53 | 7520409 | 6827471H1 | SNP00052491 | 198 | 923 | T | C | T | F293 | n/a | n/a | n/a | n/a |
| 53 | 7520409 | 6827976H1 | SNP00052495 | 444 | 1592 | G | G | A | A516 | n/a | n/a | n/a | n/a |
| 53 | 7520409 | 7637455J1 | SNP00067697 | 265 | 452 | C | C | T | H136 | n/a | n/a | n/a | n/a |
| 53 | 7520409 | 7637455J1 | SNP00067698 | 316 | 503 | C | C | A | H153 | n/a | n/a | n/a | n/a |
| 54 | 7520538 | 7626412H1 | SNP00138319 | 518 | 1175 | T | T | C | L374 | n/a | n/a | n/a | n/a |
| 55 | 7520667 | 1428210F6 | SNP00067699 | 189 | 999 | G | G | C | R318 | n/a | n/a | n/a | n/a |
| 55 | 7520667 | 1466614H1 | SNP00067699 | 137 | 996 | G | G | C | S317 | n/a | n/a | n/a | n/a |
| 55 | 7520667 | 1841407R6 | SNP00052496 | 130 | 1789 | G | G | A | Q581 | n/a | n/a | n/a | n/a |
| 55 | 7520667 | 2680928F6 | SNP00019836 | 457 | 1117 | A | C | A | R357 | 0.57 | n/a | n/a | n/a |

# Table 8

| SEQ ID NO: | PID | EST ID | SNP ID | EST SNP | CB1 SNP | EST Allele | Allele 1 | Allele 2 | Amino Acid | Caucasian Allele 1 frequency | African Allele 1 frequency | Asian Allele 1 frequency | Hispanic Allele 1 frequency |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 55 | 7520667 | 2921753H1 | SNP00099984 | 93 | 751 | G | G | A | P235 | n/d | n/a | n/a | n/a |
| 55 | 7520667 | 3512553H1 | SNP00052493 | 68 | 1526 | G | A | G | G494 | n/a | n/a | n/a | n/a |
| 55 | 7520667 | 5655746H1 | SNP00067697 | 39 | 481 | C | C | T | P145 | n/a | n/a | n/a | n/a |
| 55 | 7520667 | 5655746H1 | SNP00067698 | 89 | 532 | C | C | A | Y162 | n/a | n/a | n/a | n/a |
| 55 | 7520667 | 5865257H1 | SNP00128246 | 19 | 1208 | T | C | T | stop388 | n/a | n/a | n/a | n/a |
| 55 | 7520667 | 6827471H1 | SNP00052491 | 198 | 1010 | T | C | T | F322 | n/a | n/a | n/a | n/a |
| 55 | 7520667 | 6827976H1 | SNP00052495 | 444 | 1700 | G | G | A | A552 | n/a | n/a | n/a | n/a |
| 55 | 7520667 | 7637455H1 | SNP00099984 | 104 | 752 | A | G | A | R236 | n/d | n/a | n/a | n/a |
| 55 | 7520667 | 7637455J1 | SNP00067697 | 265 | 452 | C | C | T | H136 | n/a | n/a | n/a | n/a |
| 55 | 7520667 | 7637455J1 | SNP00067698 | 316 | 503 | C | C | A | H153 | n/a | n/a | n/a | n/a |
| 55 | 7520667 | 8027848J1 | SNP00052492 | 116 | 1408 | G | G | A | Q454 | n/a | n/a | n/a | n/a |
| 56 | 7517567 | 1395701H1 | SNP00046418 | 70 | 3944 | A | A | G | N1297 | n/d | n/d | n/d | n/d |
| 56 | 7517567 | 1404966T6 | SNP00046416 | 248 | 4589 | G | G | A | noncoding | n/a | n/a | n/a | n/a |
| 56 | 7517567 | 1443205H1 | SNP00046417 | 250 | 4368 | C | C | T | noncoding | n/a | n/a | n/a | n/a |
| 56 | 7517567 | 1807119H1 | SNP00046421 | 233 | 2340 | T | C | T | D762 | n/a | n/a | n/a | n/a |
| 56 | 7517567 | 2256848T6 | SNP00046416 | 249 | 4587 | G | G | A | noncoding | n/a | n/a | n/a | n/a |
| 56 | 7517567 | 3595946F7 | SNP00046416 | 595 | 4625 | G | G | A | noncoding | n/a | n/a | n/a | n/a |
| 56 | 7517567 | 4581582H1 | SNP00046420 | 154 | 3141 | G | G | A | P1029 | n/a | n/a | n/a | n/a |
| 56 | 7517567 | 4751423F6 | SNP00069222 | 389 | 3240 | G | A | G | R1062 | n/d | n/a | n/a | n/a |
| 56 | 7517567 | 6472082H1 | SNP00069223 | 330 | 1713 | G | A | G | E553 | n/a | n/a | n/a | n/a |
| 56 | 7517567 | 6472082H1 | SNP00124671 | 108 | 1493 | G | G | A | G480 | 0.18 | 0.48 | 0.49 | 0.42 |
| 56 | 7517567 | 7735363J1 | SNP00046421 | 138 | 2344 | C | C | T | Q764 | n/a | n/a | n/a | n/a |
| 59 | 7520005 | 1456901H1 | SNP00027692 | 199 | 752 | C | C | G | P248 | n/a | n/a | n/a | n/a |
| 59 | 7520005 | 1794980H1 | SNP00052100 | 268 | 992 | G | G | C | G328 | n/d | n/a | n/a | n/a |
| 59 | 7520005 | 3403559H1 | SNP00052099 | 208 | 453 | G | G | A | R148 | n/a | n/a | n/a | n/a |
| 59 | 7520005 | 3403559H1 | SNP00098207 | 90 | 335 | C | G | C | A109 | n/a | n/a | n/a | n/a |
| 59 | 7520005 | 7345849F6 | SNP00052099 | 542 | 454 | G | G | A | V149 | n/a | n/a | n/a | n/a |
| 59 | 7520005 | 7345849F6 | SNP00098207 | 424 | 336 | G | G | C | A109 | n/a | n/a | n/a | n/a |

# Table 8

| SEQ ID NO: | PID | EST ID | SNP ID | EST SNP | CB1 SNP | EST Allele | Allele 1 | Allele 2 | Amino Acid | Caucasian Allele 1 frequency | African Allele 1 frequency | Asian Allele 1 frequency | Hispanic Allele 1 frequency |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 60 | 7520484 | 1456901H1 | SNP00027692 | 199 | 628 | C | C | G | P203 | n/a | n/a | n/a | n/a |
| 60 | 7520484 | 1794980H1 | SNP00052100 | 268 | 868 | G | G | C | G283 | n/d | n/a | n/a | n/a |
| 60 | 7520484 | 3403559H1 | SNP00052099 | 208 | 380 | G | G | A | R120 | n/a | n/a | n/a | n/a |
| 60 | 7520484 | 3752762F6 | SNP00027692 | 464 | 629 | C | C | G | R203 | n/a | n/a | n/a | n/a |
| 60 | 7520484 | 3752762F6 | SNP00052099 | 216 | 381 | G | G | A | V121 | n/a | n/a | n/a | n/a |

What is claimed is:

1. An isolated polypeptide selected from the group consisting of:

a) a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-30,

b) a polypeptide comprising a naturally occurring amino acid sequence at least 90% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:2-3, SEQ ID NO:6, SEQ ID NO:11, SEQ ID NO:13, SEQ ID NO:17, SEQ ID NO:20, SEQ ID NO:22, and SEQ ID NO:24,

c) a polypeptide comprising a naturally occurring amino acid sequence at least 96% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:1 and SEQ ID 28-29,

d) a polypeptide comprising a naturally occurring amino acid sequence at least 91% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:4, SEQ ID NO:7, and SEQ ID NO:9,

e) a polypeptide comprising a naturally occurring amino acid sequence at least 92% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:5, SEQ ID NO:10, and SEQ ID NO:19,

f) a polypeptide comprising a naturally occurring amino acid sequence at least 98% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:18, SEQ ID NO:21, and SEQ ID NO:26,

g) a polypeptide consisting essentially of a naturally occurring amino acid sequence at least 90% identical to the amino acid sequence selected from the group consisting of SEQ ID NO:8, SEQ ID NO:12, SEQ ID NO:14, SEQ ID NO:16, SEQ ID NO:23, SEQ ID NO:25, SEQ ID NO:27, and SEQ ID NO:30,

h) a biologically active fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, and

i) an immunogenic fragment of a polypeptide having an amino acid sequence selected from the group consisting of SEQ ID NO:1-30.

2. An isolated polypeptide of claim 1 comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-30.

3. An isolated polynucleotide encoding a polypeptide of claim 1.

4. An isolated polynucleotide encoding a polypeptide of claim 2.

5. An isolated polynucleotide of claim 4 comprising a polynucleotide sequence selected from the group consisting of SEQ ID NO:31-60.

6. A recombinant polynucleotide comprising a promoter sequence operably linked to a polynucleotide of claim 3.

7. A cell transformed with a recombinant polynucleotide of claim 6.

8. A transgenic organism comprising a recombinant polynucleotide of claim 6.

9. A method of producing a polypeptide of claim 1, the method comprising:

a) culturing a cell under conditions suitable for expression of the polypeptide, wherein said cell is transformed with a recombinant polynucleotide, and said recombinant polynucleotide comprises a promoter sequence operably linked to a polynucleotide encoding the polypeptide of claim 1, and

b) recovering the polypeptide so expressed.

10. A method of claim 9, wherein the polypeptide comprises an amino acid sequence selected from the group consisting of SEQ ID NO:1-30.

11. An isolated antibody which specifically binds to a polypeptide of claim 1.

12. An isolated polynucleotide selected from the group consisting of:

a) a polynucleotide comprising a polynucleotide sequence selected from the group consisting of SEQ ID NO:31-60,

b) a polynucleotide comprising a naturally occurring polynucleotide sequence at least 90% identical to a polynucleotide sequence selected from the group consisting of SEQ ID NO:34, SEQ ID NO:36, SEQ ID NO:41, SEQ ID NO:49, and SEQ ID NO:58,

c) a polynucleotide comprising a naturally occurring polynucleotide sequence at least 99% identical to a polynucleotide sequence selected from the group consisting of SEQ ID NO:37 and SEQ ID NO:44,

d) a polynucleotide comprising a naturally occurring polynucleotide sequence at least

93% identical to a polynucleotide sequence selected from the group consisting of SEQ ID NO:42 and SEQ ID NO:57,

e) a polynucleotide comprising a naturally occurring polynucleotide sequence at least 96% identical to a polynucleotide sequence selected from the group consisting of SEQ ID NO:43 and SEQ ID NO:59,

f) a polynucleotide comprising a naturally occurring polynucleotide sequence at least 91% identical to the polynucleotide sequence of SEQ ID NO:45,

g) a polynucleotide comprising a naturally occurring polynucleotide sequence at least 97% identical to a polynucleotide sequence of SEQ ID NO:54 and SEQ ID NO:56,

h) a polynucleotide consisting essentially of a naturally occurring polynucleotide sequence at least 90% identical to a polynucleotide sequence selected from the group consisting of SEQ ID NO:31-33, SEQ ID NO:35, SEQ ID NO:38-40, SEQ ID NO:46-48, SEQ ID NO:50-53, SEQ ID NO:55, and SEQ ID NO:60,

i) a polynucleotide complementary to a polynucleotide of a),

j) a polynucleotide complementary to a polynucleotide of b),

k) a polynucleotide complementary to a polynucleotide of c),

l) a polynucleotide complementary to a polynucleotide of d),

m) a polynucleotide complementary to a polynucleotide of e),

n) a polynucleotide complementary to a polynucleotide of f),

o) a polynucleotide complementary to a polynucleotide of g),

p) a polynucleotide complementary to a polynucleotide of h), and

q) an RNA equivalent of a)-p).

13. An isolated polynucleotide comprising at least 60 contiguous nucleotides of a polynucleotide of claim 12.

14. A method of detecting a target polynucleotide in a sample, said target polynucleotide having a sequence of a polynucleotide of claim 12, the method comprising:

a) hybridizing the sample with a probe comprising at least 20 contiguous nucleotides comprising a sequence complementary to said target polynucleotide in the sample, and which probe specifically hybridizes to said target polynucleotide, under conditions whereby a hybridization complex is formed between said probe and said target polynucleotide or fragments thereof, and

b) detecting the presence or absence of said hybridization complex, and, optionally, if present, the amount thereof.

15. A method of claim 14, wherein the probe comprises at least 60 contiguous nucleotides.

16. A method of detecting a target polynucleotide in a sample, said target polynucleotide having a sequence of a polynucleotide of claim 12, the method comprising:

a) amplifying said target polynucleotide or fragment thereof using polymerase chain reaction amplification, and

b) detecting the presence or absence of said amplified target polynucleotide or fragment thereof, and, optionally, if present, the amount thereof.

17. A composition comprising a polypeptide of claim 1 and a pharmaceutically acceptable excipient.

18. A composition of claim 17, wherein the polypeptide comprises an amino acid sequence selected from the group consisting of SEQ ID NO:1-30.

19. A method for treating a disease or condition associated with decreased expression of functional CADECM, comprising administering to a patient in need of such treatment the composition of claim 17.

20. A method of screening a compound for effectiveness as an agonist of a polypeptide of claim 1, the method comprising:

a) contacting a sample comprising a polypeptide of claim 1 with a compound, and

b) detecting agonist activity in the sample.

21. A composition comprising an agonist compound identified by a method of claim 20 and a pharmaceutically acceptable excipient.

22. A method for treating a disease or condition associated with decreased expression of functional CADECM, comprising administering to a patient in need of such treatment a composition of claim 21.

23. A method of screening a compound for effectiveness as an antagonist of a polypeptide of claim 1, the method comprising:

a) contacting a sample comprising a polypeptide of claim 1 with a compound, and

b) detecting antagonist activity in the sample.

24. A composition comprising an antagonist compound identified by a method of claim 23 and a pharmaceutically acceptable excipient.

25. A method for treating a disease or condition associated with overexpression of functional CADECM, comprising administering to a patient in need of such treatment a composition of claim 24.

26. A method of screening for a compound that specifically binds to the polypeptide of claim 1, the method comprising:

a) combining the polypeptide of claim 1 with at least one test compound under suitable conditions, and

b) detecting binding of the polypeptide of claim 1 to the test compound, thereby identifying a compound that specifically binds to the polypeptide of claim 1.

27. A method of screening for a compound that modulates the activity of the polypeptide of claim 1, the method comprising:

a) combining the polypeptide of claim 1 with at least one test compound under conditions permissive for the activity of the polypeptide of claim 1,

b) assessing the activity of the polypeptide of claim 1 in the presence of the test compound, and

c) comparing the activity of the polypeptide of claim 1 in the presence of the test compound with the activity of the polypeptide of claim 1 in the absence of the test compound, wherein a change in the activity of the polypeptide of claim 1 in the presence of the test compound is indicative of a compound that modulates the activity of the polypeptide of claim 1.

28. A method of screening a compound for effectiveness in altering expression of a target polynucleotide, wherein said target polynucleotide comprises a sequence of claim 5, the method comprising:

a) contacting a sample comprising the target polynucleotide with a compound, under conditions suitable for the expression of the target polynucleotide,

b) detecting altered expression of the target polynucleotide, and

c) comparing the expression of the target polynucleotide in the presence of varying amounts of the compound and in the absence of the compound.

29. A method of screening for potential toxicity of a test compound, the method comprising:

a) treating a biological sample containing nucleic acids with the test compound,

b) hybridizing the nucleic acids of the treated biological sample with a probe comprising at least 20 contiguous nucleotides of a polynucleotide of claim 12 under conditions whereby a specific hybridization complex is formed between said probe and a target polynucleotide in the biological sample, said target polynucleotide comprising a polynucleotide sequence of a polynucleotide of claim 12 or fragment thereof,

c) quantifying the amount of hybridization complex, and

d) comparing the amount of hybridization complex in the treated biological sample with the amount of hybridization complex in an untreated biological sample, wherein a difference in the amount of hybridization complex in the treated biological sample indicates potential toxicity of the test compound.

30. A method for a diagnostic test for a condition or disease associated with the expression of CADECM in a biological sample, the method comprising:

a) combining the biological sample with an antibody of claim 11, under conditions suitable for the antibody to bind the polypeptide and form an antibody:polypeptide complex, and

b) detecting the complex, wherein the presence of the complex correlates with the presence of the polypeptide in the biological sample.

31. The antibody of claim 11, wherein the antibody is:

a) a chimeric antibody,

b) a single chain antibody,

c) a Fab fragment,

d) a $F(ab')_2$ fragment, or

e) a humanized antibody.

32. A composition comprising an antibody of claim 11 and an acceptable excipient.

33. A method of diagnosing a condition or disease associated with the expression of CADECM in a subject, comprising administering to said subject an effective amount of the composition of claim 32.

34. A composition of claim 32, further comprising a label.

35. A method of diagnosing a condition or disease associated with the expression of CADECM in a subject, comprising administering to said subject an effective amount of the composition of claim 34.

36. A method of preparing a polyclonal antibody with the specificity of the antibody of claim 11, the method comprising:

a) immunizing an animal with a polypeptide consisting of an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, or an immunogenic fragment thereof, under conditions to elicit an antibody response,

b) isolating antibodies from the animal, and

c) screening the isolated antibodies with the polypeptide, thereby identifying a polyclonal antibody which specifically binds to a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-30.

37. A polyclonal antibody produced by a method of claim 36.

38. A composition comprising the polyclonal antibody of claim 37 and a suitable carrier.

39. A method of making a monoclonal antibody with the specificity of the antibody of claim 11, the method comprising:

a) immunizing an animal with a polypeptide consisting of an amino acid sequence selected from the group consisting of SEQ ID NO:1-30, or an immunogenic fragment thereof, under conditions to elicit an antibody response,

b) isolating antibody producing cells from the animal,

c) fusing the antibody producing cells with immortalized cells to form monoclonal antibody-producing hybridoma cells,

d) culturing the hybridoma cells, and

e) isolating from the culture monoclonal antibody which specifically binds to a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-30.

40. A monoclonal antibody produced by a method of claim 39.

41. A composition comprising the monoclonal antibody of claim 40 and a suitable carrier.

42. The antibody of claim 11, wherein the antibody is produced by screening a Fab

expression library.

43. The antibody of claim 11, wherein the antibody is produced by screening a recombinant immunoglobulin library.

44. A method of detecting a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-30 in a sample, the method comprising:

a) incubating the antibody of claim 11 with the sample under conditions to allow specific binding of the antibody and the polypeptide, and
b) detecting specific binding, wherein specific binding indicates the presence of a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-30 in the sample.

45. A method of purifying a polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-30 from a sample, the method comprising:

a) incubating the antibody of claim 11 with the sample under conditions to allow specific binding of the antibody and the polypeptide, and
b) separating the antibody from the sample and obtaining the purified polypeptide comprising an amino acid sequence selected from the group consisting of SEQ ID NO:1-30.

46. A microarray wherein at least one element of the microarray is a polynucleotide of claim 13.

47. A method of generating an expression profile of a sample which contains polynucleotides, the method comprising:

a) labeling the polynucleotides of the sample,
b) contacting the elements of the microarray of claim 46 with the labeled polynucleotides of the sample under conditions suitable for the formation of a hybridization complex, and
c) quantifying the expression of the polynucleotides in the sample.

48. An array comprising different nucleotide molecules affixed in distinct physical locations on a solid substrate, wherein at least one of said nucleotide molecules comprises a first oligonucleotide or polynucleotide sequence specifically hybridizable with at least 30 contiguous nucleotides of a target polynucleotide, and wherein said target polynucleotide is a polynucleotide of

claim 12.

49. An array of claim 48, wherein said first oligonucleotide or polynucleotide sequence is completely complementary to at least 30 contiguous nucleotides of said target polynucleotide.

50. An array of claim 48, wherein said first oligonucleotide or polynucleotide sequence is completely complementary to at least 60 contiguous nucleotides of said target polynucleotide.

51. An array of claim 48, wherein said first oligonucleotide or polynucleotide sequence is completely complementary to said target polynucleotide.

52. An array of claim 48, which is a microarray.

53. An array of claim 48, further comprising said target polynucleotide hybridized to a nucleotide molecule comprising said first oligonucleotide or polynucleotide sequence.

54. An array of claim 48, wherein a linker joins at least one of said nucleotide molecules to said solid substrate.

55. An array of claim 48, wherein each distinct physical location on the substrate contains multiple nucleotide molecules, and the multiple nucleotide molecules at any single distinct physical location have the same sequence, and each distinct physical location on the substrate contains nucleotide molecules having a sequence which differs from the sequence of nucleotide molecules at another distinct physical location on the substrate.

56. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:1.

57. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:2.

58. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:3.

59. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:4.

60. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:5.

61. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:6.

62. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:7.

63. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:8.

64. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:9.

65. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:10.

66. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:11.

67. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:12.

68. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:13.

69. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:14.

70. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:15.

71. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:16.

72. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:17.

73. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:18.

74. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:19.

75. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:20.

76. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:21.

77. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:22.

78. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:23.

79. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:24.

80. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:25.

81. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:26.

82. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:27.

83. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:28.

84. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:29.

85. A polypeptide of claim 1, comprising the amino acid sequence of SEQ ID NO:30.

86. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:31.

87. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:32.

88. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:33.

89. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:34.

90. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:35.

91. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:36.

92. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:37.

93. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:38.

94. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:39.

95. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:40.

96. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:41.

97. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:42.

98. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:43.

99. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:44.

100. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:45.

101. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:46.

102. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:47.

103. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:48.

104. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:49.

105. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:50.

106. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:51.

107. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:52.

108. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:53.

109. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:54.

110. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:55.

111. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:56.

112. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:57.

113. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:58.

114. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:59.

115. A polynucleotide of claim 12, comprising the polynucleotide sequence of SEQ ID NO:60.

```
<110> INCYTE CORPORATION; SWARNAKAR, Anita;
      BECHA, Shanya D.; TRAN, Uyen K.;
      MARQUIS, Joseph P.; ELLIOTT, Vicki S.;
      RICHARDSON, Thomas W.; CHAWLA, Narinder K.;
      WANG, Jonathan T.; FAVERO, Kristin D.;
      GERA, Mili; JIANG, Xin;
      JACKSON, Alan A.

<120> CELL ADHESION AND EXTRACELLULAR MATRIX PROTEINS

<130> PF-1535 PCT

<140> To Be Assigned
<141> Herewith

<150> US 60/460,788
<151> 2003-04-04

<150> US 60/470,697
<151> 2003-05-14

<150> US 60/474,144
<151> 2003-05-28

<150> US 60/483,286
<151> 2003-06-27

<150> US 60/493,045
<151> 2003-08-05

<160> 60
<170> PERL Program

<210> 1
<211> 174
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7514594CD1

<400> 1
Met Asn Ser Ser Lys Ser Ser Glu Thr Gln Cys Thr Glu Arg Gly
  1               5                  10                  15
Cys Phe Ser Ser Gln Met Phe Leu Trp Thr Val Ala Gly Ile Pro
                 20                  25                  30
Ile Leu Phe Leu Ser Ala Cys Phe Ile Thr Arg Cys Val Val Thr
                 35                  40                  45
Phe Arg Ile Phe Gln Thr Cys Asp Glu Lys Lys Phe Gln Leu Pro
                 50                  55                  60
Glu Asn Phe Thr Glu Leu Ser Cys Tyr Asn Tyr Gly Ser Gly Ile
                 65                  70                  75
Ile Val Pro Lys Glu Phe Leu Ser Tyr Lys Lys Pro Lys Met Arg
                 80                  85                  90
Glu Val Phe Ile Gly Leu Ser Asp Gln Val Val Glu Gly Arg Trp
                 95                 100                 105
Gln Trp Val Asp Gly Thr Pro Leu Thr Lys Ser Leu Ser Phe Trp
                110                 115                 120
Asp Val Gly Glu Pro Asn Asn Ile Ala Thr Leu Glu Asp Cys Ala
                125                 130                 135
Thr Met Arg Asp Ser Ser Asn Pro Arg Gln Asn Trp Asn Asp Val
                140                 145                 150
```

```
Thr Cys Phe Leu Asn Tyr Phe Arg Ile Cys Glu Met Val Gly Ile
                155                 160                 165
Asn Pro Leu Asn Lys Gly Lys Ser Leu
                170

<210> 2
<211> 323
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7515764CD1

<400> 2
Met Val Pro Pro Pro Pro Ser Arg Gly Gly Ala Ala Arg Gly Gln
  1             5                   10                  15
Leu Gly Arg Ser Leu Gly Pro Leu Leu Leu Leu Leu Ala Leu Gly
                20                  25                  30
His Thr Trp Thr Tyr Arg Glu Glu Pro Glu Asp Gly Asp Arg Glu
                35                  40                  45
Ile Cys Ser Glu Ser Lys Ile Ala Thr Thr Lys Tyr Pro Cys Leu
                50                  55                  60
Lys Ser Ser Gly Glu Leu Thr Thr Cys Tyr Arg Lys Lys Cys Cys
                65                  70                  75
Lys Gly Tyr Lys Phe Val Leu Gly Gln Cys Ile Pro Glu Asp Tyr
                80                  85                  90
Asp Val Cys Ala Glu Ala Pro Cys Glu Gln Gln Cys Thr Asp Asn
                95                  100                 105
Phe Gly Arg Val Leu Cys Thr Cys Tyr Pro Gly Tyr Arg Tyr Asp
                110                 115                 120
Arg Glu Arg His Arg Lys Arg Glu Lys Pro Tyr Cys Leu Asp Ile
                125                 130                 135
Asp Glu Cys Ala Ser Ser Asn Gly Thr Leu Cys Ala His Ile Cys
                140                 145                 150
Ile Asn Thr Leu Gly Ser Tyr Arg Cys Glu Cys Arg Glu Gly Tyr
                155                 160                 165
Ile Arg Glu Asp Asp Gly Lys Thr Cys Thr Arg Gly Asp Lys Tyr
                170                 175                 180
Pro Asn Asp Thr Gly Ser Pro Gly Pro Lys Gly Ser Pro Gly Phe
                185                 190                 195
Pro Gly Met Pro Gly Pro Pro Gly Gln Pro Gly Pro Arg Gly Ser
                200                 205                 210
Met Gly Pro Met Gly Pro Ser Pro Asp Leu Ser His Ile Lys Gln
                215                 220                 225
Gly Arg Arg Gly Pro Val Gly Pro Pro Gly Ala Pro Gly Arg Asp
                230                 235                 240
Gly Ser Lys Gly Glu Arg Gly Ala Pro Gly Pro Arg Gly Ser Pro
                245                 250                 255
Gly Pro Leu Val Leu Ser Thr Ser Cys Tyr Leu Cys Trp Leu Thr
                260                 265                 270
Ser Ala Met Thr Ser Leu Ser Cys Arg Lys Arg Cys Ser Gly Thr
                275                 280                 285
Gly Leu Thr Leu Gln Gln Arg Ser Ser Leu Tyr Leu Arg Asn Phe
                290                 295                 300
Pro Ala Thr Gln Lys Pro Trp Thr Trp Ala Leu Glu Met Thr Ile
                305                 310                 315
Gln Glu Glu Leu Arg Gln Glu Thr
                320

<210> 3
<211> 97
<212> PRT
```

<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7514575CD1

<400> 3
```
Met Ala Pro Leu Leu Pro Ile Arg Thr Leu Pro Leu Ile Leu Ile
 1               5                  10                  15
Leu Leu Ala Leu Leu Ser Pro Gly Ala Ala Asp Phe Asn Ile Ser
                20                  25                  30
Ser Leu Ser Gly Leu Leu Ser Pro Ala Leu Thr Glu Ser Leu Leu
                35                  40                  45
Val Ala Leu Pro Pro Cys His Leu Thr Gly Gly Asn Ala Thr Leu
                50                  55                  60
Met Val Arg Arg Ala Asn Asp Ser Lys Val Val Thr Ser Ser Phe
                65                  70                  75
Val Val Pro Pro Cys Arg Gly Arg Arg Glu Leu His Phe Leu Pro
                80                  85                  90
Ser Glu Glu Gly Asp Ser His
                95
```

<210> 4
<211> 116
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7514576CD1

<400> 4
```
Met Ala Pro Leu Leu Pro Ile Arg Thr Leu Pro Leu Ile Leu Ile
 1               5                  10                  15
Leu Leu Ala Leu Leu Ser Pro Gly Ala Ala Asp Phe Asn Ile Ser
                20                  25                  30
Ser Leu Ser Gly Leu Leu Ser Pro Ala Leu Thr Glu Ser Leu Leu
                35                  40                  45
Val Ala Leu Pro Pro Cys His Leu Thr Gly Gly Asn Ala Thr Leu
                50                  55                  60
Met Val Arg Arg Ala Asn Asp Ser Lys Val Val Thr Ser Ser Phe
                65                  70                  75
Val Val Pro Pro Cys Arg Gly Arg Arg Glu Leu Val Ser Val Val
                80                  85                  90
Asp Ser Gly Ala Gly Phe Thr Val Thr Arg Leu Ser Ala Tyr Gln
                95                 100                 105
His Phe Leu Pro Ser Glu Glu Gly Asp Ser His
               110                 115
```

<210> 5
<211> 81
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7515593CD1

<400> 5
```
Met Asn Ser Ser Lys Ser Ser Glu Thr Gln Cys Thr Glu Arg Gly
 1               5                  10                  15
Cys Phe Ser Ser Gln Met Phe Leu Trp Thr Val Ala Gly Ile Pro
                20                  25                  30
```

Ile Leu Phe Leu Ser Ala Cys Phe Ile Thr Arg Cys Val Val Thr
35 40 45
Phe Arg Ile Phe Gln Thr Cys Asp Glu Lys Lys Phe Gln Leu Pro
50 55 60
Glu Asn Phe Thr Glu Leu Ser Cys Tyr Asn Tyr Gly Ser Gly Ile
65 70 75
Pro Phe Leu Gln Glu Thr
80

<210> 6
<211> 196
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7517754CD1

<400> 6
Met Leu Pro Cys Ala Ser Cys Leu Pro Gly Ser Leu Leu Leu Trp
1 5 10 15
Ala Leu Leu Leu Leu Leu Leu Gly Ser Ala Ser Pro Gln Asp Ser
20 25 30
Glu Glu Pro Asp Ser Tyr Thr Glu Cys Thr Asp Gly Tyr Glu Trp
35 40 45
Asp Pro Asp Ser Gln His Cys Arg Asp Val Asn Glu Cys Leu Thr
50 55 60
Ile Pro Glu Ala Cys Lys Gly Glu Met Lys Cys Ile Asn His Tyr
65 70 75
Gly Gly Tyr Leu Cys Leu Pro Arg Ser Ala Ala Val Ile Asn Asp
80 85 90
Leu His Gly Glu Gly Pro Pro Pro Pro Val Pro Pro Ala Gln His
95 100 105
Pro Asn Pro Cys Pro Pro Gly Tyr Glu Pro Asp Asp Gln Asp Ser
110 115 120
Cys Val Gly Pro Ala Ala Gly Gly Pro Ser Leu Cys Ala Thr Ala
125 130 135
Pro Ala Ser Ser Gly Pro Leu Pro Ser Val Ala Ala Glu Thr Leu
140 145 150
Gln Glu Val Cys Leu Pro Val Ser Val Pro Trp Glu Pro Gln Ala
155 160 165
Leu Glu Glu Gly Gln Ser Arg Asn Gln Cys Cys Trp Gly Cys Leu
170 175 180
Asp Arg Thr Leu Phe Pro Val Gln Pro Pro Thr Cys Gly Phe Arg
185 190 195
Gly

<210> 7
<211> 544
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7517819CD1

<400> 7
Met Gly Pro Gly Pro Ser Arg Ala Pro Arg Ala Pro Arg Leu Met
1 5 10 15
Leu Cys Ala Leu Ala Leu Met Val Ala Ala Gly Gly Cys Val Val
20 25 30
Ser Ala Phe Asn Leu Asp Thr Arg Phe Leu Val Val Lys Glu Ala

```
                35                       40                       45
Gly Asn Pro Gly Ser Leu Phe Gly Tyr Ser Val Ala Leu His Arg
                50                       55                       60
Gln Thr Glu Arg Gln Gln Arg Tyr Leu Leu Leu Ala Gly Ala Pro
                65                       70                       75
Arg Glu Leu Ala Val Pro Asp Gly Tyr Thr Asn Arg Thr Gly Ala
                80                       85                       90
Val Tyr Leu Cys Pro Leu Thr Ala His Lys Asp Asp Cys Glu Arg
                95                       100                      105
Met Asn Ile Thr Val Lys Asn Asp Pro Gly His His Ile Ile Glu
                110                      115                      120
Asp Met Trp Leu Gly Val Thr Val Ala Ser Gln Gly Pro Ala Gly
                125                      130                      135
Arg Val Leu Val Cys Ala His Arg Tyr Thr Gln Val Leu Trp Ser
                140                      145                      150
Gly Ser Glu Asp Gln Arg Arg Met Val Gly Lys Cys Tyr Val Arg
                155                      160                      165
Gly Asn Asp Leu Glu Leu Asp Ser Ser Asp Asp Trp Gln Thr Tyr
                170                      175                      180
His Asn Glu Met Cys Asn Ser Asn Thr Asp Tyr Leu Glu Thr Gly
                185                      190                      195
Met Cys Gln Leu Gly Thr Ser Gly Gly Phe Thr Gln Asn Thr Val
                200                      205                      210
Tyr Phe Gly Ala Pro Gly Ala Tyr Asn Trp Lys Gly Asn Ser Tyr
                215                      220                      225
Met Ile Gln Arg Lys Glu Trp Asp Leu Ser Glu Tyr Ser Tyr Lys
                230                      235                      240
Asp Pro Glu Asp Gln Gly Asn Leu Tyr Ile Gly Tyr Thr Met Gln
                245                      250                      255
Val Gly Ser Phe Ile Leu His Pro Lys Asn Ile Thr Ile Val Thr
                260                      265                      270
Gly Ala Pro Arg His Arg His Met Gly Ala Val Phe Leu Leu Ser
                275                      280                      285
Gln Glu Ala Gly Gly Asp Leu Arg Arg Arg Gln Val Leu Glu Gly
                290                      295                      300
Ser Gln Val Gly Ala Tyr Phe Gly Ser Ala Ile Ala Leu Ala Asp
                305                      310                      315
Leu Asn Asn Asp Gly Trp Gln Asp Leu Leu Val Gly Ala Pro Tyr
                320                      325                      330
Tyr Phe Glu Arg Lys Glu Glu Val Gly Gly Ala Ile Tyr Val Phe
                335                      340                      345
Met Asn Gln Ala Gly Thr Ser Phe Pro Ala His Pro Ser Leu Leu
                350                      355                      360
Leu His Gly Pro Ser Gly Ser Ala Phe Gly Leu Ser Val Ala Ser
                365                      370                      375
Ile Gly Asp Ile Asn Gln Asp Gly Phe Gln Asp Ile Ala Val Gly
                380                      385                      390
Ala Pro Phe Glu Gly Leu Gly Lys Val Tyr Ile Tyr His Ser Ser
                395                      400                      405
Ser Lys Gly Leu Leu Arg Gln Pro Gln Gln Val Ile His Gly Glu
                410                      415                      420
Lys Leu Gly Leu Pro Gly Leu Ala Thr Phe Gly Tyr Ser Leu Ser
                425                      430                      435
Gly Gln Met Asp Val Asp Glu Asn Phe Tyr Pro Asp Leu Leu Val
                440                      445                      450
Gly Ser Leu Ser Asp His Ile Val Leu Leu Arg Ala Arg Pro Val
                455                      460                      465
Ile Asn Ile Val His Lys Thr Leu Val Pro Arg Pro Ala Val Leu
                470                      475                      480
Asp Pro Ala Leu Cys Thr Ala Thr Ser Cys Thr Glu Ala Arg Ala
                485                      490                      495
Arg Ala Trp Thr Cys Val Ser His Cys Asp Pro Pro Ser Cys Thr
                500                      505                      510
```

Ser Leu Gln Gln Cys Ala Ser Gly Ala Val Leu Cys Leu Gln Pro
515 520 525
Glu Cys Arg Glu Pro Gln Leu Gln Ala Lys His His Pro Gly Leu
530 535 540
His Ser Gly Gly

<210> 8
<211> 363
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7518460CD1

<400> 8
Met Ala Pro Leu Arg Pro Leu Leu Ile Leu Ala Leu Leu Ala Trp
1 5 10 15
Val Ala Leu Ala Asp Gln Glu Ser Cys Lys Gly Arg Cys Thr Glu
20 25 30
Gly Phe Asn Val Asp Lys Lys Cys Gln Cys Asp Glu Leu Cys Ser
35 40 45
Tyr Tyr Gln Ser Cys Cys Thr Asp Tyr Thr Ala Glu Cys Lys Pro
50 55 60
Gln Val Thr Arg Gly Asp Val Phe Thr Met Pro Glu Asp Glu Tyr
65 70 75
Thr Val Tyr Asp Asp Gly Glu Glu Lys Asn Asn Ala Thr Val His
80 85 90
Glu Gln Val Gly Gly Pro Ser Leu Thr Ser Asp Leu Gln Ala Gln
95 100 105
Ser Lys Gly Asn Pro Glu Gln Thr Pro Val Leu Lys Pro Glu Glu
110 115 120
Glu Ala Pro Ala Pro Glu Val Gly Ala Ser Lys Pro Glu Gly Ile
125 130 135
Asp Ser Arg Pro Glu Thr Leu His Pro Gly Arg Pro Gln Pro Pro
140 145 150
Ala Glu Glu Glu Leu Cys Ser Gly Lys Pro Phe Asp Ala Phe Thr
155 160 165
Asp Leu Lys Asn Gly Ser Leu Phe Ala Phe Arg Gly Gln Tyr Cys
170 175 180
Tyr Glu Leu Asp Glu Lys Ala Val Arg Pro Gly Tyr Pro Lys Leu
185 190 195
Ile Arg Asp Val Trp Gly Ile Glu Gly Pro Ile Asp Ala Ala Phe
200 205 210
Thr Arg Ile Asn Cys Gln Gly Lys Thr Tyr Leu Phe Lys Gly Ser
215 220 225
Gln Tyr Trp Arg Phe Glu Asp Gly Val Leu Asp Pro Asp Tyr Pro
230 235 240
Arg Asn Ile Ser Asp Gly Phe Asp Gly Ile Pro Asp Asn Val Asp
245 250 255
Ala Ala Leu Ala Leu Pro Ala His Ser Tyr Ser Gly Arg Glu Arg
260 265 270
Val Tyr Phe Phe Lys Gly Lys Gln Tyr Trp Glu Tyr Gln Phe Gln
275 280 285
His Gln Pro Ser Gln Glu Glu Cys Glu Gly Ser Ser Leu Ser Ala
290 295 300
Val Phe Glu His Phe Ala Met Met Gln Arg Asp Ser Trp Glu Asp
305 310 315
Ile Phe Glu Leu Leu Phe Trp Gly Arg Thr Ser Asp Lys Tyr Tyr
320 325 330
Arg Val Asn Leu Arg Thr Arg Arg Val Asp Thr Val Asp Pro Pro
335 340 345

```
Tyr Pro Arg Ser Ile Ala Gln Tyr Trp Leu Gly Cys Pro Ala Pro
                350                 355                 360
Gly His Leu


<210> 9
<211> 265
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7518734CD1

<400> 9
Met Ala Leu Arg Arg Pro Pro Arg Leu Arg Leu Cys Ala Arg Leu
  1               5                  10                  15
Pro Asp Phe Phe Leu Leu Leu Leu Phe Arg Gly Cys Leu Ile Gly
                 20                  25                  30
Ala Val Asn Leu Lys Ser Ser Asn Arg Thr Pro Val Val Gln Glu
                 35                  40                  45
Phe Glu Ser Val Glu Leu Ser Cys Ile Ile Thr Asp Ser Gln Thr
                 50                  55                  60
Ser Asp Pro Arg Ile Glu Trp Lys Lys Ile Gln Asp Glu Gln Thr
                 65                  70                  75
Thr Tyr Val Phe Phe Asp Asn Lys Ile Gln Gly Asp Leu Ala Gly
                 80                  85                  90
Arg Ala Glu Ile Leu Gly Lys Thr Ser Leu Lys Ile Trp Asn Val
                 95                 100                 105
Thr Arg Arg Asp Ser Ala Leu Tyr Arg Cys Glu Val Val Ala Arg
                110                 115                 120
Asn Asp Arg Lys Glu Ile Asp Glu Ile Val Ile Glu Leu Thr Val
                125                 130                 135
Gln Val Lys Pro Val Thr Pro Val Cys Arg Val Pro Lys Ala Val
                140                 145                 150
Pro Val Gly Lys Met Ala Thr Leu His Cys Gln Glu Ser Glu Gly
                155                 160                 165
His Pro Arg Pro His Tyr Ser Trp Tyr Arg Asn Asp Val Pro Leu
                170                 175                 180
Pro Thr Asp Ser Arg Ala Asn Pro Arg Phe Arg Asn Ser Ser Phe
                185                 190                 195
His Leu Asn Ser Glu Thr Gly Thr Leu Val Phe Thr Ala Val His
                200                 205                 210
Lys Asp Asp Ser Gly Gln Tyr Tyr Cys Ile Ala Ser Asn Asp Ala
                215                 220                 225
Gly Ser Ala Arg Cys Glu Glu Gln Glu Met Glu Val Phe Thr Arg
                230                 235                 240
Thr Gln Gly Asn Gln Met Glu Leu Thr Thr Ser Ala Leu Thr Arg
                245                 250                 255
Arg Ala Thr Ser Asp Thr Ser His Arg Leu
                260                 265

<210> 10
<211> 511
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7513469CD1

<400> 10
Met Pro Pro Ala Ala Pro Ala Arg Leu Ala Leu Ala Leu Gly Leu
```

```
1               5                   10                  15
Gly Leu Trp Leu Gly Ala Leu Ala Gly Gly Pro Gly Arg Gly Cys
                20                  25                  30
Gly Pro Cys Glu Pro Pro Cys Leu Cys Gly Pro Ala Pro Gly Ala
                35                  40                  45
Ala Cys Arg Val Asn Cys Ser Gly Arg Gly Leu Arg Thr Leu Gly
                50                  55                  60
Pro Ala Leu Arg Ile Pro Ala Asp Ala Thr Ala Leu Asp Val Ser
                65                  70                  75
His Asn Leu Leu Arg Ala Leu Asp Val Gly Leu Leu Ala Asn Leu
                80                  85                  90
Ser Ala Leu Ala Glu Leu Asp Ile Ser Asn Asn Lys Ile Ser Thr
                95                  100                 105
Leu Glu Glu Gly Ile Phe Ala Asn Leu Phe Asn Leu Ser Glu Ile
                110                 115                 120
Asn Leu Ser Gly Asn Pro Phe Glu Cys Asp Cys Gly Leu Ala Trp
                125                 130                 135
Leu Pro Arg Trp Ala Glu Glu Gln Gln Val Arg Val Val Gln Pro
                140                 145                 150
Glu Ala Ala Thr Cys Ala Gly Pro Gly Ser Leu Ala Gly Gln Pro
                155                 160                 165
Leu Leu Gly Ile Pro Leu Leu Asp Ser Gly Cys Gly Glu Glu Tyr
                170                 175                 180
Val Ala Cys Leu Pro Asp Asn Ser Ser Gly Thr Val Ala Ala Val
                185                 190                 195
Ser Phe Ser Ala Ala His Glu Gly Leu Leu Gln Pro Glu Ala Cys
                200                 205                 210
Ser Ala Phe Cys Phe Ser Thr Gly Gln Gly Leu Ala Ala Leu Ser
                215                 220                 225
Glu Gln Gly Trp Cys Leu Cys Gly Ser Ala Gln Pro Ser Ser Ala
                230                 235                 240
Ser Phe Ala Cys Leu Ser Leu Cys Ser Gly Pro Pro Pro Pro Pro
                245                 250                 255
Ala Pro Thr Cys Arg Gly Pro Thr Leu Leu Gln His Val Phe Pro
                260                 265                 270
Ala Ser Pro Gly Ala Thr Leu Val Gly Pro His Gly Pro Leu Ala
                275                 280                 285
Ser Gly Gln Leu Ala Ala Phe His Ile Ala Ala Pro Leu Pro Val
                290                 295                 300
Thr Ala Thr Arg Trp Asp Phe Gly Asp Gly Ser Pro Glu Val Asp
                305                 310                 315
Ala Ala Gly Pro Ala Ala Ser His Arg Tyr Val Leu Pro Gly Arg
                320                 325                 330
Tyr His Val Thr Ala Val Leu Ala Leu Gly Thr Gly Ser Ala Leu
                335                 340                 345
Leu Gly Thr Asp Val Gln Val Glu Ala Ala Pro Ala Ala Leu Glu
                350                 355                 360
Leu Val Cys Pro Ser Ser Val Gln Ser Asp Glu Ser Leu Asp Leu
                365                 370                 375
Ser Ile Gln Asn Arg Gly Gly Ser Gly Leu Glu Ala Ala Tyr Ser
                380                 385                 390
Ile Val Ala Leu Gly Glu Glu Pro Ala Arg Ala Val His Pro Leu
                395                 400                 405
Cys Pro Ser Asp Thr Glu Ile Phe Ser Gly Asn Gly His Cys Tyr
                410                 415                 420
Arg Leu Val Val Glu Lys Ala Ala Trp Leu Gln Ala Gln Glu Gln
                425                 430                 435
Cys Gln Ala Trp Ala Gly Ala Ala Leu Ala Met Val Asp Ser Pro
                440                 445                 450
Ala Val Gln Arg Phe Leu Val Ser Arg Val Thr Arg Pro Ser Ala
                455                 460                 465
Gly Cys Arg Glu Pro Pro Arg Gly Ser Ala Gln Trp Gly Pro Val
                470                 475                 480
```

Gly Thr Pro Asp Ala Ser Gly Thr Ala Val Arg Pro Leu Ser Pro
485 490 495
Ala Arg Ala Pro Gly Gly His Gly Ile Pro Gly Pro Ala Ser Glu
500 505 510
Pro

<210> 11
<211> 161
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7518836CD1

<400> 11
Met Asp Lys Phe Trp Trp His Ala Ala Trp Gly Leu Cys Leu Val
1 5 10 15
Pro Leu Ser Leu Ala Gln Ile Gly Asp Gln Asp Thr Phe His Pro
20 25 30
Ser Gly Gly Ser His Thr Thr His Gly Ser Glu Ser Asp Gly His
35 40 45
Ser His Gly Ser Gln Glu Gly Gly Ala Asn Thr Thr Ser Gly Pro
50 55 60
Ile Arg Thr Pro Gln Ile Pro Glu Trp Leu Ile Ile Leu Ala Ser
65 70 75
Leu Leu Ala Leu Ala Leu Ile Leu Ala Val Cys Ile Ala Val Asn
80 85 90
Ser Arg Arg Arg Cys Gly Gln Lys Lys Lys Leu Val Ile Asn Ser
95 100 105
Gly Asn Gly Ala Val Glu Asp Arg Lys Pro Ser Gly Leu Asn Gly
110 115 120
Glu Ala Ser Lys Ser Gln Glu Met Val His Leu Val Asn Lys Glu
125 130 135
Ser Ser Glu Thr Pro Asp Gln Phe Met Thr Ala Asp Glu Thr Arg
140 145 150
Asn Leu Gln Asn Val Asp Met Lys Ile Gly Val
155 160

<210> 12
<211> 289
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7519868CD1

<400> 12
Met Ser Asp Ser Lys Glu Pro Arg Leu Gln Gln Leu Gly Leu Leu
1 5 10 15
Glu Glu Glu Gln Leu Arg Gly Leu Gly Phe Arg Gln Thr Arg Gly
20 25 30
Tyr Lys Ser Leu Ala Gly Cys Leu Gly His Gly Pro Leu Val Leu
35 40 45
Gln Leu Leu Ser Phe Thr Leu Leu Ala Gly Leu Leu Val Gln Val
50 55 60
Ser Lys Val Pro Ser Ser Ile Ser Gln Glu Gln Ser Arg Gln Asp
65 70 75
Ala Ile Tyr Gln Asn Leu Thr Gln Leu Lys Ala Ala Val Gly Glu
80 85 90
Leu Ser Glu Lys Ser Lys Leu Gln Glu Ile Tyr Gln Glu Leu Thr

```
                95                  100                 105
Gln Leu Lys Ala Ala Val Gly Glu Leu Pro Glu Lys Ser Lys Leu
                110                 115                 120
Gln Glu Ile Tyr Gln Glu Leu Thr Arg Leu Lys Ala Ala Val Glu
                125                 130                 135
Arg Leu Cys His Pro Cys Pro Trp Glu Trp Thr Phe Phe Gln Gly
                140                 145                 150
Asn Cys Tyr Phe Met Ser Asn Ser Gln Arg Asn Trp His Asp Ser
                155                 160                 165
Ile Thr Ala Cys Lys Glu Val Gly Ala Gln Leu Val Val Ile Lys
                170                 175                 180
Ser Ala Glu Glu Gln Asn Phe Leu Gln Leu Gln Ser Ser Arg Ser
                185                 190                 195
Asn Arg Phe Thr Trp Met Gly Leu Ser Asp Leu Asn Gln Glu Gly
                200                 205                 210
Thr Trp Gln Trp Val Asp Gly Ser Pro Leu Leu Pro Ser Phe Lys
                215                 220                 225
Gln Tyr Trp Asn Arg Gly Glu Pro Asn Asn Val Gly Glu Glu Asp
                230                 235                 240
Cys Ala Glu Phe Ser Gly Asn Gly Trp Asn Asp Asp Lys Cys Asn
                245                 250                 255
Leu Ala Lys Phe Trp Ile Cys Lys Lys Ser Ala Ala Ser Cys Ser
                260                 265                 270
Arg Asp Glu Glu Gln Phe Leu Ser Pro Ala Pro Ala Thr Pro Asn
                275                 280                 285
Pro Pro Pro Ala


&lt;210&gt; 13
&lt;211&gt; 368
&lt;212&gt; PRT
&lt;213&gt; Homo sapiens

&lt;220&gt;
&lt;221&gt; misc_feature
&lt;223&gt; Incyte ID No: 7520048CD1

&lt;400&gt; 13
Met Ser Pro Pro Leu Cys Pro Leu Leu Leu Leu Ala Val Gly Leu
  1             5                   10                  15
Arg Leu Ala Gly Thr Leu Asn Pro Ser Asp Pro Asn Thr Cys Ser
                20                  25                  30
Phe Trp Glu Ser Phe Thr Thr Thr Thr Lys Glu Ser His Ser Arg
                35                  40                  45
Pro Phe Ser Leu Leu Pro Ser Glu Pro Cys Glu Arg Pro Trp Glu
                50                  55                  60
Gly Pro His Thr Cys Pro Gln Pro Thr Val Val Tyr Arg Thr Val
                65                  70                  75
Tyr Arg Gln Val Val Lys Thr Asp His Arg Gln Arg Leu Gln Cys
                80                  85                  90
Cys His Gly Phe Tyr Glu Ser Arg Gly Phe Cys Val Pro Leu Cys
                95                  100                 105
Ala Gln Glu Cys Val His Gly Arg Cys Val Ala Pro Asn Gln Cys
                110                 115                 120
Gln Cys Val Pro Gly Trp Arg Gly Asp Asp Cys Ser Ser Glu Cys
                125                 130                 135
Ala Pro Gly Met Trp Gly Pro Gln Cys Asp Lys Pro Cys Ser Cys
                140                 145                 150
Gly Asn Asn Ser Ser Cys Asp Pro Arg Ser Gly Val Cys Ser Cys
                155                 160                 165
Pro Ser Gly Leu Gln Pro Pro Asn Cys Leu Gln Pro Cys Thr Pro
                170                 175                 180
Gly Tyr Tyr Gly Pro Ala Cys Gln Phe Arg Cys Gln Cys His Gly
```

185 190 195
Ala Pro Cys Asp Pro Gln Thr Gly Ala Cys Phe Cys Pro Ala Glu
200 205 210
Arg Thr Gly Pro Arg Ala Leu Leu Ala Ser Ser Ala Pro Ala Pro
215 220 225
Ile Leu Ala Lys Met Glu Val Ser Ser Lys Pro His Arg Ala Pro
230 235 240
Ala Ala Ala Pro Leu Ala Gly Trp Ala Pro Ser Ala Pro Cys Pro
245 250 255
Ala Gln Arg Ala Phe Thr Asp Pro Thr Ala Pro Arg Asn Val Ala
260 265 270
Ala Thr Thr Ala Ala Ser Val Thr Asp Ser Leu Gly Ser Ala Ala
275 280 285
Ala Leu Arg Val Thr Leu Gly Ile Gly Ala Gly Arg Ser Ala Arg
290 295 300
Trp Ala Ala Leu Gly Arg Thr Val Leu Arg Arg Ala Thr Ala Pro
305 310 315
Arg Thr Pro Val Ala Ser Arg Pro Thr Ala His Val Cys Ala Asn
320 325 330
Thr Ala Ser Leu Gly Thr Ala Ala Arg Ile Ala Ser Ala Pro Thr
335 340 345
Ala Ser Thr Val Ser Ala Ala Arg Pro Pro Ala Pro Ala Thr Gly
350 355 360
Ser Thr Ala Ser Ala Ala Thr Arg
365

<210> 14
<211> 688
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7520157CD1

<400> 14
Met Asn Cys Tyr Leu Leu Leu Arg Phe Met Leu Gly Ile Pro Leu
1 5 10 15
Leu Trp Pro Cys Leu Gly Ala Thr Glu Asn Ser Gln Thr Lys Lys
20 25 30
Val Lys Gln Pro Val Arg Ser His Leu Arg Val Lys Arg Gly Trp
35 40 45
Val Trp Asn Gln Phe Phe Val Pro Glu Glu Met Asn Thr Thr Ser
50 55 60
His His Ile Gly Gln Leu Arg Ser Asp Leu Asp Asn Gly Asn Asn
65 70 75
Ser Phe Gln Tyr Lys Leu Leu Gly Ala Gly Ala Gly Ser Thr Phe
80 85 90
Ile Ile Asp Glu Arg Thr Gly Asp Ile Tyr Ala Ile Gln Lys Leu
95 100 105
Asp Arg Glu Glu Arg Ser Leu Tyr Ile Leu Arg Ala Gln Val Ile
110 115 120
Asp Ile Ala Thr Gly Arg Ala Val Glu Pro Glu Ser Glu Phe Val
125 130 135
Ile Lys Val Ser Asp Ile Asn Asp Asn Glu Pro Lys Phe Leu Asp
140 145 150
Glu Pro Tyr Glu Ala Ile Val Pro Glu Met Ser Pro Glu Gly Thr
155 160 165
Leu Val Ile Gln Val Thr Ala Ser Asp Ala Asp Asp Pro Ser Ser
170 175 180
Gly Asn Asn Ala Arg Leu Leu Tyr Ser Leu Leu Gln Gly Gln Pro
185 190 195
Tyr Phe Ser Val Glu Pro Thr Thr Gly Val Ile Arg Ile Ser Ser

```
                200                         205                         210
Lys Met Asp Arg Glu Leu Gln Asp Glu Tyr Trp Val Ile Ile Gln
                215                         220                         225
Ala Lys Asp Thr Ile Gly Gln Pro Gly Ala Leu Ser Gly Thr Thr
                230                         235                         240
Ser Val Leu Ile Lys Leu Ser Asp Val Asn Asp Asn Lys Pro Ile
                245                         250                         255
Phe Lys Glu Ser Leu Tyr Arg Leu Thr Val Ser Glu Ser Ala Pro
                260                         265                         270
Thr Gly Thr Ser Ile Gly Thr Ile Met Ala Tyr Asp Asn Asp Ile
                275                         280                         285
Gly Glu Asn Ala Glu Met Asp Tyr Ser Ile Glu Glu Asp Asp Ser
                290                         295                         300
Gln Thr Phe Asp Ile Ile Thr Asn His Glu Thr Gln Glu Gly Ile
                305                         310                         315
Val Ile Leu Lys Lys Lys Val Asp Phe Glu His Gln Asn His Tyr
                320                         325                         330
Gly Ile Arg Ala Lys Val Lys Asn His His Val Pro Glu Gln Leu
                335                         340                         345
Met Lys Tyr His Thr Glu Ala Ser Thr Thr Phe Ile Lys Ile Gln
                350                         355                         360
Val Glu Asp Val Asp Glu Pro Pro Leu Phe Leu Leu Pro Tyr Tyr
                365                         370                         375
Val Phe Glu Val Phe Glu Glu Thr Pro Gln Gly Ser Phe Val Gly
                380                         385                         390
Val Val Ser Ala Thr Asp Pro Asp Asn Arg Lys Ser Pro Ile Arg
                395                         400                         405
Tyr Ser Ile Thr Arg Ser Lys Val Phe Asn Ile Asn Asp Asn Gly
                410                         415                         420
Thr Ile Thr Thr Ser Asn Ser Leu Asp Arg Glu Ile Ser Ala Trp
                425                         430                         435
Tyr Asn Leu Ser Ile Thr Ala Thr Glu Lys Tyr Asn Ile Glu Gln
                440                         445                         450
Ile Ser Ser Ile Pro Leu Tyr Val Gln Val Leu Asn Ile Asn Asp
                455                         460                         465
His Ala Pro Glu Phe Ser Gln Tyr Tyr Glu Thr Tyr Val Cys Glu
                470                         475                         480
Asn Ala Gly Ser Gly Gln Val Ile Gln Thr Ile Ser Ala Val Asp
                485                         490                         495
Arg Asp Glu Ser Ile Glu Glu His His Phe Tyr Phe Asn Leu Ser
                500                         505                         510
Val Glu Asp Thr Asn Asn Ser Ser Phe Thr Ile Ile Asp Asn Gln
                515                         520                         525
Gly Phe Ile Phe Leu Thr Leu Gly Leu Lys Gln Arg Arg Lys Gln
                530                         535                         540
Ile Leu Phe Pro Glu Lys Ser Glu Asp Phe Arg Glu Asn Ile Phe
                545                         550                         555
Gln Tyr Asp Asp Glu Gly Gly Gly Glu Glu Asp Thr Glu Ala Phe
                560                         565                         570
Asp Ile Ala Glu Leu Arg Ser Ser Thr Ile Met Arg Glu Arg Lys
                575                         580                         585
Thr Arg Lys Thr Thr Ser Ala Glu Ile Arg Ser Leu Tyr Arg Gln
                590                         595                         600
Ser Leu Gln Val Gly Pro Asp Ser Ala Ile Phe Arg Lys Phe Ile
                605                         610                         615
Leu Glu Lys Leu Glu Glu Ala Asn Thr Asp Pro Cys Ala Pro Pro
                620                         625                         630
Phe Asp Ser Leu Gln Thr Tyr Ala Phe Glu Gly Thr Gly Ser Leu
                635                         640                         645
Ala Gly Ser Leu Ser Ser Leu Glu Ser Ala Val Ser Asp Gln Asp
                650                         655                         660
Glu Ser Tyr Asp Tyr Leu Asn Glu Leu Gly Pro Arg Phe Lys Arg
                665                         670                         675
```

Leu Ala Cys Met Phe Gly Ser Ala Val Gln Ser Asn Asn
680 685

<210> 15
<211> 272
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7520485CD1

<400> 15
Met Ser Asp Ser Lys Glu Pro Arg Val Gln Gln Leu Gly Leu Leu
1 5 10 15
Glu Glu Asp Pro Thr Thr Ser Gly Ile Arg Leu Phe Pro Arg Asp
20 25 30
Phe Gln Phe Gln Gln Ile His Gly His Lys Ser Ser Thr Gly Cys
35 40 45
Leu Gly His Gly Ala Leu Val Leu Gln Leu Leu Ser Phe Met Leu
50 55 60
Leu Ala Gly Val Leu Val Ala Ile Leu Val Gln Val Ser Lys Val
65 70 75
Pro Ser Ser Leu Ser Gln Glu Gln Ser Glu Gln Asp Ala Ile Tyr
80 85 90
Gln Asn Leu Thr Gln Leu Lys Ala Ala Val Gly Glu Leu Ser Glu
95 100 105
Lys Ser Lys Pro Gln Glu Ile Tyr Gln Glu Leu Thr Gln Leu Lys
110 115 120
Ala Ala Val Gly Glu Leu Pro Glu Lys Ser Lys Leu Gln Glu Ile
125 130 135
Tyr Gln Glu Leu Thr Arg Leu Lys Ala Ala Val Gly Glu Leu Pro
140 145 150
Glu Lys Ser Lys Leu Gln Glu Ile Tyr Gln Glu Leu Thr Arg Leu
155 160 165
Lys Thr Ala Val Gly Glu Leu Pro Glu Lys Ser Lys Leu Gln Glu
170 175 180
Ile Tyr Gln Glu Leu Thr Arg Leu Lys Ala Ala Val Gly Glu Leu
185 190 195
Pro Glu Lys Ser Lys Leu Gln Glu Ile Tyr Gln Glu Leu Thr Glu
200 205 210
Leu Lys Ala Ala Val Gly Glu Leu Pro Glu Lys Ser Lys Leu Gln
215 220 225
Glu Ile Tyr Gln Glu Leu Thr Gln Leu Lys Ala Ala Val Gly Glu
230 235 240
Leu Pro Asp Gln Ser Lys Gln Gln Gln Ile Tyr Gln Glu Leu Thr
245 250 255
Asp Leu Lys Thr Ala Phe Gly Glu Phe Leu His Ile Lys Gly Pro
260 265 270
Trp Ala

<210> 16
<211> 444
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7525723CD1

<400> 16
Met Gly Ser Pro Ala His Arg Pro Ala Leu Leu Leu Leu Leu Pro

```
1               5                   10                  15
Pro Leu Leu Leu Leu Leu Leu Leu Arg Val Pro Pro Ser Arg Ser
                20                  25                  30
Phe Pro Asp Thr Pro Trp Cys Ser Pro Ile Lys Val Lys Tyr Gly
                35                  40                  45
Asp Val Tyr Cys Arg Ala Pro Gln Gly Gly Tyr Tyr Lys Thr Ala
                50                  55                  60
Leu Gly Thr Arg Cys Asp Ile Arg Cys Gln Lys Gly Tyr Glu Leu
                65                  70                  75
His Gly Ser Ser Leu Leu Ile Cys Gln Ser Asn Lys Arg Trp Ser
                80                  85                  90
Asp Lys Val Ile Cys Lys Gln Lys Arg Cys Pro Thr Leu Ala Met
                95                  100                 105
Pro Ala Asn Gly Gly Phe Lys Cys Val Asp Gly Ala Tyr Phe Asn
                110                 115                 120
Ser Arg Cys Glu Tyr Tyr Cys Ser Pro Gly Tyr Thr Leu Lys Gly
                125                 130                 135
Glu Arg Thr Val Thr Cys Met Asp Asn Lys Ala Trp Ser Gly Arg
                140                 145                 150
Pro Ala Ser Cys Val Asp Met Glu Pro Pro Arg Ile Lys Cys Pro
                155                 160                 165
Ser Val Lys Glu Arg Ile Ala Glu Pro Asn Lys Leu Thr Val Arg
                170                 175                 180
Val Ser Trp Glu Thr Pro Glu Gly Arg Asp Thr Ala Asp Gly Ile
                185                 190                 195
Leu Thr Asp Val Ile Leu Lys Gly Leu Pro Pro Gly Ser Asn Phe
                200                 205                 210
Pro Glu Gly Asp His Lys Ile Gln Tyr Thr Val Tyr Asp Arg Ala
                215                 220                 225
Glu Asn Lys Gly Thr Cys Lys Phe Arg Val Lys Val Arg Val Lys
                230                 235                 240
Arg Cys Gly Lys Leu Asn Ala Pro Glu Asn Gly Tyr Met Lys Cys
                245                 250                 255
Ser Ser Asp Gly Asp Asn Tyr Gly Ala Thr Cys Glu Phe Ser Cys
                260                 265                 270
Ile Gly Gly Tyr Glu Leu Gln Gly Ser Pro Ala Arg Val Cys Gln
                275                 280                 285
Ser Asn Leu Ala Trp Ser Gly Thr Glu Pro Thr Cys Ala Ala Met
                290                 295                 300
Asn Val Asn Val Gly Val Arg Thr Ala Ala Ala Leu Leu Asp Gln
                305                 310                 315
Phe Tyr Glu Lys Arg Arg Leu Leu Ile Val Ser Thr Pro Thr Ala
                320                 325                 330
Arg Asn Leu Leu Tyr Arg Leu Gln Leu Gly Met Leu Gln Gln Ala
                335                 340                 345
Gln Cys Gly Leu Asp Leu Arg His Ile Thr Val Val Glu Leu Val
                350                 355                 360
Gly Val Phe Pro Thr Leu Ile Gly Arg Ile Gly Ala Lys Ile Met
                365                 370                 375
Pro Pro Ala Leu Ala Leu Gln Leu Arg Leu Leu Leu Arg Ile Pro
                380                 385                 390
Leu Tyr Ser Phe Ser Met Val Leu Val Asp Lys His Gly Met Asp
                395                 400                 405
Lys Glu Arg Tyr Val Ser Leu Val Met Pro Val Ala Leu Phe Asn
                410                 415                 420
Leu Ile Asp Thr Phe Pro Leu Arg Lys Glu Glu Met Val Leu Gln
                425                 430                 435
Ala Glu Met Ser Gln Thr Cys Asn Thr
                440
```

<210> 17
<211> 357
<212> PRT

<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7525966CD1

<400> 17
Met Val Pro Pro Pro Pro Ser Arg Gly Gly Ala Ala Arg Gly Gln
1 5 10 15
Leu Gly Arg Ser Leu Gly Pro Leu Leu Leu Leu Leu Ala Leu Gly
20 25 30
His Thr Trp Thr Tyr Arg Glu Glu Pro Glu Asp Gly Asp Arg Glu
35 40 45
Ile Cys Ser Glu Ser Lys Ile Ala Thr Thr Lys Tyr Pro Cys Leu
50 55 60
Lys Ser Ser Gly Glu Leu Thr Thr Cys Tyr Arg Lys Lys Cys Cys
65 70 75
Lys Gly Tyr Lys Phe Val Leu Gly Gln Cys Val Pro Glu Asp Tyr
80 85 90
Asp Val Cys Ala Glu Ala Pro Cys Glu Gln Gln Cys Thr Asp Asn
95 100 105
Phe Gly Arg Val Leu Cys Thr Cys Tyr Pro Gly Tyr Arg Tyr Asp
110 115 120
Arg Glu Arg His Arg Lys Arg Glu Lys Pro Tyr Cys Leu Asp Ile
125 130 135
Asp Glu Cys Ala Ser Ser Asn Gly Thr Leu Cys Ala His Ile Cys
140 145 150
Ile Asn Thr Leu Gly Ser Tyr Arg Cys Glu Cys Arg Glu Gly Tyr
155 160 165
Ile Arg Glu Asp Asp Gly Lys Thr Cys Thr Arg Gly Asp Lys Tyr
170 175 180
Pro Asn Asp Thr Gly His Glu Lys Ser Glu Asn Met Val Lys Ala
185 190 195
Gly Thr Cys Cys Ala Thr Cys Lys Glu Phe Tyr Gln Met Lys Gln
200 205 210
Thr Val Leu Gln Leu Lys Gln Lys Ala His Gln Asp Gln Arg Glu
215 220 225
Ala Gln Ala Ser Pro Val Cys Gln Ala Leu Leu Gly Ser Pro Ala
230 235 240
His Gly Ala Gln Trp Asp Pro Trp Asp His Leu Leu Ile Cys Pro
245 250 255
Thr Leu Ser Lys Ala Gly Gly Ala Leu Trp Val His Gln Gly His
260 265 270
Gln Glu Glu Met Val Leu Arg Gly Arg Glu Glu Arg Leu Gly Pro
275 280 285
Glu Gly Leu Gln Asp Pro Leu Val Leu Ser Thr Ser Cys Tyr Leu
290 295 300
Cys Trp Leu Thr Ser Ala Met Thr Ser Leu Ser Cys Arg Lys Arg
305 310 315
Cys Ser Gly Thr Gly Leu Thr Leu Gln Gln Arg Ser Ser Leu Tyr
320 325 330
Leu Arg Asn Phe Pro Ala Thr Gln Lys Pro Trp Thr Trp Ala Leu
335 340 345
Glu Met Thr Ile Gln Glu Glu Leu Arg Gln Glu Thr
350 355

<210> 18
<211> 287
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature

<223> Incyte ID No: 7519667CD1

<400> 18
```
Met Ala Arg Gln Pro Pro Pro Pro Trp Val His Ala Ala Phe Leu
  1               5                  10                  15
Leu Cys Leu Leu Ser Leu Gly Gly Ala Ile Glu Ile Pro Met Asp
                 20                  25                  30
Pro Ser Ile Gln Asn Glu Leu Thr Gln Pro Pro Thr Ile Thr Lys
                 35                  40                  45
Gln Ser Ala Lys Asp His Ile Val Asp Pro Arg Asp Asn Ile Leu
                 50                  55                  60
Ile Glu Cys Glu Ala Lys Gly Asn Pro Ala Pro Ser Phe His Trp
                 65                  70                  75
Thr Arg Asn Ser Arg Phe Phe Asn Ile Ala Lys Asp Pro Arg Val
                 80                  85                  90
Ser Met Arg Arg Arg Ser Gly Thr Leu Val Ile Asp Phe Arg Ser
                 95                 100                 105
Gly Gly Arg Pro Glu Glu Tyr Glu Gly Glu Tyr Gln Cys Phe Ala
                110                 115                 120
Arg Asn Lys Phe Gly Thr Ala Leu Ser Asn Arg Ile Arg Leu Gln
                125                 130                 135
Glu Ser Pro Ala Pro Pro Asn Glu Ala Tyr Thr Asn Asn Gln Ala
                140                 145                 150
Asp Ile Ala Thr Gln Gly Trp Phe Ile Gly Leu Met Cys Ala Ile
                155                 160                 165
Ala Leu Leu Val Leu Ile Leu Leu Ile Val Cys Phe Ile Lys Arg
                170                 175                 180
Ser Arg Gly Gly Lys Tyr Pro Val Arg Glu Lys Lys Asp Val Pro
                185                 190                 195
Leu Gly Pro Glu Asp Pro Lys Glu Glu Asp Gly Ser Phe Asp Tyr
                200                 205                 210
Ser Asp Glu Asp Asn Lys Pro Leu Gln Gly Ser Gln Thr Ser Leu
                215                 220                 225
Asp Gly Thr Ile Lys Gln Gln Glu Ser Asp Asp Ser Leu Val Asp
                230                 235                 240
Tyr Gly Glu Gly Gly Glu Gly Gln Phe Asn Glu Asp Gly Ser Phe
                245                 250                 255
Ile Gly Gln Tyr Thr Val Lys Lys Asp Lys Glu Glu Thr Glu Gly
                260                 265                 270
Asn Glu Ser Ser Glu Ala Thr Ser Pro Val Asn Ala Ile Tyr Ser
                275                 280                 285
Leu Ala
```

<210> 19
<211> 572
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7520638CD1

<400> 19
```
Met Ala Val Gln Arg Ala Ala Ser Pro Arg Arg Pro Pro Ala Pro
  1               5                  10                  15
Leu Trp Pro Arg Leu Leu Leu Pro Leu Leu Leu Leu Leu Leu Pro
                 20                  25                  30
Ala Pro Ser Glu Gly Leu Gly His Ser Ala Glu Leu Ala Phe Ala
                 35                  40                  45
Val Glu Pro Ser Asp Asp Val Ala Val Pro Gly Gln Pro Ile Val
                 50                  55                  60
Leu Asp Cys Arg Val Glu Gly Thr Pro Pro Val Arg Ile Thr Trp
```

```
                65                  70                  75
Arg Lys Asn Gly Val Glu Leu Pro Glu Ser Thr His Ser Thr Leu
                80                  85                  90
Leu Ala Asn Gly Ser Leu Met Ile Arg His Phe Arg Leu Glu Pro
                95                  100                 105
Gly Gly Ser Pro Ser Asp Glu Gly Asp Tyr Glu Cys Val Ala Gln
                110                 115                 120
Asn Arg Phe Gly Leu Val Val Ser Arg Lys Ala Arg Ile Gln Ala
                125                 130                 135
Ala Thr Met Ser Asp Phe His Val His Pro Gln Ala Thr Val Gly
                140                 145                 150
Glu Glu Gly Gly Val Ala Arg Phe Gln Cys Gln Ile Arg Gly Leu
                155                 160                 165
Pro Lys Pro Leu Ile Thr Trp Glu Lys Asn Arg Val Pro Ile Asp
                170                 175                 180
Thr Asp Asn Glu Arg Tyr Thr Leu Leu Pro Lys Gly Val Leu Gln
                185                 190                 195
Ile Thr Gly Leu Arg Ala Glu Asp Gly Gly Ile Phe His Cys Val
                200                 205                 210
Ala Ser Asn Ile Ala Ser Ile Arg Ile Ser His Gly Ala Arg Leu
                215                 220                 225
Thr Val Ser Gly Ser Gly Ser Gly Ala Tyr Lys Glu Pro Ala Ile
                230                 235                 240
Leu Val Gly Pro Glu Asn Leu Thr Leu Thr Val His Gln Thr Ala
                245                 250                 255
Val Leu Glu Cys Val Ala Thr Gly Asn Pro Arg Pro Ile Val Ser
                260                 265                 270
Trp Ser Arg Leu Asp Gly Arg Pro Ile Gly Val Glu Gly Ile Gln
                275                 280                 285
Val Leu Gly Thr Gly Asn Leu Ile Ile Ser Asp Val Thr Val Gln
                290                 295                 300
His Ser Gly Val Tyr Val Cys Ala Ala Asn Arg Pro Gly Thr Arg
                305                 310                 315
Val Arg Arg Thr Ala Gln Gly Arg Leu Val Val Gln Ala Pro Ala
                320                 325                 330
Glu Phe Val Gln His Pro Gln Ser Ile Ser Arg Pro Ala Gly Thr
                335                 340                 345
Thr Ala Met Phe Thr Cys Gln Ala Gln Gly Glu Pro Pro Pro His
                350                 355                 360
Val Thr Trp Leu Lys Asn Gly Gln Val Leu Gly Pro Gly Gly His
                365                 370                 375
Val Arg Leu Lys Asn Asn Asn Ser Thr Leu Thr Ile Ser Gly Ile
                380                 385                 390
Gly Pro Glu Asp Glu Ala Ile Tyr Gln Cys Val Ala Glu Asn Ser
                395                 400                 405
Ala Gly Ser Ser Gln Ala Ser Ala Arg Leu Thr Val Leu Trp Ala
                410                 415                 420
Glu Gly Leu Pro Gly Pro Pro Arg Asn Val Arg Ala Val Ser Val
                425                 430                 435
Ser Ser Thr Glu Val Arg Val Ser Trp Ser Glu Pro Leu Ala Asn
                440                 445                 450
Thr Lys Glu Ile Ile Gly Tyr Val Leu His Ile Arg Lys Ala Ala
                455                 460                 465
Asp Pro Pro Glu Leu Glu Tyr Gln Glu Ala Val Ser Lys Ser Thr
                470                 475                 480
Phe Gln His Leu Val Ser Asp Leu Glu Pro Ser Thr Ala Tyr Ser
                485                 490                 495
Phe Tyr Ile Lys Ala Tyr Thr Pro Arg Gly Ala Ser Ser Ala Ser
                500                 505                 510
Val Pro Thr Leu Ala Ser Thr Leu Gly Glu Gly Glu Ala Trp Val
                515                 520                 525
Trp Ser Thr Lys Pro Gln Ala Leu Thr Gln Gly Ile Arg Ala Glu
                530                 535                 540
```

Pro Pro Cys Thr Leu Val Leu Val Val Arg Cys Ala Gly Cys Pro
545 550 555
Ala Leu Arg Leu Ser Gly Leu Ser Leu Leu Gly Gly Lys Gly Phe
560 565 570
Phe Phe

<210> 20
<211> 403
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7518947CD1

<400> 20
Met Val Pro Pro Pro Pro Gly Arg Gly Gly Ala Ala Arg Gly Gln
1 5 10 15
Leu Gly Arg Ser Leu Gly Pro Leu Leu Leu Leu Leu Ala Leu Gly
20 25 30
His Thr Trp Thr Tyr Arg Glu Glu Pro Glu Asp Gly Asp Arg Glu
35 40 45
Ile Cys Ser Glu Ser Lys Ile Ala Thr Thr Lys Tyr Pro Cys Leu
50 55 60
Lys Ser Ser Gly Glu Leu Thr Thr Cys Tyr Arg Lys Lys Cys Cys
65 70 75
Lys Gly Tyr Lys Phe Val Leu Gly Gln Cys Ile Pro Glu Asp Tyr
80 85 90
Asp Val Cys Ala Glu Ala Pro Cys Glu Gln Gln Cys Thr Asp Asn
95 100 105
Phe Gly Arg Val Leu Cys Thr Cys Tyr Pro Gly Tyr Arg Tyr Asp
110 115 120
Arg Glu Arg His Arg Lys Arg Glu Lys Pro Tyr Cys Leu Asp Met
125 130 135
Asp Glu Cys Ala Ser Ser Asn Gly Thr Leu Cys Ala His Ile Cys
140 145 150
Ile Asn Thr Leu Gly Ser Tyr Arg Cys Glu Cys Arg Glu Gly Tyr
155 160 165
Ile Arg Glu Asp Asp Gly Lys Thr Cys Thr Arg Gly Asp Lys Tyr
170 175 180
Pro Asn Asp Thr Gly His Glu Lys Ser Glu Asn Met Val Lys Ala
185 190 195
Gly Thr Cys Cys Ala Thr Cys Lys Glu Phe Tyr Gln Met Lys Gln
200 205 210
Thr Val Leu Gln Leu Lys Gln Lys Ile Ala Leu Leu Pro Asn Asn
215 220 225
Ala Ala Asp Leu Gly Lys Tyr Ile Thr Gly Asp Lys Val Leu Ala
230 235 240
Ser Asn Thr Tyr Leu Pro Gly Pro Pro Gly Leu Pro Gly Gly Gln
245 250 255
Gly Pro Pro Gly Ser Pro Gly Pro Lys Gly Ser Pro Gly Phe Pro
260 265 270
Gly Met Pro Gly Pro Pro Gly Gln Pro Gly Pro Arg Gly Ser Met
275 280 285
Gly Pro Met Gly Pro Ser Pro Asp Leu Ser His Ile Lys Gln Gly
290 295 300
Arg Arg Gly Pro Val Ile Leu Leu Tyr Tyr Arg Val His Gln Gly
305 310 315
His Gln Glu Glu Met Val Leu Arg Gly Arg Glu Glu Arg Leu Gly
320 325 330
Pro Glu Gly Leu Gln Asp Pro Leu Val Leu Ser Thr Ser Cys Tyr
335 340 345

Leu Cys Trp Leu Thr Ser Ala Met Thr Ser Leu Ser Cys Arg Lys
350 355 360
Arg Cys Ser Gly Thr Gly Leu Thr Leu Gln Gln Arg Ser Ser Leu
365 370 375
Tyr Leu Arg Asn Phe Pro Ala Thr Gln Lys Pro Trp Thr Trp Ala
380 385 390
Leu Glu Met Thr Ile Gln Glu Glu Leu Arg Gln Glu Thr
395 400

<210> 21
<211> 281
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7519652CD1

<400> 21
Met Ala Arg Gln Pro Pro Pro Pro Trp Val His Ala Ala Phe Leu
1 5 10 15
Leu Cys Leu Leu Ser Leu Gly Gly Ala Ile Glu Ile Pro Met Asp
20 25 30
Leu Thr Gln Pro Pro Thr Ile Thr Lys Gln Ser Ala Lys Asp His
35 40 45
Ile Val Asp Pro Arg Asp Asn Ile Leu Ile Glu Cys Glu Ala Lys
50 55 60
Gly Asn Pro Ala Pro Ser Phe His Trp Thr Arg Asn Ser Arg Phe
65 70 75
Phe Asn Ile Ala Lys Asp Pro Arg Val Ser Met Arg Arg Arg Ser
80 85 90
Gly Thr Leu Val Ile Asp Phe Arg Ser Gly Gly Arg Pro Glu Glu
95 100 105
Tyr Glu Gly Glu Tyr Gln Cys Phe Ala Arg Asn Lys Phe Gly Thr
110 115 120
Ala Leu Ser Asn Arg Ile Arg Leu Gln Glu Ser Pro Ala Pro Pro
125 130 135
Asn Glu Ala Tyr Thr Asn Asn Gln Ala Asp Ile Ala Thr Gln Gly
140 145 150
Trp Phe Ile Gly Leu Met Cys Ala Ile Ala Leu Leu Val Leu Ile
155 160 165
Leu Leu Ile Val Cys Phe Ile Lys Arg Ser Arg Gly Gly Lys Tyr
170 175 180
Pro Val Arg Glu Lys Lys Asp Val Pro Leu Gly Pro Glu Asp Pro
185 190 195
Lys Glu Glu Asp Gly Ser Phe Asp Tyr Ser Asp Glu Asp Asn Lys
200 205 210
Pro Leu Gln Gly Ser Gln Thr Ser Leu Asp Gly Thr Ile Lys Gln
215 220 225
Gln Glu Ser Asp Asp Ser Leu Val Asp Tyr Gly Glu Gly Gly Glu
230 235 240
Gly Gln Phe Asn Glu Asp Gly Ser Phe Ile Gly Gln Tyr Thr Val
245 250 255
Lys Lys Asp Lys Glu Glu Thr Glu Gly Asn Glu Ser Ser Glu Ala
260 265 270
Thr Ser Pro Val Asn Ala Ile Tyr Ser Leu Ala
275 280

<210> 22
<211> 520
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7520381CD1

<400> 22
Met Gly Ser Trp Ala Leu Leu Trp Pro Pro Leu Leu Phe Thr Gly
1 5 10 15
Leu Leu Val Arg Pro Pro Gly Thr Met Ala Gln Ala Gln Tyr Cys
20 25 30
Ser Val Asn Lys Asp Asn Phe Glu Val Glu Glu Asn Thr Asn Val
35 40 45
Thr Glu Pro Leu Val Asp Ile His Val Pro Glu Gly Gln Glu Val
50 55 60
Thr Leu Gly Ala Leu Ser Thr Pro Phe Ala Phe Arg Ile Gln Gly
65 70 75
Asn Gln Leu Phe Leu Asn Val Thr Pro Asp Tyr Glu Glu Lys Ser
80 85 90
Leu Leu Glu Ala Gln Leu Leu Cys Gln Ser Gly Gly Thr Leu Val
95 100 105
Thr Gln Leu Arg Val Phe Val Ser Val Leu Asp Val Asn Asp Asn
110 115 120
Ala Pro Glu Phe Pro Phe Lys Thr Lys Glu Ile Arg Val Glu Glu
125 130 135
Asp Thr Lys Val Asn Ser Thr Val Ile Pro Glu Thr Gln Leu Gln
140 145 150
Ala Glu Asp Arg Asp Lys Asp Asp Ile Leu Phe Tyr Thr Leu Gln
155 160 165
Glu Met Thr Ala Gly Ala Ser Asp Tyr Phe Ser Leu Val Ser Val
170 175 180
Asn Arg Pro Ala Leu Arg Leu Asp Arg Pro Leu Asp Phe Tyr Glu
185 190 195
Arg Pro Asn Met Thr Phe Trp Leu Leu Val Arg Asp Thr Pro Gly
200 205 210
Glu Asn Val Glu Pro Ser His Thr Ala Thr Ala Thr Leu Val Leu
215 220 225
Asn Val Val Pro Ala Asp Leu Arg Pro Pro Trp Phe Leu Pro Cys
230 235 240
Thr Phe Ser Asp Gly Tyr Val Cys Ile Gln Ala Gln Tyr His Gly
245 250 255
Ala Val Pro Thr Gly His Ile Leu Pro Ser Pro Leu Val Leu Arg
260 265 270
Pro Gly Pro Ile Tyr Ala Glu Asp Gly Asp Arg Gly Ile Asn Gln
275 280 285
Pro Ile Ile Tyr Ser Ile Phe Arg Gly Asn Val Asn Gly Thr Phe
290 295 300
Ile Ile His Pro Asp Ser Gly Asn Leu Thr Val Ala Arg Ser Val
305 310 315
Pro Ser Pro Met Thr Phe Leu Leu Leu Val Lys Gly Gln Gln Ala
320 325 330
Asp Leu Ala Arg Tyr Ser Val Thr Gln Val Thr Val Glu Ala Val
335 340 345
Ala Ala Ala Gly Ser Pro Pro Arg Phe Pro Gln Arg Leu Tyr Arg
350 355 360
Gly Thr Val Ala Arg Gly Ala Gly Ala Gly Val Val Val Lys Asp
365 370 375
Ala Ala Ala Pro Ser Gln Pro Leu Arg Ile Gln Ala Gln Asp Pro
380 385 390
Glu Phe Ser Asp Leu Asn Ser Ala Ile Thr Tyr Arg Ile Thr Asn
395 400 405
His Ser His Phe Arg Met Glu Gly Glu Val Val Leu Thr Thr Thr
410 415 420
Thr Leu Ala Gln Ala Gly Ala Phe Tyr Ala Glu Asp Arg Gly Gln
425 430 435

Met Trp Gly Pro Ala Asn Pro Met Ile Pro Thr Ala Ala Ala His
440 445 450
Pro Ala Ser Pro Cys Thr Ser Cys Glu Leu Leu Trp Arg Pro Leu
455 460 465
Gly Gly Gln Ala Leu Leu Gly Gly Gly Tyr Gly Gly Pro Gly Arg
470 475 480
Gly Ala Gly Cys Ala Ala Ala Ala Gly Ser Pro Trp Pro His Arg
485 490 495
Pro Cys Pro Gln Ala Leu Trp Pro Pro Ala Gln Val Leu Leu Trp
500 505 510
Gln Ser Ser Gly Ala Pro Ala Pro Arg Leu
515 520

<210> 23
<211> 674
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7520409CD1

<400> 23
Met Gly Ser Trp Ala Leu Leu Trp Pro Pro Leu Leu Phe Thr Gly
1 5 10 15
Leu Leu Val Arg Pro Pro Gly Thr Met Ala Gln Ala Gln Tyr Cys
20 25 30
Ser Val Asn Lys Asp Ile Phe Glu Val Glu Glu Asn Thr Asn Val
35 40 45
Thr Glu Pro Leu Val Asp Ile His Val Pro Glu Gly Gln Glu Val
50 55 60
Thr Leu Gly Ala Leu Ser Thr Pro Phe Ala Phe Arg Ile Gln Gly
65 70 75
Asn Gln Leu Phe Leu Asn Val Thr Pro Asp Tyr Glu Glu Lys Ser
80 85 90
Leu Leu Glu Ala Gln Leu Leu Cys Gln Ser Gly Gly Thr Leu Val
95 100 105
Thr Gln Leu Arg Val Phe Val Ser Val Leu Asp Val Asn Asp Asn
110 115 120
Ala Pro Glu Phe Pro Phe Lys Thr Lys Glu Ile Arg Val Glu Glu
125 130 135
Asp Thr Lys Val Asn Ser Thr Val Ile Pro Glu Thr Gln Leu Gln
140 145 150
Ala Glu Asp Arg Asp Lys Asp Asp Ile Leu Phe Tyr Thr Leu Gln
155 160 165
Glu Met Thr Ala Gly Ala Ser Asp Tyr Phe Ser Leu Asp Thr Pro
170 175 180
Gly Glu Asn Val Glu Pro Ser His Thr Ala Thr Ala Thr Leu Val
185 190 195
Leu Asn Val Val Pro Ala Asp Leu Arg Pro Pro Trp Phe Leu Pro
200 205 210
Cys Thr Phe Ser Asp Gly Tyr Val Cys Ile Gln Ala Gln Tyr His
215 220 225
Gly Ala Val Pro Thr Gly His Ile Leu Pro Ser Pro Leu Val Leu
230 235 240
Arg Pro Gly Pro Ile Tyr Ala Glu Asp Gly Asp Arg Gly Ile Asn
245 250 255
Gln Pro Ile Ile Tyr Ser Ile Phe Arg Gly Asn Val Asn Gly Thr
260 265 270
Phe Ile Ile His Pro Asp Ser Gly Asn Leu Thr Val Ala Arg Ser
275 280 285
Val Pro Ser Pro Met Thr Phe Leu Leu Leu Val Lys Gly Gln Gln
290 295 300

```
Ala Asp Leu Ala Arg Tyr Ser Val Thr Gln Val Thr Val Glu Ala
                305                 310                 315
Val Ala Ala Ala Gly Ser Pro Pro Arg Phe Pro Gln Ser Leu Tyr
                320                 325                 330
Arg Gly Thr Val Ala Arg Gly Ala Gly Ala Gly Val Val Val Lys
                335                 340                 345
Asp Ala Ala Ala Pro Ser Gln Pro Leu Arg Ile Gln Ala Gln Asp
                350                 355                 360
Pro Glu Phe Ser Asp Leu Asn Ser Ala Ile Thr Tyr Arg Ile Thr
                365                 370                 375
Asn His Ser His Phe Arg Met Glu Gly Glu Val Val Leu Thr Thr
                380                 385                 390
Thr Thr Leu Ala Gln Ala Gly Ala Phe Tyr Ala Glu Val Glu Ala
                395                 400                 405
His Asn Thr Val Thr Ser Gly Thr Ala Thr Thr Val Ile Glu Ile
                410                 415                 420
Gln Val Ser Glu Gln Glu Pro Pro Ser Thr Ala Gln Thr Pro Glu
                425                 430                 435
Ala Gly Thr Ser Gln Pro Met Pro Pro Gly Met Gly Thr Ser Thr
                440                 445                 450
Ser His Gln Pro Thr Thr Pro Gly Gly Gly Thr Ala Gln Thr Pro
                455                 460                 465
Glu Pro Gly Thr Ser Gln Pro Met Pro Leu Ser Lys Ser Thr Pro
                470                 475                 480
Ser Ser Gly Ala Ala Pro Ser Glu Asp Lys Arg Phe Ser Ser Val
                485                 490                 495
Asp Met Ala Ala Leu Gly Gly Val Leu Gly Ala Leu Leu Leu Leu
                500                 505                 510
Ala Leu Leu Gly Leu Ala Val Leu Val His Lys His Tyr Gly Pro
                515                 520                 525
Arg Leu Lys Cys Cys Ser Gly Lys Ala Pro Glu Pro Gln Pro Gln
                530                 535                 540
Gly Phe Asp Asn Gln Ala Phe Leu Pro Asp His Lys Ala Asn Trp
                545                 550                 555
Ala Pro Val Pro Ser Pro Thr His Asp Pro Lys Pro Ala Glu Ala
                560                 565                 570
Pro Met Pro Ala Glu Pro Ala Pro Pro Gly Pro Ala Ser Pro Gly
                575                 580                 585
Gly Ala Pro Glu Pro Pro Ala Ala Ala Arg Ala Gly Gly Ser Pro
                590                 595                 600
Thr Ala Val Arg Ser Ile Leu Thr Lys Glu Arg Arg Pro Glu Gly
                605                 610                 615
Gly Tyr Lys Ala Val Trp Phe Gly Glu Asp Ile Gly Thr Glu Ala
                620                 625                 630
Asp Val Val Val Leu Asn Ala Pro Thr Leu Asp Val Asp Gly Ala
                635                 640                 645
Ser Asp Ser Gly Ser Gly Asp Glu Gly Glu Gly Ala Gly Arg Gly
                650                 655                 660
Gly Gly Pro Tyr Asp Ala Pro Gly Gly Asp Asp Ser Tyr Ile
                665                 670

<210> 24
<211> 743
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7520538CD1

<400> 24
Met Gly Ala Ser Arg Asp Arg Gly Leu Ala Ala Leu Trp Cys Leu
  1               5                  10                  15
```

Gly Leu Leu Gly Gly Leu Ala Arg Val Ala Gly Thr His Tyr Arg
20 25 30
Tyr Leu Trp Arg Gly Cys Tyr Pro Cys His Leu Gly Gln Ala Gly
35 40 45
Tyr Pro Val Ser Ala Gly Asp Gln Arg Pro Asp Val Asp Glu Cys
50 55 60
Arg Thr His Asn Gly Gly Cys Gln His Arg Cys Val Asn Thr Pro
65 70 75
Gly Ser Tyr Leu Cys Glu Cys Lys Pro Gly Phe Arg Leu His Thr
80 85 90
Asp Ser Arg Thr Cys Leu Ala Ile Asn Ser Cys Ala Leu Gly Asn
95 100 105
Gly Gly Cys Gln His His Cys Val Gln Leu Thr Ile Thr Arg His
110 115 120
Arg Cys Gln Cys Arg Pro Gly Phe Gln Leu Gln Glu Asp Gly Arg
125 130 135
His Cys Val Pro Leu Glu Glu Pro Met Val Asp Leu Asp Gly Glu
140 145 150
Leu Pro Phe Val Arg Pro Leu Pro His Ile Ala Val Leu Gln Asp
155 160 165
Glu Leu Pro Gln Leu Phe Gln Asp Asp Asp Val Gly Ala Asp Glu
170 175 180
Glu Glu Ala Glu Leu Arg Gly Glu His Thr Leu Thr Glu Lys Phe
185 190 195
Val Cys Leu Asp Asp Ser Phe Gly His Asp Cys Ser Leu Thr Cys
200 205 210
Asp Asp Cys Arg Asn Gly Gly Thr Cys Leu Leu Gly Leu Asp Gly
215 220 225
Cys Asp Cys Pro Glu Gly Trp Thr Gly Leu Ile Cys Asn Glu Thr
230 235 240
Cys Pro Pro Asp Thr Phe Gly Lys Asn Cys Ser Phe Ser Cys Ser
245 250 255
Cys Gln Asn Gly Gly Thr Cys Asp Ser Val Thr Gly Ala Cys Arg
260 265 270
Cys Pro Pro Gly Val Ser Gly Thr Asn Cys Glu Asp Gly Cys Pro
275 280 285
Lys Gly Tyr Tyr Gly Lys His Cys Arg Lys Lys Cys Asn Cys Ala
290 295 300
Asn Arg Gly Arg Cys His Arg Leu Tyr Gly Ala Cys Leu Cys Asp
305 310 315
Pro Gly Leu Tyr Gly Arg Phe Cys His Leu Thr Cys Pro Pro Trp
320 325 330
Ala Phe Gly Pro Gly Cys Ser Glu Glu Cys Gln Cys Val Gln Pro
335 340 345
His Thr Gln Ser Cys Asp Lys Arg Asp Gly Ser Cys Ser Cys Lys
350 355 360
Ala Gly Phe Arg Gly Glu Arg Cys Gln Ala Glu Cys Glu Pro Gly
365 370 375
Tyr Phe Gly Pro Gly Cys Trp Gln Ala Cys Thr Cys Pro Val Gly
380 385 390
Val Ala Cys Asp Ser Val Ser Gly Glu Cys Gly Lys Arg Cys Pro
395 400 405
Ala Gly Phe Gln Gly Glu Asp Cys Gly Gln Glu Cys Pro Val Gly
410 415 420
Thr Phe Gly Val Asn Cys Ser Ser Ser Cys Ser Cys Gly Gly Ala
425 430 435
Pro Cys His Gly Val Thr Gly Gln Cys Arg Cys Pro Pro Gly Arg
440 445 450
Thr Gly Glu Asp Cys Glu Ala Asp Cys Pro Glu Gly Arg Trp Gly
455 460 465
Leu Gly Cys Gln Glu Ile Cys Pro Ala Cys Gln His Ala Ala Arg
470 475 480
Cys Asp Pro Glu Thr Gly Ala Cys Leu Cys Leu Pro Gly Phe Val

485 490 495
Gly Ser Arg Cys Gln Asp Val Cys Pro Ala Gly Trp Tyr Gly Pro
500 505 510
Ser Cys Gln Thr Arg Cys Ser Cys Ala Asn Asp Gly His Cys His
515 520 525
Pro Ala Thr Gly His Cys Ser Cys Ala Pro Gly Trp Thr Gly Phe
530 535 540
Ser Cys Gln Arg Ala Cys Asp Thr Gly His Trp Gly Pro Asp Cys
545 550 555
Ser His Pro Cys Asn Cys Ser Ala Gly His Gly Ser Cys Asp Ala
560 565 570
Ile Ser Gly Leu Cys Leu Cys Glu Ala Gly Tyr Val Gly Pro Arg
575 580 585
Cys Glu Gln Pro Cys Ala Lys Gly Thr Phe Gly Pro His Cys Glu
590 595 600
Gly Arg Cys Ala Cys Arg Trp Gly Gly Pro Cys His Leu Ala Thr
605 610 615
Arg Ala Cys Leu Cys Pro Pro Gly Trp Arg Gly Ala Pro Gly Val
620 625 630
Arg Met Pro Ala Cys Gly Ala Gly Leu Glu Arg Pro Val Pro Ser
635 640 645
Ala Ala Ala Ala Arg Leu Ala Leu Pro Ala Thr Thr Ser Leu Gly
650 655 660
Pro Ala Ala Val Pro Leu Ala Ser Leu Ala Pro Ala Ala Ser Arg
665 670 675
Asp Val Arg Pro Gly Gly Met Gly Gln Ala Val Asn Ser Cys Val
680 685 690
Gly Val Ser Thr Gly Ala Pro Val Met Arg Pro Arg Gly Pro Ala
695 700 705
Ala Ala Pro Leu Gly Ser Ser Gly Arg Thr Ala Thr Ser Pro Val
710 715 720
Arg Arg Ala Ala Ser Ala Pro Thr Ala Pro Thr Cys Val Gly Val
725 730 735
Gly Arg Gly Arg Pro Ala Thr Leu
740

<210> 25
<211> 710
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7520667CD1

<400> 25
Met Gly Ser Trp Ala Leu Leu Trp Pro Pro Leu Leu Phe Thr Gly
1 5 10 15
Leu Leu Val Arg Pro Pro Gly Thr Met Ala Gln Ala Gln Tyr Cys
20 25 30
Ser Val Asn Lys Asp Ile Phe Glu Val Glu Glu Asn Thr Asn Val
35 40 45
Thr Glu Pro Leu Val Asp Ile His Val Pro Glu Gly Gln Glu Val
50 55 60
Thr Leu Gly Ala Leu Ser Thr Pro Phe Ala Phe Arg Ile Gln Gly
65 70 75
Asn Gln Leu Phe Leu Asn Val Thr Pro Asp Tyr Glu Glu Lys Ser
80 85 90
Leu Leu Glu Ala Gln Leu Leu Cys Gln Ser Gly Gly Thr Leu Val
95 100 105
Thr Gln Leu Arg Val Phe Val Ser Val Leu Asp Val Asn Asp Asn
110 115 120
Ala Pro Glu Phe Pro Phe Lys Thr Lys Glu Ile Arg Val Glu Glu

```
                125                 130                 135
Asp Thr Lys Val Asn Ser Thr Val Ile Pro Glu Thr Gln Leu Gln
                140                 145                 150
Ala Glu Asp Arg Asp Lys Asp Asp Ile Leu Phe Tyr Thr Leu Gln
                155                 160                 165
Glu Met Thr Ala Gly Ala Ser Asp Tyr Phe Ser Leu Val Ser Val
                170                 175                 180
Asn Arg Pro Ala Leu Arg Leu Asp Arg Pro Leu Asp Phe Tyr Glu
                185                 190                 195
Arg Pro Asn Met Thr Phe Trp Leu Leu Val Arg Asp Thr Pro Gly
                200                 205                 210
Glu Asn Val Glu Pro Lys His Thr Gly Thr Cys Thr Leu Val Leu
                215                 220                 225
Asn Val Val Pro Ala Asp Leu Arg Pro Pro Trp Phe Leu Pro Cys
                230                 235                 240
Thr Phe Ser Asp Gly Tyr Val Cys Ile Gln Ala Gln Tyr His Gly
                245                 250                 255
Ala Val Pro Thr Gly His Ile Leu Pro Ser Pro Leu Val Leu Arg
                260                 265                 270
Pro Gly Pro Ile Tyr Ala Glu Asp Gly Asp Arg Gly Ile Asn Gln
                275                 280                 285
Pro Ile Ile Tyr Ser Ile Phe Arg Gly Asn Val Asn Gly Thr Phe
                290                 295                 300
Ile Ile His Pro Asp Ser Gly Asn Leu Thr Val Ala Arg Ser Val
                305                 310                 315
Pro Ser Pro Met Thr Phe Leu Leu Leu Val Lys Gly Gln Gln Ala
                320                 325                 330
Asp Leu Ala Arg Tyr Ser Val Thr Gln Val Thr Val Glu Ala Val
                335                 340                 345
Ala Ala Ala Gly Ser Pro Pro Arg Phe Pro Gln Arg Leu Tyr Arg
                350                 355                 360
Gly Thr Val Ala Arg Gly Ala Gly Ala Gly Val Val Val Lys Asp
                365                 370                 375
Ala Ala Ala Pro Ser Gln Pro Leu Arg Ile Gln Ala Gln Asp Pro
                380                 385                 390
Glu Phe Ser Asp Leu Asn Ser Ala Ile Thr Tyr Arg Ile Thr Asn
                395                 400                 405
His Ser His Phe Arg Met Glu Gly Glu Val Val Leu Thr Thr Thr
                410                 415                 420
Thr Leu Ala Gln Ala Gly Ala Phe Tyr Ala Glu Val Glu Ala His
                425                 430                 435
Asn Thr Val Thr Ser Gly Thr Ala Thr Thr Val Ile Glu Ile Gln
                440                 445                 450
Val Ser Glu Gln Glu Pro Pro Ser Thr Ala Thr Pro Gly Gly Gly
                455                 460                 465
Thr Ala Gln Thr Pro Glu Ala Gly Thr Ser Gln Pro Met Pro Pro
                470                 475                 480
Gly Met Gly Thr Ser Thr Ser His Gln Pro Thr Thr Pro Gly Gly
                485                 490                 495
Gly Thr Ala Gln Thr Pro Glu Pro Gly Thr Ser Gln Pro Met Pro
                500                 505                 510
Leu Ser Lys Ser Thr Pro Ser Ser Gly Gly Gly Pro Ser Glu Asp
                515                 520                 525
Lys Arg Phe Ser Val Val Asp Met Ala Ala Leu Gly Gly Val Leu
                530                 535                 540
Gly Ala Leu Leu Leu Leu Ala Leu Leu Gly Leu Thr Val Leu Val
                545                 550                 555
His Lys His Tyr Gly Pro Arg Leu Lys Cys Cys Ser Gly Lys Ala
                560                 565                 570
Pro Glu Pro Gln Pro Gln Gly Phe Asp Asn Gln Ala Phe Leu Pro
                575                 580                 585
Asp His Lys Ala Asn Trp Ala Pro Val Pro Ser Pro Thr His Asp
                590                 595                 600
```

Pro Lys Pro Ala Glu Ala Pro Met Pro Ala Glu Pro Ala Pro Pro
605 610 615
Gly Pro Ala Ser Pro Gly Gly Ala Pro Glu Pro Pro Ala Ala Ala
620 625 630
Arg Ala Gly Gly Ser Pro Thr Ala Val Arg Ser Ile Leu Thr Lys
635 640 645
Glu Arg Arg Pro Glu Gly Gly Tyr Lys Ala Val Trp Phe Gly Glu
650 655 660
Asp Ile Gly Thr Glu Ala Asp Val Val Val Leu Asn Ala Pro Thr
665 670 675
Leu Asp Val Asp Gly Ala Ser Asp Ser Gly Ser Gly Asp Glu Gly
680 685 690
Glu Gly Ala Gly Arg Gly Gly Gly Pro Tyr Asp Ala Pro Gly Gly
695 700 705
Asp Asp Ser Tyr Ile
710

<210> 26
<211> 1402
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7517567CD1

<400> 26
Met Glu Gly Asp Arg Val Ala Gly Arg Pro Val Leu Ser Ser Leu
1 5 10 15
Pro Val Leu Leu Leu Leu Gln Leu Leu Met Leu Arg Ala Ala Ala
20 25 30
Leu His Pro Asp Glu Leu Phe Pro His Gly Glu Ser Trp Gly Asp
35 40 45
Gln Leu Leu Gln Glu Gly Asp Asp Glu Ser Ser Ala Val Val Lys
50 55 60
Leu Ala Asn Pro Leu His Phe Tyr Glu Ala Arg Phe Ser Asn Leu
65 70 75
Tyr Val Gly Thr Asn Gly Ile Ile Ser Thr Gln Asp Phe Pro Arg
80 85 90
Glu Thr Gln Tyr Val Asp Tyr Asp Phe Pro Thr Asp Phe Pro Ala
95 100 105
Ile Ala Pro Phe Leu Ala Asp Ile Asp Thr Ser His Gly Arg Gly
110 115 120
Arg Val Leu Tyr Arg Glu Asp Thr Ser Pro Ala Val Leu Gly Leu
125 130 135
Ala Ala Arg Tyr Val Arg Ala Gly Phe Pro Arg Ser Ala Arg Phe
140 145 150
Thr Pro Thr His Ala Phe Leu Ala Thr Trp Glu Gln Val Gly Ala
155 160 165
Tyr Glu Glu Val Lys Arg Gly Ala Leu Pro Ser Gly Glu Leu Asn
170 175 180
Thr Phe Gln Ala Val Leu Ala Ser Asp Gly Ser Asp Ser Tyr Ala
185 190 195
Leu Phe Leu Tyr Pro Ala Asn Gly Leu Gln Phe Leu Gly Thr Arg
200 205 210
Pro Lys Glu Ser Tyr Asn Val Gln Leu Gln Leu Pro Ala Arg Val
215 220 225
Gly Phe Cys Arg Gly Glu Ala Asp Asp Leu Lys Ser Glu Gly Pro
230 235 240
Tyr Phe Ser Leu Thr Ser Thr Glu Gln Ser Val Lys Asn Leu Tyr
245 250 255
Gln Leu Ser Asn Leu Gly Ile Pro Gly Val Trp Ala Phe His Ile
260 265 270

Gly Ser Thr Ser Pro Leu Asp Asn Val Arg Pro Ala Ala Val Gly
275 280 285
Asp Leu Ser Ala Ala His Ser Ser Val Pro Leu Gly Arg Ser Phe
290 295 300
Ser His Ala Thr Ala Leu Glu Ser Asp Tyr Asn Glu Asp Asn Leu
305 310 315
Asp Tyr Tyr Asp Val Asn Glu Glu Glu Ala Glu Tyr Leu Pro Gly
320 325 330
Glu Pro Glu Glu Ala Leu Asn Gly His Ser Ser Ile Asp Val Ser
335 340 345
Phe Gln Ser Lys Val Asp Thr Lys Pro Leu Glu Gly Arg Ile Ser
350 355 360
Pro Pro Asp Ser Asp Leu Ser Ser Pro Leu His Pro Thr Pro Thr
365 370 375
Tyr Trp Pro Phe Tyr Pro Glu Thr Glu Ser Ser Thr Leu Asp Pro
380 385 390
His Thr Lys Glu Gly Thr Ser Leu Gly Glu Val Gly Gly Pro Asp
395 400 405
Leu Lys Gly Gln Val Glu Pro Trp Asp Glu Arg Glu Thr Arg Ser
410 415 420
Pro Ala Pro Pro Glu Val Asp Arg Asp Ser Leu Ala Pro Ser Trp
425 430 435
Glu Thr Pro Pro Pro Tyr Pro Glu Asn Gly Ser Ile Gln Pro Tyr
440 445 450
Pro Asp Gly Gly Pro Val Pro Ser Glu Met Asp Val Pro Pro Ala
455 460 465
His Pro Glu Glu Glu Ile Val Leu Arg Ser Tyr Pro Ala Ser Asp
470 475 480
His Thr Thr Pro Leu Ser Arg Gly Thr His Glu Val Gly Leu Glu
485 490 495
Gly Asn Ile Gly Ser Asn Thr Glu Val Phe Thr Tyr Asn Ala Ala
500 505 510
Asn Lys Glu Thr Cys Glu His Asn His Arg Gln Cys Ser Arg His
515 520 525
Ala Phe Cys Thr Asp Tyr Ala Thr Gly Phe Cys Cys His Cys Gln
530 535 540
Ser Lys Phe Tyr Gly Asn Gly Lys His Cys Leu Pro Glu Gly Ala
545 550 555
Pro His Arg Val Asn Gly Lys Val Ser Gly His Leu His Val Gly
560 565 570
His Thr Pro Val His Phe Thr Asp Val Asp Leu His Ala Tyr Ile
575 580 585
Val Gly Asn Asp Gly Arg Ala Tyr Thr Ala Ile Ser His Ile Pro
590 595 600
Gln Pro Ala Ala Gln Ala Leu Leu Pro Leu Thr Pro Ile Gly Gly
605 610 615
Leu Phe Gly Trp Leu Phe Ala Leu Glu Lys Pro Gly Ser Glu Asn
620 625 630
Gly Phe Ser Leu Ala Gly Ala Ala Phe Thr His Asp Met Glu Val
635 640 645
Thr Phe Tyr Pro Gly Glu Glu Thr Val Arg Ile Thr Gln Thr Ala
650 655 660
Glu Gly Leu Asp Pro Glu Asn Tyr Leu Ser Ile Lys Thr Asn Ile
665 670 675
Gln Gly Gln Val Pro Tyr Val Ser Ala Asn Phe Thr Ala His Ile
680 685 690
Ser Pro Tyr Lys Glu Leu Tyr His Tyr Ser Asp Ser Thr Val Thr
695 700 705
Ser Thr Ser Ser Arg Asp Tyr Ser Leu Thr Phe Gly Ala Ile Asn
710 715 720
Gln Thr Trp Ser Tyr Arg Ile His Gln Asn Ile Thr Tyr Gln Val
725 730 735
Cys Arg His Ala Pro Arg His Pro Ser Phe Pro Thr Thr Gln Gln

740 745 750
Leu Asn Val Asp Arg Val Phe Ala Leu Tyr Asn Asp Glu Glu Arg
755 760 765
Val Leu Arg Phe Ala Val Thr Asn Gln Ile Gly Pro Val Lys Glu
770 775 780
Asp Ser Asp Pro Thr Pro Val Asn Pro Cys Tyr Asp Gly Ser His
785 790 795
Met Cys Asp Thr Thr Ala Arg Cys His Pro Gly Thr Gly Val Asp
800 805 810
Tyr Thr Cys Glu Cys Ala Ser Gly Tyr Gln Gly Asp Gly Arg Asn
815 820 825
Cys Val Asp Glu Asn Glu Cys Ala Thr Gly Phe His Arg Cys Gly
830 835 840
Pro Asn Ser Val Cys Ile Asn Leu Pro Gly Ser Tyr Arg Cys Glu
845 850 855
Cys Arg Ser Gly Tyr Glu Phe Ala Asp Asp Arg His Thr Cys Ile
860 865 870
Leu Ile Thr Pro Pro Ala Asn Pro Cys Glu Asp Gly Ser His Thr
875 880 885
Cys Ala Pro Ala Gly Gln Ala Arg Cys Val Tyr His Gly Gly Ser
890 895 900
Thr Phe Ser Cys Ala Cys Leu Pro Gly Tyr Ala Gly Asp Gly His
905 910 915
Gln Cys Thr Asp Val Asp Glu Cys Ser Glu Asn Arg Cys His Pro
920 925 930
Ala Ala Thr Cys Tyr Asn Thr Pro Gly Ser Phe Ser Cys Arg Cys
935 940 945
Gln Pro Gly Tyr Tyr Gly Asp Gly Phe Gln Cys Ile Pro Asp Ser
950 955 960
Thr Ser Ser Leu Thr Pro Cys Glu Gln Gln Gln Arg His Ala Gln
965 970 975
Ala Gln Tyr Ala Tyr Pro Gly Ala Arg Phe His Ile Pro Gln Cys
980 985 990
Asp Glu Gln Gly Asn Phe Leu Pro Leu Gln Cys His Gly Ser Thr
995 1000 1005
Gly Phe Cys Trp Cys Val Asp Pro Asp Gly His Glu Val Pro Gly
1010 1015 1020
Thr Gln Thr Pro Pro Gly Ser Thr Pro Pro His Cys Gly Pro Ser
1025 1030 1035
Pro Glu Pro Thr Gln Arg Pro Pro Thr Ile Cys Glu Arg Trp Arg
1040 1045 1050
Glu Asn Leu Leu Glu His Tyr Gly Gly Thr Pro Arg Asp Asp Gln
1055 1060 1065
Tyr Val Pro Gln Cys Asp Asp Leu Gly His Phe Ile Pro Leu Gln
1070 1075 1080
Cys His Gly Lys Ser Asp Phe Cys Trp Cys Val Asp Lys Asp Gly
1085 1090 1095
Arg Glu Val Gln Gly Thr Arg Ser Gln Pro Gly Thr Thr Pro Ala
1100 1105 1110
Cys Ile Pro Thr Val Ala Pro Pro Met Val Arg Pro Thr Pro Arg
1115 1120 1125
Pro Asp Val Thr Pro Pro Ser Val Gly Thr Phe Leu Leu Tyr Thr
1130 1135 1140
Gln Gly Gln Gln Ile Gly Tyr Leu Pro Leu Asn Gly Thr Arg Leu
1145 1150 1155
Gln Lys Asp Ala Ala Lys Thr Leu Leu Ser Leu His Gly Ser Ile
1160 1165 1170
Ile Val Gly Ile Asp Tyr Asp Cys Arg Glu Arg Met Val Tyr Trp
1175 1180 1185
Thr Asp Val Ala Gly Arg Thr Ile Ser Arg Ala Gly Leu Glu Leu
1190 1195 1200
Gly Ala Glu Pro Glu Thr Ile Val Asn Ser Gly Leu Ile Ser Pro
1205 1210 1215

Glu Gly Leu Ala Ile Asp His Ile Arg Arg Thr Met Tyr Trp Thr
1220 1225 1230
Asp Ser Val Leu Asp Lys Ile Glu Ser Ala Leu Leu Asp Gly Ser
1235 1240 1245
Glu Arg Lys Val Leu Phe Tyr Thr Asp Leu Val Asn Pro Arg Ala
1250 1255 1260
Ile Ala Val Asp Pro Ile Arg Gly Asn Leu Tyr Trp Thr Asp Trp
1265 1270 1275
Asn Arg Glu Ala Pro Lys Ile Glu Thr Ser Ser Leu Asp Gly Glu
1280 1285 1290
Asn Arg Arg Ile Leu Ile Asn Thr Asp Ile Gly Leu Pro Asn Gly
1295 1300 1305
Leu Thr Phe Asp Pro Phe Ser Lys Leu Leu Cys Arg Ala Asp Ala
1310 1315 1320
Gly Thr Lys Lys Leu Glu Cys Thr Leu Pro Asp Gly Thr Gly Arg
1325 1330 1335
Arg Val Ile Gln Asn Asn Leu Lys Tyr Pro Phe Ser Ile Val Ser
1340 1345 1350
Tyr Ala Asp His Phe Tyr His Thr Asp Trp Arg Arg Asp Gly Val
1355 1360 1365
Val Ser Val Asn Lys His Ser Gly Gln Phe Thr Asp Glu Tyr Leu
1370 1375 1380
Pro Glu Gln Arg Ser His Leu Tyr Gly Ile Thr Ala Val Tyr Pro
1385 1390 1395
Tyr Cys Pro Thr Gly Arg Lys
1400

<210> 27
<211> 288
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7519867CD1

<400> 27
Met Ser Asp Ser Lys Glu Pro Arg Leu Gln Gln Leu Gly Leu Leu
1 5 10 15
Glu Glu Glu Gln Leu Arg Gly Leu Gly Phe Arg Gln Thr Arg Gly
20 25 30
Tyr Lys Ser Leu Ala Val Ser Lys Val Pro Ser Ser Ile Ser Gln
35 40 45
Glu Gln Ser Arg Gln Asp Ala Ile Tyr Gln Asn Leu Thr Gln Leu
50 55 60
Lys Ala Ala Val Gly Glu Leu Ser Glu Lys Ser Lys Leu Gln Glu
65 70 75
Ile Tyr Gln Glu Leu Thr Gln Leu Lys Ala Ala Val Gly Glu Leu
80 85 90
Pro Glu Lys Ser Lys Leu Gln Glu Ile Tyr Gln Glu Leu Thr Arg
95 100 105
Leu Lys Ala Ala Val Gly Glu Leu Pro Glu Lys Ser Lys Leu Gln
110 115 120
Glu Ile Tyr Gln Glu Leu Thr Gln Leu Lys Ala Ala Val Glu Arg
125 130 135
Leu Cys His Pro Cys Pro Trp Glu Trp Thr Phe Phe Gln Gly Asn
140 145 150
Cys Tyr Phe Met Ser Asn Ser Gln Arg Asn Trp His Asp Ser Ile
155 160 165
Thr Ala Cys Lys Glu Val Gly Ala Gln Leu Val Val Ile Lys Ser
170 175 180
Ala Glu Glu Gln Asn Phe Leu Gln Leu Gln Ser Ser Arg Ser Asn
185 190 195

```
Arg Phe Thr Trp Met Gly Leu Ser Asp Leu Asn Gln Glu Gly Thr
                200                 205                 210
Trp Gln Trp Val Asp Gly Ser Pro Leu Leu Pro Ser Phe Lys Gln
                215                 220                 225
Tyr Trp Asn Arg Gly Glu Pro Asn Asn Val Gly Glu Glu Asp Cys
                230                 235                 240
Ala Glu Phe Ser Gly Asn Gly Trp Asn Asp Asp Lys Cys Asn Leu
                245                 250                 255
Ala Lys Phe Trp Ile Cys Lys Lys Ser Ala Ala Ser Cys Ser Arg
                260                 265                 270
Asp Glu Glu Gln Phe Leu Ser Pro Ala Pro Ala Thr Pro Asn Pro
                275                 280                 285
Pro Pro Ala
```

<210> 28
<211> 481
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7519911CD1

<400> 28

```
Met Gly Lys Leu Lys Pro Gly Arg Val Glu Trp Leu Ala Ser Gly
  1               5                  10                  15
His Thr Glu Arg Pro His Leu Phe Gln Asn Leu Leu Leu Phe Leu
                 20                  25                  30
Trp Ala Leu Leu Asn Cys Gly Leu Gly Val Ser Ala Gln Gly Pro
                 35                  40                  45
Gly Glu Trp Thr Pro Trp Val Ser Trp Thr Arg Cys Ser Ser Ser
                 50                  55                  60
Cys Gly Arg Gly Val Ser Val Arg Ser Arg Arg Cys Leu Arg Leu
                 65                  70                  75
Pro Gly Glu Glu Pro Cys Trp Gly Asp Ser His Glu Tyr Arg Leu
                 80                  85                  90
Cys Gln Leu Pro Asp Cys Pro Pro Gly Ala Val Pro Phe Arg Asp
                 95                 100                 105
Leu Gln Cys Ala Leu Tyr Asn Gly Arg Pro Val Leu Gly Thr Gln
                110                 115                 120
Lys Thr Tyr Gln Trp Val Pro Phe His Gly Ala Pro Asn Gln Cys
                125                 130                 135
Asp Leu Asn Cys Leu Ala Glu Gly His Ala Phe Tyr His Ser Phe
                140                 145                 150
Gly Arg Val Leu Asp Gly Thr Ala Cys Ser Pro Gly Ala Gln Gly
                155                 160                 165
Val Cys Val Ala Gly Arg Cys Leu Ser Ala Gly Cys Asp Gly Leu
                170                 175                 180
Leu Gly Ser Gly Ala Leu Glu Asp Arg Cys Gly Arg Cys Gly Gly
                185                 190                 195
Ala Asn Asp Ser Cys Leu Phe Val Gln Arg Val Phe Arg Asp Ala
                200                 205                 210
Gly Ala Phe Ala Gly Tyr Trp Asn Val Thr Leu Ile Pro Glu Gly
                215                 220                 225
Ala Arg His Ile Arg Val Glu His Arg Ser Arg Asn His Leu Ala
                230                 235                 240
Leu Met Gly Gly Asp Gly Arg Tyr Val Leu Asn Gly His Trp Val
                245                 250                 255
Val Ser Pro Pro Gly Thr Tyr Glu Ala Ala Gly Thr His Val Val
                260                 265                 270
Tyr Thr Arg Asp Thr Gly Pro Gln Glu Thr Leu Gln Ala Ala Gly
                275                 280                 285
```

```
Pro Thr Ser His Asp Leu Leu Leu Gln Val Leu Leu Gln Glu Pro
                290                 295                 300
Asn Pro Gly Ile Glu Phe Glu Phe Trp Leu Pro Arg Glu Arg Tyr
                305                 310                 315
Ser Pro Phe Gln Ala Arg Val Gln Ala Leu Gly Trp Pro Leu Arg
                320                 325                 330
Gln Pro Gln Pro Arg Gly Val Glu Pro Gln Pro Pro Ala Ala Pro
                335                 340                 345
Ala Val Thr Pro Ala Gln Thr Pro Thr Leu Ala Pro Asp Pro Cys
                350                 355                 360
Pro Pro Cys Pro Asp Thr Arg Gly Arg Ala His Arg Leu Leu His
                365                 370                 375
Tyr Cys Gly Ser Asp Phe Val Phe Gln Ala Arg Val Leu Gly His
                380                 385                 390
His His Gln Ala Gln Glu Thr Arg Tyr Glu Val Arg Ile Gln Leu
                395                 400                 405
Val Tyr Lys Asn Arg Ser Pro Leu Arg Ala Arg Glu Tyr Val Trp
                410                 415                 420
Ala Pro Gly His Cys Pro Cys Pro Met Leu Ala Pro His Arg Asp
                425                 430                 435
Tyr Leu Met Ala Val Gln Arg Leu Val Ser Pro Asp Gly Thr Gln
                440                 445                 450
Asp Gln Leu Leu Leu Pro His Ala Gly Tyr Ala Arg Pro Trp Ser
                455                 460                 465
Pro Ala Glu Asp Ser Arg Ile Arg Leu Thr Ala Arg Arg Cys Pro
                470                 475                 480
Gly


<210> 29
<211> 460
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7520005CD1

<400> 29
Met Gly Gly Pro Arg Ala Trp Ala Leu Leu Cys Leu Gly Leu Leu
  1             5                  10                  15
Leu Pro Gly Gly Gly Ala Ala Trp Ser Ile Gly Ala Ala Pro Phe
                20                  25                  30
Ser Gly Arg Arg Asn Trp Cys Ser Tyr Val Val Thr Arg Thr Ile
                35                  40                  45
Ser Cys His Val Gln Asn Gly Thr Tyr Leu Gln Arg Val Leu Gln
                50                  55                  60
Asn Cys Pro Trp Pro Met Ser Cys Pro Gly Ser Ser Tyr Arg Thr
                65                  70                  75
Val Val Arg Pro Thr Tyr Lys Val Met Tyr Lys Val Val Thr Ala
                80                  85                  90
Arg Glu Trp Arg Cys Cys Pro Gly His Ser Gly Val Ser Cys Glu
                95                 100                 105
Glu Val Ala Ala Ser Ser Ala Ser Leu Glu Pro Met Trp Ser Gly
                110                 115                 120
Ser Thr Met Arg Arg Met Ala Leu Arg Pro Thr Ala Phe Ser Gly
                125                 130                 135
Cys Leu Asn Cys Ser Lys Val Ser Glu Leu Thr Glu Arg Leu Lys
                140                 145                 150
Val Leu Glu Ala Lys Met Thr Met Leu Thr Val Ile Glu Gln Pro
                155                 160                 165
Val Pro Pro Thr Pro Ala Thr Pro Glu Asp Pro Ala Pro Leu Trp
                170                 175                 180
```

Gly Pro Pro Pro Ala Gln Gly Ser Pro Gly Asp Gly Gly Leu Gln
185 190 195
Gly Leu Pro Gly Ala Ile Glu Ser Val Arg Val Pro Leu Leu Pro
200 205 210
Arg Asn Asp Gln Val Gly Ala Trp Gly Leu Pro Gly Pro Thr Gly
215 220 225
Pro Lys Gly Asp Ala Gly Ser Arg Gly Pro Met Gly Met Arg Gly
230 235 240
Pro Pro Gly Pro Gln Gly Pro Pro Gly Ser Pro Gly Arg Ala Gly
245 250 255
Ala Val Gly Thr Pro Gly Glu Arg Gly Pro Pro Gly Pro Pro Gly
260 265 270
Pro Pro Gly Pro Pro Gly Pro Pro Ala Pro Val Gly Pro Pro His
275 280 285
Ala Arg Ile Ser Gln His Gly Asp Pro Leu Leu Ser Asn Thr Phe
290 295 300
Thr Glu Thr Asn Asn His Trp Pro Gln Gly Pro Thr Gly Pro Pro
305 310 315
Gly Pro Pro Gly Pro Met Gly Pro Pro Gly Pro Pro Gly Pro Thr
320 325 330
Gly Val Pro Gly Ser Pro Gly His Ile Gly Pro Pro Gly Pro Thr
335 340 345
Gly Pro Lys Gly Ile Ser Gly His Pro Gly Glu Lys Gly Glu Arg
350 355 360
Gly Leu Arg Gly Glu Pro Gly Pro Gln Gly Ser Ala Gly Gln Arg
365 370 375
Gly Glu Pro Gly Pro Lys Gly Asp Pro Gly Glu Lys Ser His Trp
380 385 390
Gly Glu Gly Leu His Gln Leu Arg Glu Ala Leu Lys Ile Leu Ala
395 400 405
Glu Arg Val Leu Ile Leu Glu Thr Met Ile Gly Leu Tyr Glu Pro
410 415 420
Glu Leu Gly Ser Gly Ala Gly Pro Ala Gly Thr Gly Thr Pro Ser
425 430 435
Leu Leu Arg Gly Lys Arg Gly Gly His Ala Thr Asn Tyr Arg Ile
440 445 450
Val Ala Pro Arg Ser Arg Asp Glu Arg Gly
455 460

<210> 30
<211> 415
<212> PRT
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7520484CD1

<400> 30
Met Gly Gly Pro Arg Ala Trp Ala Leu Leu Cys Leu Gly Leu Leu
1 5 10 15
Leu Pro Gly Gly Gly Ala Ala Trp Ser Ile Gly Ala Ala Pro Phe
20 25 30
Ser Gly Arg Arg Asn Trp Cys Ser Tyr Val Val Thr Arg Thr Ile
35 40 45
Ser Cys His Val Gln Asn Gly Thr Tyr Leu Gln Arg Val Leu Gln
50 55 60
Asn Cys Pro Trp Pro Met Ser Cys Pro Gly Ser Ser Tyr Arg Thr
65 70 75
Val Val Arg Pro Thr Tyr Lys Val Met Tyr Lys Ile Val Thr Ala
80 85 90
Arg Glu Trp Arg Cys Cys Pro Gly His Ser Gly Val Ser Cys Glu
95 100 105

```
Glu Gly Cys Leu Asn Cys Ser Lys Val Ser Glu Leu Thr Glu Arg
                110                 115                 120
Leu Lys Val Leu Glu Ala Lys Met Thr Met Leu Thr Val Ile Glu
                125                 130                 135
Gln Pro Val Pro Pro Thr Pro Ala Thr Pro Glu Asp Pro Ala Pro
                140                 145                 150
Leu Trp Gly Pro Pro Pro Ala Gln Gly Ser Pro Gly Asp Gly Gly
                155                 160                 165
Leu Gln Asp Gln Val Gly Ala Trp Gly Leu Pro Gly Pro Thr Gly
                170                 175                 180
Pro Lys Gly Asp Ala Gly Ser Arg Gly Pro Met Gly Met Arg Gly
                185                 190                 195
Pro Pro Gly Pro Gln Gly Pro Arg Gly Ser Pro Gly Arg Ala Gly
                200                 205                 210
Ala Val Gly Thr Pro Gly Glu Arg Gly Pro Pro Gly Pro Pro Gly
                215                 220                 225
Pro Pro Gly Pro Pro Gly Pro Pro Ala Pro Val Gly Pro Pro His
                230                 235                 240
Ala Arg Ile Ser Gln His Gly Asp Pro Leu Leu Ser Asn Thr Phe
                245                 250                 255
Thr Glu Thr Asn Asn His Trp Pro Gln Gly Pro Thr Gly Pro Pro
                260                 265                 270
Gly Pro Pro Gly Pro Met Gly Pro Pro Gly Pro Pro Gly Pro Thr
                275                 280                 285
Gly Val Pro Gly Ser Pro Gly His Ile Gly Pro Pro Gly Pro Thr
                290                 295                 300
Gly Pro Lys Gly Ile Ser Gly His Pro Gly Glu Lys Gly Glu Arg
                305                 310                 315
Gly Leu Arg Gly Glu Pro Gly Pro Gln Gly Ser Ala Gly Gln Arg
                320                 325                 330
Gly Glu Pro Gly Pro Lys Gly Asp Pro Gly Glu Lys Ser His Trp
                335                 340                 345
Gly Glu Gly Leu His Gln Leu Arg Glu Ala Leu Lys Ile Leu Ala
                350                 355                 360
Glu Arg Val Leu Ile Leu Glu Thr Met Ile Gly Leu Tyr Glu Pro
                365                 370                 375
Glu Leu Gly Ser Gly Ala Gly Pro Ala Gly Thr Gly Thr Pro Ser
                380                 385                 390
Leu Leu Arg Gly Lys Arg Gly Gly His Ala Thr Asn Tyr Arg Ile
                395                 400                 405
Val Ala Pro Arg Ser Arg Asp Glu Arg Gly
                410                 415
```

<210> 31
<211> 833
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7514594CB1

<400> 31

```
tctacatcta tcggagctga acttcctaaa agacaaagtg tttatctttc aagattcatt 60
ctccctgaat cttaccaaca aaacactcct gaggagaaag aaagagaggg agggagagaa 120
aaagagagag agagaaacaa aaaaccaaag agagagaaaa aatgaattca tctaaatcat 180
ctgaaacaca atgcacagag agaggatgct tctcttccca aatgttctta tggactgttg 240
ctgggatccc catcctattt ctcagtgcct gtttcatcac cagatgtgtt gtgacatttc 300
gcatctttca aacctgtgat gagaaaaagt ttcagctacc tgagaatttc acagagctct 360
cctgctacaa ttatggatca ggtataatag tcccaaagga attcctttcc tacaagaaac 420
ctaaaatgag agaggttttt attggactgt cagaccaggt tgtcgagggt cggtggcaat 480
gggtggacgg cacacctttg acaaagtctc tgagcttctg ggatgtaggg gagcccaaca 540
acatagctac cctggaggac tgtgccacca tgagagactc ttcaaaccca aggcaaaatt 600
```

ggaatgatgt aacctgtttc ctcaattatt ttcggatttg tgaaatggta ggaataaatc 660
ctttgaacaa aggaaaatct ctttaagaac agaaggcaca actcaaatgt gtaaagaagg 720
aagagcaaga acatggccac acccaccgcc ccacacgaga aatttgtgcg ctgaacttca 780
aaggacttca taagtatttg ttactctgat ataaataaaa ataagtagtt tta 833

<210> 32
<211> 1026
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7515764CB1

<400> 32
tagaagccgg gagcttccct gatggtgccg ccgcctccga gccggggagg agctgccagg 60
ggccagctgg gcaggagcct gggtccgctg ctgctgctcc tggcgttggg acacacgtgg 120
acctacagag aggagccgga ggacggcgac agagaaatct gctcagagag caaaatcgcg 180
acgactaaat acccgtgtct gaagtcttca ggcgagctca ccacatgcta caggaaaaag 240
tgctgcaaag gatataaatt tgttcttgga caatgcatcc cagaagatta cgacgtttgt 300
gccgaggctc cctgtgaaca gcagtgcacg gacaactttg gccgagtgct gtgtacttgt 360
tatccgggat accgatatga ccgggagaga caccggaagc gggagaagcc atactgtctg 420
gatattgatg agtgtgccag cagcaatggg acgctgtgtg cccacatctg catcaatacc 480
ttgggcagct accgctgcga gtgccgggaa ggctacatcc gggaagatga tgggaagaca 540
tgtaccaggg gagacaaata tcccaatgac actggctcac caggaccaaa gggaagccca 600
ggcttccccg gtatgccagg ccctcctggg cagcccggcc cacggggctc aatgggaccc 660
atgggaccat ctcctgatct gtcccacatt aagcaaggcc ggaggggccc tgtgggtcca 720
ccaggggcac caggaagaga tggttctaag ggggagagag gagcgcctgg gcccagaggg 780
tctccaggac ccctggttct ttcgacttcc tgctacttat gctggctgac atccgcaatg 840
acatcactga gctgcaggaa aaggtgttcg ggcaccggac tcactcttca gcagaggagt 900
tccctttacc tcaggaattt cccagctacc cagaagccat ggacctgggc tctggagatg 960
accatccaag aagaactgag acaagagact tgagagcccc cagagacttc tacccatagc 1020
acatcc 1026

<210> 33
<211> 809
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7514575CB1

<400> 33
tagcatggca cccctgctgc ccatccggac cttgcccttg atcctgattc tgctggctct 60
gctgtcccca ggggctgcag acttcaacat ctcaagcctc tctggtctgc tgtccccggc 120
gctaacggag agcctgctgg ttgccttgcc cccctgtcac ctcacaggag gcaatgccac 180
actgatggtc cggagagcca atgacagcaa agtggtgacg tccagctttg tggtgcctcc 240
gtgccgtggg cgcagggaac tgcatttcct acctagtgaa gaaggggaca gccactgagt 300
ccagcagaga gatcccaatg tccacactcc ctcgaaggaa catggaatcc attgggctgg 360
gtatggcccg cacaggggc atggtggtca tcacggtgct gctctctgtc gccatgttcc 420
tgctggtgct gggcttcatc attgccctgg cactgggctc ccgcaagtaa ggaggtctgc 480
ccggagcagc agcttctcca ggaagcccag ggcaccatcc agctccccag cccacctgct 540
cccaggcccc aggcctgtgg ctcccttggt gccctcgcct cctcctcctg ccctcctctc 600
ccctagagcc ctctcctccc tctgtccctc tccttgcccc cagtgcctca ccttccaaca 660
ctccattatt cctctcaccc cactcctgtc agagttgact ttcctcccat tttaccactt 720
taaacacccc cataacaatt cccccatcct tcagtgaact aagtccctat aataaaggct 780
gaggctgcat ctgccacata aaaaaaaaa 809

<210> 34
<211> 1058
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7514576CB1

<400> 34
gggggatgtg ctgcaaggcg attaagttgg gtaacgccag ggttttccca gtcacgacgt 60
tgtaaaacga cggccagtga attgtgcggc catttaggtg acactataga atacagcggc 120
cgcgagctcg ggccccccaca cgtgtggtct agagctagcc taggctcgag aagcttgtcg 180
acgaattcag attagcatgg cacccctgct gcccatccgg accttgccct tgatcctgat 240
tctgctggct ctgctgtccc caggggctgc agacttcaac atctcaagcc tctctggtct 300
gctgtccccg gcgctaacgg agagcctgct ggttgccttg ccccccctgtc acctcacagg 360
aggcaatgcc acactgatgg tccggagagc caatgacagc aaagtggtga cgtccagctt 420
tgtggtgcct ccgtgccgtg ggcgcaggga actggtgagt gtggtggaca gtggtgctgg 480
cttcacagtc actcggctca gtgcatacca gcatttccta cctagtgaag aaggggacag 540
ccactgagtc cagcagagag atcccaatgt ccacactccc tcgaaggaac atggaatcca 600
ttgggctggg tatggcccgc acagggggca tggtggtcat cacggtgctg ctctctgtcg 660
ccatgttcct gctggtgctg ggcttcatca ttgctctggc actgggctcc cgcaagtaag 720
gaggtctgcc cggagcagca gcttctccag gaagcccagg gcaccatcca gctccccagc 780
ccacctgctc ccaggcccca ggcctgtggc tcccttggtg ccctcgcctc ctcctcctgc 840
cctcctctcc cctagagccc tctcctccct ctgtccctct ccttgccccc agtgcctcac 900
cttccaacac tccattattc ctctcacccc actcctgtca gagttgactt tcctcccatt 960
ttaccacttt aaacaccccc ataacaattc ccccatcctt cagtgaacta agtccctata 1020
ataaaggctg aggctgcatc tgccacataa aaaaaaaa 1058

<210> 35
<211> 534
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7515593CB1

<400> 35
tatgaattca tctaaatcat ctgaaacaca atgcacagag agaggatgct tctcttccca 60
aatgttctta tggactgttg ctgggatccc catcctattt ctcagtgcct gtttcatcac 120
cagatgtgtt gtgacatttc gcatctttca aacctgtgat gagaaaaagt ttcagctacc 180
tgagaatttc acagagctct cctgctacaa ttatggatca ggaattcctt tcctacaaga 240
aacctaaaat gagagagttt tttattggac tgtcagacca ggttgtcgag ggtcagtggc 300
aatgggtgga cggcacacct ttgacaaagt ctctgagctt ctgggatgta ggggagccca 360
acaacatagc taccctggag gactgtgcca ccatgagaga ctcttcaaac ccaaggcaaa 420
attggaatga tgtaacctgt ttcctcaatt attttcggat ttgtgaaatg gtaggaataa 480
atcctttgaa caaaggaaaa tctctttaag aacagaaggc acaactcaaa tgta 534

<210> 36
<211> 2370
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7517754CB1

<400> 36
ggtgcgagga ggtcgaggag gcgccttggc acccgactct ggcatcccgc acgtccgaca 60
tcagccctgc cctccttctc aggggcttcc attcattttg tgccaaaagg gaactgccgc 120
ccgtccgtct gcccgcaggc attgcccaag ccagccgagc cgccagagcc gcgggccgcg 180
ggggtgtcgc gggcccaacc ccaggatgct cccctgcgcc tcctgcctac ccgggtctct 240
actgctctgg gcgctgctac tgttgctctt gggatcagct tctcctcagg attctgaaga 300
gcccgacagc tacacggaat gcacagatgg ctatgagtgg gacccagaca gccagcactg 360
ccgggatgtc aacgagtgtc tgaccatccc tgaggcctgc aagggggaaa tgaagtgcat 420
caaccactac gggggctact tgtgcctgcc ccgctccgct gccgtcatca acgacctaca 480
cggcgaggga cccccgccac cagtgcctcc cgctcaacac cccaacccct gcccaccagg 540

ctatgagccc gacgatcagg acagctgtgt gggacctgcc gcaggtggcc ccagcttgtg 600
tgccactgct ccagccagct ccgggcccct cccctcggtt gctgcagaga cactgcagga 660
agtctgcctt ccagtgtctg tcccctggga gccacaggcc ttggaggaag gacagtccag 720
gaatcagtgt tgctggggat gtttggacag gaccttattt cctgtgcaac ccccaacatg 780
tggttttaga ggttaggggt gcctttcctg atggctccag gctggtgaac tcttacgcgt 840
gctgttgaga gaaccgaata caagtgttgg aactatctgt catgttgggt ggtgtgcggg 900
atcccaggca ggggctgagt ggtgtctctc gcagatgtgg acgagtgtgc ccaggccctg 960
cacgactgtc gccccagcca ggactgccat aacttgcctg gctcctatca gtgcacctgc 1020
cctgatggtt accgcaagat cgggcccgag tgtgtggaca tagacgagtg ccgctaccgc 1080
tactgccagc accgctgcgt gaacctgcct ggctccttcc gctgccagtg cgagccgggc 1140
ttccagctgg ggcctaacaa ccgctcctgt gttgatgtga acgagtgtga catgggggcc 1200
ccatgcgagc agcgctgctt caactcctat gggaccttcc tgtgtcgctg ccaccagggc 1260
tatgagctgc atcgggatgg cttctcctgc agtgatattg atgagtgtag ctactccagc 1320
tacctctgtc agtaccgctg cgtcaacgag ccaggccgtt tctcctgcca ctgcccacag 1380
ggttaccagc tgctggccac acgcctctgc caagacattg atgagtgtga gtctggtgcg 1440
caccagtgct ccgaggccca aacctgtgtc aacttccatg gggctaccg ctgcgtggac 1500
accaaccgct gcgtggagcc ctacatccag gtctctgaga accgctgtct ctgcccggcc 1560
tccaaccctc tatgtcgaga gcagccttca tccattgtgc accgctacat gaccatcacc 1620
tcggagcgga gcgtgcccgc tgacgtgttc cagatccagg cgacctccgt ctaccccggt 1680
gcctacaatg cctttcagat ccgtgctgga aactcgcagg gggactttta cattaggcaa 1740
atcaacaacg tcagcgccat gctggtcctc gcccggccgg tgacgggccc ccgggagtac 1800
gtgctggacc tggagatggt caccatgaat tccctcatga gctaccgggc cagctctgta 1860
ctgaggctca ccgtctttgt aggggcctac accttctgag gagcaggagg gagccacccт 1920
ccctgcagct accctagctg aggagcctgt tgtgaggggc agaatgagaa aggcaataaa 1980
gggagaaaga aagtcctggt ggctgaggtg ggcgggtcac actgcaggaa gcctcaggct 2040
ggggcagggt ggcacttggg ggggcaggcc aagttcacct aaatgggggt ctctatatgt 2100
tcaggcccag ggcccccat tgacaggagc tgggagctct gcaccacgag cttcagtcac 2160
cccgagagga gaggaggtaa cgaggagggc ggactccagg ccccggccca gagatttgga 2220
cttggctggc ttgcaggggt cctaagaaac tccactctgg acagcgccag gaggccctgg 2280
gttccattcc taactctgcc tcaaactgta catttggata agccctagta gttccctggg 2340
cctgtttttc tataaaacga ggcaactgga 2370

<210> 37
<211> 4695
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7517819CB1

<400> 37
aggtgaacag gtcctcacgc ccagctccgc gccctcacgc gctctcgccg ggaccccgct 60
tccgctggca gccatgggcc ccggccccag ccgcgcgccc cgcgccccac gcctgatgct 120
ctgtgcgctc gccttgatgg tggcggccgg cggctgcgtc gtctccgcct tcaacctgga 180
tacccgattc ctggtagtga aggaggccgg gaacccgggc agcctcttcg gctactcggt 240
cgccctccat cggcagacag agcggcagca gcgctacctg ctcctggctg gtgccccccg 300
ggagctcgct gtgcccgatg gctacaccaa ccggactggt gctgtgtacc tgtgcccact 360
cactgcccac aaggatgact gtgagcggat gaacatcaca gtgaaaaatg accctggcca 420
tcacattatt gaggacatgt ggcttggagt gactgtggcc agccagggcc ctgcaggcag 480
agttctggtc tgtgcccacc gctacaccca ggtgctgtgg tcagggtcag aagaccagcg 540
gcgcatggtg ggcaagtgct acgtgcgagg caatgaccta gagctggact ccagtgatga 600
ctggcagacc taccacaacg agatgtgcaa tagcaacaca gactacctgg agacgggcat 660
gtgccagctg ggcaccagcg gtggcttcac ccagaacact gtgtacttcg gcgccccccgg 720
tgcctacaac tggaaaggaa acagctacat gattcagcgc aaggagtggg acttatctga 780
gtatagttac aaggacccag aggaccaagg aaacctctat attgggtaca cgatgcaggt 840
aggcagcttc atcctgcacc ccaaaaacat caccattgtg acaggtgccc cacggcaccg 900
acatatgggc gcggtgttct tgctgagcca ggaggcaggc ggagacctgc ggaggaggca 960
ggtgctggag ggctcgcagg tgggcgccta ttttggcagc gccattgccc tggcagacct 1020
gaacaatgat gggtggcagg acctcctggt gggcgccccc tactacttcg agaggaaaga 1080
ggaagtaggg ggtgccatct atgtcttcat gaaccaggcg ggaacctcct tccctgctca 1140
cccctcactc cttcttcatg gccccagtgg ctctgccttt ggtttatctg tggccagcat 1200
tggtgacatc aaccaggatg gatttcagga tattgctgtg ggagctccgt ttgaaggctt 1260

gggcaaagtg tacatctatc acagtagctc taaggggctc cttagacagc cccagcaggt 1320
aatccatgga gagaagctgg gactgcctgg gttggccacc ttcggctatt ccctcagtgg 1380
gcagatggat gtggatgaga acttctaccc agaccttcta gtgggaagcc tgtcagacca 1440
cattgtgctg ctgcgggccc ggcccgtcat caacatcgtc cacaagacct tggtgcccag 1500
gccagctgtg ctggaccctg cactttgcac ggccacctct tgtacagagg cgagagctag 1560
ggcctggacg tgtgtgtccc actgtgaccc gccctcctgt acatccctcc aacagtgtgc 1620
aagtggagct gtgctttgct tacaaccaga gtgccgggaa ccccaactac aggcgaaaca 1680
tcaccctggc ctacactctg gaggctgaca gggaccgccg gccgccccgg ctccgctttg 1740
ccggcagtga gtccgctgtc ttccacggct tcttctccat gcccgagatg cgctgccaga 1800
agctggagct gctcctgatg gacaacctcc gtgacaaact ccgccccatc atcatctcca 1860
tgaactactc tttacctttg cggatgcccg atcgcccccg gctggggctg cggtccctgg 1920
acgcctaccc gatcctcaac caggcacagg ctctggagaa ccacactgag gtccagttcc 1980
agaaggagtg cgggcctgac aacaagtgtg agagcaactt gcagatgcgg gcagccttcg 2040
tgtcagagca gcagcagaag ctgagcaggc tccagtacag cagagacgtc cggaaattgc 2100
tcctgagcat caacgtgacg aacacccgga cctcggagcg ctccggggag gacgcccacg 2160
aggcgctgct caccctggtg gtgcctcccg ccctgctgct gtcctcagtg cgcccccccg 2220
gggcctgcca agctaatgag accatctttt gcgagctggg gaaccccttc aaacggaacc 2280
agaggatgga gctgctcatc gcctttgagg tcatcggggt gaccctgcac acaagggacc 2340
ttcaggtgca gctgcagctc tccacgtcga gtcaccagga caacctgtgg cccatgatcc 2400
tcactctgct ggtggactat acactccaga cctcgcttag catggtaaat caccggctac 2460
aaagcttctt tggggggaca gtgatgggtg agtctggcat gaaaactgtg gaggatgtag 2520
gaagcccccct caagtatgaa ttccaggtgg gcccaatggg ggaggggctg gtgggcctgg 2580
ggaccctggt cctaggtctg gagtggccct acgaagtcag caatggcaag tggctgctgt 2640
atcccacgga gatcaccgtc catggcaatg ggtcctggcc ctgccgacca cctggagacc 2700
ttatcaaccc tctcaacctc actctttctg accctgggga caggccatca tccccacagc 2760
gcaggcggcg acagctggat ccagggggag gccagggccc cccacctgtc actctggctg 2820
ctgccaaaaa agccaagtct gagactgtgc tgacctgtgc cacagggcgt gcccactgtg 2880
tgtggctaga gtgccccatc cctgatgccc ccgttgtcac caacgtgact gtgaaggcac 2940
gagtgtggaa cagcaccttc atcgaggatt acagagactt tgaccgagtc cgggtaaatg 3000
gctgggctac cctattcctc cgaaccagca tccccaccat caacatggag aacaagacca 3060
cgtggttctc tgtggacatt gactcggagc tggtggagga gctgccggcc gaaatcgagc 3120
tgtggctggt gctggtggcc gtgggtgcag ggctgctgct gctggggctg atcatcctcc 3180
tgctgtggaa gtgcggcttc ttcaagcgag cccgcactcg cgccctgtat gaagctaaga 3240
ggcagaaggc ggagatgaag agccagccgt cagagacaga gaggctgacc gacgactact 3300
gagggggcag ccccccgccc ccggcccacc tggtgtgact tctttaagcg gacccgctat 3360
tatcagatca tgcccaagta ccacgcagtg cggatccggg aggaggagcg ctacccacct 3420
ccagggagca ccctgcccac caagaagcac tgggtgacca gctggcagac tcgggaccaa 3480
tactactgac gtcctccctg atcccacccc ctcctccccc agtgtcccct ttcttcctat 3540
ttatcataag ttatgcctct gacagtccac aggggccacc acctttggct ggtagcagca 3600
ggctcaggca catacacctc gtcaagagca tgcacatgct gtctggccct ggggatcttc 3660
ccacaggagg gccagcgctg tggaccttac aacgccgagt gcactgcatt cctgtgccct 3720
agatgcacgt ggggcccact gctcgtggac tgtgctggtg catcacggat ggtgcatggg 3780
ctcgccgtgt ctcagcctct gccagcgcca aaacaagcca aagagcctcc caccagagcc 3840
gggaggaaaa ggcccctgca atgtggtgac acctcccct ttcacactgg atccatcttg 3900
agccacagtc actggattga ctttgctgtc aaaactactg acagggagca gcccccgggc 3960
cgctggctgg tgggccccca atgacaccca tgccagagag gtggggatcc tgcctaaggt 4020
tgtctacggg ggcacttgga ggacctggcg tgctcagacc caacagcaaa ggaactagaa 4080
agaaggaccc agaacggctt gctttcctgc atctctgtga agcctctctc cttggccaca 4140
gactgaactc gcagggagtg cagcaggaag gaacaaagac aggcaaacgg caacgtagcc 4200
tgggctcact gtgctggggc acggcgggat cctccacaga gaggagggga ccaattctgg 4260
acagacagat gttgggagga tacagaggag atgccacttc tcactcacca ctaccagcca 4320
gcctcagaag gccccagaga gaccctgcaa gaccacggag ggagcgacac ttgaatgtag 4380
aataggcagg gggccctgcc ccaccccatc cagccagacc ccacgctgac catgcgtcag 4440
gggcctagag gtggagttct tagctatcct tggctttcag agccagcctg gctctgcccc 4500
ctcccccatg ggctgtgtcc taaggcccat ttgagaagct gaggctagtt ccagaaaacc 4560
tctcctgacc cctgcctgtt ggcaggccca ctccccagcc ccagcccctt ccatggtact 4620
gtagcagggg aattccctcc ccctccttgt gccttctttg tatataggct tctcacggcg 4680
accaataaac agctc 4695

<210> 38
<211> 1110
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7518460CB1

<400> 38
tatggcaccc ctgagacccc ttctcatact ggccctgctg gcatgggttg ctctggctga 60
ccaagagtca tgcaagggcc gctgcactga gggcttcaac gtggacaaga agtgccagtg 120
tgacgagctc tgctcttact accagagctg ctgcacagac tatacggctg agtgcaagcc 180
ccaagtgact cgcggggatg tgttcactat gccggaggat gagtacacgg tctatgacga 240
tggcgaggag aaaaacaatg ccactgtcca tgaacaggtg gggggcccct ccctgacctc 300
tgacctccag gcccagtcca aagggaatcc tgagcagaca cctgttctga aacctgagga 360
agaggcccct gcgcctgagg tgggcgcctc taagcctgag ggatagact caaggcctga 420
gacccttcat ccagggagac ctcagccccc agcagaggag gagctgtgca gtgggaagcc 480
cttcgacgcc ttcaccgacc tcaagaacgg ttccctcttt gccttccgag ggcagtactg 540
ctatgaactg gacgaaaagg cagtgaggcc tgggtacccc aagctcatcc gagatgtctg 600
gggcatcgag ggccccatcg atgccgcctt cacccgcatc aactgtcagg ggaagaccta 660
cctcttcaag ggtagtcagt actggcgctt tgaggatggt gtcctggacc ctgattaccc 720
ccgaaatatc tctgacggct tcgatggcat cccggacaac gtggatgcag ccttggccct 780
ccctgcccat agctacagtg gccgggagcg ggtctacttc ttcaagggga aacagtactg 840
ggagtaccag ttccagcacc agcccagtca ggaggagtgt gaaggcagct ccctgtcggc 900
tgtgtttgaa cactttgcca tgatgcagcg ggacagctgg gaggacatct tcgagcttct 960
cttctggggc agaacctctg acaagtacta ccgagtcaat cttcgcacac ggcgagtgga 1020
cactgtggac cctccctacc cacgctccat cgctcagtac tggctgggct gcccagctcc 1080
tggccatctg taggagtcag agcccacata 1110

<210> 39
<211> 1787
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7518734CB1

<400> 39
tagcaacccct cgacatggcg ctgaggcggc caccgcgact ccggctctgc gctcggctgc 60
ctgacttctt cctgctgctg cttttcaggg gctgcctgat aggggctgta aatctcaaat 120
ccagcaatcg aaccccagtg gtacaggaat ttgaaagtgt ggaactgtct tgcatcatta 180
cggattcgca gacaagtgac ccccaggatcg agtggaagaa aattcaagat gaacaaacca 240
catatgtgtt ttttgacaac aaaattcagg gagacttggc gggtcgtgca gaaatactgg 300
ggaagacatc cctgaagatc tggaatgtga cacggagaga ctcagccctt tatcgctgtg 360
aggtcgttgc tcgaaatgac cgcaaggaaa ttgatgagat tgtgatcgag ttaactgtgc 420
aagtgaagcc agtgacccct gtctgtagag tgccgaaggc tgtaccagta ggcaagatgg 480
caacactgca ctgccaggag agtgagggcc acccccggcc tcactacagc tggtatcgca 540
atgatgtacc actgcccacg gattccagag ccaatcccag atttcgcaat tcttctttcc 600
acttaaactc tgaaacaggc actttggtgt tcactgctgt tcacaaggac gactctgggc 660
agtactactg cattgcttcc aatgacgcag gctcagccag gtgtgaggag caggagatgg 720
aagtctttac aagaacccag ggaaaccaga tggagttaac tacatccgca ctgacgagga 780
gggcgacttc agacacaagt catcgtttgt gatctgagac ccgcggtgtg gctgagagcg 840
cacagagcgc acgtgcacat acctctgcta gaaactcctg tcaaggcagc gagagctgat 900
gcactcggac agagctagac actcattcag aagcttttcg ttttggccaa agttgaccac 960
tactcttctt actctaacaa gccacatgaa tagaagaatt ttcctcaaga tggacccggt 1020
aaatataacc acaaggaagc gaaactgggt gcgttcactg agttgggttc ctaatctgtt 1080
tctggcctga ttcccgcatg agtattaggg tgatcttaaa gagtttgctc acgtaaacgc 1140
ccgtgctggg ccctgtgaag ccagcatgtt caccactggt cgttcagcag ccacgacagc 1200
accatgtgag atggcgaggt ggctggacag caccagcagc gcatcccggc gggaacccag 1260
aaaaggcttc ttacacagca gccttacttc atcggcccac agacaccacc gcagtttctt 1320
cttaaaggct ctgctgatcg gtgttgcagt gtccattgtg gagaagcttt ttggatcagc 1380
attttgtaaa aacaaccaaa atcaggaagg taaattggtt gctggaagag ggatcttgcc 1440
tgaggaaccc tgcttgtcca acagggtgtc aggatttaag gaaaaccttc gtcttaggct 1500
aagtctgaaa tggtactgaa atatgctttt ctatgggtct tgtttatttt ataaaatttt 1560
acatctaaat ttttgctaag gatgtatttt gattattgaa aagaaaattt ctatttaaac 1620
tgtaaatata ttgtcataca atgttaaata acctattttt ttaaaaaagt tcaacttaag 1680

gtagaagttc caagctacta gtgttaaatt ggaaaatatc aataattaag agtattttac 1740
ccaaggaatc ctctcatgga agtttactgt gatgttcctt ttctcac 1787

<210> 40
<211> 13426
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7513469CB1

<400> 40
caggtcgcgg ccgcagcccc atccagcccc gcgcccgcca tgccgtccgc cggccccgcc 60
tgagccgcgg cctccgcgcg cgggcgggcc tggggacggc ggggccatgc gcgcgctgcc 120
ctaacgatgc cgcccgccgc gcccgcccgc ctggcgctgg ccctgggcct gggcctgtgg 180
ctcggggcgc tggcgggggg ccccgggcgc ggctgcgggc cctgcgagcc cccctgcctc 240
tgcgggccag cgcccggcgc cgcctgccgc gtcaactgct cgggccgcgg gctgcggacg 300
ctcggtcccg cgctgcgcat ccccgcggac gccacagcgc tagacgtctc ccacaacctg 360
ctccgggcgc tggacgttgg gctcctggcg aacctctcgg cgctggcaga gctggatata 420
agcaacaaca agatttctac gttagaagaa ggaatatttg ctaatttatt taatttaagt 480
gaaataaacc tgagtgggaa cccgtttgag tgtgactgtg gcctggcgtg gctgccgcga 540
tgggcggagg agcagcaggt gcgggtggtg cagcccgagg cagccacgtg tgctgggcct 600
ggctccctgg ctggccagcc tctgcttggc atccccttgc tggacagtgg ctgtggtgag 660
gagtatgtcg cctgcctccc tgacaacagc tcaggcaccg tggcagcagt gtccttttca 720
gctgcccacg aaggcctgct tcagccagag gcctgcagcg ccttctgctt ctccaccggc 780
cagggcctcg cagccctctc ggagcagggc tggtgcctgt gtgggtcagc ccagccctcc 840
agtgcctcct tcgcctgcct gtccctctgc tccggccccc cgccgcctcc tgcccccacc 900
tgtaggggcc ccaccctcct ccagcacgtc ttccctgcct ccccaggggc caccctggtg 960
gggccccacg gacctctggc ctctggccag ctagcagcct tccacatcgc tgccccgctc 1020
cctgtcactg ccacacgctg ggacttcgga gacggctccc ccgaggtgga tgccgctggg 1080
ccggctgcct cgcatcgcta tgtgctgcct gggcgctatc acgtgacggc cgtgctggcc 1140
ctgggggaccg gctcagccct gctggggaca gacgtgcagg tggaagcggc acctgccgcc 1200
ctggagctcg tgtgcccgtc ctcggtgcag agtgacgaga gcctcgacct cagcatccag 1260
aaccgcggtg gttcaggcct ggaggccgcc tacagcatcg tggccctggg cgaggagccg 1320
gcccgagcgg tgcacccgct ctgcccctcg gacacggaga tcttctctgg caacgggcac 1380
tgctaccgcc tggtggtgga gaaggcggcc tggctgcagg cgcaggagca gtgtcaggcc 1440
tgggccgggg ccgccctggc aatggtggac agtcccgccg tgcagcgctt cctggtctcc 1500
cgggtcacca ggcccagtgc aggatgccga gaacctcctc gtgggagcgc ccagtgggga 1560
cctgtaggga cccctgatgc ctctggcacg gcagtacggc ctctcagccc cgcacgagcc 1620
cctggaggtc atggtattcc cgggcctgcg tctgagccgt gaagccttcc tcaccacggc 1680
cgaatttggg acccaggagc tccggcggcc cgcccagctg cggctgcagg tgtaccggct 1740
cctcagcaca gcagggaccc cggagaacgg cagcgagcct gagagcaggt ccccggacaa 1800
caggacccag ctggtccccg cgtgcatgcc agggggacgc tggtgccctg gagccaacat 1860
ctgcttgccg ctggacgcct cctgccaccc ccaggcctgc gccaatggct gcacgtcagg 1920
gccagggcta ctcgggtccc cctatgcgct atggagagag ttcctcttct ccgttcccgc 1980
ggggccccccc gcgcagtact cggtcaccct ccacggccag gatgtcctca tgctccctgg 2040
tgacctcgtt ggcttgcagc acgacgctgg ccctggcgcc ctcccgcact gctcgccggc 2100
tcccggccac cctggtcccc aggccccgta cctctccgcc aacgcctcgt catggctgcc 2160
ccacttgcca gcccagctgg agggcacttg ggcctgccct gcctgtgccc tgcggctgct 2220
tgcagccacg gaacagctca ccgtgctgct gggcctgagg cccaaccctg ggctgcggct 2280
gcctgggcgc tatgaggtcc gggcagaggt gggcaatggc gtgtccaggc acaacctgtc 2340
ctgcagcttt gacgtggtct ccccagtggc tgggctgcgg gtcatctacc ctgccccccg 2400
cgacggccgc ctctacgtgc ccaccaacgg ctcagcctcg gtgctccagg tggactctgg 2460
tgccagcgcc acggccacgg cttgctggcc tgggggcagt gtcagcgccc gctttgagaa 2520
tgcctgccct gccctggtgg ccaccttcgt gcccggctgc ccctgggaga ccaatgatac 2580
cctgttctca gtggtagcac tgccgtggct cggtgagggg gagcacgtga tggacgttgt 2640
ggtggaaaac agcgccagcc gggccaacct cagcctgcgg gtgacggcgg aggagcccat 2700
ctgtggcctc cgcgccacgc ccagccccga ggcccgtgta ctgcagggag tcccagtgag 2760
gtacagcccc gtggtggagg ccggctcgga catggtcttc tggtggacca tcaacgacaa 2820
gcagtccctg accttccaga acgtggtctt caatgtcatt tatcagagcg cggcggtctt 2880
caagctctca ctgacggcct ccaaccacgt gagcaacgtc accgtgaact acaacatcac 2940
cgtggagcgg atgaacagga tgcagggcct gcgggtctct acagtgccag ccgtgctgtc 3000

ccccaatgcc acgctggcac tgacggcggg cgtgctggtg gactcggccg tggaggtggc 3060
cttcctgtgg acctttgggg atggggagca ggccctccac cagttccagc ctccatacaa 3120
cgagtccttc ccagttccag acccctcggt ggcccaggtg ctggtggagc acaatgtcac 3180
gcacacctac gctgccccag gtgagtacct cctgaccgtg ctggcatcta atgccttcga 3240
gaacctgacg cagcaggtgc ctgtgagcgt gcgcgcctcc ctgccctccg tggctgtggg 3300
tgtgagtgac ggcgtcctgg tggccggccg gcccgtcacc ttctacccgc acccgctgcc 3360
ctcgcctggg ggtgttcttt acacgtggga cttcggggac ggctcccctg tcctgaccca 3420
gagccagccg gctgccaacc acacctatgc ctcgaggggc acctaccacg tgcgcctgga 3480
ggtcaacaac acggtgagcg gtgcggcggc ccaggcggat gtgcgcgtct ttgaggagct 3540
ccgcggactc agcgtggaca tgagcctggc cgtggagcag ggcgcccccg tggtggtcag 3600
cgccgcggtg cagacgggcg acaacatcac gtggaccttc gacatggggg acggcaccgt 3660
gctgtcgggc ccggaggcaa cagtggagca tgtgtacctg cgggcacaga actgcacagt 3720
gaccgtgggt gcgggcagcc ccgccggcca cctggcccgg agcctgcacg tgctggtctt 3780
cgtcctggag gtgctgcgcg ttgaacccgc cgcctgcatc cccacgcagc ctgacgcgcg 3840
gctcacggcc tacgtcaccg ggaacccggc ccactacctc ttcgactgga ccttcgggga 3900
tggctcctcc aacacgaccg tgcgggggtg cccgacggtg acacacaact tcacgcggag 3960
cggcacgttc cccctggcgc tggtgctgtc cagccgcgtg aacagggcgc attacttcac 4020
cagcatctgc gtggagccag aggtgggcaa cgtcaccctg cagccagaga ggcagtttgt 4080
gcagctcggg gacgaggcct ggctggtggc atgtgcctgg cccccgttcc cctaccgcta 4140
cacctgggac tttggcaccg aggaagccgc ccccacccgt gccaggggcc ctgaggtgac 4200
gttcatctac cgagacccag gctcctatct tgtgacagtc accgcgtcca acaacatctc 4260
tgctgccaat gactcagccc tggtggaggt gcaggagccc gtgctggtca ccagcatcaa 4320
ggtcaatggc tcccttgggc tggagctgca gcagccgtac ctgttctctg ctgtgggccg 4380
tgggcgcccc gccagctacc tgtgggatct gggggacggt gggtggctcg agggtccgga 4440
ggtcacccac gcttacaaca gcacaggtga cttcaccgtt agggtggccg gctggaatga 4500
ggtgagccgc agcgaggcct ggctcaatgt gacggtgaag cggcgcgtgc gggggctcgt 4560
cgtcaatgca agccgcacgg tggtgcccct gaatgggagc gtgagcttca gcacgtcgct 4620
ggaggccggc agtgatgtgc gctattcctg ggtgctctgt gaccgctgca cgcccatccc 4680
tgggggtcct accatctctt acaccttccg ctccgtgggc accttcaata tcatcgtcac 4740
ggctgagaac gaggtgggct ccgcccagga cagcatcttc gtctatgtcc tgcagctcat 4800
agaggggctg caggtggtgg gcggtggccg ctacttcccc accaaccaca cggtacagct 4860
gcaggccgtg gttagggatg gcaccaacgt ctcctacagc tggactgcct ggagggacag 4920
gggcccggcc ctggccggca gcggcaaagg cttctcgctc accgtgctcg aggccggcac 4980
ctaccatgtg cagctgcggg ccaccaacat gctgggcagc gcctgggccg actgcaccat 5040
ggacttcgtg gagcctgtgg ggtggctgat ggtggccgcc tccccgaacc cagctgccgt 5100
caacacaagc gtcaccctca gtgccgagct ggctggtggc agtggtgtcg tatacacttg 5160
gtcccttgga ggaggggctg agctgggaga cctccgagcc atttaccacc catagcttcc 5220
ccacacccgg cctgcacttg gtcaccatga cggcagggaa cccgctgggc tcagccaacg 5280
ccaccgtgga agtggatgtg caggtgcctg tgagtggcct cagcatcagg gccagcgagc 5340
ccggaggcag cttcgtggcg gccgggtcct ctgtgccctt ttgggggcag ctggccacgg 5400
gcaccaatgt gagctggtgc tgggctgtgc ccggcggcag cagcaagcgt ggccctcatg 5460
tcaccatggt cttcccggat gctggcacct tctccatccg gctcaatgcc tccaacgcag 5520
tcagctgggt ctcagccacg tacaacctca cggcggagga gcccatcgtg ggcctggtgc 5580
tgtgggccag cagcaaggtg gtggcgcccg ggcagctggt ccattttcag atcctgctgg 5640
ctgccggctc agctgtcacc ttccgcctac aggtcggcgg ggccaacccc gaggtgctcc 5700
ccgggccccg tttctcccac agcttccccc gcgtcggaga ccacgtggtg agcgtgcggg 5760
gcaaaaacca cgtgagctgg gcccaggcgc aggtgcgcat cgtggtgctg gaggccgtga 5820
gtgggctgca ggtgcccaac tgctgcgagc ctggcatcgc cacgggcact gagaggaact 5880
tcacagcccg cgtgcagcgc ggctctcggg tcgcctacgc ctggtacttc tcgctgcaga 5940
aggtccaggg cgactcgctg gtcatcctgt cgggccgcga cgtcacctac acgcccgtgg 6000
ccgcggggct gttggagatc caggtgcgcg ccttcaacgc cctgggcagt gagaaccgca 6060
cgctggtgct ggaggttcag gacgccgtcc agtatgtggc cctgcagagc ggcccctgct 6120
tcaccaaccg ctcggcgcag tttgaggccg ccaccagccc cagcccccgg cgtgtggcct 6180
accactggga ctttggggat gggtcgccag ggcaggacac agatgagccc agggccgagc 6240
actcctacct gaggcctggg gactaccgcg tgcaggtgaa cgcctccaac ctggtgagct 6300
tcttcgtggc gcaggccacg gtgaccgtcc aggtgctggc ctgccgggag ccggaggtgg 6360
acgtggtcct gcccctgcag gtgctgatgc ggcgatcaca gcgcaactac ttggaggccc 6420
acgttgacct gcgcgactgc gtcacctacc agactgagta ccgctgggag gtgtatcgca 6480
ccgccagctg ccagcggccg gggcgcccag cgcgtgtggc cctgcccggc gtggacgtga 6540
gccggcctcg gctggtgctg ccgcggctgg cgctgcctgt ggggcactac tgctttgtgt 6600
ttgtcgtgtc atttggggac acgccactga cacagagcat ccaggccaat gtgacggtgg 6660
cccccgagcg cctggtgccc atcattgagg gtggctcata ccgcgtgtgg tcagacacac 6720
gggacctggt gctggatggg agcgagtcct acgaccccaa cctggaggat ggcgaccaga 6780

cgccgctcag tttccagtgg gcctgtgtgg cttcgacaca gagggaggct ggcgggtgtg 6840
cgctgaactt tgggccccgc gggagcagca cggtcaccat tccacgggaa cggctggcgg 6900
ctggcgtgga gtacaccttc agcctcaccg tgtggaaggc cggccgcaag gaggaggcca 6960
ccaaccagac ggtgctgatc cggagtggcc gggtgcccat tgtgtccttg gagtgtgtgt 7020
cctgcaaggc acaggccgtg tacgaagtga gccgcagctc ctacgtgtac ttggagggcc 7080
gctgcctcaa ttgcagcagc ggctccaagc gagggcggtg ggctgcacgt acgttcagca 7140
acaagacgct ggtgctggat gagaccacca catccacggg cagcgcaggc atgtgactgg 7200
tgctgcggcg gggcgtgctg cgggacggcg agggatacac cttcacgctg acggtgctgg 7260
gccgctctgg cgaggaggag ggctgcgcct ccatcccct gtcccccaac cgcccgccgc 7320
tggggggctc ttgccgcctc ttcccactgg gcgctgtgca cgccctcacc accaaggtgc 7380
acttcgaatg cacgggctgg catgacgcgg aggatgctgg cgccccgctg gtgtacgccc 7440
tgctgctgca gcgctgtcgc cagggccact gcgaggagtt ctgtgtctac aagggcagcc 7500
tctccggcta cggagccgtg ctgcccccgg gtttcaggcc acacttcgag gtgggcctgg 7560
ccgtggtggt gcaggaccag ctgggagccg ctgtggtcgc cctcaacagg tctctggcca 7620
tcaccctccc agagcccaac ggcagcgcaa tggggctcac agtctggctg cacgggctca 7680
ccgctagtgt gctcccgggg ctgctgcggc aggccgatcc ccagcacgtc atcgagtact 7740
cgttggccct ggtcaccgtg ctgaacgagt acgagcgggc cctggacgtg gcggcagagc 7800
ccaagcacga gcggcagcac cgagcccaga tacgcaagaa catcacggag actctggtgt 7860
ccctgagggt ccacactgtg gatgacatcc agcagatcgc tgctgcgctg gcccagtgca 7920
tggggcccag cagggagctc gtatgccgct cgtgcctgaa gcagacgctg cacaagctgg 7980
aggccatgat gctcatcctg caggcagaga ccaccgcggg caccgtgacg cccaccgcca 8040
tcggagacag catcctcaac atcacaggag acctcatcca cctggccagc tcggacgtgc 8100
gggcaccaca gccctcagag ctgggagccg agtcaccatc tcggatggtg gcgtcccagg 8160
cctacaacct gacctctgcc ctcatgcgca tcctcatgcg ctcccgcgtg ctcaacgagg 8220
agcccctgac gctggcgggc gaggagatcg tggcccaggg caagcgctcg gacccgcgga 8280
gcctgctgtg ctatggcggc gccccagggc ctggctgcca cttctccatc cccgaggctt 8340
tcagcggggc cctggccaac ctcagtgacg tggtgcagct catctttctg gtggactcca 8400
atccctttcc ctttggctat atcagcaact acaccgtctc caccaaggtg gcctcgatgg 8460
cattccagac acaggccggc gcccagatcc ccatcgagcg gctggcctca gagcgcgcca 8520
tcaccgtgaa ggtgcccaac aactcggact gggctgcccg gggccaccgc agctccgcca 8580
actccgccaa ctccgttgtg gtccagcccc aggcctccgt cggtgctgtg gtcaccctgg 8640
acagcagcaa ccctgcggcc gggctgcatc tgcagctcaa ctatacgctg ctggacggcc 8700
actacctgtc tgaggaacct gagccctacc tggcagtcta cctacactcg gagccccggc 8760
ccaatgagca caactgctcg gctagcagga ggatccgccc agagtcactc cagggtgctg 8820
accaccggcc ctacaccttc ttcatttccc cggggagcag agacccagcg gggagttacc 8880
atctgaacct ctccagccac ttccgctggt cggcgctgca ggtgtccgtg ggcctgtaca 8940
cgtccctgtg ccagtacttc agcgaggagg acatggtgtg gcggacagag gggctgctgc 9000
ccctggagga gacctcgccc cgccaggccg tctgcctcac ccgccacctc accgccttcg 9060
gcaccagcct cttcatgccc ccaagccatg tacgctttgt gtttcctgag ccaacagcgg 9120
atgtaaacta catcgtcatg ctgacatgtg ctgtgtgcct ggtgacctac atggtcatgg 9180
ccgccatcct gcacaagctg gaccagttgg atgccagccg gggctgcgcc atccccttct 9240
gtgggcagcg gggccgcttc aagtacgaga tcctcgtcaa gacaggctgg ggccggggct 9300
caggtaccac ggcccacgtg ggcatcatgc tgtatggggt ggacagccgg agcggccacc 9360
ggcacctgga cggcgacaga gccttccacc gcaacagtct ggacatcttc cagatcgcca 9420
ccccgcacag cctgggtagc atgtggaaga tccgagtgtg gcacgacaac aaagggctca 9480
gccctgcctg gttcctgcag cacatcatcg tcagggacct gcagacggca cgcagcacct 9540
tcttcctggt caatgactgg ctttcggtgg agacggaggc caacgggggc ctggtggaga 9600
aggaggtgct ggccgcgagt cacgcagccc tgttgcgctt ccggcgcctg ctggtggctg 9660
agctgcagcg tggcttcttt gacaagcaca tctggctctc catatgggac cggccgcctc 9720
ggagctgttt cactcgcatc cagagggcca cctgctgcgt tctcctcatc tgtctcttcc 9780
tgggcgccaa cgccgtgtgg tacggggctg ttggcgactc tgcctacagc acggggcatg 9840
tgtccaggct gagcccgctg agcgtcgaca cagtcgctgt tggcctggtg tccagcgtgg 9900
ttgtctatcc cgtctacctg gccatccttt ttctcttccg gatgtcccgg agcaaggtgg 9960
ctgggagccc gagccccaca cctgccgggc agcaggtgct ggacatcgac agctgcctgg 10020
actcgtccgt gctggacagc tccttcctca cgttctcagg cctccacgct gagcaggcct 10080
ttgttggaca gatgaagagt gacttgtttc tggatgattc taagagtctg gtgtgctggc 10140
cctccggcga gggaacgctc agttggccgg acctgctcag tgacccgtcc attgtgggta 10200
gcaatctgcg gcagctggca cggggccagg cgggccatgg gctgggccca gaggaggacg 10260
gcttctccct ggccagcccc tactcgcctg ccaaatcctt ctcagcatca gatgaagacc 10320
tgatccagca ggtccttgcc gagggggtca gcagcccagc ccctacccaa gacacccaca 10380
tggaaacgga cctgctcagc agcctgtcca gcactcctgg ggagaagaca gagacgctgg 10440
cgctgcagag gctgggggag ctggggccac ccagcccagg cctgaactgg gaacagcccc 10500
aggcagcgag gctgtccagg acaggactgg tggagggtct gcggaagcgc ctgctgccgg 10560

cctggtgtgc ctccctggcc cacgggctca gcctgctcct ggtggctgtg gctgtggctg 10620
tctcagggtg ggtgggtgcg agcttccccc cgggcgtgag tgttgcgtgg ctcctgtcca 10680
gcagcgccag cttcctggcc tcattcctcg gctgggagcc actgaaggtc ttgctggaag 10740
ccctgtactt ctcactggtg gccaagcggc tgcacccgga tgaagatgac accctggtag 10800
agagcccggc tgtgacgcct gtgagcgcac gtgtgccccg cgtacggcca ccccacggct 10860
ttgcactctt cctggccaag gaagaagccc gcaaggtcaa gaggctacat ggcatgctgc 10920
ggagcctcct ggtgtacatg ctttttctgc tggtgaccct gctggccagc tatggggatg 10980
cctcatgcca tgggcacgcc taccgtctgc aaagcgccat caagcaggag ctgcacagcc 11040
gggccttcct ggccatcacg cggtctgagg agctctggcc atggatggcc cacgtgctgc 11100
tgccctacgt ccacgggaac cagtccagcc cagagctggg gcccccacgg ctgcggcagg 11160
tgcggctgca ggaagcactc tacccagacc ctcccggccc cagggtccac acgtgctcgg 11220
ccgcaggagg cttcagcacc agcgattacg acgttggctg ggagagtcct cacaatggct 11280
cggggacgtg ggcctattca gcgccggatc tgctgggggc atggtcctgg ggctcctgtg 11340
ccgtgtatga cagcgggggc tacgtgcagg agctgggcct gagcctggag gagagccgcg 11400
accggctgcg cttcctgcag ctgcacaact ggctggacaa caggagccgc gctgtgttcc 11460
tggagctcac gcgctacagc ccggccgtgg ggctgcacgc cgccgtcacg ctgcgcctcg 11520
agttcccggc ggccggccgc gccctggccg ccctcagcgt ccgccccttt gcgctgcgcc 11580
gcctcagcgc gggcctctcg ctgcctctgc tcacctcggt gtgcctgctg ctgttcgccg 11640
tgcacttcgc cgtggccgag gcccgtactt ggcacaggga agggcgctgg cgcgtgctgc 11700
ggctcggagc ctgggcgcgg tggctgctgg tggcgctgac ggcggccacg gcactggtac 11760
gcctcgccca gctgggtgcc gctgaccgcc agtggacccg tttcgtgcgc ggccgcccgc 11820
gccgcttcac tagcttcgac caggtggcgc agctgagctc cgcagcccgt ggcctggcgg 11880
cctcgctgct cttcctgctt ttggtcaagg ctgcccagca gctacgcttc gtgcgccagt 11940
ggtccgtctt tggcaagaca ttatgccgag ctctgccaga gctcctgggg gtcaccttgg 12000
gcctggtggt gctcggggta gcctacgccc agctggccgt cctgctcgtg tcttcctgtg 12060
tggactccct ctggagcgtg gcccaggccc tgttggtgct gtgccctggg actgggctct 12120
ctaccctgtg tcctgccgag tcctggcacc tgtcaccccct gctgtgtgtg gggctctggg 12180
cgctgcggct gtggggcgcc ctacggctgg gggctgttat tctccgctgg cgctaccacg 12240
ccttgcgtgg agagctgtac cggccggcct gggagcccca ggactacgag atggtggagt 12300
tgttcctgcg caggctgcgc ctctggatgg gcctcagcaa ggtcaaggag ttccgccaca 12360
aagtccgctt tgaagggatg gagccgctgc cctctcgctc ctccaggggc tccaaggtat 12420
ccccggatgt gcccccaccc agcgctggct ccgatgcctc gcacccctcc acctcctcca 12480
gccagctgga tgggctgagc gtgagcctgg gccggctggg gacaaggtgt gagcccgagc 12540
cctcccgcct ccaagccgtg ttcgaggccc tgctcaccca gtttgaccga ctcaaccagg 12600
ccacagagga cgtctaccag ctggagcagc agctgcacag cctgcaaggc cgcaggagca 12660
gccgggcgcc cgccggatct tcccgtggcc catccccggg cctgcggcca gcactgccca 12720
gccgccttgc ccgggccagt cggggtgtgg acctggccac tggccccagc aggacacccc 12780
ttcgggccaa gaacaaggtc caccccagca gcacttagtc ctccttcctg gcgggggtgg 12840
gccgtggagt cggagtggac accgctcagt attactttct gccgctgtca aggccgaggg 12900
ccaggcagaa tggctgcacg taggttcccc agagagcagg caggggcatc tgtctgtctg 12960
tgggcttcag cactttaaag aggctgtgtg gccaaccagg acccagggtc ccctccccag 13020
ctcccttggg aaggacacag cagtattgga cggtttctag cctctgagat gctaatttat 13080
ttccccgagt cctcaggtac agcgggctgt gcccggcccc acccccctggg cagatgtccc 13140
ccactgctaa ggctgctggc ttcagggagg gttagcctgc accgccgcca ccctgcccct 13200
aagttattac ctctccagtt cctaccgtac tccctgcacc gtctcactgt gtgtctcgtg 13260
tcagtaattt atatggtgtt aaaatgtgta tatttttgta tgtcactatt ttcactaggg 13320
ctgaggggcc tgcgcccaga gcctggcctc cccaacacc tgctgcgctt ggtaggtgtg 13380
gttggcgtta tggcagcccg gctgtgcttg gatgcgacat ggcctg 13426

<210> 41
<211> 699
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7518836CB1

<400> 41
tttcccaaag gggatgtgct gcaggcgatt aagttggtaa cgccgggttt tcccgtcacg 60
acgttgtaaa cgacggccgt gaattgtgcg ccatttaggt gacctgatag aatacgcgcc 120
gcgagctcgg gcccccacac gtgtggtcta gagctagcct aggctcgaga agcttgtcga 180
cgaattcaga ttaccatgga caagttttgg tggcacgcag cctggggact ctgcctcgtg 240

ccgctgagcc tggcgcagat cggagaccaa gacacattcc accccagtgg ggggtcccat 300
accactcatg gatctgaatc agatggacac tcacatggga gtcaagaagg tggagcaaac 360
acaacctctg gtcctataag gacaccccaa attccagaat ggctgatcat cttggcatcc 420
ctcttggcct tggctttgat tcttgcagtt tgcattgcag tcaacagtcg aagaaggtgt 480
gggcagaaga aaaagctagt gatcaacagt ggcaatggag ctgtggagga cagaaagcca 540
agtggactca acggagaggc cagcaagtct caggaaatgg tgcatttggt gaacaaggag 600
tcgtcagaaa ctccagacca gtttatgaca gctgatgaga caaggaacct gcagaatgtg 660
gacatgaaga ttggggtgta acacctacac cattatcta 699

<210> 42
<211> 1135
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7519868CB1

<400> 42
ttgtgggaga ctagatttag gagagtaaag atcacagggt gggaaataaa agctgtggcc 60
cccaggagtt ctggacactg ggggagagtg gggtgacatg agtgactcca aggaaccaag 120
actgcagcag ctgggcctcc tggaggagga acagctgaga ggccttggat tccgacagac 180
tcgaggatac aagagcttag cagggtgtct tggccatggt cccctggtgc tgcaactcct 240
ctccttcacg ctcttggctg ggctccttgt ccaagtgtcc aaggtcccca gctccataag 300
tcaggaacaa tccaggcaag acgcgatcta ccagaacctg acccagctta aagctgcagt 360
gggtgagctc tcagagaaat ccaagctgca ggagatctac caggagctga cccagctgaa 420
ggctgcagtg ggtgagcttc cagagaaatc taagctgcag gagatctacc aggagctgac 480
ccggctgaag gctgcagtgg aacgcctgtg ccacccctgt ccctgggaat ggacattctt 540
ccaaggaaac tgttacttca tgtctaactc ccagcggaac tggcacgact ccatcaccgc 600
ctgcaaagaa gtgggggccc agctcgtcgt aatcaaaagt gctgaggagc agaacttcct 660
acagctgcag tcttccagaa gtaaccgctt cacctggatg ggactttcag atctaaatca 720
ggaaggcacg tggcaatggg tggacggctc acctctgttg cccagcttca agcagtattg 780
gaacagagga gagcccaaca acgttgggga ggaagactgc gcggaattta gtggcaatgg 840
ctggaacgac gacaaatgta atcttgccaa attctggatc tgcaaaaagt ccgcagcctc 900
ctgctccagg gatgaagaac agtttctttc tccagcccct gccaccccaa acccccctcc 960
tgcgtagcag aacttcaccc ccttttaagc tacagttcct tctctccatc cttcgacctt 1020
cacaaatctc tgggactgtt ctttgtcaga ttcttcctcc tttagaaggc tgggtcccat 1080
tctgtccttc ttgtcatgcc tccaatttcc ctggtgtaga gcttgttttt ctgga 1135

<210> 43
<211> 3228
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7520048CB1

<400> 43
gatgcagctg gacgagctcg gccactatta cggcgcagtg tgctggaatt cgccttgccc 60
cacctggcct ctgcaatgtc accgcctctg tgtcccctcc ttctcctggc tgtgggcctg 120
cggctggctg gaactctcaa ccccagtgat cccaatacct gcagcttctg ggaaagcttc 180
actaccacca ccaaggagtc ccactcccgc cccttcagcc tgctcccctc agagccctgc 240
gagcggccct gggagggccc ccatacttgc ccccagccca cggttgtata ccggaccgtg 300
taccgtcagg tggtgaagac ggaccaccgc cagcgcctgc agtgctgcca tggcttctat 360
gagagcaggg ggttctgtgt cccgctctgt gcccaggagt gtgtccatgg ccgttgtgtg 420
gcacccaatc agtgccaatg tgtgccaggc tggcggggcg acgactgttc cagtgagtgt 480
gccccaggaa tgtgggggcc acagtgtgac aagccctgca gctgcggcaa caacagctcg 540
tgtgatccca ggagtggggt atgttcttgc ccttctggtc tgcagccccc gaactgcctt 600
cagccctgta cccctggcta ctatggccct gcctgccagt tccgctgcca gtgccacggg 660
gcaccctgcg atccccagac tggagcctgc ttctgccccg cagagagaac tgggcccagg 720
gcacttctgg cttcttctgc cccagcaccc atccttgcca aaatggaggt gtcttccaaa 780
ccccacaggg ctcctgcagc tgccccctg gctggatggg caccatctgc tccctgccct 840

gcccagaggg ctttcacgga cccaactgct cccaggaatg tcgctgccac aacggcggcc 900
tctgtgaccg attcactggg cagtgccgct gcgctccggg ttacactggg gatcggtgcc 960
gggaggagtg cccggtgggc cgctttgggc aggactgtgc tgagacgtgc gactgcgccc 1020
cggacgcccg ttgcttcccg gccaacggcg catgtctgtg cgaacacggc ttcactgggg 1080
accgctgcac ggatcgcctc tgccccgacg gcttctacgg tctcagctgc caggccccct 1140
gcacctgcga ccgggagcac agcctcagct gccacccgat gaacggggag tgctcctgcc 1200
tgccgggctg ggcgggcctc cactgcaacg agagctgccc gcaggacacg catgggccag 1260
ggtgccagga gcactgtctc tgcccgcacg gtggcgtctg ccaggctacc agcggcctct 1320
gtcagtgcgc gccgggttac acgggccctc actgtgctag tctttgtcct cctgacacct 1380
acggtgtcaa ctgttctgca cgctgctcat gtgaaaatgc catcgcctgc tcacccatcg 1440
acggcgagtg cgtctgcaag gaaggttggc agcgtggtaa ctgctctgtg ccctgcccac 1500
ccggaacctg ggcttcagt tgcaatgcca gctgccagtg tgcccatgag gcagtctgca 1560
gcccccaaac tggagcctgt acctgcaccc ctgggtggca tggggcccac tgccagctgc 1620
cctgtccgaa ggggcagttt ggagaaggtt gtgccagtcg ctgtgactgt gaccactctg 1680
atggctgtga ccctgttcat ggacgctgtc agtgccaggc tggctggatg ggtgcccgct 1740
gccacctgtc ctgccctgag ggcttatggg gagtcaactg tagcaacacc tgcacctgca 1800
agaatggggg cacctgtctc cctgagaatg gcaactgcgt gtgtgcaccc ggattccggg 1860
gcccctcctg ccagagatcc tgtcagcctg gccgctatgg caaacgctgt gtgccctgca 1920
agtgcgctaa ccactccttc tgccacccct cgaacgggac ctgctactgc ctggctggct 1980
ggacaggccc cgactgctcc cagccatgcc ctccaggaca ctggggagaa aactgtgccc 2040
agacctgcca atgtcaccat ggtgggacct gccatcccca ggatgggagc tgtatctgcc 2100
ccctaggctg gactggacac cactacttag aaggctgccc tctggggaca tttggtgcta 2160
actgctccca gccatgccag tgtggtcctg gagaaaagtg ccacccagag actggggcct 2220
gtgtatgtcc cccagggcac agtggtgcac cttgcaggat tggaatccag gagcccttta 2280
ctgtgatgcc gaccactcca gtagcgtata actcgctggg tgcagtgatt ggcattgcag 2340
tgctggggtc ccttgtggta gccctggtgg cactgttcat tggctatcgg cactggcaaa 2400
aaggcaagga gcaccaccac ctggctgtgg cttacagcag cgggcgcctg gacggcttcg 2460
agtatgtcat gccagatgtc cctccgagct acagtcacta ctactccaac cccagctacc 2520
acaccctgtc gcagtgctcc ccaaaccccc caccccctaa caaggttcca ggcccgctct 2580
ttgccagcct gcagaaccct gagcggccag gtggggccca agggcatgat aaccacacca 2640
ccctgcctgc tgactggaag caccgccggg agccccctcc agggcctctg gacaggggga 2700
gcagccacct ggaccgaagc tacagctata gctacagcaa tggcccaggc ccattctacg 2760
ataaagggct catctctgaa gaggagctcg gggccagtgt ggcttccctg agcagtgaga 2820
acccatatgc caccatccgg gacctgccca gcttgccagg gggcccccgg gagagcagct 2880
acatggagat gaaaggccct ccctcaggat ctcccccccag gcagcctcct cagttctggg 2940
acagccagag gcggcggcaa ccccagccac agagagacag tggcacctac gagcagccca 3000
gcccccctgat ccatgaccga gactctgtgg gctcccagcc ccctctgcct ccgggcctac 3060
cccccggcca ctatgactca cccaagaaca gccacatccc tggacattat gacttgcctc 3120
cagtacggca tcccccatca cctccacttc gacgccagga ccgttgagga gccaggatgg 3180
tatggcagag gccagcacac ctggctgttg ctgctcaagg ctggggac 3228

<210> 44
<211> 2265
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7520157CB1

<400> 44
gatcagctgg cgagctccct cagctttacg gcgcagtgtg ctggaatcgc cctgtgacgt 60
cctgtttcct ctgaagagta atagttggaa tcaaaagagt caacgcaatg aactgttatt 120
tactgctgcg ttttatgttg ggaattcctc tcctatggcc ttgtcttgga gcaacagaaa 180
actctcaaac aaagaaagtc aagcagccag tgcgatctca tttgagagtg aagcgtggct 240
gggtgtggaa ccaatttttt gtaccagagg aaatgaatac gactagtcat cacatcggcc 300
agctaagatc tgatttagac aatggaaaca attctttcca gtacaagctt ttgggggctg 360
gagctggaag tacttttatc attgatgaaa gaacaggtga catatatgcc atacagaagc 420
ttgatagaga ggagcgatcc ctctacatct taagagccca ggtaatagac atcgctactg 480
gaagggctgt ggaacctgag tctgagtttg tcatcaaagt ttcggatatc aatgacaatg 540
aaccaaaatt cctagatgaa ccttatgagg ccattgtacc agagatgtct ccagaaggaa 600
cattagttat ccaggtgaca gcaagtgatg ctgacgatcc ctcaagtggt aataatgctc 660
gtctcctcta cagcttactt caaggccagc catatttttc tgttgaacca acaacaggag 720

tcataagaat atcttctaaa atggatagag aactgcaaga tgagtattgg gtaatcattc 780
aagccaagga cacgattggt cagccaggag cgttgtctgg aacaacaagt gtattaatta 840
aactttcaga tgttaatgac aataagccta tatttaaaga aagtttatac cgcttgactg 900
tctctgaatc tgcacccact gggacttcta taggaacaat catggcatat gataatgaca 960
taggagagaa tgcagaaatg gattacagca ttgaagagga tgattcgcaa acatttgaca 1020
ttattactaa tcatgaaact caagaaggaa tagttatatt aaaaaagaaa gtggattttg 1080
agcaccagaa ccactacggt attagagcaa aagttaaaaa ccatcatgtt cctgagcagc 1140
tcatgaagta ccacactgag gcttccacca ctttcattaa gatccaggtg gaagatgttg 1200
atgagcctcc tcttttcctc cttccatatt atgtatttga agtttttgaa gaaaccccac 1260
agggatcatt tgtaggcgtg gtgtctgcca cggacccaga caataggaaa tctcctatca 1320
ggtattctat tactaggagc aaagtgttca atatcaatga taatggtaca atcactacaa 1380
gtaactcact ggatcgtgaa atcagtgctt ggtacaacct aagtattaca gccacagaaa 1440
aatacaatat agaacagatc tcttcgatcc cactgtatgt gcaagttctt aacatcaatg 1500
atcatgctcc tgagttctct caatactatg agacttatgt ttgtgaaaat gcaggctctg 1560
gtcaggtaat tcagactatc agtgcagtgg atagagatga atccatagaa gagcaccatt 1620
tttactttaa tctatctgta gaagacacta acaattcaag ttttacaatc atagataatc 1680
aagggtttat ttttttgact ttgggtttaa aacaacggag aaaacagatt ctatttcctg 1740
agaaaagtga agatttcaga gagaatatat tccaatatga tgatgaaggg ggtggagaag 1800
aagatacaga ggcctttgat atagcagagc tgaggagtag taccataatg cgggaacgca 1860
agactcggaa aaccacaagc gctgagatca ggagcctata caggcagtct ttgcaagttg 1920
gccccgacag tgccatattc aggaaattca ttctggaaaa gctcgaagaa gctaatactg 1980
atccgtgtgc ccctcctttt gattccctcc agacctacgc ttttgaggga acagggtcat 2040
tagctggatc cctgagctcc ttagaatcag cagtctctga tcaggatgaa agctatgatt 2100
accttaatga gttgggacct cgctttaaaa gattagcatg catgtttggt tctgcagtgc 2160
agtcaaataa ttagggcttt ttaccatcaa aatttttaaa agtgctaatg tgtattcgaa 2220
cccaatggta gtcttaaaga gttttgtgcc ctggctctat ggcgg 2265

<210> 45
<211> 2093
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7520485CB1

<400> 45
gtttgtcttc ctgagagaca gtagatttag aaagtgagga tcagagggtg gaaaataaaa 60
gctgtggtcc ccaggagtcc tgaacatctg gggacagcgg gaaaacatga gtgactccaa 120
ggaaccaagg gtgcagcagc tgggcctcct ggaagaagat ccaacaacca gtggcatcag 180
actttttcca agagactttc aattccagca gatacatggc cacaagagct ctacagggtg 240
tcttggccat ggcgccctgg tgctgcaact cctctccttc atgctcttgg ctggggtcct 300
ggtggccatc cttgtccaag tgtccaaggt ccccagctcc ctaagtcagg aacaatccga 360
gcaagacgca atctaccaga acctgaccca gcttaaagct gcagtgggtg agctctcaga 420
gaaatccaag ccgcaggaga tctaccagga gctgacccag ctgaaggctg cagtgggtga 480
gttgccagag aaatccaagc tgcaggagat ctaccaggag ctgacccggc tgaaggctgc 540
agtgggtgag ttgccagaga aatccaagct gcaggagatc taccaggagc tgacccggct 600
gaagactgca gtgggtgagt tgccagagaa atccaagctg caggagatct accaggagct 660
gacccggctg aaggctgcag tgggtgagtt gccagagaaa tccaagctgc aggagatcta 720
ccaggagctg acggagctga aggctgcagt gggtgagttg ccagagaaat ccaagctgca 780
ggagatctac caggagctga cccagctgaa ggctgcagtg ggtgagttgc cagaccagtc 840
caagcagcag caaatctatc aagaactgac cgatttgaag actgcatttg gtgagttcct 900
gcacatcaag ggtccttggg cctgaggtgg tctctgtgtg atgtgacttt acttgagtta 960
ccaaccctgc ctgagcctca gtttcctccc ctgtgaaatg agaacacggg gtaactgcga 1020
tcattgcact tggtgttcac agtgggccag gcacacagta gacactcagt tcaaccaatc 1080
cgctgccttc tgttctgagc tctaaagctg ctaattctcc cagtctggaa gagatggagc 1140
caggactcct gggttctcct ggggatcagg gagagtcttg ggaagtaggg agaggaatgg 1200
tctctcccca ttccaagcgg tccacccaac atggacagtc aaggaagggc cctgattttc 1260
aagggcttga aggctgggca gatatgggaa tatggacaac ctaatctgag cccagccctg 1320
accagcctcc cccaacagaa cgcctgtgcc gccactgtcc caaggactgg acattcttcc 1380
aaggaaactg ttacttcatg tctaactccc agcggaactg gcacgactcc gtcaccgcct 1440
gccaggaagt gagggcccag ctcgtcgtaa tcaaaactgc tgaggagcag aacttcctac 1500
agctgcagac ttccaggagt aaccgcttct cctggatggg actttcagac ctaaatcagg 1560

aaggcacgtg gcaatgggtg gacggctcac ctctgtcacc cagcttccag cggtactgga 1620
acagtggaga acccaacaat agcgggaatg aagactgtgc ggaatttagt ggcagtggct 1680
ggaacgacaa tcgatgtgac gttgacaatt actggatctg caaaaagccc gcagcctgct 1740
tcagagacga atagttgttt ccctgctggc ctcagcctcc attgtggtat agcagaactt 1800
cacccacttc tacaccccgt gcactctttt gactggggac ttgctggttg aaggagctca 1860
tcttgcaggc tggaagcacc agggaattaa ttcccccagt caaccaatgg catccagaga 1920
gggcatggag gctccataca acctcttcca cccccacatc tttctttgtc ctatacatgt 1980
cttccatttg gctgtttctg agttgtagcc tttataataa agtggtaaat gttgtaactt 2040
gtcctataag aatataggag gaaccgaaat tttgcccagg ccatcatctt ggg 2093

<210> 46
<211> 1384
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7525723CB1

<400> 46
tagctaagcg cccgctcgcc atggggagcc ccgcacatcg gcccgcgctg ctgctgctgc 60
tgccgcctct gctgctgctg ctgctgctgc gcgtcccgcc cagccgcagc ttcccagata 120
ccccgtggtg ctcccccatc aaggtgaagt atggggatgt gtactgcagg gcccctcaag 180
gaggatacta caaaacagcc ctgggaacca ggtgcgacat tcgctgccag aagggctacg 240
agctgcatgg ctcttcccta ctgatctgcc agtcaaacaa acgatggtct gacaaggtca 300
tctgcaaaca aaagcgatgt cctacccttg ccatgccagc aaatggaggg tttaagtgtg 360
tagatggtgc ctactttaac tcccggtgtg agtattattg ttcaccagga tacacgttga 420
aaggggagcg gaccgtcaca tgtatggaca acaaggcctg gagcggccgg ccagcctcct 480
gtgtggatat ggaacctcct agaatcaagt gcccaagtgt gaaggaacgc attgcagaac 540
ccaacaaact gacagtccgg gtgtcctggg agacacccga aggaagagac acagcagatg 600
gaattcttac tgatgtcatt ctaaaaggcc tcccccaggg ctccaacttt ccagaaggag 660
accacaagat ccagtacaca gtctatgaca gagctgagaa taagggcact tgcaaatttc 720
gagttaaagt aagagtcaaa cgctgtggca aactcaatgc cccagagaat ggttacatga 780
agtgctccag cgacggtgat aattatggag ccacctgtga gttctcctgc atcggcggct 840
atgagctcca gggtagccct gcccgagtat gtcaatccaa cctggcttgg tctggcacgg 900
agcccacctg tgcagccatg aacgtcaatg tgggtgtcag aacggcagct gcacttctgg 960
atcagtttta tgagaaaagg agactcctca ttgtgtccac acccacagcc cgaaacctcc 1020
tttaccggct ccagctagga atgctgcagc aagcacagtg tggccttgat cttcgacaca 1080
tcaccgtggt ggagctggtg ggtgtgttcc cgactctcat tggcaggata ggagcaaaga 1140
ttatgcctcc agccctagcg ctgcagctca ggctgttgct gcgaatccca ctctactcct 1200
tcagtatggt gctagtggat aagcatggca tggacaaaga gcgctatgtc tccctggtga 1260
tgcctgtggc cctgttcaac ctgattgaca cttttccctt gagaaaagaa gagatggtcc 1320
tacaagccga aatgagccag acctgtaaca cctgacatga tggttcctct cttggcaatt 1380
ccta 1384

<210> 47
<211> 1128
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7525966CB1

<400> 47
tagaagccgg gagcttccct gatggtgccg ccgcctccga gccggggagg agctgccagg 60
ggccagctgg gcaggagcct gggtccgctg ctgctgctcc tggcgttggg acacacgtgg 120
acctacagag aggagccgga ggacggcgac agagaaatct gctcagagag caaaatcgcg 180
acgactaaat acccgtgtct gaagtcttca ggcgagctca ccacatgcta caggaaaaag 240
tgctgcaaag gatataaatt tgttcttgga caatgcgtcc cagaagatta cgacgtttgt 300
gccgaggctc cctgtgaaca gcagtgcacg gacaactttg gccgagtgct gtgtacttgt 360
tatccgggat accgatatga ccgggagaga caccggaagc gggagaagcc atactgtctg 420
gatattgatg agtgtgccag cagcaatggg acgctgtgtg cccacatctg catcaatacc 480

ttgggcagct accgctgcga gtgccgggaa ggctacatcc gggaagatga tgggaagaca 540
tgtaccaggg gagacaaata tcccaatgac actggccatg agaagtctga gaacatggtg 600
aaagccggaa cttgctgtgc cacatgcaag gagttctacc agatgaagca gaccgtgctg 660
cagctgaagc aaaaggctca ccaggaccaa agggaagccc aggcttcccc ggtatgccag 720
gccctcctgg gcagcccggc ccacggggct caatgggacc catgggacca tctcctgatc 780
tgtcccacat taagcaaggc cggaggggcc ctgtgggtcc accaggggca ccaggaagag 840
atggttctaa gggggagaga ggagcgcctg ggcccagagg gtctccagga cccccctggtt 900
ctttcgactt cctgctactt atgctggctg acatccgcaa tgacatcact gagctgcagg 960
aaaaggtgtt cgggcaccgg actcactctt cagcagagga gttcccttta cctcaggaat 1020
ttcccagcta cccagaagcc atggacctgg gctctggaga tgaccatcca agaagaactg 1080
agacaagaga cttgagagcc cccagagact tctacccata gcacatcc 1128

<210> 48
<211> 1066
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7519667CB1

<400> 48
aggttgattg acttatgtgc aatttgggac gctggagttt accttccctc cgcagcctgg 60
aacagagcct cctctggtgt tgcaaggaag aggctgaatg aggcagagaa gctgagtgct 120
gtccaggagg cccagttaaa gcggctcgag gtgacaagac cccgagtgct ggggagcagg 180
gagcagggcc aggtgccgag gatggccagg cagccaccgc cgccctgggt ccatgcagcc 240
ttcctcctct gcctcctcag tcttggcgga gccatcgaaa ttcctatgga tccaagcatt 300
cagaatgagc tgacgcagcc gccaaccatc accaagcagt cagcgaagga tcacatcgtg 360
gacccccgtg ataacatcct gattgagtgt gaagcaaaag ggaaccctgc cccccagcttc 420
cactggacac gaaacagcag attcttcaac atcgccaagg acccccgggt gtccatgagg 480
aggaggtctg ggaccctggt gattgacttc cgcagtggcg ggcggccgga ggaatatgag 540
ggggaatatc agtgcttcgc ccgcaacaaa tttggcacgg ccctgtccaa taggatccgc 600
ctgcaggagt caccagcacc cccgaatgaa gcttacacca acaaccaagc ggacatcgcc 660
acccagggct ggttcattgg gcttatgtgc gccatcgccc tcctggtgct gatcctgctc 720
atcgtctgtt tcatcaagag gagtcgcggc ggcaagtacc cagtacgaga aaagaaggat 780
gttccccttg gccctgaaga ccccaaggaa gaggatggct catttgacta tagtgatgag 840
gacaacaagc ccctgcaggg cagtcagaca tctctggacg gcaccatcaa gcagcaggag 900
agtgacgaca gcctggtgga ctatggcgag ggtggcgagg gtcagttcaa tgaagacggc 960
tccttcatcg gccagtacac ggtcaaaaag gacaaggagg aaacagaggg caacgaaagc 1020
tcagaggcca cgtcacctgt caatgctatc tactctctgg cctaac 1066

<210> 49
<211> 2870
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7520638CB1

<400> 49
tgccgagccg agctgaccga gagccccatg gctgtgcagc gcgccgcgtc tccgcgccgc 60
ccgcccgccc cgctctggcc ccggctcctg ctgccgctgc tgttgctgct gctgcccgcg 120
ccgagcgagg gtcttggcca ctctgctgaa ctggcatttg ctgtggagcc aagtgatgat 180
gttgccgtcc ccgggcagcc tatagtgctg gactgcaggg tggaggggac ccctccagtg 240
cgaatcacct ggaggaagaa tggggtagag ctgccagaga gtacccactc caccttgctg 300
gccaatgggt ccttgatgat ccgtcacttc aggctggagc cgggaggcag cccttcggat 360
gaaggtgact atgagtgtgt ggcccagaac cgctttgggc tggtggtcag ccggaaggct 420
cgcatccaag ctgcaaccat gtcggacttc cacgtgcatc cccaggccac cgtgggtgag 480
gagggtggtg tggcccgctt ccagtgccaa atccgtgggc ttcccaaacc cctgatcact 540
tgggagaaga acagagtccc aattgacacg gacaatgaga ggtacacatt gctgcccaag 600
gggtcctgc agatcacagg acttcgagct gaggacggtg gcatcttcca ctgtgtggcc 660
tcaaacatcg ccagtatccg gatcagccac ggggccaggc tcactgtgtc aggctcgggc 720

tctggggcct acaaggagcc agccatcctc gtggggcctg agaacctcac cctgacagtg 780
caccagaccg cggtgcttga gtgtgtcgcc acgggcaacc cgcgccccat tgtgtcctgg 840
agccgcctgg atggtcgccc tatcggggtg gagggcatcc aggtgctggg cacaggaaac 900
ctcatcatct cagacgtgac ggtccagcac tctggcgtct acgtctgtgc agccaacaga 960
cctggcaccc gggtgaggag aacggcacag ggccggctgg tggtgcaagc cccagctgag 1020
tttgtgcagc atccccagtc catctccagg ccagctggga ccacagccat gttcacctgc 1080
caagcccagg gtgagccacc gcctcatgtc acgtggctga aaaatggaca ggtgctgggg 1140
ccaggaggcc acgtcaggct caagaataac aacagcacac tgaccatttc tggaatcggt 1200
cctgaggatg aagccattta tcagtgtgtg gccgagaaca gtgcgggctc atcacaggcc 1260
agtgccaggc tgaccgtact gtgggctgag ggctccccg ggcctccccg caatgtgcgg 1320
gcagtctctg tgtcttccac tgaggtgcgt gtgtcctgga gtgagccgct ggccaacacc 1380
aaggagatca tcggctacgt cctgcacatc aggaaggctg ctgacccacc ggagctggag 1440
tatcaggagg cagtcagcaa gagcaccttt cagcacctgg tcagcgacct ggagccctcc 1500
acagcctaca gtttctacat caaggcctac acaccaaggg gggccagctc agcctctgtg 1560
cccaccctag ctagcaccct gggtgaaggt gaggcctggg tgtggagcac aaaaccacag 1620
gcactcactc agggtatcag agcagaacca ccatgcacac ttgttctggt tgtgcgctgt 1680
gcagggtgcc cggccctgag gctgtcaggg ctgagtctac ttggaggaaa gggcttcttt 1740
ttctaattcc taggaaggtg ctgctttcat aagcacatgc ccttggccag aggccagagg 1800
gctgtcaggc ttttgggggt cggggtcaca ggtggaaggt agaaagatag agcttccact 1860
gggtcctcac atcgtggcct ccttctctac ctgcagcccc tgccccaccc ccactgtcag 1920
tgcgagtcct gggcagctcc tccttgcagc tgctgtggga gccttggccc cggctggccc 1980
agcacgaggg cggcttcaag ctgttttacc gcccagcaag caagacctcc ttcaccggcc 2040
ccatcctgct gcctggaacc gtctcctcct acaacctcag ccagctcgac cccactgcag 2100
tgtatgaggt gaagctgctc gcctacaacc agcatggaga tggcaatgcc acagtccgct 2160
ttgtgtcttt gaggggagca tctgagagga cagccttgag ccccaccatg tgactgccgg 2220
gaggaggagg ccgccaacca gacgtccacc acaggcatcg tcatcggcat ccacatcggg 2280
gtcacttgca tcatcttctg tgtcctcttc ctcctgttcg gccaaagggg cagggccctc 2340
ctgtgtaaag atgtggaaaa ccagctgtcc cctccacagg gtccccggag ccagagggac 2400
ctggcattct ggccctaaat ggggcgagac ggggacagcg gggccagctg ggccgagacg 2460
agaaacgtgt ggatatgaag gagctggagc agctgttccc cccggccagc gcagcagggc 2520
agccggaccc cagacccaca gatcctgcag cccccgctcc gtgtgaggag acccagctct 2580
ccgtgctgcc acttcagggg tgcggcctga tggaggggaa gacgacggag gcgaagacca 2640
cagaggccac ggctccctgc gccggcctgg cggctgcccc accaccccca gatggaggcc 2700
ctggcctcct cagtgaaggc caggcttcca ggcctgcagc ggcccgggtt acccagccag 2760
ctcactcgga acagtagcca gtgtctggca ggctccagag ggtggacgga gcggggccca 2820
ttctcaggtc aaaagcaaga tttctactgt catgtgggat ttggatggta 2870

<210> 50
<211> 2280
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7518947CB1

<400> 50
ggacaacagg aaggaagctt attaacccct ctcttccccc tccctcttct ccccggtgga 60
gaatcagctc ctcctcctcc cggtcgtgag accctggtgc aaaatgcaaa caaccacccg 120
ctttccccac cctgcgcgct tctcccagag gagggagagg gaagaggcga gggcgcagcc 180
ccaagccgag tccagcaagt ctgtcaatcg gacgtgaacg tcccgacgag acaacttgaa 240
cctcagggcc ctcacgccag ccggacaccg cagcccccata atattccgga cccctgctcc 300
ccgtgcaggt ggtgcaaacg cccctgctct ccggcgcccc ctgccgggaa gagggaaggc 360
ggcggcggcc gcggaggagc aggacgcttg gtccggacgg agctcggcgc tgggaagaag 420
ccgggagctt ccctgatggt gccgccgcct ccgggccggg gaggagctgc caggggccag 480
ctgggcagga gcctgggtcc gctgctgctg ctcctggcgt tgggacacac gtggacctac 540
agagaggagc cggaggacgg cgacagagaa atctgctcag agagcaaaat cgcgacgact 600
aaatacccgt gtctgaagtc ttcaggcgag ctcaccacat gctacaggaa aaagtgctgc 660
aaaggatata aatttgttct tggacaatgc atcccagaag attacgacgt ttgtgccgag 720
gctccctgtg aacagcagtg cacggacaac tttggccgag tgctgtgtac ttgttatccg 780
ggataccgat atgaccggga gagacaccgg aagcgggaga agccatactg tctggatatg 840
gatgagtgtg ccagcagcaa tgggacgctg tgtgcccaca tctgcatcaa taccttgggc 900
agctaccgct gcgagtgccg ggaaggctac atccgggaag atgatgggaa gacatgtacc 960

aggggagaca aatatcccaa tgacactggc cacgagaagt ctgagaacat ggtgaaagcc 1020
ggaacttgct gtgccacatg caaggagttc taccagatga agcagaccgt gctgcagctg 1080
aagcaaaaga ttgctctgct ccccaacaat gcagctgacc tgggcaagta tatcactggt 1140
gacaaggtgc tggcctcaaa cacctacctt ccaggacctc ctggcctgcc tgggggccag 1200
ggccctcccg gctcaccagg accaaaggga agcccaggct tccccggtat gccaggccct 1260
cctgggcagc ccggcccacg gggctcaatg ggacccatgg gaccatctcc tgatctgtcc 1320
cacattaagc aaggccggag gggccctgtg atccttttgt attacagggt ccaccagggg 1380
caccaggaag agatggttct aagggggaga gaggagcgcc tgggcccaga gggtctccag 1440
gaccccctgg ttctttcgac ttcctgctac ttatgctggc tgacatccgc aatgacatca 1500
ctgagctgca ggaaaaggtg ttcgggcacc ggactcactc ttcagcagag gagttccctt 1560
tacctcagga atttcccagc tacccagaag ccatggacct gggctctgga gatgaccatc 1620
caagaagaac tgagacaaga gacttgagag cccccagaga cttctaccca tagcacatcc 1680
caacaccgtc acgccaaagg aagagaaaga tcaactcacc tgcagttaaa ccatctaaag 1740
agaagaaaga ccactggaga cctagaaaac atacattttt ctcttctctt ctcctgacgt 1800
ctctccactc ctcttcttcc aaatacgatg ctattttcag agtcccctcc taggcctgca 1860
gacatgaggg agtgaatgat tgatttacct gcttctcact aagagtccat tggggtggtt 1920
tgcattgtaa cttttctttt acatcctatt tttccaggaa ctttggattt aagtactctc 1980
acagtgtctt aaatcataaa ttcttgaagt taaatttggc agagtatcaa aagggggaaa 2040
atgacaaagt gagctctaag aaaatgtgag gctacttcta agatgtgtgt tcacaataga 2100
ccataactcc tctagtatca aaattggggc tcttcagtta aaaaggggtg gggaggacaa 2160
acgtgtcgat gtgctttggt ggagaatttt ttccttgtgc ttctagtaga ctttaaatat 2220
tgtatccctt tgtcaaacct tgtttcccaa attcaattaa agagaggaga gaattgaatg 2280

<210> 51
<211> 1050
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7519652CB1

<400> 51
aggttgattg acttatgtgc aatttgggac gctggagttt accttccctc cgcagcctgg 60
aacagagcct cctctggtgt tgcaaggaag aggctgaatg aggcagagaa gctgagtgct 120
gtccaggagg cccagttaaa gcggctcgag gtgacaagac cccgagtgct ggggagcagg 180
gagcagggcc aggtgccgag gatggccagg cagccaccgc cgccctgggt ccatgcagcc 240
ttcctcctct gcctcctcag tcttggcgga gccatcgaaa ttcctatgga tctgacgcag 300
ccgccaacca tcaccaagca gtcagcgaag gatcacatcg tggacccccg tgataacatc 360
ctgattgagt gtgaagcaaa agggaaccct gcccccagct tccactggac acgaaacagc 420
agattcttca acatcgccaa ggacccccgg gtgtccatga ggaggaggtc tgggaccctg 480
gtgattgact tccgcagtgg cgggcggccg gaggaatatg agggggaata tcagtgcttc 540
gcccgcaaca aatttggcac ggccctgtcc aataggatcc gcctgcagga gtcaccagca 600
cccccgaatg aagcttacac caacaaccaa gcggacatcg ccacccaggg ctggttcatt 660
gggcttatgt gcgccatcgc cctcctggtg ctgatcctgc tcatcgtctg tttcatcaag 720
aggagtcgcg gcggcaagta cccagtacga gaaaagaagg atgttcccct tggccctgaa 780
gaccccaagg aagaggatgg ctcatttgac tatagtgatg aggacaacaa gcccctgcag 840
ggcagtcaga catctctgga cggcaccatc aagcagcagg agagtgacga cagcctggtg 900
gactatggcg agggtggcga gggtcagttc aatgaagacg gctccttcat cggccagtac 960
acggtcaaaa aggacaagga ggaaacagag ggcaacgaaa gctcagaggc cacgtcacct 1020
gtcaatgcta tctactctct ggcctaacgg 1050

<210> 52
<211> 2055
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7520381CB1

<400> 52
tacgccagac ccgctcagac cctcctgcca ggtgacagcc gccaagatgg ggtcttgggc 60

cctgctgtgg cctcccctgc tgttcaccgg gctgctcgtc cgaccccccgg ggaccatggc 120
ccaggcccag tactgctctg tgaacaagga caactttgaa gtagaggaga acacaaatgt 180
caccgagccg ctggtggaca tccacgtccc ggagggccag gaggtgaccc tcggagcctt 240
gtccaccccc tttgcatttc ggatccaggg aaaccagctg tttctcaacg tgactcctga 300
ttacgaggag aagtcactgc ttgaggctca gctgctgtgt cagagcggag gcacattggt 360
gacccagcta agggtgttcg tgtcagtgct ggacgtcaat gacaatgccc ccgaattccc 420
ctttaagacc aaggagataa gggtggagga ggacacgaaa gtgaactcca ccgtcatccc 480
cgagacgcaa ctgcaggctg aggaccgcga caaggacgac attctgttct acaccctcca 540
ggaaatgaca gcaggtgcca gtgactactt ctccctggtg agtgtaaacc gtcccgccct 600
gaggctggac cggcccctgg acttctacga gcggccgaac atgaccttct ggctgctggt 660
gcgggacact ccgggggaga atgtggaacc cagccacact gccaccgcca cactagtgct 720
gaacgtggtg cccgccgacc tgcggccccc gtggttcctg ccctgcacct tctcagatgg 780
ctacgtctgc attcaagctc agtaccacgg agctgtcccc acggggcaca tactgccatc 840
tcccctcgtc ctgcgtcccg gacccatcta cgctgaggac ggagaccgcg gcatcaacca 900
gcccatcatc tacagcatct ttaggggaaa cgtgaatggt acattcatca tccacccaga 960
ctcgggcaac ctcaccgtgg ccaggagtgt ccccagcccc atgaccttcc ttctgctggt 1020
gaagggccaa caggccgacc ttgcccgcta ctcagtgacc caggtcaccg tggaggctgt 1080
ggctgcggcc gggagcccgc cccgcttccc ccagagactg tatcgtggca ccgtggcgcg 1140
tggcgctgga gcgggcgttg tggtcaagga tgcagctgcc ccttctcagc ctctgaggat 1200
ccaggctcag gacccggagt tctcggacct caactcggcc atcacatatc gaattaccaa 1260
ccactcacac ttccggatgg agggagaggt tgtgctgacc accaccacac tggcacaggc 1320
gggagccttc tacgcagagg acagaggcca gatgtggggt ccagcgaatc ccatgattcc 1380
cacggctgcc gctcaccctg caagtccctg cacctcttgt gagctcctgt ggcggcccct 1440
cggaggacaa gcgcttctcg gtggtggata tggcggccct gggcggggtg ctgggtgcgc 1500
tgctgctgct ggctctcctt ggcctcaccg tccttgtcca caagcactat ggcccccggc 1560
tcaagtgctg ctctggcaaa gctccggagc cccagcccca aggctttgac aaccaggcgt 1620
tcctccctga ccacaaggcc aactgggcgc ccgtccccag ccccacgcac gaccccaagc 1680
ccgcggaggc accgatgccc gcagagcccg caccccccgg ccctccctcc ccaggcggtg 1740
cccctgagcc ccccgcagcg gcccgagctg gcggaagccc cacggcggtg aggtccatcc 1800
tgaccaagga gcggcggccg gagggcgggt acaaggccgt ctggtttggc gaggacatcg 1860
ggacggaggc agacgtggtc gttctcaacg cgcccaccct ggacgtggat ggcgccagtg 1920
actccggcag cggcgacgag ggcgagggcg cggggagggg tgggggtccc tacgacgcac 1980
ccggtggtga tgactcctac atctaagtgg cccctccacc ctctccccca gccgcacggg 2040
cactggaggt ctcga 2055

<210> 53
<211> 2120
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7520409CB1

<400> 53
tacgccagac ccgctcagac cctcctgcca ggtgacagcc gccaagatgg ggtcttgggc 60
cctgctgtgg cctcccctgc tgttcaccgg gctgctcgtc cgaccccccgg ggaccatggc 120
ccaggcccag tactgctctg tgaacaagga catctttgaa gtagaggaga acacaaatgt 180
caccgagccg ctggtggaca tccacgtccc ggagggccag gaggtgaccc tcggagcctt 240
gtccaccccc tttgcatttc ggatccaggg aaaccagctg tttctcaacg tgactcctga 300
ttacgaggag aagtcactgc ttgaggctca gctgctgtgt cagagcggag gcacattggt 360
gacccagcta agggtgttcg tgtcagtgct ggacgtcaat gacaatgccc ccgaattccc 420
ctttaagacc aaggagataa gggtggagga ggacacgaaa gtgaactcca ccgtcatccc 480
cgagacgcaa ctgcaggctg aggaccgcga caaggacgac attctgttct acaccctcca 540
ggaaatgaca gcaggtgcca gtgactactt ctccctggac actccggggg agaatgtgga 600
acccagccac actgccaccg ccacactagt gctgaacgtg gtgcccgccg acctgcggcc 660
cccgtggttc ctgccctgca ccttctcaga tggctacgtc tgcattcaag ctcagtacca 720
cggggctgtc cccacggggc acatactgcc atctcccctc gtcctgcgtc ccggacccat 780
ctacgctgag gacggagacc gcggcatcaa ccagcccatc atctacagca tctttagggg 840
aaacgtgaat ggtacattca tcatccaccc agactcgggc aacctcaccg tggccaggag 900
tgtccccagc cccatgacct tccttctgct ggtgaagggc caacaggccg accttgcccg 960
ctactcagtg acccaggtca ccgtggaggc tgtggctgcg gccgggagcc cgccccgctt 1020
cccccagagc ctgtatcgtg gcaccgtggc gcgtggcgct ggagcgggcg ttgtggtcaa 1080

ggatgcagct gccccttctc agcctctgag gatccaggct caggacccgg agttctcgga 1140
cctcaactcg gccatcacat atcgaattac caaccactca cacttccgga tggagggaga 1200
ggttgtgctg accaccacca cactggcaca ggcgggagcc ttctacgcag aggttgaggc 1260
ccacaacacg gtgacctctg gcaccgcaac cacagtcatt gagatacaag tttccgaaca 1320
ggagcccccc tccacagcac agaccccaga ggcaggaacc tctcagccga tgcccccccgg 1380
tatgggaacc agcacctccc accaaccaac cacacctggt gggggcacag cacagacccc 1440
agagccagga acctctcagc cgatgcccct cagcaagagc accccatctt caggtgcggc 1500
cccctcggag gacaagcgct tctcgtcggt ggatatggcg gccctgggcg gggtgctggg 1560
tgcgctgctg ctgctggctc tccttggcct cgccgtcctt gtccacaagc actatggccc 1620
ccggctcaag tgctgctctg gcaaagctcc ggagccccag ccccaaggct ttgacaacca 1680
ggcgttcctc cctgaccaca aggccaactg ggcgcccgtc cccagcccca cgcacgaccc 1740
caagcccgcg gaggcaccga tgcccgcaga gcccgcaccc cccggccctg cctccccagg 1800
cggtgcccct gagcccccccg cagcggcccg agctggcgga agccccacgg cggtgaggtc 1860
catcctgacc aaggagcggc ggccagaggg cgggtacaag gctgtctggt ttggcgagga 1920
catcgggacg gaggcagacg tggtcgttct caacgcgccc accctggacg tggatggcgc 1980
cagtgactcc ggcagcggcg atgagggcga gggcgcgggg aggggtgggg gtccctacga 2040
tgcgcccggt ggtgatgact cctacatcta agtggcccct ccaccctctc cccagccgc 2100
acgggcactg gaggtctcga 2120

<210> 54
<211> 2852
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7520538CB1

<400> 54
tagggcgcag cgcggttggt cctcgccccc tcccgcccgc cggcctacca ggccatgggg 60
gcgtcccggg accgcgggct ggccgcgctc tggtgccttg ggctcctggg gggcctggcg 120
cgcgtcgcgg gcacgcacta ccgctacctc tggaggggct gctacccatg tcacctgggc 180
caggccggct accccgtgag cgccggtgac cagaggccag atgtggacga atgccgaacc 240
cacaacggtg gctgccagca ccggtgcgtg aacaccccag gctcctacct ctgtgagtgc 300
aagcccggct tccggctcca cactgacagc aggacctgcc tggccattaa ctcctgcgcc 360
ctgggcaatg gcggctgcca gcaccactgt gtccagctca caatcactcg gcatcgctgc 420
cagtgccggc ccgggttcca gctccaggag gacggcaggc attgtgtccc cctggaggag 480
ccgatggtgg acctggacgg cgagctgcct ttcgtgcggc ccctgcccca cattgccgtg 540
ctccaggacg agctgccgca actcttccag gatgacgacg tcggggccga tgaggaagag 600
gcagagttgc ggggcgaaca cacgctcaca gagaagtttg tctgcctgga tgactccttt 660
ggccatgact gcagcttgac ctgtgatgac tgcaggaacg gagggacctg tctcctgggc 720
ctggatggct gtgattgccc cgagggctgg actgggctca tctgcaatga gacttgtcct 780
ccggacacct ttgggaagaa ctgcagcttc tcctgcagct gtcagaatgg tgggacctgc 840
gactctgtca cgggggcctg ccgctgcccc ccgggtgtca gtggaactaa ctgtgaggat 900
ggctgcccca agggctacta tggcaagcac tgtcgcaaga aatgcaactg tgccaaccgg 960
ggccggtgcc accgcctcta cggggcctgc ctctgcgacc cagggctcta cggccgcttc 1020
tgccacctca cctgcccgcc gtgggccttt gggccgggct gctcggagga gtgccagtgt 1080
gtgcagcccc acacgcagtc ctgtgacaag agggatggca gctgctcctg caaggctggc 1140
ttccggggcg agcgctgtca ggcagagtgt gagccgggct actttgggcc ggggtgctgg 1200
caggcatgca cctgcccagt gggcgtggcc tgtgactccg tgagcggcga gtgtgggaag 1260
cggtgtcctg ctggcttcca gggagaggac tgtggccaag agtgcccggt ggggacgttt 1320
ggcgtgaact gctcgagctc ctgctcctgt ggggggcccc cctgccacgg ggtcacgggg 1380
cagtgccggt gtccgccggg gaggactggg gaagactgtg aggcagattg tcccgagggc 1440
cgctgggggc tgggctgcca ggagatctgc ccagcatgcc agcacgctgc ccgctgcgac 1500
cctgagaccg gagcctgcct gtgcctccct ggcttcgtcg gcagccgctg ccaggacgtg 1560
tgcccagcag gctggtatgg tcccagctgc cagacaaggt gctcttgtgc caatgatggg 1620
cactgccacc cagccaccgg acactgcagc tgtgcccccg ggtggaccgg ctttagctgc 1680
cagagagcct gtgatactgg gcactgggga cctgactgca gccacccctg caactgcagc 1740
gctggccacg ggagctgtga tgccatcagc ggcctgtgtc tgtgtgaggc tggctacgtg 1800
ggcccgcggt gcgagcagcc ctgtgccaag ggcacattcg ggcctcactg tgaggggcgc 1860
tgtgcctgcc ggtggggagg cccctgccac cttgccacca gggcctgcct ctgccctccg 1920
gggtggcggg gggctccagg tgtgagaatg cctgcctgcg gggctggttt ggagaggcct 1980
gtgcccagcg ctgcagctgc ccgcctggcg ctgcctgcca ccacgtcact ggggcctgcc 2040

gctgtccccc tggcttcact ggctccggct gcgagcaggg atgtccgccc gggcggtatg 2100
ggccaggctg tgaacagctg tgtgggtgtc tcaacggggg ctcctgtgat gcggccacgg 2160
gggcctgccg ctgccccact gggttcctcg ggacggactg caacctcacc tgtccgcagg 2220
gccgcttcgg ccccaactgc acccacgtgt gtgggtgtgg gcagggggcg gcctgcgacc 2280
ctgtgaccgg cacctgcctc tgcccccggg agagagccgg cgtccgctgt gagcgaggct 2340
gcccccagaa ccggtttggc gtgggctgcg agcacacctg ctcctgcaga aatgggggcc 2400
tgtgccacgc cagcaacggc agctgctcct gtggcctggg ctggacgggg cggcactgcg 2460
agctggcctg tccccctggg cgctacggag ccgcctgcca tctggagtgc tcctgccaca 2520
acaacagcac gtgtgagcct gccacgggca cctgccgctg cggccccggc ttctatggcc 2580
aggcctgcga gcacccctgt cccctggctt ccacggggct ggctgccagg ggttgtgctg 2640
gtgtcaacat ggagcccccct gcgaccccat cagtggccga tgcctctgcc ctgccggctt 2700
ccacggccac ttctgtgaga ggggtgggcc cctccggctc cccgagaacc cgtccttagc 2760
ccagggctca gcgggcacac tgcccgcctc cagcagaccc acatcccgga cgctggacca 2820
gcgaggcact agtagagcag tcccgtggag ca 2852

<210> 55
<211> 2227
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7520667CB1

<400> 55
tacgccagac ccgctcagac cctcctgcca ggtgacagcc gccaagatgg ggtcttgggc 60
cctgctgtgg cctcccctgc tgttcaccgg gctgctcgtc cgacccccgg ggaccatggc 120
ccaggcccag tactgctctg tgaacaagga catctttgaa gtagaggaga acacaaatgt 180
caccgagccg ctggtggaca tccacgtccc ggagggccag gaggtgaccc tcggagcctt 240
gtccacccccc tttgcatttc ggatccaggg aaaccagctg tttctcaacg tgactcctga 300
ttacgaggag aagtcactgc ttgaggctca gctgctgtgt cagagcggag gcacattggt 360
gacccagcta agggtgttcg tgtcagtgct ggacgtcaat gacaatgccc ccgaattccc 420
ctttaagacc aaggagataa gggtggagga ggacacgaaa gtgaactcca ccgtcatccc 480
tgagacgcaa ctgcaggctg aggaccgcga caaggacgac attctgttct acaccctcca 540
ggaaatgaca gcaggtgcca gtgactactt ctccctggtg agtgtaaacc gtcccgccct 600
gaggctggac cggcccctgg acttctacga gcggccgaac atgaccttct ggctgctggt 660
gcgggacact ccgggggaga atgtggaacc caagcacact ggcacctgca cactagtgct 720
gaacgtggtg cccgccgacc tgcggccccc gtggttcctg ccctgcacct tctcagatgg 780
ctacgtctgc attcaagctc agtaccacgg ggctgtcccc acggggcaca tactgccatc 840
tcccctcgtc ctgcgtcccg gacccatcta cgctgaggac ggagaccgcg gcatcaacca 900
gcccatcatc tacagcatct ttaggggaaa cgtgaatggt acattcatca tccacccaga 960
ctcgggcaac ctcaccgtgg ccaggagtgt ccccagcccc atgaccttcc ttctgctggt 1020
gaagggccaa caggccgacc ttgcccgcta ctcagtgacc caggtcaccg tggaggctgt 1080
ggctgcggcc gggagcccgc cccgcttccc ccagagactg tatcgtggca ccgtggcgcg 1140
tggcgctgga gcgggcgttg tggtcaagga tgcagctgcc ccttctcagc ctctgaggat 1200
ccaggctcag gacccggagt tctcggacct caactcggcc atcacatatc gaattaccaa 1260
ccactcacac ttccggatgg agggagaggt tgtgctgacc accaccacac tggcacaggc 1320
gggagccttc tacgcagagg ttgaggccca caacacggtg acctctggca ccgcaaccac 1380
agtcattgag atacaagttt ccgaacagga gcccccctcc acagccacac ccggtggggg 1440
cacagcacag accccagagg caggaacctc tcagccgatg ccccccggta tgggaaccag 1500
cacctcccac caaccaacca cgcccggtgg gggcacagca cagaccccag agccaggaac 1560
ctctcagccg atgcccctca gcaagagcac cccatcttca ggtggcggcc cctcggagga 1620
caagcgcttc tcggtggtgg atatggcggc cctgggcggg gtgctgggtg cgctgctgct 1680
gctggctctc cttggcctca ccgtccttgt ccacaagcac tatggccccc ggctcaagtg 1740
ctgctctggc aaagctccgg agccccagcc ccaaggcttt gacaaccagg cgttcctccc 1800
tgaccacaag gccaactggg cgcccgtccc cagccccacg cacgacccca agcccgcgga 1860
ggcaccgatg cccgcagagc ccgcaccccc cggccctgcc tccccaggcg gtgcccctga 1920
gcccccccgca gcggcccgag ctggcggaag ccccacggcg gtgaggtcca tcctgaccaa 1980
ggagcggcgg ccagagggcg ggtacaaggc tgtctggttt ggcgaggaca tcgggacgga 2040
ggcagacgtg gtcgttctca acgcgcccac cctggacgtg gatggcgcca gtgactccgg 2100
cagcggcgat gagggcgagg gcgcggggag gggtgggggt ccctacgatg cgcccggtgg 2160
tgatgactcc tacatctaag tggcccctcc accctctccc ccagccgcac gggcactgga 2220
ggtctcg 2227

<210> 56
<211> 4953
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7517567CB1

<400> 56
ggaggccggg gtcgggacgc gttgtgcagc gggtaagcgc acggccgagc gagcatggag 60
ggggaccggg tggccgggcg gccggtgctg tcgtcgttac cagtgctact gctgctgcag 120
ttgctaatgt tgcgggccgc ggcgctgcac ccagacgagc tcttcccaca cggggagtcg 180
tggggggacc agctcctgca ggaaggcgac gacgaaagct cagccgtggt gaagctggcg 240
aatcccctgc acttctacga agcccgattc agcaacctct acgtgggcac caacggcatc 300
atctccactc aggacttccc cagggaaacg cagtatgtgg actatgattt ccccaccgac 360
ttcccggcca tcgccccttt tctggcggac atcgacacga gccacggcag aggccgagtc 420
ctgtaccgag aggacacctc ccccgcagtg ctgggcctgg ccgccccgcta tgtgcgcgct 480
ggcttcccgc gctctgcgcg ctttaccccc acccacgcct tcctggccac ctgggagcag 540
gtaggcgctt acgaggaggt caagcgcggg gcgctgccct cgggagagct gaacactttc 600
caggcagttt tggcatctga tgggtctgat agctacgccc tctttcttta tcctgccaac 660
ggcctgcagt tccttggaac ccgccccaaa gagtcttaca atgtccagct tcagcttcca 720
gctcgggtgg gcttctgccg aggggaggct gatgatctga agtcagaagg accatatttc 780
agcttgacta gcactgaaca gtctgtgaaa aatctctatc aactaagcaa cctggggatc 840
cctggagtgt gggctttcca tatcggcagc acttccccgt tggacaatgt caggccagct 900
gcagttggag acctttccgc tgcccactct tctgttcccc tgggacgttc cttcagccat 960
gctacagccc tggaaagtga ctataatgag gacaatttgg attactacga tgtgaatgag 1020
gaggaagctg aataccttcc gggtgaacca gaggaggcat tgaatggcca cagcagcatt 1080
gatgtttcct tccaatccaa agtggataca aagcctttag agggtaggat ctcccctcca 1140
gattctgatc tgtcctcccc cttgcatcca acacctactt attggccatt ctatcctgaa 1200
acagaatctt ccaccttgga tcctcacacc aaagaaggaa catctctggg agaggtaggg 1260
ggcccagatt taaaaggcca agttgagccc tgggatgaga gagagaccag aagcccagct 1320
ccaccagagg tagacagaga ttcactggct ccttcctggg aaaccccacc accgtacccc 1380
gaaaacggaa gcatccagcc ctacccagat ggagggccag tgccttcgga aatggatgtt 1440
cccccagctc atcctgaaga agaaattgtt cttcgaagtt accctgcttc agatcacact 1500
acacccttaa gtcgagggac gcatgaggtg ggactggaag gcaacatagg ttccaacacc 1560
gaggtcttca cgtataatgc tgccaacaag gaaacctgtg aacacaacca cagacaatgc 1620
tcccggcatg ccttctgcac ggactatgcc actggcttct gctgccactg ccaatccaag 1680
ttttatggaa atgggaagca ctgtctgcct gaaggggcac ctcaccgagt gaatgggaaa 1740
gtgagtggcc acctccacgt gggccataca cccgtgcact tcactgatgt ggacctgcat 1800
gcgtatatcg tgggcaatga tggcagagcc tacacggcca tcagccacat cccacagcca 1860
gcagcccagg ccctcctccc cctcacacca attggaggcc tgtttggctg gctctttgct 1920
ttagaaaaac ctggctctga gaacggcttc agcctcgcag gtgctgcctt tacccatgac 1980
atggaagtta cattctaccc gggagaggag acggttcgta tcactcaaac tgctgaggga 2040
cttgacccag agaactacct gagcattaag accaacattc aaggccaggt gccttacgtc 2100
tcagcaaatt tcacagccca catctctccc tacaaggagc tgtaccacta ctccgactcc 2160
actgtgacct ctacaagttc cagagactac tctctgactt ttggtgcaat caaccaaaca 2220
tggtcctacc gcatccacca gaacatcact taccaggtgt gcaggcacgc cccccagacac 2280
ccgtccttcc ccaccaccca gcagctgaac gtggaccggg tctttgcctt gtataatgac 2340
gaagaaagag tgcttagatt tgctgtgacc aatcaaattg gcccggtcaa agaagattca 2400
gaccccactc cggtgaatcc ttgctatgat gggagccaca tgtgtgacac aacagcacgg 2460
tgccatccag ggacaggtgt agattacacc tgtgagtgcg catctgggta ccagggagat 2520
ggacggaact gtgtggatga aaatgaatgt gcaactggct ttcatcgctg tggccccaac 2580
tctgtatgta tcaacttgcc tggaagctac aggtgtgagt gccggagtgg ttatgagttt 2640
gcagatgacc ggcatacttg catcttgatc accccacctg ccaacccctg tgaggatggc 2700
agtcatacct gtgctcctgc tgggcaggcc cggtgtgttt accatggagg cagcacgttc 2760
agctgtgcct gcctgcctgg ttatgccggc gatgggcacc agtgcactga tgtagatgaa 2820
tgctcagaaa acagatgtca ccctgcagct acctgctaca atactcctgg ttccttctcc 2880
tgccgttgtc aacccggata ttatggggat ggatttcagt gcatacctga ctccacctca 2940
agcctgacac cctgtgaaca acagcagcgc catgcccagg cccagtatgc ctaccctggg 3000
gcccggttcc acatccccca atgcgacgag cagggcaact tcctgcccct acagtgtcat 3060
ggcagcactg gtttctgctg gtgcgtggac cctgatggtc atgaagttcc tggtacccag 3120
actccacctg gctccacccc gcctcactgt ggaccatcac cagagcccac ccagaggccc 3180

ccgaccatct gtgagcgctg gagggaaaac ctgctggagc actacggtgg caccccccga 3240
gatgaccagt acgtgcccca gtgcgatgac ctgggccact tcatccccct gcagtgccac 3300
ggaaagagcg acttctgctg gtgtgtggac aaagatggca gagaggtgca gggcacccgc 3360
tcccagccag gcaccacccc tgcgtgtata cccaccgtcg ctccacccat ggtccggccc 3420
acgcccccggc cagatgtgac ccctccatct gtgggcacct tcctgctcta tactcagggc 3480
cagcagattg gctacttacc cctcaatggc accaggcttc agaaggatgc agctaagacc 3540
ctgctgtctc tgcatggctc cataatcgtg ggaattgatt acgactgccg ggagaggatg 3600
gtgtactgga cagatgttgc tggacggaca atcagccgtg ccggtctgga actgggagca 3660
gagcctgaga cgatcgtgaa ttcaggtctg ataagccctg aaggacttgc catagaccac 3720
atccgcagaa caatgtactg gacggacagt gtcctggata agatagagag cgccctgctg 3780
gatggctctg agcgcaaggt cctcttctac acagatctgg tgaatccccg tgccatcgct 3840
gtggatccaa tccgaggcaa cttgtactgg acagactgga atagagaagc tcctaaaatt 3900
gaaacgtcat ctttagatgg agaaaacaga agaattctga tcaatacaga cattggattg 3960
cccaatggct taacctttga ccctttctct aaactgctct gccgggcaga tgcaggaacc 4020
aaaaaactgg agtgtacact acctgatgga actggacggc gtgtcattca aaacaacctc 4080
aagtacccct tcagcatcgt aagctatgca gatcacttct accacacaga ctggaggagg 4140
gatggtgttg tatcagtaaa taaacatagt ggccagttta ctgatgagta tctcccagaa 4200
caacgatctc acctctacgg gataactgca gtctacccct actgcccaac aggaagaaag 4260
taagtacagt aatgtaaagg aagacttgga gtttacaatc agaacctgga ccctaaagaa 4320
cagtgactgc aaaggcaaag aaagtaaaaa aggaattggc cattagacgt tcctgagcat 4380
ccaagatgaa cattttgtag tgcaaaaaga cttttgtgaa aagctgatac ctcaatcttt 4440
actactgtat ttttaaaaat gaaggttgtt attgcaagtt taaaaaggta acagaatttt 4500
aactgttgct tattaaagca acttcttgta aacatttatc attaatattt aaaagatcaa 4560
attcattcaa ctaagaatta gagtttaaga ctctaaacct gatttttgcc atggattcct 4620
tctggccaag aaattaaagc acatgtgatc aatataacaa tataatccta aaccttgaca 4680
gttggagaag ccaatgcaga actgatggga aaggaccaat tatttatagt ttcccaacaa 4740
aagttctaag atttttttacc tctgcatcag tgcatttcta tttatatcaa aaggtgctaa 4800
aatgattcaa tttgcatttt ctgatcctgt agtgcctcta tagaagtacc cacagaaagt 4860
aaagtatcac atttataaat accaaagatg taacaatttt aaaattttct agattactcc 4920
aataaagtat aagggccgga ttccagcaca ctg 4953

<210> 57
<211> 1133
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7519867CB1

<400> 57
ttgtgggaga ctagatttag gaagtaaaga tcacagggtg ggaaataaaa gctgtggccc 60
ccaggagttc tggacactgg gggagagtgg ggtgacatga gtgactccaa ggaaccaaga 120
ctgcagcagc tgggcctcct ggaggaggaa cagctgagag gccttggatt ccgacagact 180
cgaggataca agagcttagc agtgtccaag gtccccagct ccataagtca ggaacaatcc 240
aggcaagacg cgatctacca gaacctgacc cagcttaaag ctgcagtggg tgagctctca 300
gagaaatcca agctgcagga gatctaccag gagctgaccc agctgaaggc tgcagtgggt 360
gagcttccag agaaatctaa gctgcaggag atctaccagg agctgacccg gctgaaggct 420
gcagtgggtg agcttccaga gaaatctaag ctgcaggaga tctaccagga gctgacccag 480
ctgaaggctg cagtggaacg cctgtgccac ccctgtccct gggaatggac attcttccaa 540
ggaaactgtt acttcatgtc taactcccag cggaactggc acgactccat caccgcctgc 600
aaagaagtgg gggcccagct cgtcgtaatc aaaagtgctg aggagcagaa cttcctacag 660
ctgcagtctt ccagaagtaa ccgcttcacc tggatgggac tttcagatct aaatcaggaa 720
ggcacgtggc aatgggtgga cggctcacct ctgttgccca gcttcaagca gtattggaac 780
agaggagagc ccaacaacgt tggggaggaa gactgcgcgg aatttagtgg caatggctgg 840
aacgacgaca aatgtaatct tgccaaattc tggatctgca aaaagtccgc agcctcctgc 900
tccagggatg aagaacagtt tctttctcca gcccctgcca ccccaaaccc ccctcctgcg 960
tagcagaact tcaccccctt ttaagctaca gttccttctc tccatccttc gaccttcaca 1020
aaatctctgg gactgttctt tgtcagattc ttcctccttt agaaggctgg gtcccattct 1080
gtccttcttg tcatgcctcc aatttccact ggtgtagagc ttgtttttct gga 1133

<210> 58
<211> 1786

<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7519911CB1

<400> 58
tgccccggag ctcagtccca cccccctccgc cgatgaggcc atgaggcacc gaacggacct 60
gggccaggtc accccctccc tgtcccatcc ccaatccggg ggactccctg ctgtctggga 120
gctgagggag ccctggggcc ggggtgcagc tgagggcctc ttctagggac ccaactcctg 180
atccagcccc tgccggccgc accttccgca gggtgtcaca gaatcatgga ctgagtccac 240
ttggccccca ctgcacagat ggggaaactg aagcctggga gagtggagtg gctggcctca 300
ggccacacgg aaaggcccca cctcttccag aacctcctgc tcttcctgtg ggccctgctg 360
aactgtggtt tgggggtcag tgctcagggt ccgggcgagt ggaccccgtg ggtgtcctgg 420
acccgctgct ccagctcctg cgggcgtggc gtctccgtgc gcagccggcg ctgcctccgg 480
cttcctgggg aagaaccgtg ctggggagac tcccatgagt accgcctctg ccagttgcca 540
gactgccccc caggggctgt gcccttccga gacctacagt gtgccctgta caatggccgc 600
cctgtcctgg gcacccagaa gacctaccag tgggtgccct tccatggggc gcccaaccag 660
tgcgacctca actgcctggc tgaggggcac gccttctacc acagcttcgg ccgcgtcctg 720
gacggcaccg cctgcagccc gggtgcccag gggtctgcg tggctggccg ctgccttagc 780
gccggctgtg atgggttgtt gggctcgggt gccctcgagg accgctgtgg ccgctgcgga 840
ggcgccaacg actcgtgcct tttcgtgcag cgcgtgtttc gtgacgccgg tgccttcgct 900
gggtactgga acgtgaccct gatccccgag ggcgccagac acatccgcgt ggaacacagg 960
agccgcaacc acctggcact gatgggggc gatgggcgct acgtgcttaa tgggcactgg 1020
gtggtcagcc caccagggac ctacgaggcg gccggcacgc atgtggtcta cacccgagac 1080
acagggcccc aggagacatt gcaagcagcc gggcccacct cccatgacct gctcctacag 1140
gtcctcctgc aggagcccaa ccctggcatc gagtttgagt tctggctccc tcgggagcgc 1200
tacagcccct tccaggctcg tgtgcaggcc ctgggctggc ccctgaggca gcctcagccc 1260
cggggggtgg agcctcagcc ccccgcagcc cctgctgtca cccctgcaca gaccccaacg 1320
ctggccccag acccctgccc accctgccct gacacccgcg gccgcgccca ccgactactc 1380
cactattgcg gcagtgactt tgtgttccag gcccgagtgc tgggccacca ccaccaggcc 1440
caggagaccc gctatgaggt gcgcatccag ctcgtctaca agaaccgctc gccactgcgg 1500
gcacgcgagt acgtgtgggc gccaggccac tgcccctgcc cgatgctggc accccaccgg 1560
gactacctga tggctgtcca gcgtcttgtc agccccgacg gcacacagga ccagctgctg 1620
ctgccccacg ccggctacgc ccggccctgg agccctgcgg aggacagccg catacgcctg 1680
actgcccggc gctgtcctgg ctgagcccct gcaggagccc cggccacaca cagcaagaaa 1740
gatacatctg accagcctca acgtcaacgt atttcccctc tcaccc 1786

<210> 59
<211> 1483
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7520005CB1

<400> 59
tggtgcagca tgggcggccc gcgggcctgg gcgctactct gcctcgggct cctgctcccg 60
ggaggcggcg ctgcgtggag catcggggca gctccgttct ccggacgcag gaactggtgc 120
tcctatgtgg tgacccgcac catctcatgc catgtgcaga atggcaccta ccttcagcga 180
gtgctgcaga actgccctg gcccatgagc tgtccgggga gcagctacag aactgtggtg 240
agacccacat acaaggtgat gtacaaggta gtgaccgccc gtgagtggag gtgctgccct 300
gggcactcag gagtgagctg cgaggaagtt gcagcttcct ctgcctcctt ggagcccatg 360
tggtcgggca gtaccatgcg gcggatggcg cttcggccca cagccttctc aggttgtctc 420
aactgcagca aagtgtcaga gctgacagag cggctgaagg tgctggaggc caagatgacc 480
atgctgactg tcatagagca gccagtacct ccaacaccag ctacccctga ggaccctgcc 540
ccgctctggg gtccccctcc tgcccagggc agccccggag atggaggcct ccaggggctg 600
ccaggagcca tagagagtgt gagggtcccg ctgcttcccc gaaatgacca agtcggtgct 660
tgggggcttc ccgggcccac cggccccaag ggagatgccg gcagtcgggg cccaatgggg 720
atgagaggcc caccaggtcc acagggcccc ccagggagcc ctggccgggc tggagctgtg 780
ggcacccctg gagagagggg acctcctggg ccaccagggc ctcctggccc ccctgggccc 840

ccagcccctg ttgggccacc ccatgcccgg atctcccagc atggagaccc attgctgtcc 900
aacaccttca ctgagaccaa caaccactgg ccccagggac ccactgggcc tccaggccct 960
ccagggccca tgggtccccc tgggcctcct ggccccacag gtgtccctgg gagtcctggt 1020
cacataggac ccccaggccc cactggaccc aaaggaatct ctggccaccc aggagagaag 1080
ggcgagagag gactgcgtgg ggagcctggc cccaaggct ctgctgggca gcgggggggaa 1140
cctggcccta agggagaccc tggtgagaag agccactggg gggaggggtt gcaccagcta 1200
cgcgaggctt tgaagatttt agctgagagg gttttaatct tggaaacaat gattgggctc 1260
tatgaaccag agctggggtc tggggcgggc cctgccggca caggcacccc cagcctcctt 1320
cggggcaaga ggggcggaca tgcaaccaac taccggatcg tggcccccag gagccgggac 1380
gagagaggct gagggtggtg gcggcccctg aggcagacca ggccaggctt cccctcctac 1440
ctggactcgg ccagctgaag aaccgaattc gagcacactg cgg 1483

<210> 60
<211> 1306
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<223> Incyte ID No: 7520484CB1

<400> 60
cggagggcgc ctggtgcagc atgggcggcc cgcgggcttg ggcgctgctc tgcctcgggc 60
tcctgctccc gggaggcggc gctgcgtgga gcatcggggc agctccgttc tccggacgca 120
ggaactggtg ctcctatgtg gtgacccgca ccatctcatg ccatgtgcag aatggcacct 180
accttcagcg agtgctgcag aactgcccct ggcccatgag ctgtccgggg agcagctaca 240
gaactgtggt gagacccaca tacaaggtga tgtacaagat agtgaccgcc cgtgagtgga 300
ggtgctgccc tgggcactca ggagtgagct gcgaggaagg ttgtctcaac tgcagcaaag 360
tgtcagagct gacagagcgg ctgaaggtgc tggaggccaa gatgaccatg ctgactgtca 420
tagagcagcc agtacctcca acaccagcta cccctgagga ccctgccccg ctctggggtc 480
cccctcctgc ccagggcagc cccggagatg gaggcctcca ggaccaagtc ggtgcttggg 540
ggcttcccgg gcccaccggc cccaagggag atgccggcag tcggggccca atggggatga 600
gaggcccacc aggtccacag ggcccccgag ggagccctgg ccgggctgga gctgtgggca 660
cccctggaga gaggggacct cctgggccac cagggcctcc tggccccccct gggcccccag 720
cccctgttgg gccaccccat gcccggatct cccagcatgg agacccattg ctgtccaaca 780
ccttcactga gaccaacaac cactggcccc agggacccac tgggcctcca ggccctccag 840
ggcccatggg tcccccctggg cctcctggcc ccacaggtgt ccctgggagt cctggtcaca 900
taggaccccc aggccccact ggacccaaag gaatctctgg ccacccagga gagaagggcg 960
agagaggact gcgtggggag cctggccccc aaggctctgc tgggcagcgg ggggaacctg 1020
gccctaaggg agaccctggt gagaagagcc actgggggga ggggttgcac cagctacgcg 1080
aggctttgaa gattttagct gagagggttt taatcttgga aacaatgatt gggctctatg 1140
aaccagagct ggggtctggg gcgggccctg ccggcacagg cacccccagc ctccttcggg 1200
gcaagagggg cggacatgca accaactacc ggatcgtggc cccccaggagc cgggacgaga 1260
gaggctgagg gtggtggcgg cccctgaggc agaccaggcc aggcta 1306